# Solutions Manual to
# Pattern Recognition and Machine Learning

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# Contents

# 1 Introduction

## 1.1

Let

$$E(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}\left(y(x_n,\mathbf{w}) - t_n\right)^2. \tag{1.1}$$

Setting the derivative of $E(\mathbf{w})$ with respect to $\mathbf{w}$ to zero gives

$$\mathbf{0} = \sum_{n=1}^{N}\frac{\partial y(x_n,\mathbf{w})}{\partial \mathbf{w}}\left(y(x_n,\mathbf{w}) - t_n\right). \tag{1.2}$$

If

$$y(x_n,\mathbf{w}) = \mathbf{w}^\intercal \boldsymbol{\phi}(x_n), \tag{1.3}$$

then

$$\mathbf{0} = \sum_{n=1}^{N}\boldsymbol{\phi}(x_n)\left(\mathbf{w}^\intercal\boldsymbol{\phi}(x_n) - t_n\right). \tag{1.4}$$

Then,

$$\left(\sum_{n=1}^{N}\boldsymbol{\phi}(x_n)\boldsymbol{\phi}(x_n)^\intercal\right)\mathbf{w} = \sum_{n=1}^{N} t_n\boldsymbol{\phi}(x_n). \tag{1.5}$$

Therefore,

$$\operatorname*{argmin}_{\mathbf{w}} E(\mathbf{w}) = \mathbf{A}^{-1}\mathbf{v}, \tag{1.6}$$

where

$$\begin{aligned}\mathbf{A} &= \sum_{n=1}^{N}\boldsymbol{\phi}(x_n)\boldsymbol{\phi}(x_n)^\intercal,\\ \mathbf{v} &= \sum_{n=1}^{N} t_n\boldsymbol{\phi}(x_n).\end{aligned} \tag{1.7}$$

If

$$\boldsymbol{\phi}(x_n) = \begin{bmatrix} 1 \\ x_n \\ \vdots \\ x_n^M \end{bmatrix},$$

then

$$
\begin{aligned}
A_{mm'} &= \sum_{n=1}^{N} x_n^{m+m'}, \\
v_m &= \sum_{n=1}^{N} t_n x_n^m.
\end{aligned}
\tag{1.8}
$$

## 1.2

Let

$$
E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^{N} (y(x_n, \mathbf{w}) - t_n)^2 + \frac{\lambda}{2} \|\mathbf{w}\|^2. \tag{1.9}
$$

Setting the derivative of $E(\mathbf{w})$ with respect to $\mathbf{w}$ to zero gives

$$
\mathbf{0} = \sum_{n=1}^{N} \frac{\partial y(x_n, \mathbf{w})}{\partial \mathbf{w}} (y(x_n, \mathbf{w}) - t_n) + \lambda \mathbf{w}. \tag{1.10}
$$

If

$$
y(x_n, \mathbf{w}) = \mathbf{w}^\intercal \boldsymbol{\phi}(x_n), \tag{1.11}
$$

then

$$
\mathbf{0} = \sum_{n=1}^{N} \boldsymbol{\phi}(x_n) \left(\mathbf{w}^\intercal \boldsymbol{\phi}(x_n) - t_n\right) + \lambda \mathbf{w}. \tag{1.12}
$$

Then,

$$
\left( \sum_{n=1}^{N} \boldsymbol{\phi}(x_n) \boldsymbol{\phi}(x_n)^\intercal + \lambda \mathbf{I} \right) \mathbf{w} = \sum_{n=1}^{N} t_n \boldsymbol{\phi}(x_n). \tag{1.13}
$$

Therefore,

$$
\operatorname*{argmin}_{\mathbf{w}} E(\mathbf{w}) = \mathbf{A}^{-1} \mathbf{v}, \tag{1.14}
$$

where

$$
\begin{aligned}
\mathbf{A} &= \sum_{n=1}^{N} \boldsymbol{\phi}(x_n) \boldsymbol{\phi}(x_n)^\intercal + \lambda \mathbf{I}, \\
\mathbf{v} &= \sum_{n=1}^{N} t_n \boldsymbol{\phi}(x_n).
\end{aligned}
\tag{1.15}
$$

If

$$\boldsymbol{\phi}(x_n) = \begin{bmatrix} 1 \\ x_n \\ \vdots \\ x_n^M \end{bmatrix},$$

then

$$\begin{aligned} A_{mm'} &= \sum_{n=1}^{N} x_n^{m+m'} + \lambda I_{mm'}, \\ v_m &= \sum_{n=1}^{N} t_n x_n^m. \end{aligned} \tag{1.16}$$

## 1.3

Let

$$\begin{aligned} &p(a|r) = \frac{3}{10}, p(o|r) = \frac{2}{5}, p(l|r) = \frac{3}{10}, \\ &p(a|b) = \frac{1}{2}, p(o|b) = \frac{1}{2}, \\ &p(a|g) = \frac{3}{10}, p(o|g) = \frac{3}{10}, p(l|g) = \frac{2}{5}. \end{aligned} \tag{1.17}$$

Let

$$p(r) = \frac{1}{5}, p(b) = \frac{1}{5}, p(g) = \frac{3}{5}. \tag{1.18}$$

### (a)

By Bayes' theorem,

$$p(a) = p(a|r)p(r) + p(a|b)p(b) + p(a|g)p(g). \tag{1.19}$$

Therefore,

$$p(a) = \frac{17}{50}. \tag{1.20}$$

### (b)

By Bayes' theorem,

$$p(g|o) = \frac{p(g, o)}{p(o)}. \tag{1.21}$$

By Bayes' theorem, the right hand side can be written as

$$\frac{p(o|g)p(g)}{p(o|r)p(r) + p(o|b)p(b) + p(o|g)p(g)}. \tag{1.22}$$

Therefore,

$$p(g|o) = \frac{1}{2}. \tag{1.23}$$

## 1.4

Let $x$ and $y$ be variables such that

$$x = g(y). \tag{1.24}$$

Let $\hat{x}$ and $\hat{y}$ be the locations of the maximum of $p_x$ and $p_y$. Let us assume that

$$g'(y) \neq 0, \tag{1.25}$$

in a neighbourhood of $\hat{y}$.

### (a)

Since

$$\int p_x(x)dx = \int p_x\left(g(y)\right)|g'(y)|dy, \tag{1.26}$$

we have

$$p_y(y) = p_x\left(g(y)\right)|g'(y)|. \tag{1.27}$$

Taking the derivative and substituting

$$y = \hat{y} \tag{1.28}$$

gives

$$0 = g'(\hat{y})p'_x\left(g\left(\hat{y}\right)\right) + p_x\left(g\left(\hat{y}\right)\right)g''\left(\hat{y}\right). \tag{1.29}$$

Therefore, in general,

$$\hat{x} \neq g\left(\hat{y}\right). \tag{1.30}$$

### (b)

By (a), if

$$g(y) = ay + b, \tag{1.31}$$

then

$$0 = ap'_x\left(g\left(\hat{y}\right)\right). \tag{1.32}$$

Therefore,

$$\hat{x} = g\left(\hat{y}\right). \tag{1.33}$$

## 1.5

We have

$$\operatorname{var} f(x) = \mathrm{E}\left(f(x) - \mathrm{E}\, f(x)\right)^2. \tag{1.34}$$

The right hand side can be written as

$$\mathrm{E}\left(\left(f(x)\right)^2 - 2f(x)\,\mathrm{E}\, f(x) + \left(\mathrm{E}\, f(x)\right)^2\right) = \mathrm{E}\left(f(x)\right)^2 - \left(\mathrm{E}\, f(x)\right)^2. \tag{1.35}$$

Therefore,

$$\operatorname{var} f(x) = \mathrm{E}\left(f(x)\right)^2 - \left(\mathrm{E}\, f(x)\right)^2. \tag{1.36}$$

## 1.6

We have

$$\operatorname{cov}(x, y) = \mathrm{E}\left(\left(x - \mathrm{E}\, x\right)\left(y - \mathrm{E}\, y\right)\right). \tag{1.37}$$

The right hand side can be written as

$$\mathrm{E}\, xy - \mathrm{E}\left(x\,\mathrm{E}\, y\right) - \mathrm{E}\left(y\,\mathrm{E}\, x\right) + \mathrm{E}\left(\mathrm{E}\, x\,\mathrm{E}\, y\right) = \mathrm{E}\, xy - \mathrm{E}\, x\,\mathrm{E}\, y. \tag{1.38}$$

The right hand side can be written as

$$\int xyp(x, y)dxdy - \int xp(x)dx \int yp(y)dy. \tag{1.39}$$

If $x$ and $y$ are independent, then

$$p(x, y) = p(x)p(y). \tag{1.40}$$

Then,

$$\int xyp(x, y)dxdy = \int p(x)dx \int p(y)dy. \tag{1.41}$$

Therefore,

$$\operatorname{cov}(x, y) = 0. \tag{1.42}$$

## 1.7

### (a)

Let

$$I = \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2\sigma^2}x^2\right) dx. \tag{1.43}$$

Then,

$$I^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \exp\left(-\frac{1}{2\sigma^2}\left(x^2+y^2\right)\right) dxdy. \tag{1.44}$$

By the transformation

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} r\cos\theta \\ r\sin\theta \end{bmatrix}, \tag{1.45}$$

the right hand side can be written as

$$\begin{aligned} &\int_0^{\infty}\int_0^{2\pi} \exp\left(-\frac{1}{2\sigma^2}r^2\right) \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} drd\theta \\ =&2\pi\int_0^{\infty} r\exp\left(-\frac{1}{2\sigma^2}r^2\right) dr. \end{aligned} \tag{1.46}$$

By the transformation

$$s = \frac{r}{\sigma}. \tag{1.47}$$

the right hand side can be written as

$$2\pi\sigma^2\int_0^{\infty} s\exp\left(-\frac{1}{2}s^2\right) ds = 2\pi\sigma^2\left[-\exp\left(-\frac{1}{2}s^2\right)\right]_0^{\infty}. \tag{1.48}$$

Therefore,

$$I = \left(2\pi\sigma^2\right)^{\frac{1}{2}}. \tag{1.49}$$

### (b)

We have

$$\mathcal{N}\left(x|\mu,\sigma^2\right) = \left(2\pi\sigma^2\right)^{-\frac{1}{2}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right). \tag{1.50}$$

Then,

$$\int_{-\infty}^{\infty} \mathcal{N}\left(x|\mu,\sigma^2\right) dx = \left(2\pi\sigma^2\right)^{-\frac{1}{2}} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right) dx. \tag{1.51}$$

By the transformation $t = x - \mu$, the integral of the right hand side can be written as

$$\int_{-\infty}^{\infty} \exp\left(-\frac{1}{2\sigma^2}t^2\right) dt. \tag{1.52}$$

By 1.7(a), it can be written as $(2\pi\sigma^2)^{\frac{1}{2}}$. Therefore,

$$\int_{-\infty}^{\infty} \mathcal{N}\left(x|\mu,\sigma^2\right) dx = 1. \tag{1.53}$$

## 1.8

### (a)

Let $x$ be a variable such that

$$p(x) = \mathcal{N}(x|\mu,\sigma^2). \tag{1.54}$$

Then,

$$\mathrm{E}\, x = \int_{-\infty}^{\infty} x\mathcal{N}\left(x|\mu,\sigma^2\right) dx. \tag{1.55}$$

The right hand side can be written as

$$\left(2\pi\sigma^2\right)^{-\frac{1}{2}} \int_{-\infty}^{\infty} x \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right) dx. \tag{1.56}$$

By the transformation

$$y = x - \mu, \tag{1.57}$$

the integral can be written as

$$\begin{aligned} &\int_{-\infty}^{\infty} (y+\mu) \exp\left(-\frac{1}{2\sigma^2}y^2\right) dy \\ &= \int_{-\infty}^{\infty} y \exp\left(-\frac{1}{2\sigma^2}y^2\right) dy + \mu \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2\sigma^2}y^2\right) dy. \end{aligned} \tag{1.58}$$

By 1.7(a), the right hand side can be written as

$$\mu\left(2\pi\sigma^2\right)^{\frac{1}{2}}. \tag{1.59}$$

Therefore,

$$\mathrm{E}\, x = \mu. \tag{1.60}$$

### (b)

By 1.7(b),

$$\int_{-\infty}^{\infty} \mathcal{N}\left(x|\mu, \sigma^2\right) dx = 1, \tag{1.61}$$

so that

$$\left(2\pi\sigma^2\right)^{-\frac{1}{2}} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right) dx = 1. \tag{1.62}$$

Taking the derivative with respect to $\sigma^2$ gives

$$\begin{aligned}
&(2\pi)^{-\frac{1}{2}} \left(-\frac{1}{2}\right) \left(\sigma^2\right)^{-\frac{3}{2}} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right) dx \\
&+ \left(2\pi\sigma^2\right)^{-\frac{1}{2}} \int_{-\infty}^{\infty} \frac{1}{2} \left(\sigma^2\right)^{-2} (x-\mu)^2 \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right) dx = 0.
\end{aligned} \tag{1.63}$$

The left hand side can be written as

$$\begin{aligned}
&-\frac{1}{2}\left(\sigma^2\right)^{-1} \int_{-\infty}^{\infty} \mathcal{N}\left(x|\mu, \sigma^2\right) dx \\
&+\frac{1}{2}\left(\sigma^2\right)^{-2} \int_{-\infty}^{\infty} (x-\mu)^2 \mathcal{N}\left(x|\mu, \sigma^2\right) dx \\
=&-\frac{1}{2}\left(\sigma^2\right)^{-1} + \frac{1}{2}\left(\sigma^2\right)^{-2} \operatorname{var} x.
\end{aligned} \tag{1.64}$$

Therefore,

$$\operatorname{var} x = \sigma^2. \tag{1.65}$$

## 1.9

### (a)

Let $x$ be a variable such that

$$p(x) = \mathcal{N}\left(x|\mu, \sigma^2\right). \tag{1.66}$$

Setting the derivative of the right hand side with respect to $x$ to zero gives

$$0 = \left(2\pi\sigma^2\right)^{-\frac{1}{2}} \left(-\frac{1}{\sigma^2}(x-\mu)\right) \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right). \tag{1.67}$$

Therefore,

$$\operatorname{mode} x = \mu. \tag{1.68}$$

### (b)

Let $\mathbf{x}$ be a variable in $D$ dimensions such that

$$p(\mathbf{x}) = \mathcal{N}\left(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}\right). \tag{1.69}$$

Setting the derivative of the right hand side with respect to $\mathbf{x}$ to zero gives

$$\begin{aligned}\mathbf{0} = &-\left(2\pi\right)^{-\frac{D}{2}}\left(\det\boldsymbol{\Sigma}\right)^{-\frac{1}{2}}\left(\boldsymbol{\Sigma}^{-1}+\left(\boldsymbol{\Sigma}^{-1}\right)^{\intercal}\right)\left(\mathbf{x}-\boldsymbol{\mu}\right)\\ &\exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\intercal}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right).\end{aligned} \tag{1.70}$$

Therefore,

$$\operatorname{mode}\mathbf{x} = \boldsymbol{\mu}. \tag{1.71}$$

## 1.10

### (a)

We have

$$\mathrm{E}(x+y) = \iint (x+y)p(x,y)dxdy. \tag{1.72}$$

The right hand side can be written as

$$\begin{aligned}&\int x\left(\int p(x,y)dy\right)dx + \int y\left(\int p(x,y)dx\right)dy\\ =&\int xp(x)dx + \int yp(y)dy.\end{aligned} \tag{1.73}$$

The right hand side can be written as

$$\mathrm{E}\,x + \mathrm{E}\,y. \tag{1.74}$$

Therefore,

$$\mathrm{E}(x+y) = \mathrm{E}\,x + \mathrm{E}\,y. \tag{1.75}$$

### (b)

We have

$$\operatorname{var}(x+y) = \mathrm{E}\left(x+y-\mathrm{E}(x+y)\right)^2 \tag{1.76}$$

The right hand side can be written as

$$\begin{aligned}&\mathrm{E}\left(x-\mathrm{E}\,x\right)^2+2\,\mathrm{E}\left(\left(x-\mathrm{E}\,x\right)\left(y-\mathrm{E}\,y\right)\right)+\mathrm{E}\left(y-\mathrm{E}\,y\right)^2\\ &=\operatorname{var}x+2\operatorname{cov}(x,y)+\operatorname{var}y.\end{aligned}\tag{1.77}$$

By 1.6, if $x$ and $y$ are independent, then

$$\operatorname{cov}(x,y)=0.\tag{1.78}$$

Therefore,

$$\operatorname{var}(x+y)=\operatorname{var}x+\operatorname{var}y.\tag{1.79}$$

## 1.11

Let $x_1,\cdots,x_N$ be independent variables such that

$$p(x_n)=\mathcal{N}\left(x_n|\mu,\sigma^2\right).\tag{1.80}$$

Then,

$$\ln p(\mathbf{x})=-\frac{N}{2}\ln\left(2\pi\sigma^2\right)-\frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2.\tag{1.81}$$

Setting the derivatives of $\ln p(\mathbf{x})$ with respect to $\mu$ and $\sigma^2$ to zero gives

$$\begin{aligned}0&=\frac{1}{\sigma^2}\sum_{n=1}^{N}(x_n-\mu),\\ 0&=-\frac{N}{2\sigma^2}+\frac{1}{2\left(\sigma^2\right)^2}\sum_{n=1}^{N}(x_n-\mu)^2.\end{aligned}\tag{1.82}$$

Therefore, the maximum likelihood solutions to $\mu$ and $\sigma^2$ are given by

$$\begin{aligned}\mu_{\mathrm{ML}}&=\frac{1}{N}\sum_{n=1}^{N}x_n,\\ \sigma^2_{\mathrm{ML}}&=\frac{1}{N}\sum_{n=1}^{N}(x_n-\mu_{\mathrm{ML}})^2.\end{aligned}\tag{1.83}$$

## 1.12

### (a)

Let $x_n$ and $x_{n'}$ be independent variables such that

$$p(x_n) = \mathcal{N}\left(x_n|\mu, \sigma^2\right), \quad p(x_{n'}) = \mathcal{N}\left(x_{n'}|\mu, \sigma^2\right). \tag{1.84}$$

Then,

$$\mathrm{E}\, x_n x_{n'} = \mu^2. \tag{1.85}$$

By the property

$$\mathrm{E}\, x_n^2 = \mathrm{var}\, x_n + (\mathrm{E}\, x_n)^2, \tag{1.86}$$

we have

$$\mathrm{E}\, x_n^2 = \sigma^2 + \mu^2. \tag{1.87}$$

Therefore,

$$\mathrm{E}\, x_n x_{n'} = \mu^2 + I_{nn'}\sigma^2. \tag{1.88}$$

### (b)

Let $x_1, \cdots, x_N$ be independent variables such that

$$p(x_n) = \mathcal{N}\left(x_n|\mu, \sigma^2\right). \tag{1.89}$$

By 1.11, the maximum likelihood solution to $\mu$ is given by

$$\mu_{\mathrm{ML}} = \frac{1}{N}\sum_{n=1}^{N} x_n. \tag{1.90}$$

Then,

$$\mathrm{E}\, \mu_{\mathrm{ML}} = \frac{1}{N}\sum_{n=1}^{N} \mathrm{E}\, x_n. \tag{1.91}$$

Therefore,

$$\mathrm{E}\, \mu_{\mathrm{ML}} = \mu. \tag{1.92}$$

**(c)**

Let $x_1, \cdots, x_N$ be independent variables such that

$$p(x_n) = \mathcal{N}\left(x_n|\mu, \sigma^2\right). \tag{1.93}$$

By 1.11, the maximum likelihood solution to $\sigma^2$ is given by

$$\sigma^2_{\mathrm{ML}} = \frac{1}{N}\sum_{n=1}^{N}\left(x_n - \mu_{\mathrm{ML}}\right)^2. \tag{1.94}$$

Then,

$$\mathrm{E}\,\sigma^2_{\mathrm{ML}} = \frac{1}{N}\sum_{n=1}^{N}\mathrm{E}\left(x_n - \mu_{\mathrm{ML}}\right)^2. \tag{1.95}$$

The right hand side can be written as

$$\begin{aligned}&\frac{1}{N}\sum_{n=1}^{N}\mathrm{E}\left(x_n^2 - 2\mu_{\mathrm{ML}}x_n + \mu_{\mathrm{ML}}^2\right)\\ =&\frac{1}{N}\sum_{n=1}^{N}\mathrm{E}\,x_n^2 - \frac{2}{N}\mathrm{E}\left(\mu_{\mathrm{ML}}\left(\sum_{n=1}^{N}x_n\right)\right) + \mathrm{E}\,\mu_{\mathrm{ML}}^2.\end{aligned} \tag{1.96}$$

The first term of the right hand side can be written as

$$\frac{1}{N}\sum_{n=1}^{N}\left(\mu^2 + \sigma^2\right) = \mu^2 + \sigma^2, \tag{1.97}$$

while, by 1.11, the second and third terms can be written as

$$-\frac{2}{N}\mathrm{E}\left(N\left(\frac{1}{N}\sum_{n=1}^{N}x_n\right)^2\right) + \mathrm{E}\left(\frac{1}{N}\sum_{n=1}^{N}x_n\right)^2 = -\mathrm{E}\left(\frac{1}{N}\sum_{n=1}^{N}x_n\right)^2. \tag{1.98}$$

By (a), the right hand side can be written as

$$\begin{aligned}&-\frac{1}{N^2}\sum_{n=1}^{N}\mathrm{E}\,x_n^2 - \frac{2}{N^2}\sum_{1\le n<n'\le N}\mathrm{E}\,x_n x_{n'}\\ =&-\frac{1}{N^2}N\left(\mu^2 + \sigma^2\right) - \frac{2}{N^2}\frac{N(N-1)}{2}\mu^2.\end{aligned} \tag{1.99}$$

The right hand side can be written as

$$-\frac{1}{N}\left(\mu^2+\sigma^2\right)-\frac{N-1}{N}\mu^2=-\mu^2-\frac{1}{N}\sigma^2. \tag{1.100}$$

Then,

$$\mathrm{E}\,\sigma_{\mathrm{ML}}^2=\mu^2+\sigma^2-\mu^2-\frac{1}{N}\sigma^2. \tag{1.101}$$

Therefore,

$$\mathrm{E}\,\sigma_{\mathrm{ML}}^2=\frac{N-1}{N}\sigma^2. \tag{1.102}$$

## 1.13

Let $x_1,\cdots,x_N$ be variables such that

$$\begin{aligned}\mathrm{E}\,x_n&=\mu,\\ \operatorname{var}x_n&=\sigma^2.\end{aligned} \tag{1.103}$$

We have

$$\mathrm{E}\left(\frac{1}{N}\sum_{n=1}^{N}\left(x_n-\mu\right)^2\right)=\frac{1}{N^2}\sum_{n=1}^{N}\mathrm{E}\left(x_n-\mu\right)^2. \tag{1.104}$$

The right hand side can be written as

$$\frac{1}{N^2}\sum_{n=1}^{N}\operatorname{var}x_n=\frac{\sigma^2}{N}. \tag{1.105}$$

Therefore,

$$\mathrm{E}\left(\frac{1}{N}\sum_{n=1}^{N}\left(x_n-\mu\right)^2\right)=\frac{\sigma^2}{N}. \tag{1.106}$$

## 1.14

Let

$$\begin{aligned}w_{dd'}^{\mathrm{S}}&=\frac{1}{2}(w_{dd'}+w_{d'd}),\\ w_{dd'}^{\mathrm{A}}&=\frac{1}{2}(w_{dd'}-w_{d'd}).\end{aligned} \tag{1.107}$$

**(a)**

We have

$$
\begin{aligned}
w_{dd'} &= w^{\mathrm{S}}_{dd'} + w^{\mathrm{A}}_{dd'}, \\
w^{\mathrm{S}}_{dd'} &= w^{\mathrm{S}}_{d'd}, \\
w^{\mathrm{A}}_{dd'} &= -w^{\mathrm{A}}_{d'd}.
\end{aligned}
\tag{1.108}
$$

**(b)**

We have

$$
\sum_{d=1}^{D}\sum_{d'=1}^{D} w^{\mathrm{A}}_{dd'} x_d x_{d'} = \frac{1}{2}\sum_{d=1}^{D}\sum_{d'=1}^{D} (w_{dd'} - w_{d'd}) x_d x_{d'}. \tag{1.109}
$$

The right hand side can be written as

$$
\frac{1}{2}\left(\sum_{d=1}^{D}\sum_{d'=1}^{D} w_{dd'} x_d x_{d'} - \sum_{d=1}^{D}\sum_{d'=1}^{D} w_{d'd} x_d x_{d'}\right) = 0. \tag{1.110}
$$

Therefore,

$$
\sum_{d=1}^{D}\sum_{d'=1}^{D} w^{\mathrm{A}}_{dd'} x_d x_{d'} = 0. \tag{1.111}
$$

**(c)**

We have

$$
\sum_{d=1}^{D}\sum_{d'=1}^{D} w_{dd'} x_d x_{d'} = \sum_{d=1}^{D}\sum_{d'=1}^{D} \left(w^{\mathrm{S}}_{dd'} + w^{\mathrm{A}}_{dd'}\right) x_d x_{d'}. \tag{1.112}
$$

By (b), the right hand side can be written as

$$
\sum_{d=1}^{D}\sum_{d'=1}^{D} w^{\mathrm{S}}_{dd'} x_d x_{d'} + \sum_{d=1}^{D}\sum_{d'=1}^{D} w^{\mathrm{A}}_{dd'} x_d x_{d'} = \sum_{d=1}^{D}\sum_{d'=1}^{D} w^{\mathrm{S}}_{dd'} x_d x_{d'}, \tag{1.113}
$$

Therefore,

$$
\sum_{d=1}^{D}\sum_{d'=1}^{D} w_{dd'} x_d x_{d'} = \sum_{d=1}^{D}\sum_{d'=1}^{D} w^{\mathrm{S}}_{dd'} x_d x_{d'}. \tag{1.114}
$$

### (d)

Since $\mathbf{W}^{\mathrm{S}}$ is a $D \times D$ symmetric matrix, its number of independent parameters is $\frac{D(D+1)}{2}$.

## 1.15

### (a)

Let $n(D, M)$ be the number of independent parameters of a polynomial in $D$ dimensions and $M$ orders. Then

$$n(1, M) = n(1, M-1) = 1. \tag{1.115}$$

Let us assume that

$$n(D, M) = \sum_{d=1}^{D} n(d, M-1). \tag{1.116}$$

The independent terms of a polynomial in $D+1$ dimensions and $M$ orders can be split into 1. the ones of a polynomial in $D$ dimensions and $M$ orders and 2. the ones generated by multiplying the ones in $D+1$ dimensions and $M$ orders by the $D+1$th variable. Then,

$$n(D+1, M) = n(D, M) + n(D+1, M-1), \tag{1.117}$$

so that

$$n(D+1, M) = \sum_{d=1}^{D+1} n(d, M-1). \tag{1.118}$$

Therefore, the assumption is proved by induction on $D$.

### (b)

We have

$$\sum_{d=1}^{1} \frac{(d+M-2)!}{(d-1)!(M-1)!} = 1. \tag{1.119}$$

Let us assume that

$$\sum_{d=1}^{D} \frac{(d+M-2)!}{(d-1)!(M-1)!} = \frac{(D+M-1)!}{(D-1)!M!}. \tag{1.120}$$

Then,

$$\sum_{d=1}^{D+1} \frac{(d+M-2)!}{(d-1)!(M-1)!} = \frac{(D+M-1)!}{(D-1)!M!} + \frac{(D+M-1)!}{D!(M-1)!}. \tag{1.121}$$

The right hand side can be written as

$$\frac{D(D+M-1)! + M(D+M-1)!}{D!M!} = \frac{(D+M)!}{D!M!}. \tag{1.122}$$

Therefore, the assumption is proved by induction on $D$.

### (c)

By 1.14(d),

$$n(D, 2) = \frac{D(D+1)}{2}. \tag{1.123}$$

Let us assume that

$$n(D, M) = \frac{(D+M-1)!}{(D-1)!M!}. \tag{1.124}$$

By (a),

$$n(D, M+1) = \sum_{d=1}^{D} n(d, M). \tag{1.125}$$

By the assumption and (b), the right hand side can be written as

$$\sum_{d=1}^{D} \frac{(d+M-1)!}{(d-1)!M!} = \frac{(D+M)!}{(D-1)!(M+1)!}. \tag{1.126}$$

Therefore, the assumption is proved by induction on $M$.

## 1.16

### (a)

Let $N(D, M)$ be the number of independent parameters in all of the terms up to and including the ones of $D$ dimensions and $M$ orders. By 1.15,

$$N(D, M) = \sum_{m=0}^{M} n(D, m), \tag{1.127}$$

where

$$n(D, m) = \frac{(D+m-1)!}{(D-1)!m!}. \tag{1.128}$$

**(b)**

By (a),

$$N(D,0) = 1. \tag{1.129}$$

Let us assume that

$$\sum_{m=0}^{M} n(D,m) = \frac{(D+M)!}{D!M!}. \tag{1.130}$$

Then,

$$\sum_{m=0}^{M+1} n(D,m) = \frac{(D+M)!}{D!M!} + \frac{(D+M)!}{(D-1)!(M+1)!}. \tag{1.131}$$

The right hand side can be written as

$$\frac{(M+1)(D+M)! + D(D+M)!}{D!(M+1)!} = \frac{(D+M+1)!}{D!(M+1)!}. \tag{1.132}$$

Then, the assumption is proved by induction on $M$. Therefore,

$$N(D,M) = \frac{(D+M)!}{D!M!}. \tag{1.133}$$

**(c)**

By the approximation

$$n! \simeq n^n \exp(-n), \tag{1.134}$$

we have

$$\frac{(D+M)!}{D!M!} \simeq \frac{(D+M)^{D+M}}{D^D M^M}. \tag{1.135}$$

The right hand side can be written as

$$D^M \left(1 + \frac{M}{D}\right)^D \left(\frac{1}{M} + \frac{1}{D}\right)^M = M^D \left(1 + \frac{D}{M}\right)^M \left(\frac{1}{D} + \frac{1}{M}\right)^D. \tag{1.136}$$

Therefore,

$$N(D,M) \simeq \begin{cases} D^M, & D \gg M, \\ M^D, & M \gg D. \end{cases} \tag{1.137}$$

### (d)

By (b),

$$\begin{aligned} N(10,3) &= 286, \\ N(100,3) &= 176851, \\ N(1000,3) &= 167668501. \end{aligned} \tag{1.138}$$

## 1.17

Let

$$\Gamma(x) = \int_0^\infty u^{x-1}\exp(-u)du. \tag{1.139}$$

### (a)

We have

$$\Gamma(x+1) = \int_0^\infty u^x \exp(-u)du. \tag{1.140}$$

The right hand side can be written as

$$[-u^x \exp(-u)]_{u=0}^{u=\infty} + \int_0^\infty x u^{x-1}\exp(-u)du = x\Gamma(x). \tag{1.141}$$

Therefore,

$$\Gamma(x+1) = x\Gamma(x). \tag{1.142}$$

### (b)

We have

$$\Gamma(1) = \int_0^\infty \exp(-u)du, \tag{1.143}$$

so that

$$\Gamma(1) = 0!. \tag{1.144}$$

For a positive integer $x$, let us assume that

$$\Gamma(x) = (x-1)!. \tag{1.145}$$

By (a),

$$\Gamma(x+1) = x\Gamma(x), \tag{1.146}$$

so that

$$\Gamma(x+1) = x!. \tag{1.147}$$

Therefore, the assumption is proved by induction on $x$.

## 1.18

### (a)

Let

$$\prod_{d=1}^{D} \int_{-\infty}^{\infty} \exp\left(-x_d^2\right) dx_d = S_D \int_0^{\infty} \exp\left(-r^2\right) r^{D-1} dr, \tag{1.148}$$

where $S_D$ is the surface area of a sphere of unit radius in $D$ dimensions. By 1.7, the left hand side can be written as $\pi^{\frac{D}{2}}$. By the transformation

$$s = r^2, \tag{1.149}$$

the right hand side can be written as

$$\frac{S_D}{2} \int_0^{\infty} \exp(-s) s^{\frac{D-1}{2}} s^{-\frac{1}{2}} ds = \frac{S_D}{2} \Gamma\left(\frac{D}{2}\right). \tag{1.150}$$

Therefore,

$$S_D = \frac{2\pi^{\frac{D}{2}}}{\Gamma\left(\frac{D}{2}\right)}. \tag{1.151}$$

### (b)

The volume of the sphere can be written as

$$V_D = S_D \int_0^1 r^{D-1} dr. \tag{1.152}$$

Therefore,

$$V_D = \frac{S_D}{D}. \tag{1.153}$$

### (c)

By (a) and (b),

$$\begin{aligned} S_2 &= 2\pi, \\ V_2 &= \pi. \end{aligned} \tag{1.154}$$

Similarly,

$$\begin{aligned} S_3 &= 4\pi, \\ V_3 &= \frac{4}{3}\pi. \end{aligned} \tag{1.155}$$

## 1.19

### (a)

The volume of a cube of side 2 in $D$ dimensions is $2^D$. By 1.18, the ratio of the volume of the cocentric sphere of radius 1 divided by the volume of the cube is given by

$$\frac{V_D}{2^D} = \frac{\pi^{\frac{D}{2}}}{D 2^{D-1}\Gamma\left(\frac{D}{2}\right)}. \tag{1.156}$$

### (b)

By (a) and the Stering's formula

$$\Gamma(x+1) \simeq (2\pi)^{\frac{1}{2}} \exp(-x) x^{\frac{x+1}{2}}, \tag{1.157}$$

we have

$$\frac{V_D}{2^D} \simeq \frac{\pi^{\frac{D}{2}}}{D 2^{D-1}(2\pi)^{\frac{1}{2}} \exp\left(1-\frac{D}{2}\right)\left(\frac{D}{2}-1\right)^{\frac{D}{4}}}. \tag{1.158}$$

The right hand side can be written as

$$\frac{1}{2e(2\pi)^{\frac{1}{2}}} \frac{1}{D} \left(\frac{e^2\pi^2}{8D-16}\right)^{\frac{D}{4}}. \tag{1.159}$$

Therefore,

$$\lim_{D\to\infty} \frac{V_D}{2^D} = 0. \tag{1.160}$$

### (c)

The ratio of the distance from the center of the cube to one of the corners divided by the perpendicular distance to one of the sides is given by

$$\frac{\sqrt{\sum_{i=1}^{D} 1^2}}{1} = \sqrt{D}. \tag{1.161}$$

Therefore, the ratio goes to $\infty$ as $D \to \infty$.

## 1.20

Let $\mathbf{x}$ be a variable in $D$ dimensions such that

$$p(\mathbf{x}) = (2\pi\sigma^2)^{-\frac{D}{2}} \exp\left(-\frac{\|\mathbf{x}\|^2}{2\sigma^2}\right). \tag{1.162}$$

### (a)

We have

$$\int_{r\le\|\mathbf{x}\|\le r+\epsilon} p(\mathbf{x})d\mathbf{x} = \int_r^{r+\epsilon}\int (2\pi\sigma^2)^{-\frac{D}{2}} \exp\left(-\frac{r'^2}{2\sigma^2}\right) J dr' d\boldsymbol{\phi}, \tag{1.163}$$

where $\boldsymbol{\phi}$ is the vector of the angular components of the polar corrdinate and $J$ is the Jacobian of the transformation from the Cartesian to polar coordinate. For a sufficiently small $\epsilon$, the right hand side can be approximated as

$$\begin{aligned}&(2\pi\sigma^2)^{-\frac{D}{2}} \exp\left(-\frac{r^2}{2\sigma^2}\right)\int_r^{r+\epsilon}\int J dr' d\boldsymbol{\phi}\\ =&(2\pi\sigma^2)^{-\frac{D}{2}} \exp\left(-\frac{r^2}{2\sigma^2}\right)\int_{r\le\|\mathbf{x}\|\le r+\epsilon} d\mathbf{x}.\end{aligned} \tag{1.164}$$

Therefore,

$$\int_{r\le\|\mathbf{x}\|\le r+\epsilon} p(\mathbf{x})d\mathbf{x} \simeq p(r)\epsilon, \tag{1.165}$$

where

$$p(r) = (2\pi\sigma^2)^{-\frac{D}{2}} S_D r^{D-1} \exp\left(-\frac{r^2}{2\sigma^2}\right), \tag{1.166}$$

and $S_D$ is the surface area of a unit sphere in $D$ dimensions.

**(b)**

Setting the derivative of $p(r)$ to zero gives

$$0 = (2\pi\sigma^2)^{-\frac{D}{2}} S_D \left( (D-1)r^{D-2} - \frac{r^D}{\sigma^2} \right) \exp\left( -\frac{r^2}{2\sigma^2} \right). \tag{1.167}$$

Therefore, $p(r)$ is maximised at a sigle stationary point

$$\hat{r} = \sqrt{D-1}\sigma. \tag{1.168}$$

**(c)**

We have

$$\frac{p(\hat{r}+\epsilon)}{p(\hat{r})} = \left( \frac{\hat{r}+\epsilon}{\hat{r}} \right)^{D-1} \exp\left( -\frac{2\hat{r}\epsilon + \epsilon^2}{2\sigma^2} \right). \tag{1.169}$$

By (b), the right hand side can be written as

$$\begin{aligned} &\exp\left( (D-1)\ln\left(1+\frac{\epsilon}{\hat{r}}\right) - \frac{2\hat{r}\epsilon+\epsilon^2}{2\sigma^2} \right) \\ =&\exp\left( \frac{\hat{r}^2}{\sigma^2}\ln\left(1+\frac{\epsilon}{\hat{r}}\right) - \frac{2\hat{r}\epsilon+\epsilon^2}{2\sigma^2} \right). \end{aligned} \tag{1.170}$$

By the Taylor series

$$\ln(1+x) = x - \frac{1}{2}x^2 + o\left(x^3\right), \tag{1.171}$$

the right hand side can be approximated as

$$\exp\left( \frac{\hat{r}^2}{\sigma^2}\left( \frac{\epsilon}{\hat{r}} - \frac{\epsilon^2}{2\hat{r}^2} \right) - \frac{2\hat{r}\epsilon+\epsilon^2}{2\sigma^2} \right) = \exp\left( -\frac{\epsilon^2}{\sigma^2} \right). \tag{1.172}$$

Therefore,

$$p(\hat{r}+\epsilon) \simeq p(\hat{r}) \exp\left( -\frac{\epsilon^2}{\sigma^2} \right). \tag{1.173}$$

### (d)

Let $\hat{\mathbf{r}}$ be a vector of length $\hat{r}$. We have

$$\frac{p(\mathbf{0})}{p\left(\hat{\mathbf{r}}\right)} = \exp\left(\frac{\hat{r}^2}{2\sigma^2}\right). \tag{1.174}$$

By (b), the right hand side can be written as

$$\exp\left(\frac{D-1}{2}\right). \tag{1.175}$$

Therefore,

$$\frac{p(\mathbf{0})}{p\left(\hat{\mathbf{r}}\right)} = \exp\left(\frac{D-1}{2}\right). \tag{1.176}$$

## 1.21

### (a)

If $0 \leq a \leq b$, then

$$0 \leq a(b-a). \tag{1.177}$$

Therefore,

$$a \leq (ab)^{\frac{1}{2}}. \tag{1.178}$$

### (b)

For a two-class classification problem of $\mathbf{x}$, let the classes be $\mathcal{C}_1$ and $\mathcal{C}_2$ and let the decision regions be $\mathcal{R}_1$ and $\mathcal{R}_2$. Let us choose the decision regions to minimise the probability of misclassification. Then,

$$\begin{aligned} p(\mathbf{x}, \mathcal{C}_1) > p(\mathbf{x}, \mathcal{C}_2) \Rightarrow \mathbf{x} \in \mathcal{C}_1, \\ p(\mathbf{x}, \mathcal{C}_2) > p(\mathbf{x}, \mathcal{C}_1) \Rightarrow \mathbf{x} \in \mathcal{C}_2. \end{aligned} \tag{1.179}$$

By (a),

$$\begin{aligned} \int_{\mathcal{R}_1} p(\mathbf{x}, \mathcal{C}_2) d\mathbf{x} \leq \int_{\mathcal{R}_1} \left(p(\mathbf{x}, \mathcal{C}_1) p(\mathbf{x}, \mathcal{C}_2)\right)^{\frac{1}{2}} d\mathbf{x}, \\ \int_{\mathcal{R}_2} p(\mathbf{x}, \mathcal{C}_1) d\mathbf{x} \leq \int_{\mathcal{R}_2} \left(p(\mathbf{x}, \mathcal{C}_1) p(\mathbf{x}, \mathcal{C}_2)\right)^{\frac{1}{2}} d\mathbf{x}. \end{aligned} \tag{1.180}$$

Therefore,

$$\int_{\mathcal{R}_1} p(\mathbf{x}, \mathcal{C}_2) d\mathbf{x} + \int_{\mathcal{R}_2} p(\mathbf{x}, \mathcal{C}_1) d\mathbf{x} \leq \int \left(p(\mathbf{x}, \mathcal{C}_1) p(\mathbf{x}, \mathcal{C}_2)\right)^{\frac{1}{2}} d\mathbf{x}. \tag{1.181}$$

## 1.22

Let

$$\mathrm{E}\, L = \sum_k \sum_j \int_{\mathcal{R}_j} L_{kj} p(\mathbf{x}, \mathcal{C}_k) d\mathbf{x}, \tag{1.182}$$

where

$$L_{kj} = 1 - I_{kj}. \tag{1.183}$$

The right hand side can be written as

$$\begin{aligned} &\sum_k \sum_j \int_{\mathcal{R}_j} \left(p(\mathbf{x}, \mathcal{C}_k) - p(\mathbf{x}, \mathcal{C}_j)\right) d\mathbf{x} \\ =&\sum_j \int_{\mathcal{R}_j} \left( \sum_k p(\mathbf{x}, \mathcal{C}_k) - p(\mathbf{x}, \mathcal{C}_j) \right) d\mathbf{x}. \end{aligned} \tag{1.184}$$

The right hand side can be written as

$$\sum_j \int_{\mathcal{R}_j} \left(p(\mathbf{x}) - p(\mathbf{x}, \mathcal{C}_j)\right) d\mathbf{x} = 1 - \sum_j \int_{\mathcal{R}_j} p(\mathbf{x}, \mathcal{C}_j) d\mathbf{x}. \tag{1.185}$$

Then,

$$\mathrm{E}\, L = 1 - \sum_j \int_{\mathcal{R}_j} p(\mathcal{C}_j|\mathbf{x}) p(\mathbf{x}) d\mathbf{x}. \tag{1.186}$$

Therefore, minimising $\mathrm{E}\, L$ reduces to choosing the criterion to maximise the posterior probatility $p(\mathcal{C}_j|\mathbf{x})$.

## 1.23

Let

$$\mathrm{E}\, L = \sum_k \sum_j \int_{\mathcal{R}_j} L_{kj} p(\mathbf{x}, \mathcal{C}_k) d\mathbf{x}. \tag{1.187}$$

The right hand side can be written as

$$\sum_j \int_{\mathcal{R}_j} \sum_k L_{kj} p(\mathbf{x}, \mathcal{C}_k) d\mathbf{x} = \sum_j \int_{\mathcal{R}_j} \left( \sum_k L_{kj} p(\mathcal{C}_k|\mathbf{x}) \right) p(\mathbf{x}) d\mathbf{x}. \tag{1.188}$$

Then,

$$\mathrm{E}\, L = \sum_j \int_{\mathcal{R}_j} \left( \sum_k L_{kj} p(\mathcal{C}_k|\mathbf{x}) \right) p(\mathbf{x}) d\mathbf{x}. \tag{1.189}$$

Therefore, minimising $\mathrm{E}\, L$ reduces to minimising $\sum_k L_{kj} p(\mathcal{C}_k|\mathbf{x})$.

## 1.24 (Incomplete)

Let

$$\mathrm{E}\,L = \sum_k \sum_j \int_{\mathcal{R}_j} L_{kj} p(\mathbf{x}, \mathcal{C}_k) d\mathbf{x} + \lambda \int_{\forall k p(\mathcal{C}_k|\mathbf{x})<\theta} p(\mathbf{x}) d\mathbf{x}. \tag{1.190}$$

## 1.25

Let

$$\mathrm{E}\,L\left(\mathbf{t}, \mathbf{y}(\mathbf{x})\right) = \iint \|\mathbf{y}(\mathbf{x}) - \mathbf{t}\|^2 p(\mathbf{x}, \mathbf{t}) d\mathbf{x} d\mathbf{t}. \tag{1.191}$$

Setting the derivative of $\mathrm{E}\,L(\mathbf{t}, \mathbf{y}(\mathbf{x}))$ with respect to $\mathbf{y}(\mathbf{x})$ to zero gives

$$\mathbf{0} = 2 \int \left(\mathbf{y}(\mathbf{x}) - \mathbf{t}\right) p(\mathbf{x}, \mathbf{t}) d\mathbf{t}. \tag{1.192}$$

The integral of the right hand side can be written as

$$\mathbf{y}(\mathbf{x}) \int p(\mathbf{x}, \mathbf{t}) d\mathbf{t} - \int \mathbf{t} p(\mathbf{x}, \mathbf{t}) d\mathbf{t} = \mathbf{y}(\mathbf{x}) p(\mathbf{x}) - p(\mathbf{x}) \int \mathbf{t} p(\mathbf{t}|\mathbf{x}) d\mathbf{t}. \tag{1.193}$$

The integral in the second term of the right hand side can be written as $\mathrm{E}_{\mathbf{t}}(\mathbf{t}|\mathbf{x})$. Then, the right hand side can be written as

$$\mathbf{0} = p(\mathbf{x}) \left(\mathbf{y}(\mathbf{x}) - \mathrm{E}_{\mathbf{t}}(\mathbf{t}|\mathbf{x})\right). \tag{1.194}$$

Therefore,

$$\operatorname*{argmin}_{\mathbf{y}(\mathbf{x})} \mathrm{E}\,L(\mathbf{t}, \mathbf{y}(\mathbf{x})) = \mathrm{E}_{\mathbf{t}}(\mathbf{t}|\mathbf{x}). \tag{1.195}$$

For a single target variable $t$, it reduces to

$$\operatorname*{argmin}_{\mathbf{y}(\mathbf{x})} \mathrm{E}\,L(t, \mathbf{y}(\mathbf{x})) = \mathrm{E}_t(t|\mathbf{x}). \tag{1.196}$$

## 1.26

Let

$$\mathrm{E}\,L(\mathbf{t}, \mathbf{y}(\mathbf{x})) = \iint \|\mathbf{y}(\mathbf{x}) - \mathbf{t}\|^2 p(\mathbf{x}, \mathbf{t}) d\mathbf{x} d\mathbf{t}. \tag{1.197}$$

The right hand side can be written as

$$\iint \|\mathbf{y}(\mathbf{x}) - \mathrm{E}(\mathbf{t}|\mathbf{x}) + \mathrm{E}(\mathbf{t}|\mathbf{x}) - \mathbf{t}\|^2 p(\mathbf{x},\mathbf{t}) d\mathbf{x} d\mathbf{t} = I_1 + 2I_2 + I_3, \tag{1.198}$$

where

$$\begin{aligned} I_1 &= \iint \|\mathbf{y}(\mathbf{x}) - \mathrm{E}(\mathbf{t}|\mathbf{x})\|^2 p(\mathbf{x},\mathbf{t}) d\mathbf{x} d\mathbf{t}, \\ I_2 &= \iint (\mathbf{y}(\mathbf{x}) - \mathrm{E}(\mathbf{t}|\mathbf{x}))^\intercal (\mathrm{E}(\mathbf{t}|\mathbf{x}) - \mathbf{t})\, p(\mathbf{x},\mathbf{t}) d\mathbf{x} d\mathbf{t}, \\ I_3 &= \iint \|\mathrm{E}(\mathbf{t}|\mathbf{x}) - \mathbf{t}\|^2 p(\mathbf{x},\mathbf{t}) d\mathbf{x} d\mathbf{t}. \end{aligned} \tag{1.199}$$

The expression of $I_1$ can be written as

$$\begin{aligned} &\int \|\mathbf{y}(\mathbf{x}) - \mathrm{E}(\mathbf{t}|\mathbf{x})\|^2 \left( \int p(\mathbf{x},\mathbf{t}) d\mathbf{t} \right) d\mathbf{x} \\ =& \int \|\mathbf{y}(\mathbf{x}) - \mathrm{E}(\mathbf{t}|\mathbf{x})\|^2 p(\mathbf{x}) d\mathbf{x}. \end{aligned} \tag{1.200}$$

The expression of $I_2$ can be written as

$$\int (\mathbf{y}(\mathbf{x}) - \mathrm{E}(\mathbf{t}|\mathbf{x}))^\intercal \left( \int (\mathrm{E}(\mathbf{t}|\mathbf{x}) - \mathbf{t})\, p(\mathbf{t}|\mathbf{x}) d\mathbf{t} \right) p(\mathbf{x}) d\mathbf{x}. \tag{1.201}$$

We have

$$\int (\mathrm{E}(\mathbf{t}|\mathbf{x}) - \mathbf{t})\, p(\mathbf{t}|\mathbf{x}) d\mathbf{t} = \mathrm{E}(\mathbf{t}|\mathbf{x}) \int p(\mathbf{t}|\mathbf{x}) d\mathbf{t} - \int \mathbf{t} p(\mathbf{t}|\mathbf{x}) d\mathbf{t}. \tag{1.202}$$

The right hand side can be written as

$$\mathrm{E}(\mathbf{t}|\mathbf{x}) - \mathrm{E}(\mathbf{t}|\mathbf{x}) = 0. \tag{1.203}$$

The expression of $I_3$ can be written as

$$\int \left( \int \|\mathbf{t} - \mathrm{E}(\mathbf{t}|\mathbf{x})\|^2 p(\mathbf{t}|\mathbf{x}) d\mathbf{t} \right) p(\mathbf{x}) d\mathbf{x} = \int \mathrm{var}(\mathbf{t}|\mathbf{x}) p(\mathbf{x}) d\mathbf{x}. \tag{1.204}$$

Then,

$$\mathrm{E}\, L(\mathbf{t}, \mathbf{y}(\mathbf{x})) = \int \|\mathbf{y}(\mathbf{x}) - \mathrm{E}(\mathbf{t}|\mathbf{x})\|^2 p(\mathbf{x}) d\mathbf{x} + \int \mathrm{var}(\mathbf{t}|\mathbf{x}) p(\mathbf{x}) d\mathbf{x}. \tag{1.205}$$

Therefore,

$$\operatorname*{argmin}_{\mathbf{y}(\mathbf{x})} \mathrm{E}\, L(\mathbf{t}, \mathbf{y}(\mathbf{x})) = \mathrm{E}(\mathbf{t}|\mathbf{x}). \tag{1.206}$$

## 1.27 (Incomplete)

### (a)

Let

$$\mathrm{E}\, L_q = \iint |y(\mathbf{x}) - t|^q p(\mathbf{x}, t) d\mathbf{x} dt. \tag{1.207}$$

Setting the derivative of $\mathrm{E}\, L_q$ with respect to $y(\mathbf{x})$ to zero gives

$$0 = q p(\mathbf{x}) \int |y(\mathbf{x}) - t|^{q-1} \operatorname{sign}(y(\mathbf{x}) - t) p(t|\mathbf{x}) dt. \tag{1.208}$$

Therefore,

$$\begin{aligned} &\operatorname*{argmin}_{y(\mathbf{x})} \mathrm{E}\, L_q \\ &= \left\{ y(\mathbf{x}) \mid \int |y(\mathbf{x}) - t|^{q-1} \operatorname{sign}(y(\mathbf{x}) - t) p(t|\mathbf{x}) dt = 0 \right\}. \end{aligned} \tag{1.209}$$

### (b)

We have

$$\mathrm{E}\, L_1 = \int \left( \int \operatorname{sign}(y(\mathbf{x}) - t) p(t|\mathbf{x}) dt \right) p(\mathbf{x}) d\mathbf{x}. \tag{1.210}$$

The integral of the right hand side with respect to $t$ can be written as

$$\int_{y(\mathbf{x})}^{\infty} p(t|\mathbf{x}) dt - \int_{-\infty}^{y(\mathbf{x})} p(t|\mathbf{x}) dt. \tag{1.211}$$

Therefore,

$$\operatorname*{argmin}_{y(\mathbf{x})} \mathrm{E}\, L_1 = \operatorname{median}(t|\mathbf{x}). \tag{1.212}$$

### (c)

We have

$$\lim_{q \to 0} \left( \operatorname*{argmin}_{y(\mathbf{x})} \mathrm{E}\, L_q \right) = \operatorname{mode}(t|\mathbf{x})? \tag{1.213}$$

## 1.28

### (a)

Let us assume that

$$p(x, y) = p(x)p(y) \Rightarrow h(x, y) = h(x) + h(y). \tag{1.214}$$

Then,

$$h\left(p^2\right) = 2h(p). \tag{1.215}$$

Let us assume that, for a positive integer $n$,

$$h\left(p^n\right) = nh(p). \tag{1.216}$$

Then, by the first assumption,

$$h\left(p^{n+1}\right) = h\left(p^n\right) + h(p), \tag{1.217}$$

so that

$$h\left(p^{n+1}\right) = (n + 1)h(p). \tag{1.218}$$

Therefore, the second assumption is proved by induction on $n$.

### (b)

For positive integers $m$ and $n$,

$$h\left(p^n\right) = h\left(p^{\frac{n}{m}m}\right). \tag{1.219}$$

By the second assumption in (a), the left hand side can be written as $nh(p)$. By the first assumption in (a), the right hand side can be written as $mh\left(p^{\frac{n}{m}}\right)$. Therefore,

$$h\left(p^{\frac{n}{m}}\right) = \frac{n}{m}h(p). \tag{1.220}$$

### (c)

By the continuity, for a positive real number $a$,

$$h\left(p^a\right) = ah(p). \tag{1.221}$$

Taking the derivative with respect to $a$ and substituting $a = 1$ gives

$$(p \ln p)h'(p) = h(p). \tag{1.222}$$

Then,

$$\int \frac{h'(p)}{h(p)} dp = \int \frac{1}{p \ln p} dp + \text{const} \,. \tag{1.223}$$

Ignorting the constants, the left hand side can be written as $\ln h(p)$ and the right hand side can be written as $\ln(\ln p)$. Therefore,

$$h(p) \propto \ln p. \tag{1.224}$$

## 1.29

Let $x$ be an $M$-state discrete random variable. Then, the entropy is given by

$$\mathrm{H}(x) = -\sum_{m=1}^{M} p(x_m) \ln p(x_m), \tag{1.225}$$

where

$$\sum_{m=1}^{M} p(x_m) = 1. \tag{1.226}$$

By Jensen's inequality,

$$\sum_{m=1}^{M} p(x_i) \ln \frac{1}{p(x_m)} \leq \ln \left( \sum_{m=1}^{M} 1 \right). \tag{1.227}$$

Therefore,

$$\mathrm{H}(x) \leq \ln M. \tag{1.228}$$

## 1.30

Let

$$\begin{aligned} p(x) &= \mathcal{N}\left(x|\mu, \sigma^2\right), \\ q(x) &= \mathcal{N}\left(x|m, s^2\right). \end{aligned} \tag{1.229}$$

Then, the Kullback-Leibler divergence is given by

$$\mathrm{KL}\left(p||q\right) = -\int p(x) \ln \frac{q(x)}{p(x)} dx. \tag{1.230}$$

The right hand side can be written as

$$
\begin{aligned}
&-\int_{-\infty}^{\infty} p(x) \ln \frac{(2\pi s^2)^{-\frac{1}{2}} \exp\left(-\frac{(x-m)^2}{2s^2}\right)}{(2\pi \sigma^2)^{-\frac{1}{2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)} dx \\
=&-\int_{-\infty}^{\infty} p(x) \left(-\frac{1}{2} \ln \frac{s^2}{\sigma^2} - \frac{(x-m)^2}{2s^2} + \frac{(x-\mu)^2}{2\sigma^2}\right) dx.
\end{aligned}
\tag{1.231}
$$

The right hand side can be written as

$$
\ln \frac{s}{\sigma} \int_{-\infty}^{\infty} p(x) dx + \frac{1}{2s^2} \int_{-\infty}^{\infty} (x-m)^2 p(x) dx - \frac{1}{2\sigma^2} \int_{-\infty}^{\infty} (x-\mu)^2 p(x) dx. \tag{1.232}
$$

The integral of the second term can be written as

$$
\begin{aligned}
&\int_{-\infty}^{\infty} (x-\mu+\mu-m)^2 p(x) dx \\
=&\int_{-\infty}^{\infty} (x-\mu)^2 p(x) dx + 2(\mu - m) \int_{-\infty}^{\infty} (x-\mu) p(x) dx \\
&+ (\mu - m)^2 \int_{-\infty}^{\infty} p(x) dx.
\end{aligned}
\tag{1.233}
$$

Therefore,

$$
\mathrm{KL}\left(p||q\right) = \ln \frac{s}{\sigma} + \frac{\sigma^2 + (\mu - m)^2}{2s^2} - \frac{1}{2}. \tag{1.234}
$$

## 1.31

Let $\mathbf{x}$ and $\mathbf{y}$ be two variables. We have

$$
\begin{aligned}
\mathrm{H}(\mathbf{x}) &= -\int p(\mathbf{x}) \ln p(\mathbf{x}) d\mathbf{x}, \\
\mathrm{H}(\mathbf{y}) &= -\int p(\mathbf{y}) \ln p(\mathbf{y}) d\mathbf{y}, \\
\mathrm{H}(\mathbf{x}, \mathbf{y}) &= -\iint p(\mathbf{x}, \mathbf{y}) \ln p(\mathbf{x}, \mathbf{y}) d\mathbf{x} d\mathbf{y}.
\end{aligned}
\tag{1.235}
$$

The expressions of $\mathrm{H}(\mathbf{x})$ and $\mathrm{H}(\mathbf{y})$ can be written as

$$
\begin{aligned}
&-\int \left(\int p(\mathbf{x}, \mathbf{y}) d\mathbf{y}\right) \ln p(\mathbf{x}) d\mathbf{x}, \\
&-\int \left(\int p(\mathbf{x}, \mathbf{y}) d\mathbf{x}\right) \ln p(\mathbf{y}) d\mathbf{y}.
\end{aligned}
\tag{1.236}
$$

Then,

$$\mathrm{H}(\mathbf{x}) + \mathrm{H}(\mathbf{y}) - \mathrm{H}(\mathbf{x}, \mathbf{y}) = -\iint p(\mathbf{x}, \mathbf{y}) \ln\left(\frac{p(\mathbf{x})p(\mathbf{y})}{p(\mathbf{x}, \mathbf{y})}\right) d\mathbf{x} d\mathbf{y}. \tag{1.237}$$

We have

$$\iint p(\mathbf{x}, \mathbf{y}) d\mathbf{x} d\mathbf{y} = 1. \tag{1.238}$$

Then, by Jensen's inequality,

$$-\iint p(\mathbf{x}, \mathbf{y}) \ln\left(\frac{p(\mathbf{x})p(\mathbf{y})}{p(\mathbf{x}, \mathbf{y})}\right) d\mathbf{x} d\mathbf{y} \geq -\ln\left(\iint p(\mathbf{x})p(\mathbf{y}) d\mathbf{x} d\mathbf{y}\right). \tag{1.239}$$

The right hand side can be written as

$$-\ln\left(\int p(\mathbf{x}) d\mathbf{x} \int p(\mathbf{y}) d\mathbf{y}\right) = 0. \tag{1.240}$$

Therefore,

$$\mathrm{H}(\mathbf{x}, \mathbf{y}) \leq \mathrm{H}(\mathbf{x}) + \mathrm{H}(\mathbf{y}). \tag{1.241}$$

## 1.32

Let $\mathbf{x}$ and $\mathbf{y}$ be variables such that

$$\mathbf{y} = \mathbf{A}\mathbf{x}, \tag{1.242}$$

where $\mathbf{A}$ is a nonsingular matrix. We have

$$\int p_x(\mathbf{x}) d\mathbf{x} = \int p_x\left(\mathbf{A}^{-1}\mathbf{y}\right) \left|\det \mathbf{A}^{-1}\right| d\mathbf{y}. \tag{1.243}$$

Then,

$$p_y(\mathbf{y}) = p_x\left(\mathbf{A}^{-1}\mathbf{y}\right) \left|\det \mathbf{A}^{-1}\right|. \tag{1.244}$$

We have

$$\mathrm{H}(\mathbf{y}) = -\int p_y(\mathbf{y}) \ln p_y(\mathbf{y}) d\mathbf{y}. \tag{1.245}$$

The right hand side can be written as

$$\begin{aligned} &-\int p_y(\mathbf{y}) \ln\left(p_x\left(\mathbf{A}^{-1}\mathbf{y}\right) \left|\det \mathbf{A}^{-1}\right|\right) d\mathbf{y} \\ &= -\int p_y(\mathbf{y}) \ln p_x\left(\mathbf{A}^{-1}\mathbf{y}\right) d\mathbf{y} + \ln\left|\det \mathbf{A}\right| \int p_y(\mathbf{y}) d\mathbf{y}. \end{aligned} \tag{1.246}$$

The first term can be written as

$$-\left|\det \mathbf{A}^{-1}\right| \int p_x\left(\mathbf{A}^{-1}\mathbf{y}\right) \ln p_x\left(\mathbf{A}^{-1}\mathbf{y}\right) d\mathbf{y}. \tag{1.247}$$

By the transformation

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{y}, \tag{1.248}$$

it can be written as

$$-\int p_x(\mathbf{x}) \ln p_x(\mathbf{x}) d\mathbf{x} = \mathrm{H}(\mathbf{x}). \tag{1.249}$$

Therefore,

$$\mathrm{H}(\mathbf{y}) = \mathrm{H}(\mathbf{x}) + \ln|\det \mathbf{A}|. \tag{1.250}$$

## 1.33

Let $x$ and $y$ be two discrete variables with $K$ and $L$ states. Then,

$$\mathrm{H}(y|x) = -\sum_{k=1}^{K}\sum_{l=1}^{L} p(x_k, y_l) \ln p(y_l|x_k). \tag{1.251}$$

If

$$\mathrm{H}(y|x) = 0, \tag{1.252}$$

then

$$0 = -\sum_{k=1}^{K} p(x_k) \left(\sum_{l=1}^{L} p(y_l|x_k) \ln p(y_l|x_k)\right). \tag{1.253}$$

We have

$$\begin{aligned} p(x_k) &\geq 0, \\ p(y_l|x_k) \ln p(y_l|x_k) &\leq 0. \end{aligned} \tag{1.254}$$

Then, the equation reduces to

$$p(y_l|x_k) \ln p(y_l|x_k) = 0. \tag{1.255}$$

Then,

$$p(y_l|x_k) = \begin{cases} 1, \\ 0. \end{cases} \tag{1.256}$$

We have

$$\sum_{l=1}^{L} p(y_l|x_k) = 1. \tag{1.257}$$

Therefore,

$$p(y_l|x_k) = \begin{cases} 1, & \text{if } K = 1, \\ 0, & \text{otherwise.} \end{cases} \tag{1.258}$$

## 1.34

Let $x$ be a variable. We have

$$\mathrm{H}(x) = -\int_{-\infty}^{\infty} p(x) \ln p(x) dx. \tag{1.259}$$

In order to maximise $\mathrm{H}(x)$ with the constratints

$$\begin{aligned} \int_{-\infty}^{\infty} p(x) dx &= 1, \\ \int_{-\infty}^{\infty} x p(x) dx &= \mu, \\ \int_{-\infty}^{\infty} (x-\mu)^2 p(x) dx &= \sigma^2, \end{aligned} \tag{1.260}$$

let

$$\begin{aligned} L(p) = &\mathrm{H}(x) + \lambda_1 \left( \int_{-\infty}^{\infty} p(x) dx - 1 \right) + \lambda_2 \left( \int_{-\infty}^{\infty} x p(x) dx - \mu \right) \\ &+ \lambda_3 \left( \int_{-\infty}^{\infty} (x-\mu)^2 p(x) dx - \sigma^2 \right). \end{aligned} \tag{1.261}$$

Setting the variation with respect to $p$ to zero gives

$$0 = -\ln p - 1 + \lambda_1 + \lambda_2 x + \lambda_3 (x-\mu)^2. \tag{1.262}$$

Then,

$$p(x) = \exp\left(-1 + \lambda_1 + \lambda_2 x + \lambda_3 (x-\mu)^2\right), \tag{1.263}$$

so that

$$p(x) = c \exp\left(-q(x)\right), \tag{1.264}$$

where

$$
\begin{aligned}
c &= \exp\left(-1+\lambda_1-\frac{\lambda_2^2}{4\lambda_3}\right),\\
q(x) &= -\lambda_3\left(x-\left(\mu-\frac{\lambda_2}{2\lambda_3}\right)\right)^2.
\end{aligned} \tag{1.265}
$$

Substituting it to the constraints gives

$$
\begin{aligned}
c\int_{-\infty}^{\infty}\exp\left(-q(x)\right)dx &= 1,\\
c\int_{-\infty}^{\infty}x\exp\left(-q(x)\right)dx &= \mu,\\
c\int_{-\infty}^{\infty}(x-\mu)^2\exp\left(-q(x)\right)dx &= \sigma^2.
\end{aligned} \tag{1.266}
$$

By the transformation

$$
y^2 = q(x), \tag{1.267}
$$

they can be written as

$$
\begin{aligned}
c\int_{-\infty}^{\infty}\exp\left(-y^2\right)(-\lambda_3)^{-\frac{1}{2}}dy &= 1,\\
c\int_{-\infty}^{\infty}\left((-\lambda_3)^{-\frac{1}{2}}y+\mu-\frac{\lambda_2}{2\lambda_3}\right)\exp\left(-y^2\right)(-\lambda_3)^{-\frac{1}{2}}dy &= \mu,\\
c\int_{-\infty}^{\infty}\left((-\lambda_3)^{-\frac{1}{2}}y-\frac{\lambda_2}{2\lambda_3}\right)^2\exp\left(-y^2\right)(-\lambda_3)^{-\frac{1}{2}}dy &= \sigma^2.
\end{aligned} \tag{1.268}
$$

We have

$$
\begin{aligned}
\int_{-\infty}^{\infty}\exp\left(-y^2\right)dy &= \Gamma\left(\frac{1}{2}\right),\\
\int_{-\infty}^{\infty}y\exp\left(-y^2\right)dy &= 0,\\
\int_{-\infty}^{\infty}y^2\exp\left(-y^2\right)dy &= \Gamma\left(\frac{3}{2}\right),
\end{aligned} \tag{1.269}
$$

Then,

$$
\begin{aligned}
c(-\lambda_3)^{-\frac{1}{2}}\Gamma\left(\frac{1}{2}\right) &= 1, \\
c\left(\mu - \frac{\lambda_2}{2\lambda_3}\right)(-\lambda_3)^{-\frac{1}{2}}\Gamma\left(\frac{1}{2}\right) &= \mu, \\
c\left((-\lambda_3)^{-\frac{3}{2}}\Gamma\left(\frac{3}{2}\right) + (-\lambda_3)^{-\frac{1}{2}}\frac{\lambda_2^2}{4\lambda_3^2}\Gamma\left(\frac{1}{2}\right)\right) &= \sigma^2.
\end{aligned} \tag{1.270}
$$

Then,

$$
\begin{aligned}
\lambda_1 &= 1 - \frac{1}{2}\ln\left(2\pi\sigma^2\right), \\
\lambda_2 &= 0, \\
\lambda_3 &= -\frac{1}{2\sigma^2}.
\end{aligned} \tag{1.271}
$$

Therefore,

$$
p(x) = \left(2\pi\sigma^2\right)^{-\frac{1}{2}}\exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right). \tag{1.272}
$$

## 1.35

Let $x$ be a variable such that

$$
p(x) = \mathcal{N}\left(x|\mu,\sigma^2\right). \tag{1.273}
$$

Then,

$$
\mathrm{H}(x) = -\int_{-\infty}^{\infty}\mathcal{N}\left(x|\mu,\sigma^2\right)\ln\mathcal{N}\left(x|\mu,\sigma^2\right)dx. \tag{1.274}
$$

The right hand side can be written as

$$
\begin{aligned}
&-\int_{-\infty}^{\infty}\mathcal{N}\left(x|\mu,\sigma^2\right)\left(-\frac{1}{2}\ln\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}(x-\mu)^2\right)dx \\
=&\frac{1}{2}\ln\left(2\pi\sigma^2\right)\int_{-\infty}^{\infty}\mathcal{N}\left(x|\mu,\sigma^2\right)dx \\
&+\frac{1}{2\sigma^2}\int_{-\infty}^{\infty}(x-\mu)^2\mathcal{N}\left(x|\mu,\sigma^2\right)dx.
\end{aligned} \tag{1.275}
$$

Therefore,

$$
\mathrm{H}(x) = \frac{1}{2}\left(1 + \ln\left(2\pi\sigma^2\right)\right). \tag{1.276}
$$

## 1.36 (Incomplete)

Let $f$ be a strictly convex function. Then,

$$f(\lambda a + (1-\lambda)b) \leq \lambda f(a) + (1-\lambda)f(b), \tag{1.277}$$

where $a \leq b$ and $0 \leq \lambda \leq 1$. Let

$$x = \lambda a + (1-\lambda)b. \tag{1.278}$$

Then, the inequality can be written as

$$f(x) \leq \frac{b-x}{b-a}f(a) + \frac{x-a}{b-a}f(b). \tag{1.279}$$

Let

$$g(x) = \frac{b-x}{b-a}f(a) + \frac{x-a}{b-a}f(b) - f(x). \tag{1.280}$$

Then,

$$g(x) \geq 0. \tag{1.281}$$

Additionally, for $x > a$,

$$g(x) = (x-a)\left(\frac{f(b)-f(a)}{b-a} - \frac{f(x)-f(a)}{x-a}\right). \tag{1.282}$$

By the mean value theorem, there exists $c$ and $y$ such that $a \leq c \leq b$, $a \leq y \leq x$ and

$$\begin{aligned} f'(c) &= \frac{f(b)-f(a)}{b-a}, \\ f'(y) &= \frac{f(x)-f(a)}{x-a}. \end{aligned} \tag{1.283}$$

Then, for $x > a$, the inequality reduces to

$$f'(y) \leq f'(c). \tag{1.284}$$

## 1.37

Let $\mathbf{x}$ and $\mathbf{y}$ be two variables. We have

$$\mathrm{H}(\mathbf{x}, \mathbf{y}) = -\iint p(\mathbf{x}, \mathbf{y}) \ln p(\mathbf{x}, \mathbf{y}) d\mathbf{x} d\mathbf{y}. \tag{1.285}$$

The right hand side can be written as

$$\begin{aligned} &-\iint p(\mathbf{x},\mathbf{y})\left(\ln p(\mathbf{y}|\mathbf{x}) + \ln p(\mathbf{x})\right) d\mathbf{x}d\mathbf{y} \\ =&-\iint p(\mathbf{x},\mathbf{y}) \ln p(\mathbf{y}|\mathbf{x}) d\mathbf{x}d\mathbf{y} - \int \left(\int p(\mathbf{x},\mathbf{y})d\mathbf{y}\right) \ln p(\mathbf{x}) d\mathbf{x}. \end{aligned} \tag{1.286}$$

We have

$$\begin{aligned} \mathrm{H}(\mathbf{y}|\mathbf{x}) &= -\iint p(\mathbf{x},\mathbf{y}) \ln p(\mathbf{y}|\mathbf{x}) d\mathbf{x}d\mathbf{y}, \\ \mathrm{H}(\mathbf{x}) &= -\int p(\mathbf{x}) \ln p(\mathbf{x}) d\mathbf{x}. \end{aligned} \tag{1.287}$$

Therefore,

$$\mathrm{H}(\mathbf{x},\mathbf{y}) = \mathrm{H}(\mathbf{y}|\mathbf{x}) + \mathrm{H}(\mathbf{x}). \tag{1.288}$$

## 1.38

Let $f$ be a strictly convex function. Then,

$$f\left(\lambda x_1 + (1-\lambda)x_2\right) \leq \lambda f(x_1) + (1-\lambda) f(x_2), \tag{1.289}$$

where

$$0 \leq \lambda \leq 1. \tag{1.290}$$

Let us assume that

$$f\left(\sum_{m=1}^{M} \lambda_m x_m\right) \leq \sum_{m=1}^{M} \lambda_m f(x_m), \tag{1.291}$$

where

$$\begin{aligned} \sum_{m=1}^{M} \lambda_m &= 1, \\ \lambda_m &\geq 0. \end{aligned} \tag{1.292}$$

We have

$$\begin{aligned} &f\left(\sum_{m=1}^{M+1} \lambda_m x_m\right) \\ \leq&\lambda_{M+1} f(x_{M+1}) + (1-\lambda_{M+1}) f\left(\sum_{m=1}^{M} \frac{\lambda_m}{1-\lambda_{M+1}} x_m\right), \end{aligned} \tag{1.293}$$

where

$$\sum_{m=1}^{M+1} \lambda_m = 1, \qquad \lambda_m \geq 0. \tag{1.294}$$

By the assumption,

$$f\left(\sum_{m=1}^{M} \frac{\lambda_m}{1-\lambda_{M+1}} x_m\right) \leq \sum_{m=1}^{M} \frac{\lambda_m}{1-\lambda_{M+1}} f(x_m). \tag{1.295}$$

Then,

$$f\left(\sum_{m=1}^{M+1} \lambda_m x_m\right) \leq \lambda_{M+1} f(x_{M+1}) + (1-\lambda_{M+1}) \sum_{m=1}^{M} \frac{\lambda_m}{1-\lambda_{M+1}} f(x_m), \tag{1.296}$$

so that

$$f\left(\sum_{m=1}^{M+1} \lambda_m x_m\right) \leq \sum_{m=1}^{M+1} \lambda_m f(x_m). \tag{1.297}$$

Therefore, the assumption is proved by induction on $M$.

## 1.39

Let $x$ and $y$ be two binary variables where

$$\begin{aligned} p(x=0, y=0) &= \frac{1}{3}, \\ p(x=0, y=1) &= \frac{1}{3}, \\ p(x=1, y=0) &= 0, \\ p(x=1, y=1) &= \frac{1}{3}. \end{aligned} \tag{1.298}$$

### (a)

We have

$$\mathrm{H}(x) = -\sum_x p(x) \ln p(x). \tag{1.299}$$

We have

$$\begin{aligned} p(x=0) &= p(x=0, y=0) + p(x=0, y=1), \\ p(x=1) &= p(x=1, y=0) + p(x=1, y=1). \end{aligned} \tag{1.300}$$

Then,

$$\begin{aligned} p(x=0) &= \frac{2}{3}, \\ p(x=1) &= \frac{1}{3}. \end{aligned} \tag{1.301}$$

Therefore,

$$\mathrm{H}(x) = \ln 3 - \frac{2}{3} \ln 2. \tag{1.302}$$

**(b)**

We have

$$\mathrm{H}(y) = -\sum_{y} p(y) \ln p(y). \tag{1.303}$$

We have

$$\begin{aligned} p(y=0) &= p(x=0, y=0) + p(x=1, y=0), \\ p(y=1) &= p(x=0, y=1) + p(x=1, y=1). \end{aligned} \tag{1.304}$$

Then,

$$\begin{aligned} p(y=0) &= \frac{1}{3}, \\ p(y=1) &= \frac{2}{3}. \end{aligned} \tag{1.305}$$

Therefore,

$$\mathrm{H}(y) = \ln 3 - \frac{2}{3} \ln 2. \tag{1.306}$$

**(c)**

We have

$$\mathrm{H}(y|x) = -\sum_{x,y} p(x, y) \ln p(y|x). \tag{1.307}$$

By Bayes' theorem,

$$\begin{aligned}
p(y=0|x=0) &= \frac{p(x=0,y=0)}{p(x=0)}, \\
p(y=0|x=1) &= \frac{p(x=1,y=0)}{p(x=1)}, \\
p(y=1|x=0) &= \frac{p(x=0,y=1)}{p(x=0)}, \\
p(y=1|x=1) &= \frac{p(x=1,y=1)}{p(x=1)}.
\end{aligned} \tag{1.308}$$

Then,

$$\begin{aligned}
p(y=0|x=0) &= \frac{1}{2}, \\
p(y=0|x=1) &= 0, \\
p(y=1|x=0) &= \frac{1}{2}, \\
p(y=1|x=1) &= 1.
\end{aligned} \tag{1.309}$$

Therefore,

$$\mathrm{H}(y|x) = \frac{2}{3}\ln 2. \tag{1.310}$$

**(d)**

We have

$$\mathrm{H}(x|y) = -\sum_{x,y} p(x,y)\ln p(x|y). \tag{1.311}$$

By Bayes' theorem,

$$\begin{aligned}
p(x=0|y=0) &= \frac{p(x=0,y=0)}{p(y=0)}, \\
p(x=0|y=1) &= \frac{p(x=0,y=1)}{p(y=1)}, \\
p(x=1|y=0) &= \frac{p(x=1,y=0)}{p(y=0)}, \\
p(x=1|y=1) &= \frac{p(x=1,y=1)}{p(y=1)}.
\end{aligned} \tag{1.312}$$

Then,

$$
\begin{aligned}
p(x=0|y=0) &= 1, \\
p(x=0|y=1) &= \frac{1}{2}, \\
p(x=1|y=0) &= 0, \\
p(x=1|y=1) &= \frac{1}{2}.
\end{aligned} \tag{1.313}
$$

Therefore,

$$
\mathrm{H}(x|y) = \frac{2}{3}\ln 2. \tag{1.314}
$$

**(e)**

We have

$$
\mathrm{H}(x,y) = -\sum_{x,y} p(x,y)\ln p(x,y). \tag{1.315}
$$

Therefore,

$$
\mathrm{H}(x,y) = \ln 3. \tag{1.316}
$$

**(f)**

We have

$$
\mathrm{I}(x,y) = -\sum_{x,y} p(x,y)\ln \frac{p(x)p(y)}{p(x,y)}. \tag{1.317}
$$

The right hand side can be written as

$$
\mathrm{H}(x) + \mathrm{H}(y) - \mathrm{H}(x,y). \tag{1.318}
$$

Therefore,

$$
\mathrm{I}(x,y) = \ln 3 - \frac{4}{3}\ln 2. \tag{1.319}
$$

## 1.40

Let $\lambda_1, \cdots, \lambda_M$ and $x_1, \cdots, x_M$ be numbers such that

$$
\begin{aligned}
\sum_{m=1}^{M} \lambda_m &= 1, \\
\lambda_m &\geq 0, \\
x_m &> 0.
\end{aligned} \tag{1.320}
$$

By Jensen's inequality,

$$\sum_{m=1}^{M} \lambda_m \ln x_m \leq \ln \left( \sum_{m=1}^{M} \lambda_m x_m \right), \tag{1.321}$$

so that

$$\prod_{m=1}^{M} x_m^{\lambda_m} \leq \sum_{m=1}^{M} \lambda_m x_m. \tag{1.322}$$

Substituting

$$\lambda_m = \frac{1}{M} \tag{1.323}$$

to the inequality gives

$$\left( \prod_{m=1}^{M} x_m \right)^{\frac{1}{M}} \leq \frac{1}{M} \sum_{m=1}^{M} x_m. \tag{1.324}$$

## 1.41

Let $\mathbf{x}$ and $\mathbf{y}$ be variables.

### (a)

We have

$$\mathrm{I}(\mathbf{x}, \mathbf{y}) = -\iint p(\mathbf{x}, \mathbf{y}) \ln \left( \frac{p(\mathbf{x}) p(\mathbf{y})}{p(\mathbf{x}, \mathbf{y})} \right) d\mathbf{x} d\mathbf{y}. \tag{1.325}$$

The right hand side can be written as

$$\begin{aligned} & -\iint p(\mathbf{x}, \mathbf{y}) \left( \ln p(\mathbf{x}) + \ln \left( \frac{p(\mathbf{y})}{p(\mathbf{x}, \mathbf{y})} \right) \right) d\mathbf{x} d\mathbf{y} \\ = & -\iint p(\mathbf{x}, \mathbf{y}) \left( \ln p(\mathbf{x}) - \ln p(\mathbf{x}|\mathbf{y}) \right) d\mathbf{x} d\mathbf{y}. \end{aligned} \tag{1.326}$$

The right hand side can be written as

$$-\int \left( \int p(\mathbf{x}, \mathbf{y}) d\mathbf{y} \right) \ln p(\mathbf{x}) d\mathbf{x} + \iint p(\mathbf{x}, \mathbf{y}) \ln p(\mathbf{x}|\mathbf{y}) d\mathbf{x} d\mathbf{y}. \tag{1.327}$$

Therefore,

$$\mathrm{I}(\mathbf{x}, \mathbf{y}) = \mathrm{H}(\mathbf{x}) - \mathrm{H}(\mathbf{x}|\mathbf{y}). \tag{1.328}$$

**(b)**

We have

$$\mathrm{I}(\mathbf{x}, \mathbf{y}) = \mathrm{I}(\mathbf{y}, \mathbf{x}). \tag{1.329}$$

By (a), the right hand side can be written as

$$\mathrm{H}(\mathbf{y}) - \mathrm{H}(\mathbf{y}|\mathbf{x}). \tag{1.330}$$

Therefore,

$$\mathrm{I}(\mathbf{x}, \mathbf{y}) = \mathrm{H}(\mathbf{y}) - \mathrm{H}(\mathbf{y}|\mathbf{x}). \tag{1.331}$$

# 2 Probability Distributions

## 2.1

Let $x$ be a binary variable such that

$$p(x) = \mu^x (1-\mu)^{1-x}. \tag{2.1}$$

### (a)

We have

$$\sum_x p(x) = 1 - \mu + \mu. \tag{2.2}$$

Therefore,

$$\sum_x p(x) = 1. \tag{2.3}$$

### (b)

We have

$$\begin{aligned} \operatorname{E} x &= \sum_x x p(x), \\ \operatorname{E} x^2 &= \sum_x x^2 p(x), \end{aligned} \tag{2.4}$$

Then,

$$\begin{aligned} \operatorname{E} x &= \mu, \\ \operatorname{E} x^2 &= \mu. \end{aligned} \tag{2.5}$$

We have

$$\operatorname{var} x = \operatorname{E} x^2 - \left(\operatorname{E} x\right)^2. \tag{2.6}$$

Therefore,

$$\operatorname{var} x = \mu(1-\mu). \tag{2.7}$$

### (c)

We have

$$\operatorname{H}(x) = -\sum_x p(x) \ln p(x). \tag{2.8}$$

Therefore,

$$\operatorname{H}(x) = -\mu \ln \mu - (1-\mu) \ln(1-\mu). \tag{2.9}$$

## 2.2

Let $x$ be a variable from $\{-1, 1\}$ such that

$$p(x) = \left(\frac{1-\mu}{2}\right)^{\frac{1-x}{2}} \left(\frac{1+\mu}{2}\right)^{\frac{1+x}{2}}. \tag{2.10}$$

### (a)

We have

$$\sum_x p(x) = \frac{1-\mu}{2} + \frac{1+\mu}{2}. \tag{2.11}$$

Therefore,

$$\sum_x p(x) = 1. \tag{2.12}$$

### (b)

We have

$$\begin{aligned} \operatorname{E} x &= \sum_x x p(x), \\ \operatorname{E} x^2 &= \sum_x x^2 p(x). \end{aligned} \tag{2.13}$$

The right hand sides can be written as

$$\begin{aligned} -\frac{1-\mu}{2} + \frac{1+\mu}{2} &= \mu, \\ \frac{1-\mu}{2} + \frac{1+\mu}{2} &= 1. \end{aligned} \tag{2.14}$$

Then,

$$\begin{aligned} \operatorname{E} x &= \mu, \\ \operatorname{E} x^2 &= 1. \end{aligned} \tag{2.15}$$

We have

$$\operatorname{var} x = \operatorname{E} x^2 - (\operatorname{E} x)^2. \tag{2.16}$$

Therefore,

$$\operatorname{var} x = 1 - \mu^2. \tag{2.17}$$

**(c)**

We have

$$\mathrm{H}(x) = -\sum_x p(x|\mu) \ln p(x|\mu). \tag{2.18}$$

Therefore,

$$\mathrm{H}(x) = -\frac{1-\mu}{2} \ln \frac{1-\mu}{2} - \frac{1+\mu}{2} \ln \frac{1+\mu}{2}. \tag{2.19}$$

## 2.3

**(a)**

We have

$$\begin{aligned} \binom{N}{n} &= \frac{N!}{n!(N-n)!}, \\ \binom{N}{n-1} &= \frac{N!}{(n-1)!(N-n+1)!} \end{aligned} \tag{2.20}$$

Then,

$$\binom{N}{n} + \binom{N}{n-1} = \frac{(N-n+1)N! + nN!}{n!(N-n+1)!}. \tag{2.21}$$

The right hand side can be written as

$$\frac{(N+1)!}{n!(N+1-n)!} = \binom{N+1}{n}. \tag{2.22}$$

Therefore,

$$\binom{N}{n} + \binom{N}{n-1} = \binom{N+1}{n}. \tag{2.23}$$

**(b)**

We have

$$1 + x = \sum_{n=0}^{1} \binom{1}{n} x^n. \tag{2.24}$$

Let us assume that

$$(1+x)^N = \sum_{n=0}^{N} \binom{N}{n} x^n. \tag{2.25}$$

Then,

$$(1+x)^{N+1} = \sum_{n=0}^{N} \binom{N}{n} x^n + \sum_{n=0}^{N} \binom{N}{n} x^{n+1}. \tag{2.26}$$

By (a), the right hand side can be written as

$$\sum_{n=0}^{N} \binom{N}{n} x^n + \sum_{n=1}^{N+1} \binom{N}{n-1} x^n = 1 + x^{N+1} + \sum_{n=1}^{N} \binom{N+1}{n} x^n. \tag{2.27}$$

Then,

$$(1+x)^{N+1} = \sum_{n=0}^{N+1} \binom{N+1}{n} x^n. \tag{2.28}$$

Therefore, the assumption is proved by induction on $N$.

**(c)**

Let $n$ be a variable such that

$$p(n) = \binom{N}{n} \mu^n (1-\mu)^{N-n}. \tag{2.29}$$

Then,

$$\sum_{n=0}^{N} p(n) = \sum_{n=0}^{N} \binom{N}{n} \mu^n (1-\mu)^{N-n}. \tag{2.30}$$

By (b), the right hand side can be written as

$$(1-\mu)^N \sum_{n=0}^{N} \binom{N}{n} \left(\frac{\mu}{1-\mu}\right)^n = (1-\mu)^N \left(1 + \frac{\mu}{1-\mu}\right)^N. \tag{2.31}$$

Therefore,

$$\sum_{n=0}^{N} p(n) = 1. \tag{2.32}$$

## 2.4

Let $n$ be a variable such that

$$p(n) = \binom{N}{n} \mu^n (1-\mu)^{N-n}. \tag{2.33}$$

**(a)**

We have

$$\mathrm{E}\,n = \sum_{n=0}^{N} n \binom{N}{n} \mu^n (1-\mu)^{N-n}. \tag{2.34}$$

By 2.3(c),

$$\sum_{n=0}^{N} \binom{N}{n} \mu^n (1-\mu)^{N-n} = 1. \tag{2.35}$$

Taking the derivative with respect to $\mu$ gives

$$\sum_{n=0}^{N} n \binom{N}{n} \mu^{n-1} (1-\mu)^{N-n} - \sum_{n=0}^{N} (N-n) \binom{N}{n} \mu^n (1-\mu)^{N-n-1} = 0. \tag{2.36}$$

The first term of the left hand side can be written as

$$\frac{1}{\mu} \sum_{n=0}^{N} np(n) = \frac{1}{\mu}\,\mathrm{E}\,n. \tag{2.37}$$

Since

$$(N-n)\binom{N}{n} = N\binom{N-1}{n}, \tag{2.38}$$

the second term can be written as

$$-N \sum_{n=0}^{N-1} \binom{N-1}{n} \mu^n (1-\mu)^{N-n-1} = -N. \tag{2.39}$$

Therefore,

$$\mathrm{E}\,n = N\mu. \tag{2.40}$$

**(b)**

By 2.3(c),

$$\sum_{n=0}^{N} \binom{N}{n} \mu^n (1-\mu)^{N-n} = 1. \tag{2.41}$$

Taking the second derivative with respect to $\mu$ gives

$$
\begin{aligned}
&\sum_{n=0}^{N} n(n-1)\binom{N}{n}\mu^{n-2}(1-\mu)^{N-n} \\
&-2\sum_{n=0}^{N} n(N-n)\binom{N}{n}\mu^{n-1}(1-\mu)^{N-n-1} \\
&+\sum_{n=0}^{N}(N-n)(N-n-1)\binom{N}{n}\mu^{n}(1-\mu)^{N-n-2} = 0.
\end{aligned}
\tag{2.42}
$$

The first term of the left hand side can be written as

$$
\frac{1}{\mu^2}\sum_{n=0}^{N} n(n-1)p(n) = \frac{1}{\mu^2}\,\mathrm{E}\,n(n-1). \tag{2.43}
$$

Since

$$
\begin{aligned}
n(N-n)\binom{N}{n} &= N(N-1)\binom{N-2}{n-1}, \\
(N-n)(N-n-1)\binom{N}{n} &= N(N-1)\binom{N-2}{n},
\end{aligned}
\tag{2.44}
$$

the second and third terms can be written as

$$
\begin{aligned}
-2N(N-1)\sum_{n=1}^{N-1}\binom{N-2}{n-1}\mu^{n-1}(1-\mu)^{N-n-1} &= -2N(N-1), \\
N(N-1)\sum_{n=0}^{N}\binom{N-2}{n}\mu^{n}(1-\mu)^{N-n-2} &= N(N-1).
\end{aligned}
\tag{2.45}
$$

Then,

$$
\mathrm{E}\,n(n-1) = N(N-1)\mu^2. \tag{2.46}
$$

We have

$$
\operatorname{var} n = \mathrm{E}\,n(n-1) + \mathrm{E}\,n - (\mathrm{E}\,n)^2. \tag{2.47}
$$

Therefore,

$$
\operatorname{var} n = N\mu(1-\mu). \tag{2.48}
$$

## 2.5

We have

$$\Gamma(a)\Gamma(b) = \int_0^\infty x^{a-1}\exp(-x)dx \int_0^\infty y^{b-1}\exp(-y)dy. \tag{2.49}$$

By the transformation

$$t = x + y, \tag{2.50}$$

the right hand side can be written as

$$\begin{aligned}
&\int_0^\infty x^{a-1}\left(\int_x^\infty (t-x)^{b-1}\exp(-t)dt\right)dx \\
&= \int_0^\infty \left(\int_0^t x^{a-1}(t-x)^{b-1}dx\right)\exp(-t)dt.
\end{aligned} \tag{2.51}$$

By the transformation

$$x = t\mu, \tag{2.52}$$

the right hand side can be written as

$$\begin{aligned}
&\int_0^\infty \left(\int_0^1 (t\mu)^{a-1}t^{b-1}(1-\mu)^{b-1}td\mu\right)\exp(-t)dt \\
&= \int_0^1 \mu^{a-1}(1-\mu)^{b-1}d\mu \int_0^\infty t^{a+b-1}\exp(-t)dt.
\end{aligned} \tag{2.53}$$

Then,

$$\Gamma(a)\Gamma(b) = \Gamma(a+b)\int_0^1 \mu^{a-1}(1-\mu)^{b-1}d\mu. \tag{2.54}$$

Therefore,

$$\int_0^1 \mu^{a-1}(1-\mu)^{b-1}d\mu = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}. \tag{2.55}$$

## 2.6

Let $\mu$ be a variable such that

$$p(\mu) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\mu^{a-1}(1-\mu)^{b-1}. \tag{2.56}$$

**(a)**

We have

$$\mathrm{E}\,\mu = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \int_0^1 \mu^a (1-\mu)^{b-1} d\mu, \\ \mathrm{E}\,\mu^2 = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \int_0^1 \mu^{a+1} (1-\mu)^{b-1} d\mu. \tag{2.57}$$

By 2.5,

$$\int_0^1 \mu^a (1-\mu)^{b-1} d\mu = \frac{\Gamma(a+1)\Gamma(b)}{\Gamma(a+b+1)}, \\ \int_0^1 \mu^{a+1} (1-\mu)^{b-1} d\mu = \frac{\Gamma(a+2)\Gamma(b)}{\Gamma(a+b+2)}. \tag{2.58}$$

Therefore,

$$\mathrm{E}\,\mu = \frac{a}{a+b}, \\ \mathrm{E}\,\mu^2 = \frac{a(a+1)}{(a+b)(a+b+1)}. \tag{2.59}$$

**(b)**

We have

$$\operatorname{var}\mu = \mathrm{E}\,\mu^2 - (\mathrm{E}\mu)^2. \tag{2.60}$$

By (a), the right hand side can be written as

$$\frac{a(a+1)}{(a+b)(a+b+1)} - \left(\frac{a}{a+b}\right)^2 = \frac{a(a+1)(a+b) - a^2(a+b+1)}{(a+b)^2(a+b+1)}. \tag{2.61}$$

Therefore,

$$\operatorname{var}\mu = \frac{ab}{(a+b)^2(a+b+1)}. \tag{2.62}$$

**(c)**

We have

$$\ln p(\mu) = \ln\Gamma(a+b) - \ln\Gamma(a) - \ln\Gamma(b) + (a-1)\ln\mu + (b-1)\ln(1-\mu). \tag{2.63}$$

Setting the derivative of $\ln p$ with respect to $\mu$ to zero gives

$$0 = \frac{a-1}{\mu} - \frac{b-1}{1-\mu}. \tag{2.64}$$

Therefore,

$$\operatorname{mode}\mu = \begin{cases} \frac{a-1}{a+b-2}, & \text{if } a \geq 1 \text{and } b \geq 1, \\ \text{none}, & \text{otherwise.} \end{cases} \tag{2.65}$$

## 2.7

Let $m$ and $l$ be variables such that

$$\begin{aligned} p(m, l|\mu) &= \binom{m+l}{m} \mu^m (1-\mu)^l, \\ p(\mu) &= \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \mu^{a-1} (1-\mu)^{b-1}. \end{aligned} \tag{2.66}$$

By 2.6,

$$\mathrm{E}\,\mu = \frac{a}{a+b}. \tag{2.67}$$

Setting the derivative of $p(m, l|\mu)$ with respect to $\mu$ to zero gives

$$0 = \binom{m+l}{m} \mu^m (1-\mu)^l \left( \frac{m}{\mu} + \frac{l}{1-\mu} \right). \tag{2.68}$$

Then, the maximum likelihood solution to $\mu$ is given by

$$\mu_{\mathrm{ML}} = \frac{m}{m+l}. \tag{2.69}$$

By Bayes' thereorem,

$$p(\mu|m, l) = \frac{p(m, l|\mu) p(\mu)}{p(m, l)}. \tag{2.70}$$

By 2.5, the right hand side can be written as

$$p(\mu|m, l) = \frac{\Gamma(m+l+a+b)}{\Gamma(m+a)\Gamma(l+b)} \mu^{m+a-1} (1-\mu)^{l+b-1}. \tag{2.71}$$

By 2.6,

$$\mathrm{E}(\mu|m, l) = \frac{m+a}{m+l+a+b}. \tag{2.72}$$

Therefore,

$$\mathrm{E}(\mu|m, l) = \lambda \mu_{\mathrm{ML}} + (1-\lambda)\,\mathrm{E}\,\mu, \tag{2.73}$$

where

$$\lambda = \frac{m+l}{m+l+a+b}. \tag{2.74}$$

## 2.8

Let $x$ and $y$ be variables.

### (a)

We have

$$\mathrm{E}\, x = \int x p(x) dx. \tag{2.75}$$

The right hand side can be written as

$$\int x \left( \int p(x, y) dy \right) dx = \int \left( \int x p(x|y) dx \right) p(y) dy. \tag{2.76}$$

Therefore,

$$\mathrm{E}\, x = \mathrm{E}\left(\mathrm{E}(x|y)\right). \tag{2.77}$$

### (b)

We have

$$\operatorname{var} x = \mathrm{E}\left(x - \mathrm{E}\, x\right)^2. \tag{2.78}$$

By (a), the right hand side can be written as

$$\begin{aligned} &\mathrm{E}\left(\mathrm{E}\left((x - \mathrm{E}\, x)^2 | y\right)\right) \\ =&\mathrm{E}\left(\mathrm{E}\left(\left(f(x, y) + g(y)\right)^2 | y\right)\right), \end{aligned} \tag{2.79}$$

where

$$\begin{aligned} f(x, y) &= x - \mathrm{E}(x|y), \\ g(y) &= \mathrm{E}(x|y) - \mathrm{E}\, x. \end{aligned} \tag{2.80}$$

The right hand side can be written as $I_1 + 2I_2 + I_3$, where

$$\begin{aligned} I_1 &= \mathrm{E}\left(\mathrm{E}\left(f^2(x, y)|y\right)\right), \\ I_2 &= \mathrm{E}\left(\mathrm{E}\left(f(x, y)g(y)|y\right)\right), \\ I_3 &= \mathrm{E}\left(g^2(y)\right). \end{aligned} \tag{2.81}$$

We have

$$\operatorname{var}(x|y) = \mathrm{E}\left(f^2(x, y)|y\right). \tag{2.82}$$

Then, the expression of $I_1$ can be written as

$$\mathrm{E}\left(\operatorname{var}(x|y)\right). \tag{2.83}$$

We have

$$\mathrm{E}\left(f(x,y)|y\right) = 0. \tag{2.84}$$

Then, the expression of $I_2$ can be written as

$$\mathrm{E}\left(\mathrm{E}\left(f(x,y)|y\right)g(y)\right) = 0. \tag{2.85}$$

By (a),

$$\mathrm{E}\left(\mathrm{E}(x|y)\right) = \mathrm{E}\,x. \tag{2.86}$$

Then, the expression of $I_3$ can be written as

$$\operatorname{var}\left(\mathrm{E}(x|y)\right). \tag{2.87}$$

Therefore,

$$\operatorname{var} x = \mathrm{E}\left(\operatorname{var}(x|y)\right) + \operatorname{var}\left(\mathrm{E}(x|y)\right). \tag{2.88}$$

## 2.9

For a vector $\boldsymbol{\mu}$ in 2 dimensions, by 2.5,

$$\int_{\substack{\mu_1+\mu_2=1\\ \mu_1\geq 0,\mu_2\geq 0}} \mu_1^{\alpha_1-1}\mu_2^{\alpha_2-1}d\boldsymbol{\mu} = \frac{\Gamma(\alpha_1)\Gamma(\alpha_2)}{\Gamma(\alpha_1+\alpha_2)}. \tag{2.89}$$

For a vector $\boldsymbol{\mu}$ in $K$ dimensions, let us assume that

$$\int_{\substack{\sum_{k=1}^K \mu_k=1\\ \mu_k\geq 0}} \prod_{k=1}^K \mu_k^{\alpha_k-1}d\boldsymbol{\mu} = \frac{\prod_{k=1}^K\Gamma(\alpha_k)}{\Gamma(\sum_{k=1}^K\alpha_k)}. \tag{2.90}$$

For a vector $\boldsymbol{\mu}$ in $K+1$ dimensions, let

$$\boldsymbol{\mu}' = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_{K-1} \\ \mu_K + \mu_{K+1} \end{bmatrix}. \tag{2.91}$$

Then,

$$\begin{aligned} &\int_{\substack{\sum_{k=1}^{K+1} \mu_k=1\\ \mu_k\geq 0}} \prod_{k=1}^{K+1} \mu_k^{\alpha_k-1}d\boldsymbol{\mu} \\ =&\int_{\substack{\sum_{k=1}^{K} \mu'_k=1\\ \mu'_k\geq 0}} \left(\prod_{k=1}^{K-1} {\mu'_k}^{\alpha_k-1}\right) I(\mu'_K)d\boldsymbol{\mu}', \end{aligned} \tag{2.92}$$

where

$$I(\mu'_K) = \int_0^{\mu'_K} (\mu'_K - \mu_{K+1})^{\alpha_K - 1} \mu_{K+1}^{\alpha_{K+1}-1} d\mu_{K+1}. \tag{2.93}$$

By the transformation

$$x = \frac{\mu_{K+1}}{\mu'_K}, \tag{2.94}$$

the expression of $I(\mu'_K)$ can be written as

$$\begin{aligned} &\int_0^1 (\mu'_K(1-x))^{\alpha_K - 1} (\mu'_K x)^{\alpha_{K+1}-1} \mu'_K dx \\ =&{\mu'}_K^{\alpha_K + \alpha_{K+1} - 1} \int_0^1 x^{\alpha_{K+1}-1} (1-x)^{\alpha_K - 1} dx. \end{aligned} \tag{2.95}$$

By 2.5, the integral of the right hand side can be written as

$$\frac{\Gamma(\alpha_K)\Gamma(\alpha_{K+1})}{\Gamma(\alpha_K + \alpha_{K+1})}. \tag{2.96}$$

Then,

$$\begin{aligned} &\int_{\substack{\sum_{k=1}^{K+1} \mu_k = 1 \\ \mu_k \geq 0}} \prod_{k=1}^{K+1} {\mu_k}^{\alpha_k - 1} d\boldsymbol{\mu} \\ =&\frac{\Gamma(\alpha_K)\Gamma(\alpha_{K+1})}{\Gamma(\alpha_K + \alpha_{K+1})} \int_{\substack{\sum_{k=1}^{K} \mu'_k = 1 \\ \mu'_k \geq 0}} \left( \prod_{k=1}^{K-1} {\mu'_k}^{\alpha_k - 1} \right) {\mu'}_K^{\alpha_K + \alpha_{K+1} - 1} d\boldsymbol{\mu}'. \end{aligned} \tag{2.97}$$

By 2.5, the integral of the right hand side can be written as

$$\frac{(\prod_{k=1}^{K-1} \Gamma(\alpha_k))\Gamma(\alpha_K + \alpha_{K+1})}{\Gamma(\sum_{k=1}^{K+1} \alpha_k)}. \tag{2.98}$$

Then,

$$\int_{\substack{\sum_{k=1}^{K+1} \mu_k = 1 \\ \mu_k \geq 0}} \prod_{k=1}^{K+1} {\mu_k}^{\alpha_k - 1} d\boldsymbol{\mu} = \frac{\prod_{k=1}^{K+1} \Gamma(\alpha_k)}{\Gamma(\sum_{k=1}^{K+1} \alpha_k)}. \tag{2.99}$$

Therefore, the assumption is proved by induction on $K$.

## 2.10

Let $\boldsymbol{\mu}$ be a variable such that

$$p(\boldsymbol{\mu}) = \frac{\Gamma(\sum_{k=1}^K \alpha_k)}{\prod_{k=1}^K \Gamma(\alpha_k)} \prod_{k=1}^K \mu_k^{\alpha_k - 1}. \tag{2.100}$$

We have

$$\begin{aligned} \mathrm{E}\, \mu_k &= \int \mu_k p(\boldsymbol{\mu}) d\boldsymbol{\mu}, \\ \mathrm{E}\, \mu_k^2 &= \int \mu_k^2 p(\boldsymbol{\mu}) d\boldsymbol{\mu}, \\ \mathrm{E}\, \mu_k \mu_{k'} &= \int \mu_k \mu_{k'} p(\boldsymbol{\mu}) d\boldsymbol{\mu}. \end{aligned} \tag{2.101}$$

Let $k \neq k'$. By 2.9, the right hand sides can be written as

$$\begin{aligned} \frac{\Gamma(\sum_{k=1}^K \alpha_k)}{\prod_{k=1}^K \Gamma(\alpha_k)} \frac{\frac{\Gamma(\alpha_k+1)}{\Gamma(\alpha_k)} \prod_{k=1}^K \Gamma(\alpha_k)}{\Gamma(\sum_{k=1}^K \alpha_k + 1)} &= \frac{\alpha_k}{\sum_{k=1}^K \alpha_k}, \\ \frac{\Gamma(\sum_{k=1}^K \alpha_k)}{\prod_{k=1}^K \Gamma(\alpha_k)} \frac{\frac{\Gamma(\alpha_k+2)}{\Gamma(\alpha_k)} \prod_{k=1}^K \Gamma(\alpha_k)}{\Gamma\left(\sum_{k=1}^K \alpha_k + 2\right)} &= \frac{\alpha_k(\alpha_k + 1)}{\sum_{k=1}^K \alpha_k (\sum_{k=1}^K \alpha_k + 1)}, \\ \frac{\Gamma\left(\sum_{k=1}^K \alpha_k\right)}{\prod_{k=1}^K \Gamma(\alpha_k)} \frac{\frac{\Gamma(\alpha_k+1)\Gamma(\alpha_{k'}+1)}{\Gamma(\alpha_k)\Gamma(\alpha_{k'})} \prod_{k=1}^K \Gamma(\alpha_k)}{\Gamma\left(\sum_{k=1}^K \alpha_k + 2\right)} &= \frac{\alpha_k \alpha_{k'}}{\sum_{k=1}^K \alpha_k (\sum_{k=1}^K \alpha_k + 1)}. \end{aligned} \tag{2.102}$$

Then,

$$\begin{aligned} \mathrm{E}\, \mu_k &= \frac{\alpha_k}{\sum_{k=1}^K \alpha_k}. \\ \mathrm{E}\, \mu_k^2 &= \frac{\alpha_k(\alpha_k + 1)}{\sum_{k=1}^K \alpha_k \left(\sum_{k=1}^K \alpha_k + 1\right)}, \\ \mathrm{E}\, \mu_k \mu_{k'} &= \frac{\alpha_k \alpha_{k'}}{\sum_{k=1}^K \alpha_k \left(\sum_{k=1}^K \alpha_k + 1\right)}. \end{aligned} \tag{2.103}$$

We have

$$\begin{aligned} \operatorname{var} \mu_k &= \mathrm{E}\, \mu_k^2 - (\mathrm{E}\, \mu_k)^2, \\ \operatorname{cov}(\mu_k, \mu_{k'}) &= \mathrm{E}\, \mu_k \mu_{k'} - \mathrm{E}\, \mu_k\, \mathrm{E}\, \mu_{k'}. \end{aligned} \tag{2.104}$$

Therefore,

$$\begin{aligned} \operatorname{var}\mu_k &= \frac{\alpha_k \left( \left( \sum_{k=1}^K \alpha_k \right) - \alpha_k \right)}{(\sum_{k=1}^K \alpha_k)^2 (\sum_{k=1}^K \alpha_k + 1)}, \\ \operatorname{cov}(\mu_k, \mu_{k'}) &= -\frac{\alpha_k \alpha_{k'}}{(\sum_{k=1}^K \alpha_k)^2 (\sum_{k=1}^K \alpha_k + 1)}. \end{aligned} \tag{2.105}$$

## 2.11

Let $\boldsymbol{\mu}$ be a variable such that

$$p(\boldsymbol{\mu}) = \frac{\Gamma\left(\sum_{k=1}^K \alpha_k\right)}{\prod_{k=1}^K \Gamma(\alpha_k)} \prod_{k=1}^K \mu_k^{\alpha_k - 1}. \tag{2.106}$$

We have

$$\operatorname{E}\ln\mu_k = \int (\ln\mu_k)\, p(\boldsymbol{\mu}) d\boldsymbol{\mu}. \tag{2.107}$$

We have

$$\frac{\partial}{\partial\alpha_k} p(\boldsymbol{\mu}) = \left( \frac{\Gamma'\left(\sum_{k=1}^K \alpha_k\right)}{\Gamma\left(\sum_{k=1}^K \alpha_k\right)} - \frac{\Gamma'(\alpha_k)}{\Gamma(\alpha_k)} + \ln\mu_k \right) p(\boldsymbol{\mu}). \tag{2.108}$$

Then,

$$\operatorname{E}\ln\mu_k = \frac{\partial}{\partial\alpha_k} \int p(\boldsymbol{\mu}) d\boldsymbol{\mu} + \left( \psi(\alpha_k) - \psi\left( \sum_{k=1}^K \alpha_k \right) \right) \int p(\boldsymbol{\mu}) d\boldsymbol{\mu}, \tag{2.109}$$

where

$$\psi(a) = \frac{d}{da} \ln\Gamma(a). \tag{2.110}$$

Therefore,

$$\operatorname{E}\ln\mu_k = \psi(\alpha_k) - \psi\left( \sum_{k=1}^K \alpha_k \right). \tag{2.111}$$

## 2.12

Let $x$ be a variable such that

$$p(x) = \frac{1}{b-a}, \tag{2.112}$$

where $a < b$. Then

$$\int_a^b p(x)dx = 1. \tag{2.113}$$

We have

$$\begin{aligned} \mathrm{E}\, x &= \frac{1}{b-a}\int_a^b x dx, \\ \mathrm{E}\, x^2 &= \frac{1}{b-a}\int_a^b x^2 dx. \end{aligned} \tag{2.114}$$

Then,

$$\begin{aligned} \mathrm{E}\, x &= \frac{1}{2}(a+b), \\ \mathrm{E}\, x^2 &= \frac{1}{3}\left(a^2+ab+b^2\right). \end{aligned} \tag{2.115}$$

We have

$$\operatorname{var} x = \mathrm{E}\, x^2 - (\mathrm{E}\, x)^2. \tag{2.116}$$

Therefore,

$$\operatorname{var} x = \frac{1}{12}(b-a)^2. \tag{2.117}$$

## 2.13

Let $\mathbf{x}$ be a variable in $D$ dimensions and let

$$\begin{aligned} p(\mathbf{x}) &= \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}), \\ q(\mathbf{x}) &= \mathcal{N}(\mathbf{x}|\mathbf{m}, \mathbf{L}). \end{aligned} \tag{2.118}$$

We have

$$\mathrm{KL}(p||q) = -\int \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) \ln \frac{\mathcal{N}(\mathbf{x}|\mathbf{m}, \mathbf{L})}{\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma})} d\mathbf{x}. \tag{2.119}$$

The logarithm in the right hand side can be written as

$$\begin{aligned} &\ln \frac{(2\pi)^{-\frac{D}{2}}|\det \mathbf{L}|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\mathbf{m})^\intercal \mathbf{L}^{-1}(\mathbf{x}-\mathbf{m})\right)}{(2\pi)^{-\frac{D}{2}}|\det \boldsymbol{\Sigma}|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\intercal \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)} \\ =&\frac{1}{2}\ln\left|\frac{\det \boldsymbol{\Sigma}}{\det \mathbf{L}}\right| + \frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\intercal \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) - \frac{1}{2}(\mathbf{x}-\mathbf{m})^\intercal \mathbf{L}^{-1}(\mathbf{x}-\mathbf{m}). \end{aligned} \tag{2.120}$$

Then, the integral can be written as

$$\frac{1}{2}\ln\left|\frac{\det\mathbf{\Sigma}}{\det\mathbf{L}}\right| + \frac{1}{2}I_1 - \frac{1}{2}I_2, \tag{2.121}$$

where

$$\begin{aligned} I_1 &= \int (\mathbf{x}-\boldsymbol{\mu})^\intercal \mathbf{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\mathcal{N}(\mathbf{x}|\boldsymbol{\mu},\mathbf{\Sigma})d\mathbf{x}, \\ I_2 &= \int (\mathbf{x}-\mathbf{m})^\intercal \mathbf{L}^{-1}(\mathbf{x}-\mathbf{m})\mathcal{N}(\mathbf{x}|\boldsymbol{\mu},\mathbf{\Sigma})d\mathbf{x}. \end{aligned} \tag{2.122}$$

We have

$$(\mathbf{x}-\mathbf{m})^\intercal \mathbf{L}^{-1}(\mathbf{x}-\mathbf{m}) = (\mathbf{x}-\boldsymbol{\mu}+\boldsymbol{\mu}-\mathbf{m})^\intercal \mathbf{L}^{-1}(\mathbf{x}-\boldsymbol{\mu}+\boldsymbol{\mu}-\mathbf{m}). \tag{2.123}$$

Then, the expression of $I_2$ can be written as

$$\begin{aligned} &\int (\mathbf{x}-\boldsymbol{\mu})^\intercal \mathbf{L}^{-1}(\mathbf{x}-\boldsymbol{\mu})\mathcal{N}(\mathbf{x}|\boldsymbol{\mu},\mathbf{\Sigma})d\mathbf{x} \\ &+ 2(\boldsymbol{\mu}-\mathbf{m})^\intercal \mathbf{L}^{-1}\int (\mathbf{x}-\boldsymbol{\mu})\mathcal{N}(\mathbf{x}|\boldsymbol{\mu},\mathbf{\Sigma})d\mathbf{x} \\ &+ (\boldsymbol{\mu}-\mathbf{m})^\intercal \mathbf{L}^{-1}(\boldsymbol{\mu}-\mathbf{m})\int \mathcal{N}(\mathbf{x}|\boldsymbol{\mu},\mathbf{\Sigma})d\mathbf{x} \\ =& I_2' + (\boldsymbol{\mu}-\mathbf{m})^\intercal \mathbf{L}^{-1}(\boldsymbol{\mu}-\mathbf{m}), \end{aligned} \tag{2.124}$$

where

$$I_2' = \int (\mathbf{x}-\boldsymbol{\mu})^\intercal \mathbf{L}^{-1}(\mathbf{x}-\boldsymbol{\mu})\mathcal{N}(\mathbf{x}|\boldsymbol{\mu},\mathbf{\Sigma})d\mathbf{x}. \tag{2.125}$$

We have

$$\int (\mathbf{x}-\boldsymbol{\mu})(\mathbf{x}-\boldsymbol{\mu})^\intercal \mathcal{N}(\mathbf{x}|\boldsymbol{\mu},\mathbf{\Sigma})d\mathbf{x} = \mathbf{\Sigma}, \tag{2.126}$$

so that

$$\int (\mathbf{x}-\boldsymbol{\mu})^\intercal(\mathbf{x}-\boldsymbol{\mu})\mathcal{N}(\mathbf{x}|\boldsymbol{\mu},\mathbf{\Sigma})d\mathbf{x} = \operatorname{tr}\mathbf{\Sigma}. \tag{2.127}$$

Then,

$$\begin{aligned} I_1 &= \operatorname{tr}\left(\mathbf{\Sigma}^{-1}\mathbf{\Sigma}\right), \\ I_2' &= \operatorname{tr}\left(\mathbf{L}^{-1}\mathbf{\Sigma}\right). \end{aligned} \tag{2.128}$$

Therefore,

$$\begin{aligned} &\mathrm{KL}(p||q) \\ =&\frac{1}{2}\left(\ln\left|\frac{\det\mathbf{L}}{\det\mathbf{\Sigma}}\right| - D + \operatorname{tr}\left(\mathbf{L}^{-1}\mathbf{\Sigma}\right) + (\boldsymbol{\mu}-\mathbf{m})^\intercal \mathbf{L}^{-1}(\boldsymbol{\mu}-\mathbf{m})\right). \end{aligned} \tag{2.129}$$

## 2.14

Let $\mathbf{x}$ be a variable in $D$ dimensions. We have

$$\mathrm{H}(\mathbf{x}) = -\int p(\mathbf{x}) \ln p(\mathbf{x}) d\mathbf{x}. \tag{2.130}$$

In order to maximise $\mathrm{H}(x)$ with the constratints

$$\begin{aligned} \int p(\mathbf{x}) d\mathbf{x} &= 1, \\ \int \mathbf{x} p(\mathbf{x}) d\mathbf{x} &= \boldsymbol{\mu}, \\ \int (\mathbf{x}-\boldsymbol{\mu})(\mathbf{x}-\boldsymbol{\mu})^\intercal p(\mathbf{x}) d\mathbf{x} &= \boldsymbol{\Sigma}, \end{aligned} \tag{2.131}$$

let

$$\begin{aligned} L(p) =& \mathrm{H}(\mathbf{x}) + \lambda \left( \int p(\mathbf{x}) d\mathbf{x} - 1 \right) + \mathbf{l}^\intercal \left( \int \mathbf{x} p(\mathbf{x}) d\mathbf{x} - \boldsymbol{\mu} \right) \\ &+ \mathbf{m}^\intercal \left( \int (\mathbf{x}-\boldsymbol{\mu})(\mathbf{x}-\boldsymbol{\mu})^\intercal p(\mathbf{x}) d\mathbf{x} - \boldsymbol{\Sigma} \right) \mathbf{m}. \end{aligned} \tag{2.132}$$

Setting the variation with respect to $p$ to zero gives

$$0 = -\ln p(\mathbf{x}) - 1 + \lambda + \mathbf{l}^\intercal \mathbf{x} + \|\mathbf{m}^\intercal(\mathbf{x}-\boldsymbol{\mu})\|^2. \tag{2.133}$$

Then,

$$p(\mathbf{x}) = \exp\left(-1 + \lambda + \mathbf{l}^\intercal \mathbf{x} + \|\mathbf{m}^\intercal(\mathbf{x}-\boldsymbol{\mu})\|^2\right). \tag{2.134}$$

We have

$$\begin{aligned} &\mathbf{l}^\intercal \mathbf{x} + \|\mathbf{m}^\intercal(\mathbf{x}-\boldsymbol{\mu})\|^2 \\ =& -(\mathbf{x}-\boldsymbol{\mu}-\mathbf{M}\mathbf{l})^\intercal \mathbf{M}^{-1} (\mathbf{x}-\boldsymbol{\mu}-\mathbf{M}\mathbf{l}) + \mathbf{l}^\intercal \mathbf{M}\mathbf{l} + \mathbf{l}^\intercal \boldsymbol{\mu}. \end{aligned} \tag{2.135}$$

where

$$\mathbf{M} = -\left(\mathbf{m}\mathbf{m}^\intercal\right)^{-1}. \tag{2.136}$$

Then,

$$p(\mathbf{x}) = c \exp\left(-(\mathbf{x}-\boldsymbol{\mu}-\mathbf{M}\mathbf{l})^\intercal \mathbf{M}^{-1} (\mathbf{x}-\boldsymbol{\mu}-\mathbf{M}\mathbf{l})\right), \tag{2.137}$$

where

$$c = \exp\left(-1 + \lambda + \mathbf{l}^\intercal \mathbf{M} \mathbf{l} + \mathbf{l}^\intercal \boldsymbol{\mu}\right). \tag{2.138}$$

Substituting it to the constraints and the transformation

$$\mathbf{y} = \mathbf{x} - \boldsymbol{\mu} - \mathbf{M}\mathbf{l} \tag{2.139}$$

gives

$$\begin{aligned} c\int \exp\left(-\mathbf{y}^\intercal \mathbf{M}^{-1}\mathbf{y}\right) d\mathbf{y} &= 1, \\ c\int \left(\mathbf{y} + \boldsymbol{\mu} + \mathbf{M}\mathbf{l}\right) \exp\left(-\mathbf{y}^\intercal \mathbf{M}^{-1}\mathbf{y}\right) d\mathbf{y} &= \boldsymbol{\mu}, \\ c\int \left(\mathbf{y} + \mathbf{M}\mathbf{l}\right)\left(\mathbf{y} + \mathbf{M}\mathbf{l}\right)^\intercal \exp\left(-\mathbf{y}^\intercal \mathbf{M}^{-1}\mathbf{y}\right) d\mathbf{y} &= \boldsymbol{\Sigma}. \end{aligned} \tag{2.140}$$

We have

$$\begin{aligned} \int \exp\left(-\mathbf{y}^\intercal \mathbf{y}\right) d\mathbf{y} &= \left(\Gamma\left(\frac{1}{2}\right)\right)^D, \\ \int \mathbf{y} \exp\left(-\mathbf{y}^\intercal \mathbf{y}\right) d\mathbf{y} &= \mathbf{0}, \\ \int \mathbf{y}\mathbf{y}^\intercal \exp\left(-\mathbf{y}^\intercal \mathbf{y}\right) d\mathbf{y} &= \Gamma\left(\frac{3}{2}\right)\left(\Gamma\left(\frac{1}{2}\right)\right)^{D-1} \mathbf{I}, \end{aligned} \tag{2.141}$$

so that

$$\begin{aligned} \int \exp\left(-\mathbf{y}^\intercal \mathbf{M}^{-1}\mathbf{y}\right) d\mathbf{y} &= \pi^{\frac{D}{2}} |\det \mathbf{M}|^{\frac{1}{2}}, \\ \int \mathbf{y} \exp\left(-\mathbf{y}^\intercal \mathbf{M}^{-1}\mathbf{y}\right) d\mathbf{y} &= \mathbf{0}, \\ \int \mathbf{y}\mathbf{y}^\intercal \exp\left(-\mathbf{y}^\intercal \mathbf{M}^{-1}\mathbf{y}\right) d\mathbf{y} &= \frac{1}{2}\pi^{\frac{D}{2}} (\det \mathbf{M})^{\frac{1}{2}} \mathbf{M}. \end{aligned} \tag{2.142}$$

Then,

$$\begin{aligned} c\pi^{\frac{D}{2}} |\det \mathbf{M}|^{\frac{1}{2}} &= 1, \\ c\pi^{\frac{D}{2}} |\det \mathbf{M}|^{\frac{1}{2}} (\boldsymbol{\mu} + \mathbf{M}\mathbf{l}) &= \boldsymbol{\mu}, \\ c\pi^{\frac{D}{2}} |\det \mathbf{M}|^{\frac{1}{2}} \left(\frac{1}{2}\mathbf{M} + \mathbf{M}\mathbf{l}(\mathbf{M}\mathbf{l})^\intercal\right) &= \boldsymbol{\Sigma}, \end{aligned} \tag{2.143}$$

so that

$$\begin{aligned} c &= (2\pi)^{-\frac{D}{2}} |\det \boldsymbol{\Sigma}|^{-\frac{1}{2}}, \\ \mathbf{l} &= \mathbf{0}, \\ \mathbf{M} &= 2\boldsymbol{\Sigma}. \end{aligned} \tag{2.144}$$

Therefore,

$$p(\mathbf{x}) = (2\pi)^{-\frac{D}{2}} |\det \mathbf{\Sigma}|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\intercal \mathbf{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right). \tag{2.145}$$

## 2.15

Let $\mathbf{x}$ be a variable in $D$ dimensions such that

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \mathbf{\Sigma}). \tag{2.146}$$

Then, by the definition, the entropy is given by

$$\mathrm{H}(\mathbf{x}) = -\int \mathcal{N}(\mathbf{x}|\mu, \mathbf{\Sigma}) \ln \mathcal{N}(\mathbf{x}|\mu, \mathbf{\Sigma}) d\mathbf{x}. \tag{2.147}$$

We have

$$\ln \mathcal{N}(\mathbf{x}|\mu, \mathbf{\Sigma}) = -\frac{D}{2}\ln(2\pi) - \frac{1}{2}\ln|\det \mathbf{\Sigma}| - \frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\intercal \mathbf{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}). \tag{2.148}$$

Then, the expression of $\mathrm{H}(\mathbf{x})$ can be written as

$$\begin{aligned} &\left(\frac{D}{2}\ln(2\pi) + \frac{1}{2}\ln|\det \mathbf{\Sigma}|\right)\int \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \mathbf{\Sigma}) d\mathbf{x} \\ &+ \frac{1}{2}\int (\mathbf{x}-\boldsymbol{\mu})^\intercal \mathbf{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \mathbf{\Sigma}) d\mathbf{x}. \end{aligned} \tag{2.149}$$

We have

$$\int (\mathbf{x}-\boldsymbol{\mu})(\mathbf{x}-\boldsymbol{\mu})^\intercal \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \mathbf{\Sigma}) d\mathbf{x} = \mathbf{\Sigma}, \tag{2.150}$$

so that

$$\int (\mathbf{x}-\boldsymbol{\mu})^\intercal (\mathbf{x}-\boldsymbol{\mu}) \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \mathbf{\Sigma}) d\mathbf{x} = \operatorname{tr} \mathbf{\Sigma}. \tag{2.151}$$

Then,

$$\int (\mathbf{x}-\boldsymbol{\mu})^\intercal \mathbf{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \mathbf{\Sigma}) d\mathbf{x} = \operatorname{tr}\left(\mathbf{\Sigma}^{-1}\mathbf{\Sigma}\right), \tag{2.152}$$

where the right hand side can be written as $D$. Therefore,

$$\mathrm{H}(\mathbf{x}) = \frac{D}{2}(1 + \ln(2\pi)) + \frac{1}{2}\ln|\det \mathbf{\Sigma}|. \tag{2.153}$$

## 2.16

Let $x$ be a variable such that

$$x = x_1 + x_2, \tag{2.154}$$

where

$$\begin{aligned} p(x_1) &= \mathcal{N}\left(x_1 | \mu_1, \tau_1^{-1}\right), \\ p(x_2) &= \mathcal{N}\left(x_2 | \mu_2, \tau_2^{-1}\right). \end{aligned} \tag{2.155}$$

By marginalisation,

$$p(x) = \int_{-\infty}^{\infty} p(x|x_2) p(x_2) dx_2. \tag{2.156}$$

The integrand of the right hand side can be written as

$$\mathcal{N}\left(x | \mu_1 + x_2, \tau_1^{-1}\right) \mathcal{N}\left(x_2 | \mu_2, \tau_2^{-1}\right). \tag{2.157}$$

The logarithm of the integrand except the terms independent of $x$ and $z$ is given by

$$-\frac{\tau_1}{2}(x - \mu_1 - x_2)^2 - \frac{\tau_2}{2}(x_2 - \mu_2)^2 = -\frac{1}{2}\mathbf{v}^\intercal \mathbf{M} \mathbf{v}, \tag{2.158}$$

where

$$\mathbf{v} = \begin{bmatrix} x_2 - \mu_2 \\ x - \mu_1 - \mu_2 \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \tau_1 + \tau_2 & -\tau_1 \\ -\tau_1 & \tau_1 \end{bmatrix}. \tag{2.159}$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix} \tau_2^{-1} & \tau_2^{-1} \\ \tau_2^{-1} & \tau_1^{-1} + \tau_2^{-1} \end{bmatrix}. \tag{2.160}$$

Then,

$$p(x) = \mathcal{N}\left(x | \mu_1 + \mu_2, \tau_1^{-1} + \tau_2^{-1}\right). \tag{2.161}$$

Therefore, by 1.35,

$$\mathrm{H}(x) = \frac{1}{2}\left(1 + \ln(2\pi) + \ln\left(\tau_1^{-1} + \tau_2^{-1}\right)\right). \tag{2.162}$$

## 2.17

Let $\boldsymbol{\Sigma}$ be a matrix and let

$$\begin{aligned}\mathbf{S} &= \frac{1}{2}\left(\boldsymbol{\Sigma}^{-1} + \left(\boldsymbol{\Sigma}^{-1}\right)^{\intercal}\right), \\ \mathbf{A} &= \frac{1}{2}\left(\boldsymbol{\Sigma}^{-1} - \left(\boldsymbol{\Sigma}^{-1}\right)^{\intercal}\right).\end{aligned} \tag{2.163}$$

Then,

$$\boldsymbol{\Sigma}^{-1} = \mathbf{S} + \mathbf{A}, \tag{2.164}$$

so that

$$(\mathbf{x}-\boldsymbol{\mu})^{\intercal}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) = (\mathbf{x}-\boldsymbol{\mu})^{\intercal}\mathbf{S}(\mathbf{x}-\boldsymbol{\mu}) + (\mathbf{x}-\boldsymbol{\mu})^{\intercal}\mathbf{A}(\mathbf{x}-\boldsymbol{\mu}). \tag{2.165}$$

The second term of the right hand side can be written as

$$\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\intercal}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) - \frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\intercal}\left(\boldsymbol{\Sigma}^{-1}\right)^{\intercal}(\mathbf{x}-\boldsymbol{\mu}). \tag{2.166}$$

The second term of the right hand side can be written as

$$-\frac{1}{2}\left(\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)^{\intercal}(\mathbf{x}-\boldsymbol{\mu}) = -\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\intercal}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}). \tag{2.167}$$

Then,

$$(\mathbf{x}-\boldsymbol{\mu})^{\intercal}\mathbf{A}(\mathbf{x}-\boldsymbol{\mu}) = 0. \tag{2.168}$$

Therefore,

$$(\mathbf{x}-\boldsymbol{\mu})^{\intercal}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) = (\mathbf{x}-\boldsymbol{\mu})^{\intercal}\mathbf{S}(\mathbf{x}-\boldsymbol{\mu}). \tag{2.169}$$

## 2.18

### (a)

Let $\boldsymbol{\Sigma}$ be a $D \times D$ real symmetric matrix such that

$$\boldsymbol{\Sigma}\mathbf{u}_d = \lambda_d \mathbf{u}_d, \tag{2.170}$$

where $\mathbf{u}_1, \cdots, \mathbf{u}_D$ are unit eigenvectors. Then,

$$\overline{\mathbf{u}_d}^{\intercal}\boldsymbol{\Sigma}\mathbf{u}_d = \lambda_d, \tag{2.171}$$

where $\overline{\mathbf{u}_d}$ is the conjugate of $\mathbf{u}_d$. Since $\mathbf{\Sigma}$ is real and symmetric, the left hand side can be written as

$$\overline{\mathbf{u}_d}^{\mathsf{T}}\overline{\mathbf{\Sigma}}^{\mathsf{T}}\mathbf{u}_d = \left(\overline{\mathbf{\Sigma}\mathbf{u}_d}\right)^{\mathsf{T}}\mathbf{u}_d. \tag{2.172}$$

The right hand side can be writtet as

$$\overline{\lambda_d}\,\overline{\mathbf{u}_d}^{\mathsf{T}}\mathbf{u}_d = \overline{\lambda_d}. \tag{2.173}$$

Therefore,

$$\lambda_d = \overline{\lambda_d}. \tag{2.174}$$

### (b)

For $d \neq d'$, taking the inner product with $\mathbf{u}_d'$ on both sides of

$$\mathbf{\Sigma}\mathbf{u}_d = \lambda_d\mathbf{u}_d \tag{2.175}$$

gives

$$\mathbf{u}_{d'}^{\mathsf{T}}\mathbf{\Sigma}\mathbf{u}_d = \lambda_d\mathbf{u}_{d'}^{\mathsf{T}}\mathbf{u}_d. \tag{2.176}$$

Since $\mathbf{\Sigma}$ is symmetric, the left hand side can be written as

$$\mathbf{u}_{d'}^{\mathsf{T}}\mathbf{\Sigma}^{\mathsf{T}}\mathbf{u}_d = \left(\mathbf{\Sigma}\mathbf{u}_{d'}\right)^{\mathsf{T}}\mathbf{u}_d, \tag{2.177}$$

where the right hand side can be written as $\lambda_{d'}\mathbf{u}_{d'}^{\mathsf{T}}\mathbf{u}_d$. Then,

$$\lambda_d\mathbf{u}_{d'}^{\mathsf{T}}\mathbf{u}_d = \lambda_{d'}\mathbf{u}_{d'}^{\mathsf{T}}\mathbf{u}_d. \tag{2.178}$$

Therefore, if $\lambda_d \neq \lambda_{d'}$, then

$$\mathbf{u}_{d'}^{\mathsf{T}}\mathbf{u}_d = 0. \tag{2.179}$$

## 2.19

Let $\mathbf{\Sigma}$ be a $D \times D$ real symmetric matrix such that

$$\mathbf{\Sigma}\mathbf{u}_d = \lambda_d\mathbf{u}_d, \tag{2.180}$$

where $\mathbf{u}_1, \cdots, \mathbf{u}_D$ are unit eigenvectors. Let

$$\begin{aligned}\mathbf{\Lambda} &= \mathrm{diag}(\lambda_1, \cdots, \lambda_D),\\ \mathbf{U} &= [\mathbf{u}_1 \cdots \mathbf{u}_D].\end{aligned} \tag{2.181}$$

Then,

$$\mathbf{\Sigma U} = \mathbf{U\Lambda}. \tag{2.182}$$

We have

$$\mathbf{U}^\intercal \mathbf{U} = \mathbf{I}. \tag{2.183}$$

Then,

$$\mathbf{\Sigma} = \mathbf{U\Lambda U}^\intercal, \tag{2.184}$$

so that

$$\mathbf{\Sigma}^{-1} = \mathbf{U\Lambda}^{-1}\mathbf{U}^\intercal. \tag{2.185}$$

Therefore,

$$\begin{aligned} \mathbf{\Sigma} &= \sum_{d=1}^{D} \lambda_d \mathbf{u}_d \mathbf{u}_d^\intercal, \\ \mathbf{\Sigma}^{-1} &= \sum_{d=1}^{D} \frac{1}{\lambda_d} \mathbf{u}_d \mathbf{u}_d^\intercal. \end{aligned} \tag{2.186}$$

## 2.20

Let $\mathbf{\Sigma}$ be a $D \times D$ real symmetric matrix such that

$$\mathbf{\Sigma u}_d = \lambda_d \mathbf{u}_d, \tag{2.187}$$

where $\mathbf{u}_1, \cdots, \mathbf{u}_D$ are unit eigenvectors. Let

$$\begin{aligned} \mathbf{\Lambda} &= \mathrm{diag}(\lambda_1, \cdots, \lambda_D), \\ \mathbf{U} &= [\mathbf{u}_1 \cdots \mathbf{u}_D]. \end{aligned} \tag{2.188}$$

By 2.19,

$$\mathbf{\Sigma} = \mathbf{U\Lambda U}^\intercal. \tag{2.189}$$

Then,

$$\mathbf{v}^\intercal \mathbf{\Sigma v} = \mathbf{w}^\intercal \mathbf{\Lambda w}, \tag{2.190}$$

where

$$\mathbf{w} = \mathbf{U}^\intercal \mathbf{v}. \tag{2.191}$$

The right hand side can be written as $\sum_{d=1}^{D} \lambda_d w_d^2$. Therefore, the necessary and sufficient condition for

$$\mathbf{v}^\intercal \mathbf{\Sigma v} > 0 \tag{2.192}$$

for any real vector $\mathbf{v}$ is

$$\lambda_d > 0. \tag{2.193}$$

## 2.21

Let $\mathbf{\Sigma}$ be a $D \times D$ real symmetric matrix. Then, the number of independent parameters is $\frac{D(D+1)}{2}$.

## 2.22

Let $\mathbf{\Sigma}$ be a $D \times D$ symmetric matrix such that

$$\mathbf{\Sigma\Lambda} = \mathbf{I}. \tag{2.194}$$

Taking the transpose of the both sides gives

$$\mathbf{\Lambda}^\intercal \mathbf{\Sigma} = \mathbf{I}. \tag{2.195}$$

Therefore,

$$\mathbf{\Lambda}^\intercal = \mathbf{\Lambda}. \tag{2.196}$$

## 2.23

Let $\mathbf{\Sigma}$ be a $D \times D$ real symmetric matrix such that

$$\mathbf{\Sigma u}_d = \lambda_d \mathbf{u}_d, \tag{2.197}$$

where $\mathbf{u}_1, \cdots, \mathbf{u}_D$ are unit eigenvectors. Let

$$\begin{aligned} \mathbf{\Lambda}' &= \operatorname{diag}\left(\lambda_1^{-\frac{1}{2}}, \cdots, \lambda_D^{-\frac{1}{2}}\right), \\ \mathbf{U} &= [\mathbf{u}_1 \cdots \mathbf{u}_D]. \end{aligned} \tag{2.198}$$

By 2.19,

$$\mathbf{\Sigma} = \mathbf{U}\mathbf{\Lambda}'^{-2}\mathbf{U}^\intercal. \tag{2.199}$$

Then,

$$\int_{(\mathbf{x}-\boldsymbol{\mu})^\intercal \mathbf{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})=\Delta} d\mathbf{x} = \int_{(\mathbf{x}-\boldsymbol{\mu})^\intercal \mathbf{U}\mathbf{\Lambda}'^2\mathbf{U}^\intercal(\mathbf{x}-\boldsymbol{\mu})=\Delta} d\mathbf{x}. \tag{2.200}$$

By the transformation

$$\mathbf{y} = \mathbf{\Lambda}'\mathbf{U}^\intercal(\mathbf{x}-\boldsymbol{\mu}), \tag{2.201}$$

the right hand side can be written as

$$\int_{\|\mathbf{y}\|^2=\Delta} \left|\det\left(\mathbf{U}\boldsymbol{\Lambda}'^{-1}\right)\right| d\mathbf{y}. \tag{2.202}$$

We have

$$\left|\det\left(\mathbf{U}\boldsymbol{\Lambda}'^{-1}\right)\right| = |\det \boldsymbol{\Sigma}|^{\frac{1}{2}}. \tag{2.203}$$

Therefore,

$$\int_{(\mathbf{x}-\boldsymbol{\mu})^\intercal \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})=\Delta} d\mathbf{x} = |\det \boldsymbol{\Sigma}|^{\frac{1}{2}} \Delta^D V_D, \tag{2.204}$$

where

$$V_D = \int_{\|\mathbf{x}\|=1} d\mathbf{x}. \tag{2.205}$$

## 2.24

Let $\mathbf{A}$ be a square matrix and $\mathbf{D}$ be an invertible matrix. We have

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix} = \begin{bmatrix} \mathbf{I} & \mathbf{B}\mathbf{D}^{-1} \\ \mathbf{O} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{A}-\mathbf{B}\mathbf{D}^{-1}\mathbf{C} & \mathbf{O} \\ \mathbf{C} & \mathbf{D} \end{bmatrix}. \tag{2.206}$$

The right hand side can be written as

$$\begin{bmatrix} \mathbf{I} & \mathbf{B}\mathbf{D}^{-1} \\ \mathbf{O} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{A}-\mathbf{B}\mathbf{D}^{-1}\mathbf{C} & \mathbf{O} \\ \mathbf{O} & \mathbf{D} \end{bmatrix} \begin{bmatrix} \mathbf{I} & \mathbf{O} \\ \mathbf{D}^{-1}\mathbf{C} & \mathbf{I} \end{bmatrix}. \tag{2.207}$$

Then,

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{I} & \mathbf{O} \\ -\mathbf{D}^{-1}\mathbf{C} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{M} & \mathbf{O} \\ \mathbf{O} & \mathbf{D}^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{I} & -\mathbf{B}\mathbf{D}^{-1} \\ \mathbf{O} & \mathbf{I} \end{bmatrix}, \tag{2.208}$$

where

$$\mathbf{M} = \left(\mathbf{A}-\mathbf{B}\mathbf{D}^{-1}\mathbf{C}\right)^{-1}. \tag{2.209}$$

Therefore,

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{M} & -\mathbf{B}\mathbf{D}^{-1}\mathbf{M} \\ -\mathbf{D}^{-1}\mathbf{C}\mathbf{M} & \mathbf{D}^{-1}+\mathbf{D}^{-1}\mathbf{C}\mathbf{M}\mathbf{B}\mathbf{D}^{-1} \end{bmatrix}. \tag{2.210}$$

## 2.25

Let $\mathbf{x}$ be a variable such that

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}), \tag{2.211}$$

where

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_a \\ \mathbf{x}_b \\ \mathbf{x}_c \end{bmatrix}, \boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_a \\ \boldsymbol{\mu}_b \\ \boldsymbol{\mu}_c \end{bmatrix}, \boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{aa} & \boldsymbol{\Sigma}_{ab} & \boldsymbol{\Sigma}_{ac} \\ \boldsymbol{\Sigma}_{ba} & \boldsymbol{\Sigma}_{bb} & \boldsymbol{\Sigma}_{bc} \\ \boldsymbol{\Sigma}_{ca} & \boldsymbol{\Sigma}_{cb} & \boldsymbol{\Sigma}_{cc} \end{bmatrix}. \tag{2.212}$$

Let

$$\boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1}, \tag{2.213}$$

where

$$\boldsymbol{\Lambda} = \begin{bmatrix} \boldsymbol{\Lambda}_{aa} & \boldsymbol{\Lambda}_{ab} & \boldsymbol{\Lambda}_{ac} \\ \boldsymbol{\Lambda}_{ba} & \boldsymbol{\Lambda}_{bb} & \boldsymbol{\Lambda}_{bc} \\ \boldsymbol{\Lambda}_{ca} & \boldsymbol{\Lambda}_{cb} & \boldsymbol{\Lambda}_{cc} \end{bmatrix}. \tag{2.214}$$

By marginalisation,

$$p(\mathbf{x}_a|\mathbf{x}_b) = \int p(\mathbf{x}_a|\mathbf{x}_b, \mathbf{x}_c) p(\mathbf{x}_c) d\mathbf{x}_c. \tag{2.215}$$

The logarithm of $p(\mathbf{x})$ except the terms independent of $\mathbf{x}_a$ can be written as

$$\begin{aligned} & -\frac{1}{2}(\mathbf{x}_a - \boldsymbol{\mu}_a)^\intercal \boldsymbol{\Lambda}_{aa}(\mathbf{x}_a - \boldsymbol{\mu}_a) - \frac{1}{2}(\mathbf{x}_a - \boldsymbol{\mu}_a)^\intercal \boldsymbol{\Lambda}_{ab}(\mathbf{x}_b - \boldsymbol{\mu}_b) \\ & -\frac{1}{2}(\mathbf{x}_a - \boldsymbol{\mu}_a)^\intercal \boldsymbol{\Lambda}_{ac}(\mathbf{x}_c - \boldsymbol{\mu}_c) - \frac{1}{2}(\mathbf{x}_b - \boldsymbol{\mu}_b)^\intercal \boldsymbol{\Lambda}_{ba}(\mathbf{x}_a - \boldsymbol{\mu}_a) \\ & -\frac{1}{2}(\mathbf{x}_c - \boldsymbol{\mu}_c)^\intercal \boldsymbol{\Lambda}_{ca}(\mathbf{x}_a - \boldsymbol{\mu}_a) \\ = & -\frac{1}{2}(\mathbf{x}_a - \boldsymbol{\mu}_{a|b,c})^\intercal \boldsymbol{\Lambda}_{aa}(\mathbf{x}_a - \boldsymbol{\mu}_{a|b,c}) + \frac{1}{2}\boldsymbol{\mu}_{a|b,c}^\intercal \boldsymbol{\Lambda}_{aa} \boldsymbol{\mu}_{a|b,c}, \end{aligned} \tag{2.216}$$

where

$$\boldsymbol{\mu}_{a|b,c} = \boldsymbol{\mu}_a - \boldsymbol{\Lambda}_{aa}^{-1}\boldsymbol{\Lambda}_{ab}(\mathbf{x}_b - \boldsymbol{\mu}_b) - \boldsymbol{\Lambda}_{aa}^{-1}\boldsymbol{\Lambda}_{ac}(\mathbf{x}_c - \boldsymbol{\mu}_c). \tag{2.217}$$

Then,

$$\begin{aligned} p(\mathbf{x}_a|\mathbf{x}_b, \mathbf{x}_c) &= \mathcal{N}\left(\mathbf{x}_a|\boldsymbol{\mu}_{a|b,c}, \boldsymbol{\Lambda}_{aa}^{-1}\right), \\ p(\mathbf{x}_c) &= \mathcal{N}\left(\mathbf{x}_c|\boldsymbol{\mu}_c, \boldsymbol{\Lambda}_{cc}^{-1}\right), \end{aligned} \tag{2.218}$$

Then, the logarithm of the integrand except the terms independent of $\mathbf{x}_a$, $\mathbf{x}_b$ and $\mathbf{x}_c$ can be written as

$$\begin{aligned} &-\frac{1}{2}(\mathbf{x}_a-\boldsymbol{\mu}_{a|b,c})^\intercal\boldsymbol{\Lambda}_{aa}(\mathbf{x}_a-\boldsymbol{\mu}_{a|b,c})-\frac{1}{2}(\mathbf{x}_c-\boldsymbol{\mu}_c)^\intercal\boldsymbol{\Lambda}_{cc}(\mathbf{x}_c-\boldsymbol{\mu}_c) \\ =&-\frac{1}{2}\mathbf{v}^\intercal\mathbf{M}\mathbf{v}, \end{aligned} \tag{2.219}$$

where

$$\begin{aligned} \mathbf{v}&=\begin{bmatrix}\mathbf{x}_c-\boldsymbol{\mu}_c \\ \mathbf{x}_a-\boldsymbol{\mu}_a+\boldsymbol{\Lambda}_{aa}^{-1}\boldsymbol{\Lambda}_{ab}(\mathbf{x}_b-\boldsymbol{\mu}_b)\end{bmatrix}, \\ \mathbf{M}&=\begin{bmatrix}\boldsymbol{\Lambda}_{ca}\boldsymbol{\Lambda}_{aa}^{-1}\boldsymbol{\Lambda}_{ac}+\boldsymbol{\Lambda}_{cc} & \boldsymbol{\Lambda}_{ca} \\ \boldsymbol{\Lambda}_{ac} & \boldsymbol{\Lambda}_{aa}\end{bmatrix}. \end{aligned} \tag{2.220}$$

By 2.24,

$$\mathbf{M}^{-1}=\begin{bmatrix}\boldsymbol{\Lambda}_{cc}^{-1} & -\boldsymbol{\Lambda}_{cc}^{-1}\boldsymbol{\Lambda}_{ca}\boldsymbol{\Lambda}_{aa}^{-1} \\ -\boldsymbol{\Lambda}_{aa}^{-1}\boldsymbol{\Lambda}_{ac}\boldsymbol{\Lambda}_{cc}^{-1} & \boldsymbol{\Lambda}_{aa}^{-1}+\boldsymbol{\Lambda}_{aa}^{-1}\boldsymbol{\Lambda}_{ac}\boldsymbol{\Lambda}_{cc}^{-1}\boldsymbol{\Lambda}_{ca}\boldsymbol{\Lambda}_{aa}^{-1}\end{bmatrix}. \tag{2.221}$$

Therefore,

$$p(\mathbf{x}_a|\mathbf{x}_b)=\mathcal{N}\left(\mathbf{x}_a|\boldsymbol{\mu}_{a|b},\boldsymbol{\Sigma}_{a|b}\right), \tag{2.222}$$

where

$$\begin{aligned} \boldsymbol{\mu}_{a|b}&=\boldsymbol{\mu}_a-\boldsymbol{\Lambda}_{aa}^{-1}\boldsymbol{\Lambda}_{ab}(\mathbf{x}_b-\boldsymbol{\mu}_b), \\ \boldsymbol{\Sigma}_{a|b}&=\boldsymbol{\Lambda}_{aa}^{-1}+\boldsymbol{\Lambda}_{aa}^{-1}\boldsymbol{\Lambda}_{ac}\boldsymbol{\Lambda}_{cc}^{-1}\boldsymbol{\Lambda}_{ca}\boldsymbol{\Lambda}_{aa}^{-1}. \end{aligned} \tag{2.223}$$

## 2.26

Let $\mathbf{A}$ be a square matrix and $\mathbf{C}$ be an invertible matrix. By 2.24,

$$\begin{bmatrix}\mathbf{A} & \mathbf{B} \\ \mathbf{D} & -\mathbf{C}^{-1}\end{bmatrix}^{-1}=\begin{bmatrix}\mathbf{M} & \mathbf{MBC} \\ \mathbf{CDM} & -\mathbf{C}+\mathbf{CDMBC}\end{bmatrix}, \tag{2.224}$$

where

$$\mathbf{M}=(\mathbf{A}+\mathbf{BCD})^{-1}. \tag{2.225}$$

By 2.24,

$$\begin{bmatrix}-\mathbf{C}^{-1} & \mathbf{D} \\ \mathbf{B} & \mathbf{A}\end{bmatrix}^{-1}=\begin{bmatrix}-\mathbf{N} & \mathbf{NDA}^{-1} \\ \mathbf{A}^{-1}\mathbf{BN} & \mathbf{A}^{-1}-\mathbf{A}^{-1}\mathbf{BNDA}^{-1}\end{bmatrix}, \tag{2.226}$$

where

$$\mathbf{N}=\left(\mathbf{C}^{-1}+\mathbf{DA}^{-1}\mathbf{B}\right)^{-1}. \tag{2.227}$$

Therefore,

$$\mathbf{M}=\mathbf{A}^{-1}-\mathbf{A}^{-1}\mathbf{BNDA}^{-1}. \tag{2.228}$$

## 2.27

### (a)

Let $\mathbf{x}$ and $\mathbf{z}$ be two variables. We have

$$\mathrm{E}(\mathbf{x}+\mathbf{z}) = \iint (\mathbf{x}+\mathbf{z})p(\mathbf{x},\mathbf{z})d\mathbf{x}d\mathbf{z}. \tag{2.229}$$

The right hand side can be written as

$$\begin{aligned} &\int \mathbf{x}\left(\int p(\mathbf{x},\mathbf{z})d\mathbf{z}\right)d\mathbf{x} + \int \mathbf{z}\left(\int p(\mathbf{x},\mathbf{z})d\mathbf{x}\right)d\mathbf{z} \\ =&\int \mathbf{x}p(\mathbf{x})d\mathbf{x} + \int \mathbf{z}p(\mathbf{z})d\mathbf{z}. \end{aligned} \tag{2.230}$$

Therefore,

$$\mathrm{E}(\mathbf{x}+\mathbf{z}) = \mathrm{E}\,\mathbf{x} + \mathrm{E}\,\mathbf{z}. \tag{2.231}$$

### (b)

Let $\mathbf{x}$ and $\mathbf{z}$ be two independent variables in the same dimension. We have

$$\begin{aligned} &\mathrm{var}(\mathbf{x}+\mathbf{z}) \\ =&\iint (\mathbf{x}+\mathbf{z}-\mathrm{E}(\mathbf{x}+\mathbf{z}))(\mathbf{x}+\mathbf{z}-\mathrm{E}(\mathbf{x}+\mathbf{z}))^\intercal p(\mathbf{x},\mathbf{z})d\mathbf{x}d\mathbf{z}. \end{aligned} \tag{2.232}$$

The right hand side can be written as $I_1 + I_2 + I_3 + I_4$, where

$$\begin{aligned} I_1 &= \iint (\mathbf{x}-\mathrm{E}\,\mathbf{x})(\mathbf{x}-\mathrm{E}\,\mathbf{x})^\intercal p(\mathbf{x},\mathbf{z})d\mathbf{x}d\mathbf{z}, \\ I_2 &= \iint (\mathbf{x}-\mathrm{E}\,\mathbf{x})(\mathbf{z}-\mathrm{E}\,\mathbf{z})^\intercal p(\mathbf{x},\mathbf{z})d\mathbf{x}d\mathbf{z}, \\ I_3 &= \iint (\mathbf{z}-\mathrm{E}\,\mathbf{z})(\mathbf{x}-\mathrm{E}\,\mathbf{x})^\intercal p(\mathbf{x},\mathbf{z})d\mathbf{x}d\mathbf{z}, \\ I_4 &= \iint (\mathbf{z}-\mathrm{E}\,\mathbf{z})(\mathbf{z}-\mathrm{E}\,\mathbf{z})^\intercal p(\mathbf{x},\mathbf{z})d\mathbf{x}d\mathbf{z}. \end{aligned} \tag{2.233}$$

The expression of $I_1$ can be written as

$$\begin{aligned} &\int (\mathbf{x}-\mathrm{E}\,\mathbf{x})(\mathbf{x}-\mathrm{E}\,\mathbf{x})^\intercal \left(\int p(\mathbf{x},\mathbf{z})d\mathbf{z}\right)d\mathbf{x} \\ =&\int (\mathbf{x}-\mathrm{E}\,\mathbf{x})(\mathbf{x}-\mathrm{E}\,\mathbf{x})^\intercal p(\mathbf{x})d\mathbf{x}. \end{aligned} \tag{2.234}$$

The expression of $I_2$ can be written as

$$\int (\mathbf{x} - \mathrm{E}\,\mathbf{x}) p(\mathbf{x}) d\mathbf{x} \left( \int (\mathbf{z} - \mathrm{E}\,\mathbf{z}) p(\mathbf{z}) d\mathbf{z} \right)^{\intercal} = \mathbf{O}. \tag{2.235}$$

The expression of $I_3$ can be written as

$$\int (\mathbf{z} - \mathrm{E}\,\mathbf{z}) p(\mathbf{z}) d\mathbf{z} \left( \int (\mathbf{x} - \mathrm{E}\,\mathbf{x}) p(\mathbf{x}) d\mathbf{x} \right)^{\intercal} = \mathbf{O}. \tag{2.236}$$

The expression of $I_4$ can be written as

$$\begin{aligned} &\int (\mathbf{z} - \mathrm{E}\,\mathbf{z}) (\mathbf{z} - \mathrm{E}\,\mathbf{z})^{\intercal} \left( \int p(\mathbf{x}, \mathbf{z}) d\mathbf{x} \right) d\mathbf{z} \\ = &\int (\mathbf{z} - \mathrm{E}\,\mathbf{z}) (\mathbf{z} - \mathrm{E}\,\mathbf{z})^{\intercal} p(\mathbf{z}) d\mathbf{z}. \end{aligned} \tag{2.237}$$

Therefore,

$$\operatorname{var}(\mathbf{x} + \mathbf{z}) = \operatorname{var}\mathbf{x} + \operatorname{var}\mathbf{z}. \tag{2.238}$$

## 2.28

Let $\mathbf{x}$ and $\mathbf{y}$ be variables such that

$$\begin{aligned} p(\mathbf{x}) &= \mathcal{N}\left(\mathbf{x} | \boldsymbol{\mu}, \boldsymbol{\Lambda}^{-1}\right), \\ p(\mathbf{y}) &= \mathcal{N}\left(\mathbf{y} | \mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{L}^{-1} + \mathbf{A}\boldsymbol{\Lambda}^{-1}\mathbf{A}^{\intercal}\right), \\ \operatorname{cov}(\mathbf{x}, \mathbf{y}) &= \boldsymbol{\Lambda}^{-1}\mathbf{A}^{\intercal}. \end{aligned} \tag{2.239}$$

Then,

$$p(\mathbf{z}) = \mathcal{N}(\mathbf{z} | \mathbf{m}, \mathbf{S}), \tag{2.240}$$

where

$$\mathbf{z} = \begin{bmatrix} \mathbf{y} \\ \mathbf{x} \end{bmatrix}, \mathbf{m} = \begin{bmatrix} \mathbf{A}\boldsymbol{\mu} + \mathbf{b} \\ \boldsymbol{\mu} \end{bmatrix}, \mathbf{S} = \begin{bmatrix} \mathbf{L}^{-1} + \mathbf{A}\boldsymbol{\Lambda}^{-1}\mathbf{A}^{\intercal} & \mathbf{A}\boldsymbol{\Lambda}^{-1} \\ \boldsymbol{\Lambda}^{-1}\mathbf{A}^{\intercal} & \boldsymbol{\Lambda}^{-1} \end{bmatrix}. \tag{2.241}$$

By 2.24,

$$\mathbf{S}^{-1} = \begin{bmatrix} \mathbf{L} & -\mathbf{L}\mathbf{A} \\ -\mathbf{A}^{\intercal}\mathbf{L} & \mathbf{M} \end{bmatrix}, \tag{2.242}$$

where

$$\mathbf{M} = \boldsymbol{\Lambda} + \mathbf{A}^{\intercal}\mathbf{L}\mathbf{A}. \tag{2.243}$$

We have

$$\mathbf{A}\boldsymbol{\mu} + \mathbf{b} + \mathbf{A}(\mathbf{x} - \boldsymbol{\mu}) = \mathbf{A}\mathbf{x} + \mathbf{b}, \tag{2.244}$$

and

$$\boldsymbol{\mu} + \mathbf{M}^{-1}\mathbf{A}^\intercal\mathbf{L}(\mathbf{y} - \mathbf{A}\boldsymbol{\mu} - \mathbf{b}) = \mathbf{M}^{-1}\left(\boldsymbol{\Lambda}\boldsymbol{\mu} + \mathbf{A}^\intercal\mathbf{L}(\mathbf{y} - \mathbf{b})\right). \tag{2.245}$$

Therefore,

$$\begin{aligned} p(\mathbf{y}|\mathbf{x}) &= \mathcal{N}\left(\mathbf{y}|\mathbf{A}\mathbf{x} + \mathbf{b}, \mathbf{L}^{-1}\right), \\ p(\mathbf{x}|\mathbf{y}) &= \mathcal{N}\left(\mathbf{x}|\mathbf{M}^{-1}\left(\boldsymbol{\Lambda}\boldsymbol{\mu} + \mathbf{A}^\intercal\mathbf{L}(\mathbf{y} - \mathbf{b})\right), \mathbf{M}^{-1}\right). \end{aligned} \tag{2.246}$$

## 2.29

Let

$$\mathbf{R} = \begin{bmatrix} \boldsymbol{\Lambda} + \mathbf{A}^\intercal\mathbf{L}\mathbf{A} & -\mathbf{A}^\intercal\mathbf{L} \\ -\mathbf{L}\mathbf{A} & \mathbf{L} \end{bmatrix}. \tag{2.247}$$

By 2.24,

$$\mathbf{R}^{-1} = \begin{bmatrix} \boldsymbol{\Lambda}^{-1} & \boldsymbol{\Lambda}^{-1}\mathbf{A}^\intercal \\ \mathbf{A}\boldsymbol{\Lambda}^{-1} & \mathbf{L}^{-1} + \mathbf{A}\boldsymbol{\Lambda}^{-1}\mathbf{A}^\intercal \end{bmatrix}, \tag{2.248}$$

so that

$$\mathbf{R}^{-1}\begin{bmatrix} \boldsymbol{\Lambda}\boldsymbol{\mu} - \mathbf{A}^\intercal\mathbf{L}\mathbf{b} \\ \mathbf{L}\mathbf{b} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\mu} \\ \mathbf{A}\boldsymbol{\mu} + \mathbf{b} \end{bmatrix}. \tag{2.249}$$

## 2.30

Refer to 2.29.

## 2.31

Let $\mathbf{y}$ be a variable such that

$$\mathbf{y} = \mathbf{x} + \mathbf{z}, \tag{2.250}$$

where

$$\begin{aligned} p(\mathbf{x}) &= \mathcal{N}\left(\mathbf{x}|\boldsymbol{\mu}_{\mathbf{x}}, \boldsymbol{\Sigma}_{\mathbf{x}}\right), \\ p(\mathbf{z}) &= \mathcal{N}\left(\mathbf{z}|\boldsymbol{\mu}_{\mathbf{z}}, \boldsymbol{\Sigma}_{\mathbf{z}}\right). \end{aligned} \tag{2.251}$$

By marginalisation,

$$p(\mathbf{y}) = \int p(\mathbf{y}|\mathbf{x})p(\mathbf{x})d\mathbf{x}. \tag{2.252}$$

The right hand side can be written as

$$\int \mathcal{N}\left(\mathbf{y}|\mathbf{x}+\boldsymbol{\mu}_{\mathbf{z}}, \boldsymbol{\Sigma}_{\mathbf{z}}\right) \mathcal{N}\left(\mathbf{x}|\boldsymbol{\mu}_{\mathbf{x}}, \boldsymbol{\Sigma}_{\mathbf{x}}\right) d\mathbf{x}. \tag{2.253}$$

The logarithm of the integrand except the terms independent of $\mathbf{x}$ and $\mathbf{y}$ is given by

$$\begin{aligned} &-\frac{1}{2}\left(\mathbf{y}-\mathbf{x}-\boldsymbol{\mu}_{\mathbf{z}}\right)^{\intercal} \boldsymbol{\Sigma}_{\mathbf{z}}^{-1}\left(\mathbf{y}-\mathbf{x}-\boldsymbol{\mu}_{\mathbf{z}}\right)-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_{\mathbf{x}})^{\intercal}\boldsymbol{\Sigma}_{\mathbf{x}}^{-1}(\mathbf{x}-\boldsymbol{\mu}_{\mathbf{x}}) \\ =&-\frac{1}{2}\mathbf{u}^{\intercal}\mathbf{M}\mathbf{u}+\mathbf{u}^{\intercal}\mathbf{v}-\frac{1}{2}\boldsymbol{\mu}_{\mathbf{z}}^{\intercal}\boldsymbol{\Sigma}_{\mathbf{z}}^{-1}\boldsymbol{\mu}_{\mathbf{z}}-\frac{1}{2}\boldsymbol{\mu}_{\mathbf{x}}^{\intercal}\boldsymbol{\Sigma}_{\mathbf{x}}^{-1}\boldsymbol{\mu}_{\mathbf{x}}, \end{aligned} \tag{2.254}$$

where

$$\mathbf{u}=\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix}, \mathbf{M}=\begin{bmatrix}\boldsymbol{\Sigma}_{\mathbf{x}}^{-1}+\boldsymbol{\Sigma}_{\mathbf{z}}^{-1} & -\boldsymbol{\Sigma}_{\mathbf{z}}^{-1}\\ -\boldsymbol{\Sigma}_{\mathbf{z}}^{-1} & \boldsymbol{\Sigma}_{\mathbf{z}}^{-1}\end{bmatrix}, \mathbf{v}=\begin{bmatrix}\boldsymbol{\Sigma}_{\mathbf{x}}^{-1}\boldsymbol{\mu}_{\mathbf{x}}-\boldsymbol{\Sigma}_{\mathbf{z}}^{-1}\boldsymbol{\mu}_{\mathbf{z}}\\ \boldsymbol{\Sigma}_{\mathbf{z}}^{-1}\boldsymbol{\mu}_{\mathbf{z}}\end{bmatrix}. \tag{2.255}$$

By 2.24,

$$\mathbf{M}^{-1}=\begin{bmatrix}\boldsymbol{\Sigma}_{\mathbf{x}} & \boldsymbol{\Sigma}_{\mathbf{x}}\\ \boldsymbol{\Sigma}_{\mathbf{x}} & \boldsymbol{\Sigma}_{\mathbf{x}}+\boldsymbol{\Sigma}_{\mathbf{z}}\end{bmatrix}, \tag{2.256}$$

so that

$$\mathbf{M}^{-1}\mathbf{v}=\begin{bmatrix}\boldsymbol{\mu}_{\mathbf{x}}\\ \boldsymbol{\mu}_{\mathbf{x}}+\boldsymbol{\mu}_{\mathbf{z}}\end{bmatrix}. \tag{2.257}$$

Therefore,

$$p(\mathbf{y})=\mathcal{N}\left(\mathbf{y}|\boldsymbol{\mu}_{\mathbf{x}}+\boldsymbol{\mu}_{\mathbf{z}}, \boldsymbol{\Sigma}_{\mathbf{x}}+\boldsymbol{\Sigma}_{\mathbf{z}}\right). \tag{2.258}$$

## 2.32

Let $\mathbf{x}$ and $\mathbf{y}$ be variables such that

$$\begin{aligned} p(\mathbf{x})&=\mathcal{N}\left(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Lambda}^{-1}\right), \\ p(\mathbf{y}|\mathbf{x})&=\mathcal{N}\left(\mathbf{y}|\mathbf{A}\mathbf{x}+\mathbf{b}, \mathbf{L}^{-1}\right). \end{aligned} \tag{2.259}$$

By Bayes' theorem,

$$p(\mathbf{x}|\mathbf{y})p(\mathbf{y})=p(\mathbf{y}|\mathbf{x})p(\mathbf{x}). \tag{2.260}$$

The logarithm of the right hand side except the terms independent of $\mathbf{x}$ and $\mathbf{y}$ can be written as

$$\begin{aligned} &-\frac{1}{2}\left(\mathbf{y}-\mathbf{A}\mathbf{x}-\mathbf{b}\right)^{\intercal}\mathbf{L}\left(\mathbf{y}-\mathbf{A}\mathbf{x}-\mathbf{b}\right)-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\intercal}\boldsymbol{\Lambda}(\mathbf{x}-\boldsymbol{\mu}) \\ =&-\frac{1}{2}\left(\mathbf{y}-\mathbf{A}(\mathbf{x}-\boldsymbol{\mu})-\mathbf{A}\boldsymbol{\mu}-\mathbf{b}\right)^{\intercal}\mathbf{L}\left(\mathbf{y}-\mathbf{A}(\mathbf{x}-\boldsymbol{\mu})-\mathbf{A}\boldsymbol{\mu}-\mathbf{b}\right) \\ &-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\intercal}\boldsymbol{\Lambda}(\mathbf{x}-\boldsymbol{\mu}). \end{aligned} \tag{2.261}$$

The right hand side can be written as

$$\begin{aligned}
&-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\intercal\left(\boldsymbol{\Lambda}+\mathbf{A}^\intercal\mathbf{L}\mathbf{A}\right)(\mathbf{x}-\boldsymbol{\mu})+(\mathbf{x}-\boldsymbol{\mu})^\intercal\mathbf{A}^\intercal\mathbf{L}(\mathbf{y}-\mathbf{A}\boldsymbol{\mu}-\mathbf{b})\\
&-\frac{1}{2}(\mathbf{y}-\mathbf{A}\boldsymbol{\mu}-\mathbf{b})^\intercal\mathbf{L}(\mathbf{y}-\mathbf{A}\boldsymbol{\mu}-\mathbf{b})\\
=&-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}-\mathbf{z})^\intercal\mathbf{M}(\mathbf{x}-\boldsymbol{\mu}-\mathbf{z})+\frac{1}{2}\mathbf{z}^\intercal\mathbf{M}\mathbf{z}\\
&-\frac{1}{2}(\mathbf{y}-\mathbf{A}\boldsymbol{\mu}-\mathbf{b})^\intercal\mathbf{L}(\mathbf{y}-\mathbf{A}\boldsymbol{\mu}-\mathbf{b}),
\end{aligned} \tag{2.262}$$

where

$$\begin{aligned}
\mathbf{z}&=\mathbf{M}^{-1}\mathbf{A}^\intercal\mathbf{L}(\mathbf{y}-\mathbf{A}\boldsymbol{\mu}-\mathbf{b}),\\
\mathbf{M}&=\boldsymbol{\Lambda}+\mathbf{A}^\intercal\mathbf{L}\mathbf{A}.
\end{aligned} \tag{2.263}$$

The right hand side can be written as

$$-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}-\mathbf{z})^\intercal\mathbf{M}(\mathbf{x}-\boldsymbol{\mu}-\mathbf{z})-\frac{1}{2}(\mathbf{y}-\mathbf{A}\boldsymbol{\mu}-\mathbf{b})^\intercal\mathbf{N}(\mathbf{y}-\mathbf{A}\boldsymbol{\mu}-\mathbf{b}), \tag{2.264}$$

where

$$\mathbf{N}=\mathbf{L}-\mathbf{L}\mathbf{A}\mathbf{M}^{-1}\mathbf{A}^\intercal\mathbf{L}. \tag{2.265}$$

Then,

$$\begin{aligned}
p(\mathbf{x}|\mathbf{y})&=\mathcal{N}\left(\mathbf{x}|\boldsymbol{\mu}+\mathbf{z},\mathbf{M}^{-1}\right),\\
p(\mathbf{y})&=\mathcal{N}\left(\mathbf{y}|\mathbf{A}\boldsymbol{\mu}+\mathbf{b},\mathbf{N}^{-1}\right).
\end{aligned} \tag{2.266}$$

We have

$$\boldsymbol{\mu}+\mathbf{z}=\mathbf{M}^{-1}\left(\mathbf{M}\boldsymbol{\mu}+\mathbf{A}^\intercal\mathbf{L}(\mathbf{y}-\mathbf{A}\boldsymbol{\mu}-\mathbf{b})\right). \tag{2.267}$$

The right hand side can be written as

$$\mathbf{M}^{-1}\left(\boldsymbol{\Lambda}\boldsymbol{\mu}+\mathbf{A}^\intercal\mathbf{L}(\mathbf{y}-\mathbf{b})\right). \tag{2.268}$$

By 2.26,

$$\mathbf{N}^{-1}=\mathbf{L}^{-1}+\mathbf{A}\boldsymbol{\Lambda}^{-1}\mathbf{A}^\intercal. \tag{2.269}$$

Therefore,

$$\begin{aligned}
p(\mathbf{x}|\mathbf{y})&=\mathcal{N}\left(\mathbf{x}|\mathbf{M}^{-1}\left(\boldsymbol{\Lambda}\boldsymbol{\mu}+\mathbf{A}^\intercal\mathbf{L}(\mathbf{y}-\mathbf{b})\right),\mathbf{M}^{-1}\right),\\
p(\mathbf{y})&=\mathcal{N}\left(\mathbf{y}|\mathbf{A}\boldsymbol{\mu}+\mathbf{b},\mathbf{L}^{-1}+\mathbf{A}\boldsymbol{\Lambda}^{-1}\mathbf{A}^\intercal\right).
\end{aligned} \tag{2.270}$$

## 2.33

Refer to 2.32, while a different approach is presented below.

Let $\mathbf{x}$ and $\mathbf{y}$ be variables such that

$$
\begin{aligned}
p(\mathbf{x}) &= \mathcal{N}\left(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Lambda}^{-1}\right), \\
p(\mathbf{y}|\mathbf{x}) &= \mathcal{N}\left(\mathbf{y}|\mathbf{A}\mathbf{x} + \mathbf{b}, \mathbf{L}^{-1}\right).
\end{aligned} \tag{2.271}
$$

By Bayes' theorem,

$$
p(\mathbf{x}|\mathbf{y})p(\mathbf{y}) = p(\mathbf{y}|\mathbf{x})p(\mathbf{x}). \tag{2.272}
$$

The logarithm of the right hand side except the terms independent of $\mathbf{x}$ and $\mathbf{y}$ can be written as

$$
\begin{aligned}
&-\frac{1}{2}\left(\mathbf{y} - \mathbf{A}\mathbf{x} - \mathbf{b}\right)^{\intercal} \mathbf{L}\left(\mathbf{y} - \mathbf{A}\mathbf{x} - \mathbf{b}\right) - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^{\intercal}\boldsymbol{\Lambda}(\mathbf{x} - \boldsymbol{\mu}) \\
=& -\frac{1}{2}\mathbf{v}^{\intercal}\mathbf{S}^{-1}\mathbf{v} + \mathbf{v}^{\intercal}\mathbf{u} - \frac{1}{2}\mathbf{b}^{\intercal}\mathbf{L}\mathbf{b} - \frac{1}{2}\boldsymbol{\mu}^{\intercal}\boldsymbol{\Lambda}\boldsymbol{\mu},
\end{aligned} \tag{2.273}
$$

where

$$
\mathbf{v} = \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}, \mathbf{S}^{-1} = \begin{bmatrix} \mathbf{M} & -\mathbf{A}^{\intercal}\mathbf{L} \\ -\mathbf{L}\mathbf{A} & \mathbf{L} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} -\mathbf{A}^{\intercal}\mathbf{L}\mathbf{b} + \boldsymbol{\Lambda}\boldsymbol{\mu} \\ \mathbf{L}\mathbf{b} \end{bmatrix}, \tag{2.274}
$$

and

$$
\mathbf{M} = \boldsymbol{\Lambda} + \mathbf{A}^{\intercal}\mathbf{L}\mathbf{A}. \tag{2.275}
$$

We have

$$
\mathbf{M}^{-1}\left(-\mathbf{A}^{\intercal}\mathbf{L}\mathbf{b} + \boldsymbol{\Lambda}\boldsymbol{\mu} + \mathbf{A}^{\intercal}\mathbf{L}\mathbf{y}\right) = \mathbf{M}^{-1}\left(\boldsymbol{\Lambda}\boldsymbol{\mu} + \mathbf{A}^{\intercal}\mathbf{L}(\mathbf{y} - \mathbf{b})\right). \tag{2.276}
$$

Then,

$$
p(\mathbf{x}|\mathbf{y}) = \mathcal{N}\left(\mathbf{x}|\mathbf{M}^{-1}\left(\boldsymbol{\Lambda}\boldsymbol{\mu} + \mathbf{A}^{\intercal}\mathbf{L}(\mathbf{y} - \mathbf{b})\right), \mathbf{M}^{-1}\right). \tag{2.277}
$$

By 2.24,

$$
\mathbf{S} = \begin{bmatrix} \boldsymbol{\Lambda}^{-1} & \boldsymbol{\Lambda}^{-1}\mathbf{A}^{\intercal} \\ \mathbf{A}\boldsymbol{\Lambda}^{-1} & \mathbf{L}^{-1} + \mathbf{A}\boldsymbol{\Lambda}^{-1}\mathbf{A}^{\intercal} \end{bmatrix}, \tag{2.278}
$$

so that

$$
\mathbf{S}\mathbf{u} = \begin{bmatrix} \boldsymbol{\mu} \\ \mathbf{A}\boldsymbol{\mu} + \mathbf{b} \end{bmatrix}. \tag{2.279}
$$

Then,

$$
p(\mathbf{y}) = \mathcal{N}\left(\mathbf{y}|\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{L}^{-1} + \mathbf{A}\boldsymbol{\Lambda}^{-1}\mathbf{A}^{\intercal}\right). \tag{2.280}
$$

## 2.34

Let $\mathbf{x}_1, \cdots, \mathbf{x}_N$ be independent variables such that

$$p(\mathbf{x}_n) = \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}, \boldsymbol{\Sigma}). \tag{2.281}$$

Then,

$$\ln p(\mathbf{X}) = -\frac{ND}{2}\ln(2\pi) - \frac{N}{2}\ln|\det\boldsymbol{\Sigma}| - \frac{1}{2}\sum_{n=1}^{N}(\mathbf{x}_n - \boldsymbol{\mu})^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}_n - \boldsymbol{\mu}). \tag{2.282}$$

By 3.21, setting the derivatives of $\ln p(\mathbf{X})$ with respect to $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ to zero gives

$$\begin{aligned}
\mathbf{0} &= \sum_{n=1}^{N}\left(\boldsymbol{\Sigma}^{-1} + \left(\boldsymbol{\Sigma}^{-1}\right)^\intercal\right)(\mathbf{x}_n - \boldsymbol{\mu}), \\
\mathbf{O} &= -\frac{N}{2}\left(\boldsymbol{\Sigma}^{-1}\right)^\intercal + \frac{1}{2}\left(\boldsymbol{\Sigma}^{-1}\right)^2\sum_{n=1}^{N}(\mathbf{x}_n - \boldsymbol{\mu})(\mathbf{x}_n - \boldsymbol{\mu})^\intercal.
\end{aligned} \tag{2.283}$$

Therefore, the maximum likelihood solutions to $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are given by

$$\begin{aligned}
\boldsymbol{\mu}_{\mathrm{ML}} &= \frac{1}{N}\sum_{n=1}^{N}\mathbf{x}_n, \\
\boldsymbol{\Sigma}_{\mathrm{ML}} &= \frac{1}{N}\sum_{n=1}^{N}(\mathbf{x}_n - \boldsymbol{\mu}_{\mathrm{ML}})(\mathbf{x}_n - \boldsymbol{\mu}_{\mathrm{ML}})^\intercal.
\end{aligned} \tag{2.284}$$

## 2.35

### (a)

Let $\mathbf{x}$ be a variable such that

$$p(\mathbf{x}) = \mathcal{N}\left(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}\right). \tag{2.285}$$

Then,

$$\mathrm{E}\,\mathbf{x}\mathbf{x}^\intercal = \mathrm{E}(\mathbf{x} - \boldsymbol{\mu} + \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu} + \boldsymbol{\mu})^\intercal. \tag{2.286}$$

The right hand side can be written as

$$\mathrm{E}(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\intercal + \boldsymbol{\mu}\,\mathrm{E}(\mathbf{x} - \boldsymbol{\mu})^\intercal + \mathrm{E}(\mathbf{x} - \boldsymbol{\mu})\boldsymbol{\mu}^\intercal + \boldsymbol{\mu}\boldsymbol{\mu}^\intercal. \tag{2.287}$$

We have

$$\begin{aligned} \mathrm{E}\,\mathbf{x} &= \boldsymbol{\mu}, \\ \operatorname{var}\mathbf{x} &= \boldsymbol{\Sigma}. \end{aligned} \tag{2.288}$$

Therefore,

$$\mathrm{E}\,\mathbf{x}\mathbf{x}^\intercal = \boldsymbol{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^\intercal. \tag{2.289}$$

**(b)**

Let $\mathbf{x}_n$ and $\mathbf{x}_m$ be independent variables such that

$$\begin{aligned} p(\mathbf{x}_n) &= \mathcal{N}\left(\mathbf{x_n}|\boldsymbol{\mu}, \boldsymbol{\Sigma}\right), \\ p(\mathbf{x}_m) &= \mathcal{N}\left(\mathbf{x_m}|\boldsymbol{\mu}, \boldsymbol{\Sigma}\right). \end{aligned} \tag{2.290}$$

If $n \neq m$, then

$$\mathrm{E}\,\mathbf{x}_n\mathbf{x}_m^\intercal = \mathrm{E}\,\mathbf{x}_n\,\mathrm{E}\,\mathbf{x}_m^\intercal. \tag{2.291}$$

Therefore, by (a),

$$\mathrm{E}\,\mathbf{x}_n\mathbf{x}_m^\intercal = I_{nm}\boldsymbol{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^\intercal. \tag{2.292}$$

**(c)**

Let $\mathbf{x}_1, \cdots, \mathbf{x}_N$ be independent variables such that

$$p(\mathbf{x}_n) = \mathcal{N}\left(\mathbf{x}_n|\boldsymbol{\mu}, \boldsymbol{\Sigma}\right). \tag{2.293}$$

By 2.34, the maximum likelihood solutions to $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are given by

$$\begin{aligned} \boldsymbol{\mu}_{\mathrm{ML}} &= \frac{1}{N}\sum_{n=1}^{N}\mathbf{x}_n, \\ \boldsymbol{\Sigma}_{\mathrm{ML}} &= \frac{1}{N}\sum_{n=1}^{N}(\mathbf{x}_n - \boldsymbol{\mu}_{\mathrm{ML}})(\mathbf{x}_n - \boldsymbol{\mu}_{\mathrm{ML}})^\intercal. \end{aligned} \tag{2.294}$$

Then,

$$\mathrm{E}\,\boldsymbol{\Sigma}_{\mathrm{ML}} = \frac{1}{N}\sum_{n=1}^{N}\mathrm{E}(\mathbf{x}_n - \boldsymbol{\mu}_{\mathrm{ML}})(\mathbf{x}_n - \boldsymbol{\mu}_{\mathrm{ML}})^\intercal. \tag{2.295}$$

The right hand side can be written as

$$\frac{1}{N}S_1 - \frac{1}{N^2}S_2 - \frac{1}{N^2}S_3 + \frac{1}{N^3}S_4, \tag{2.296}$$

where

$$\begin{aligned}
S_1 &= \sum_{n=1}^{N} \mathrm{E}\, \mathbf{x}_n \mathbf{x}_n^\intercal, \\
S_2 &= \sum_{n=1}^{N} \mathrm{E} \left( \sum_{n'=1}^{N} \mathbf{x}'_{\mathbf{n}} \right) \mathbf{x}_n^\intercal, \\
S_3 &= \sum_{n=1}^{N} \mathrm{E}\, \mathbf{x}_n \left( \sum_{n'=1}^{N} \mathbf{x}'_{\mathbf{n}} \right)^\intercal, \\
S_4 &= \sum_{n=1}^{N} \mathrm{E} \left( \sum_{n'=1}^{N} \mathbf{x}'_{\mathbf{n}} \right) \left( \sum_{n''=1}^{N} \mathbf{x}''_{\mathbf{n}} \right)^\intercal .
\end{aligned} \tag{2.297}$$

By (b),

$$S_1 = N \left( \mathbf{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^\intercal \right). \tag{2.298}$$

By (b),

$$S_2 = S_3 = (\mathbf{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^\intercal) + (N-1)\boldsymbol{\mu}\boldsymbol{\mu}^\intercal. \tag{2.299}$$

The right hand side can be written as

$$N\mathbf{\Sigma} + N^2 \boldsymbol{\mu}\boldsymbol{\mu}^\intercal. \tag{2.300}$$

By (b),

$$S_4 = N \left( N \left( \mathbf{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^\intercal \right) + N(N-1)\boldsymbol{\mu}\boldsymbol{\mu}^\intercal \right). \tag{2.301}$$

The right hand side can be written as

$$N^2\mathbf{\Sigma} + N^3 \boldsymbol{\mu}\boldsymbol{\mu}^\intercal. \tag{2.302}$$

Then,

$$\mathrm{E}\, \mathbf{\Sigma}_{\mathrm{ML}} = \mathbf{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^\intercal - 2 \left( \frac{1}{N} \mathbf{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^\intercal \right) + \frac{1}{N} \mathbf{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^\intercal. \tag{2.303}$$

Therefore,

$$\mathrm{E}\, \mathbf{\Sigma}_{\mathrm{ML}} = \frac{N-1}{N} \mathbf{\Sigma}. \tag{2.304}$$

## 2.36

Let $x_1, \cdots, x_N$ be independent variables such that

$$p(x_n) = \mathcal{N}\left(x_n|\mu, \sigma^2\right). \tag{2.305}$$

where $\mu$ is known. By 2.34, the maximum likelihood solution to $\sigma^2$ is given by

$$\sigma^2_{\mathrm{ML}}{}^{(N)} = \frac{1}{N}\sum_{n=1}^{N}(x_n - \mu)^2. \tag{2.306}$$

The right hand side can be written as

$$\frac{1}{N}(x_N - \mu)^2 + \frac{1}{N}\sum_{n=1}^{N-1}(x_n - \mu)^2 = \frac{1}{N}(x_N - \mu)^2 + \frac{N-1}{N}\sigma^2_{\mathrm{ML}}{}^{(N-1)}. \tag{2.307}$$

Then,

$$\sigma^2_{\mathrm{ML}}{}^{(N)} = \sigma^2_{\mathrm{ML}}{}^{(N-1)} + \frac{1}{N}\left((x_N - \mu)^2 - \sigma^2_{\mathrm{ML}}{}^{(N-1)}\right). \tag{2.308}$$

We have

$$\frac{\partial}{\partial \sigma^2}\left(-\ln p(x_n)\right) = \frac{1}{2\sigma^2} - \frac{1}{2\left(\sigma^2\right)^2}(x_n - \mu)^2. \tag{2.309}$$

Therefore,

$$\sigma^2_{\mathrm{ML}}{}^{(N)} = \sigma^2_{\mathrm{ML}}{}^{(N-1)} - \frac{2\left(\sigma^2_{\mathrm{ML}}{}^{(N-1)}\right)^2}{N}\delta, \tag{2.310}$$

where

$$\delta = \left.\left(\frac{\partial}{\partial \sigma^2}\left(-\ln p(x_N)\right)\right)\right|_{\sigma^2 = \sigma^2_{\mathrm{ML}}{}^{(N-1)}}. \tag{2.311}$$

## 2.37

Let $\mathbf{x}_1, \cdots, \mathbf{x}_N$ be independent variables such that

$$p(\mathbf{x}_n) = \mathcal{N}\left(\mathbf{x}_n|\boldsymbol{\mu}, \boldsymbol{\Sigma}\right), \tag{2.312}$$

where $\boldsymbol{\mu}$ is known. By 2.34, the maximum likelihood solution to $\boldsymbol{\Sigma}$ is given by

$$\boldsymbol{\Sigma}^{(N)}_{\mathrm{ML}} = \frac{1}{N}\sum_{n=1}^{N}(\mathbf{x}_n - \boldsymbol{\mu})(\mathbf{x}_n - \boldsymbol{\mu})^{\intercal}. \tag{2.313}$$

The right hand side can be written as

$$\begin{aligned}
&\frac{1}{N}(\mathbf{x}_N-\boldsymbol{\mu})(\mathbf{x}_N-\boldsymbol{\mu})^\intercal+\frac{1}{N}\sum_{n=1}^{N-1}(\mathbf{x}_n-\boldsymbol{\mu})(\mathbf{x}_n-\boldsymbol{\mu})^\intercal \\
=&\frac{1}{N}(\mathbf{x}_N-\boldsymbol{\mu})(\mathbf{x}_N-\boldsymbol{\mu})^\intercal+\frac{N-1}{N}\boldsymbol{\Sigma}_{\mathrm{ML}}^{(N-1)}.
\end{aligned} \tag{2.314}$$

Then,

$$\boldsymbol{\Sigma}_{\mathrm{ML}}^{(N)}=\boldsymbol{\Sigma}_{\mathrm{ML}}^{(N-1)}+\frac{1}{N}\left((\mathbf{x}_N-\boldsymbol{\mu})(\mathbf{x}_N-\boldsymbol{\mu})^\intercal-\boldsymbol{\Sigma}_{\mathrm{ML}}^{(N-1)}\right). \tag{2.315}$$

By 3.21, we have

$$\frac{\partial}{\partial\boldsymbol{\Sigma}}\left(-\ln p(x_n)\right)=-\frac{1}{2}\left(\boldsymbol{\Sigma}^{-1}\right)^\intercal+\frac{1}{2}\left(\boldsymbol{\Sigma}^{-1}\right)^2(\mathbf{x}_N-\boldsymbol{\mu})(\mathbf{x}_N-\boldsymbol{\mu})^\intercal. \tag{2.316}$$

Therefore,

$$\boldsymbol{\Sigma}_{\mathrm{ML}}^{(N)}=\boldsymbol{\Sigma}_{\mathrm{ML}}^{(N-1)}-\frac{\left(\boldsymbol{\Sigma}_{\mathrm{ML}}^{(N-1)}\right)^2}{N}\left(\frac{\partial}{\partial\boldsymbol{\Sigma}}\left(-\ln p\left(\mathbf{x}_N\right)\right)\right)\Bigg|_{\boldsymbol{\Sigma}=\boldsymbol{\Sigma}_{\mathrm{ML}}^{(N-1)}}. \tag{2.317}$$

## 2.38

Let $x_1,\cdots,x_N$ be conditionally independent variables such that

$$\begin{aligned}
p(x_n|\mu)&=\mathcal{N}\left(x_n|\mu,\sigma^2\right), \\
p(\mu)&=\mathcal{N}\left(\mu|\mu_0,\sigma_0^2\right).
\end{aligned} \tag{2.318}$$

By Bayes' theorem,

$$p(\mu|\mathbf{x})p(\mathbf{x})=p(\mathbf{x}|\mu)p(\mu). \tag{2.319}$$

The logarithm of the right hand side excpt the terms independent of $\mathbf{x}$ and $\mu$ can be written as

$$-\frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2-\frac{1}{2\sigma_0^2}(\mu-\mu_0)^2=-\frac{1}{2}\mathbf{v}^\intercal\mathbf{M}\mathbf{v}+\mathbf{v}^\intercal\mathbf{u}-\frac{\mu_0^2}{2\sigma_0^2}, \tag{2.320}$$

where

$$\mathbf{v}=\begin{bmatrix}\mu\\ \mathbf{x}\end{bmatrix},\mathbf{M}=\begin{bmatrix}s^{-2} & -\sigma^{-2}\mathbf{1}^\intercal\\ -\sigma^{-2}\mathbf{1} & \sigma^{-2}\mathbf{I}\end{bmatrix},\mathbf{u}=\begin{bmatrix}\sigma_0^{-2}\mu_0\\ \mathbf{0}\end{bmatrix}, \tag{2.321}$$

and

$$s^2 = \left(N\sigma^{-2} + \sigma_0^{-2}\right)^{-1}. \tag{2.322}$$

We have

$$s^2 \left(\sigma_0^{-2}\mu_0 + \sigma^{-2}\mathbf{1}^\intercal\mathbf{x}\right) = m, \tag{2.323}$$

where

$$\begin{aligned} m &= s^2 \left(N\sigma^{-2}\bar{x} + \sigma_0^{-2}\mu_0\right), \\ \bar{x} &= \frac{1}{N}\sum_{n=1}^{N} x_n. \end{aligned} \tag{2.324}$$

Therefore,

$$p(\mu|\mathbf{x}) = \mathcal{N}(\mu|m, s^2). \tag{2.325}$$

## 2.39

Let $x_1, \cdots, x_N$ be conditionally independent variables such that

$$\begin{aligned} p(x_n|\mu) &= \mathcal{N}\left(x_n|\mu, \sigma^2\right), \\ p(\mu) &= \mathcal{N}\left(\mu|\mu_0, \sigma_0^2\right). \end{aligned} \tag{2.326}$$

### (a)

By 2.38,

$$p(\mu|\mathbf{x}) = \mathcal{N}\left(\mu|m, s^2\right), \tag{2.327}$$

where

$$\begin{aligned} m &= s^2 \left(N\sigma^{-2}\bar{x} + \sigma_0^{-2}\mu_0\right), \\ s^2 &= \left(N\sigma^{-2} + \sigma_0^{-2}\right)^{-1}, \end{aligned} \tag{2.328}$$

where

$$\bar{x} = \frac{1}{N}\sum_{n=1}^{N} x_n. \tag{2.329}$$

Let

$$p(\mu|\mathbf{x}') = \mathcal{N}(\mu|m', s'^2). \tag{2.330}$$

where

$$\mathbf{x}' = \begin{bmatrix} x_1 \\ \vdots \\ x_{N-1} \end{bmatrix}. \tag{2.331}$$

We have

$$s^{-2}m = \sigma^{-2}\sum_{n=1}^{N} x_n + \sigma_0^{-2}\mu_0, \tag{2.332}$$

so that

$$s'^{-2}m' = \sigma^{-2}\sum_{n=1}^{N-1} x_n + \sigma_0^{-2}\mu_0. \tag{2.333}$$

Therefore,

$$s^{-2}m = s'^{-2}m' + \sigma^{-2}x_N. \tag{2.334}$$

We have

$$s^{-2} = (N-1)\sigma^{-2} + \sigma_0^{-2} + \sigma^2. \tag{2.335}$$

Therefore,

$$s^{-2} = s'^{-2} + \sigma^{-2}. \tag{2.336}$$

**(b)**

By Bayes' theorem,

$$p(\mu|\mathbf{x})p(\mathbf{x}) = p(\mathbf{x}|\mu)p(\mu). \tag{2.337}$$

We have

$$\begin{aligned} p(\mathbf{x}) &= p(x_N)p(\mathbf{x}'), \\ p(\mathbf{x}|\mu) &= p(x_N|\mu)p(\mathbf{x}'|\mu), \end{aligned} \tag{2.338}$$

where

$$\mathbf{x}' = \begin{bmatrix} x_1 \\ \vdots \\ x_{N-1} \end{bmatrix}. \tag{2.339}$$

Then,

$$p(\mu|\mathbf{x})p(x_N)p(\mathbf{x}') = p(x_N|\mu)p(\mathbf{x}'|\mu)p(\mu). \tag{2.340}$$

By Bayes' theorem,

$$p(\mathbf{x}'|\mu)p(\mu) = p(\mu|\mathbf{x}')p(\mathbf{x}'). \tag{2.341}$$

Then,

$$p(\mu|\mathbf{x})p(x_N) = p(\mu|\mathbf{x}')p(x_N|\mu). \tag{2.342}$$

Let

$$p(\mu|\mathbf{x}') = \mathcal{N}(\mu|m', s'^2). \tag{2.343}$$

Then, the logarithm of the right hand side except the terms independent of $\mu$ or $x_N$ can be written as

$$-\frac{1}{2s'^2}(\mu - m')^2 - \frac{1}{2\sigma^2}(x_N - \mu)^2 = -\frac{1}{2}\mathbf{v}^\intercal\mathbf{M}\mathbf{v} + \mathbf{v}^\intercal\mathbf{u} - \frac{m'^2}{2s'^2}, \tag{2.344}$$

where

$$\mathbf{v} = \begin{bmatrix} \mu \\ x_N \end{bmatrix}, \mathbf{M} = \begin{bmatrix} s^{-2} & -\sigma^{-2} \\ -\sigma^{-2} & \sigma^{-2} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} s'^{-2}m' \\ 0 \end{bmatrix}, \tag{2.345}$$

and

$$s^2 = \left(s'^{-2} + \sigma^{-2}\right)^{-1}. \tag{2.346}$$

Then,

$$p(\mu|\mathbf{x}) = \mathcal{N}\left(\mu|m, s^2\right), \tag{2.347}$$

where

$$m = s^2\left(s'^{-2}m' + \sigma^{-2}x_N\right). \tag{2.348}$$

Therefore,

$$\begin{aligned} s^{-2}m &= s'^{-2}m' + \sigma^{-2}x_N, \\ s^{-2} &= s'^2 + \sigma^{-2}. \end{aligned} \tag{2.349}$$

## 2.40

Let $\mathbf{x}_1, \cdots, \mathbf{x}_N$ be conditionally independent variables such that

$$\begin{aligned} p(\mathbf{x}_n|\boldsymbol{\mu}) &= \mathcal{N}\left(\mathbf{x}_n|\boldsymbol{\mu}, \boldsymbol{\Sigma}\right), \\ p(\boldsymbol{\mu}) &= \mathcal{N}\left(\boldsymbol{\mu}|\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0\right). \end{aligned} \tag{2.350}$$

By Bayes' theorem,

$$p(\boldsymbol{\mu}|\mathbf{X})p(\mathbf{X}) = p(\mathbf{X}|\boldsymbol{\mu})p(\boldsymbol{\mu}). \tag{2.351}$$

The logarithm of the right hand side excpt the terms independent of $\mathbf{X}$ and $\boldsymbol{\mu}$ can be written as

$$-\frac{1}{2}\sum_{n=1}^{N}(\mathbf{x}_n - \boldsymbol{\mu})^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}_n - \boldsymbol{\mu}) - \frac{1}{2}(\boldsymbol{\mu} - \boldsymbol{\mu}_0)^\intercal\boldsymbol{\Sigma}_0^{-1}(\boldsymbol{\mu} - \boldsymbol{\mu}_0). \tag{2.352}$$

We have

$$\begin{aligned}&\sum_{n=1}^{N}(\mathbf{x}_n-\boldsymbol{\mu})^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}_n-\boldsymbol{\mu})\\ =&\sum_{n=1}^{N}(\mathbf{x}_n-\bar{\mathbf{x}})^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}_n-\bar{\mathbf{x}})+N(\bar{\mathbf{x}}-\boldsymbol{\mu})^\intercal\boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}}-\boldsymbol{\mu}).\end{aligned} \tag{2.353}$$

where

$$\bar{\mathbf{x}}=\frac{1}{N}\sum_{n=1}^{N}\mathbf{x}_n. \tag{2.354}$$

Then, the logarithm can be written as

$$-\frac{1}{2}\mathbf{v}^\intercal\mathbf{M}\mathbf{v}+\frac{1}{2}\mathbf{v}^\intercal\mathbf{u}-\frac{1}{2}\sum_{n=1}^{N}(\mathbf{x}_n-\bar{\mathbf{x}})^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}_n-\bar{\mathbf{x}})-\frac{1}{2}\boldsymbol{\mu}_0^\intercal\boldsymbol{\Sigma}_0^{-1}\boldsymbol{\mu}_0, \tag{2.355}$$

where

$$\mathbf{v}=\begin{bmatrix}\boldsymbol{\mu}\\ \bar{\mathbf{x}}\end{bmatrix},\mathbf{M}=\begin{bmatrix}\mathbf{S}^{-1} & -N\boldsymbol{\Sigma}^{-1}\\ -N\boldsymbol{\Sigma}^{-1} & N\boldsymbol{\Sigma}^{-1}\end{bmatrix},\mathbf{u}=\begin{bmatrix}\boldsymbol{\Sigma}_0^{-1}\boldsymbol{\mu}_0\\ \mathbf{0}\end{bmatrix}, \tag{2.356}$$

and

$$\mathbf{S}=\left(N\boldsymbol{\Sigma}^{-1}+\boldsymbol{\Sigma}_0^{-1}\right)^{-1}. \tag{2.357}$$

Therefore,

$$p(\boldsymbol{\mu}|\mathbf{X})=\mathcal{N}(\boldsymbol{\mu}|\mathbf{m},\mathbf{S}), \tag{2.358}$$

where

$$\mathbf{m}=\mathbf{S}\left(N\boldsymbol{\Sigma}^{-1}\bar{\mathbf{x}}+\boldsymbol{\Sigma}_0^{-1}\boldsymbol{\mu}_0\right). \tag{2.359}$$

## 2.41

We have

$$\mathrm{Gam}(\lambda|a,b)=\frac{b^a}{\Gamma(a)}\lambda^{a-1}\exp(-b\lambda). \tag{2.360}$$

Then,

$$\int_0^\infty \mathrm{Gam}(\lambda|a,b)d\lambda=\frac{b^a}{\Gamma(a)}\int_0^\infty \lambda^{a-1}\exp(-b\lambda)d\lambda. \tag{2.361}$$

By the transformation
$$\lambda' = b\lambda, \tag{2.362}$$
the right hand side can be written as
$$\frac{b^a}{\Gamma(a)} \int_0^\infty \left(\frac{\lambda'}{b}\right)^{a-1} \exp(-\lambda') \frac{1}{b} d\lambda' = \frac{1}{\Gamma(a)} \int_0^\infty \lambda'^{a-1} \exp(-\lambda') d\lambda'. \tag{2.363}$$
The intgral of the right hand side can be written as $\Gamma(a)$. Therefore,
$$\int_0^\infty \mathrm{Gam}(\lambda|a, b) d\lambda = 1. \tag{2.364}$$

## 2.42

Let $\lambda$ be a variable such that
$$p(\lambda) = \mathrm{Gam}(\lambda|a, b). \tag{2.365}$$
We have
$$\mathrm{Gam}(\lambda|a, b) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} \exp(-b\lambda). \tag{2.366}$$

### (a)

We have
$$\mathrm{E}\,\lambda = \frac{b^a}{\Gamma(a)} \int_0^\infty \lambda^a \exp\left(-b\lambda\right) d\lambda. \tag{2.367}$$
By the transformation
$$\lambda' = b\lambda, \tag{2.368}$$
the right hand side can be written as
$$\frac{b^a}{\Gamma(a)} \int_0^\infty \left(\frac{\lambda'}{b}\right)^{a} \exp(-\lambda') \frac{1}{b} d\lambda' = \frac{1}{b\Gamma(a)} \int_0^\infty \lambda'^{a} \exp(-\lambda') d\lambda'. \tag{2.369}$$
The integral of the right hand side can be written as $\Gamma(a+1)$. Therefore,
$$\mathrm{E}\,\lambda = \frac{a}{b}. \tag{2.370}$$

**(b)**

We have

$$\mathrm{E}\,\lambda^2 = \frac{b^a}{\Gamma(a)} \int_0^\infty \lambda^{a+1} \exp\left(-b\lambda\right) d\lambda. \tag{2.371}$$

By the transformation

$$\lambda' = b\lambda, \tag{2.372}$$

the right hand side can be written as

$$\frac{b^a}{\Gamma(a)} \int_0^\infty \left(\frac{\lambda'}{b}\right)^{a+1} \exp(-\lambda') \frac{1}{b} d\lambda' = \frac{1}{b^2\Gamma(a)} \int_0^\infty \lambda'^{a+1} \exp(-\lambda') d\lambda'. \tag{2.373}$$

The integral of the right hand side can be written as $\Gamma(a+2)$. Then,

$$\mathrm{E}\,\lambda^2 = \frac{a(a+1)}{b^2}. \tag{2.374}$$

We have

$$\mathrm{var}\,\lambda = \mathrm{E}\,\lambda^2 - (\mathrm{E}\,\lambda)^2. \tag{2.375}$$

Therefore, by (a),

$$\mathrm{var}\,\lambda = \frac{a}{b^2}. \tag{2.376}$$

**(c)**

Setting the derivative of $\ln \mathrm{Gam}(\lambda|a, b)$ with respect to $\lambda$ to zero gives

$$0 = \frac{a-1}{\lambda} - b. \tag{2.377}$$

Therefore,

$$\mathrm{mode}\,\lambda = \frac{a-1}{b}. \tag{2.378}$$

## 2.43

Let

$$p\left(x|\sigma^2, q\right) = \frac{q}{2\Gamma(\frac{1}{q})} \left(2\sigma^2\right)^{-\frac{1}{q}} \exp\left(-\frac{|x|^q}{2\sigma^2}\right). \tag{2.379}$$

**(a)**

We have

$$\int_{-\infty}^{\infty} p\left(x|\sigma^2, q\right) dx = \frac{q}{\Gamma(\frac{1}{q})}\left(2\sigma^2\right)^{-\frac{1}{q}} \int_{0}^{\infty} \exp\left(-\frac{x^q}{2\sigma^2}\right) dx. \tag{2.380}$$

By the transformation

$$x' = \frac{x^q}{2\sigma^2}, \tag{2.381}$$

the integral of the right hand side can be written as

$$\begin{aligned} &\int_0^{\infty} \exp(-x')\left(2\sigma^2\right)^{\frac{1}{q}} \frac{1}{q} x^{\frac{1}{q}-1} dx' \\ &= \left(2\sigma^2\right)^{\frac{1}{q}} \frac{1}{q} \int_0^{\infty} x^{\frac{1}{q}-1} \exp(-x') dx'. \end{aligned} \tag{2.382}$$

The integral of the right hand side can be written as $\Gamma(\frac{1}{q})$. Therefore,

$$\int_{-\infty}^{\infty} p\left(x|\sigma^2, q\right) dx = 1. \tag{2.383}$$

**(b)**

We have

$$p\left(x|\sigma^2, 2\right) = \frac{1}{\Gamma(\frac{1}{2})}\left(2\sigma^2\right)^{-\frac{1}{2}} \exp\left(-\frac{x^2}{2\sigma^2}\right). \tag{2.384}$$

Therefore,

$$p\left(x|\sigma^2, 2\right) = \mathcal{N}\left(x|0, \sigma^2\right). \tag{2.385}$$

**(c)**

Let $\mathbf{t}$ be a vector such that

$$t_n = y(\mathbf{x}_n, \mathbf{w}) + \epsilon_n, \tag{2.386}$$

where

$$p(\epsilon_n) = p\left(\epsilon_n|\sigma^2, q\right). \tag{2.387}$$

Then, the logarithm of $p(\epsilon_n)$ except the terms independent of $\mathbf{w}$ and $\sigma^2$ can be written as

$$-\frac{|\epsilon_n|^q}{2\sigma^2} - \frac{1}{q}\ln\left(2\sigma^2\right). \tag{2.388}$$

Therefore, the logarithm of $p\left(\mathbf{t}|\mathbf{X},\mathbf{w},\sigma^2\right)$ except the terms independent of $\mathbf{w}$ and $\sigma^2$ can be written as

$$-\frac{1}{2\sigma^2}\sum_{n=1}^{N}|y(\mathbf{x}_n,\mathbf{w})-t_n|^q-\frac{N}{q}\ln\left(2\sigma^2\right). \tag{2.389}$$

## 2.44

Let $x_1,\cdots,x_N$ be conditionally independent variables such that

$$\begin{aligned} p(x_n|\mu,\tau) &= \mathcal{N}\left(x_n|\mu,\tau^{-1}\right), \\ p(\mu|\tau) &= \mathcal{N}\left(\mu|\mu_0,(\beta\tau)^{-1}\right), \\ p(\tau) &= \mathrm{Gam}(\tau|a,b). \end{aligned} \tag{2.390}$$

By Bayes' theorem,

$$p(\mu,\tau|\mathbf{x})=p(\mu|\tau,\mathbf{x})p(\tau|\mathbf{x}). \tag{2.391}$$

By Bayes' theorem,

$$p(\mu|\tau,\mathbf{x})p(\tau|\mathbf{x})p(\mathbf{x})=p(\mathbf{x}|\mu,\tau)p(\mu|\tau)p(\tau). \tag{2.392}$$

The logarithm of the right hand side except the terms independent of $\mathbf{x}$, $\mu$ and $\tau$ can be written as

$$\begin{aligned} l(\mu,\tau)=&-\frac{N}{2}\ln\tau^{-1}-\frac{\tau}{2}\sum_{n=1}^{N}(x_n-\mu)^2-\frac{1}{2}\ln\tau^{-1}-\frac{\beta\tau}{2}(\mu-\mu_0)^2 \\ &+(a-1)\ln\tau-b\tau. \end{aligned} \tag{2.393}$$

We have

$$\sum_{n=1}^{N}(x_n-\mu)^2=\sum_{n=1}^{N}(x_n-\bar{x})^2+N(\bar{x}-\mu)^2, \tag{2.394}$$

where

$$\bar{x}=\frac{1}{N}\sum_{n=1}^{N}x_n. \tag{2.395}$$

Then,

$$l(\mu,\tau)=l_1(\tau)+l_2(\mu,\tau), \tag{2.396}$$

where

$$
\begin{aligned}
l_1(\tau) &= \left(a + \frac{N+1}{2} - 1\right)\ln\tau - \left(b + \frac{1}{2}\sum_{n=1}^{N}(x_n - \bar{x})^2\right)\tau, \\
l_2(\mu, \tau) &= -\frac{N\tau}{2}(\bar{x} - \mu)^2 - \frac{\beta\tau}{2}(\mu - \mu_0)^2.
\end{aligned}
\tag{2.397}
$$

We have

$$
l_2(\mu, \tau) = -\frac{1}{2}\mathbf{v}^\intercal \mathbf{M}\mathbf{v} + \mathbf{v}^\intercal \mathbf{u} - \frac{\beta\tau}{2}\mu_0^2, \tag{2.398}
$$

where

$$
\mathbf{v} = \begin{bmatrix} \mu \\ \bar{x} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} (N+\beta)\tau & -N\tau \\ -N\tau & N\tau \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \beta\tau\mu_0 \\ 0 \end{bmatrix}. \tag{2.399}
$$

Then,

$$
p(\mu|\tau, \mathbf{x}) = \mathcal{N}\left(\mu|m, s^2\right), \tag{2.400}
$$

where

$$
\begin{aligned}
m &= (N+\beta)^{-1}\left(N\bar{x} + \beta\mu_0\right), \\
s^2 &= (N+\beta)^{-1}\tau^{-1}.
\end{aligned}
\tag{2.401}
$$

Then, $\ln p(\mu|\tau, \mathbf{x})$ except the terms independent of $\tau$ or dependent on $\mu$ can be written as

$$
l_3(\tau) = -\frac{1}{2}\ln\tau^{-1}. \tag{2.402}
$$

By 2.24,

$$
\mathbf{M}^{-1} = \begin{bmatrix} (\beta\tau)^{-1} & (\beta\tau)^{-1} \\ (\beta\tau)^{-1} & (N\tau)^{-1} + (\beta\tau)^{-1} \end{bmatrix}, \tag{2.403}
$$

so that

$$
\mathbf{M}^{-1}\mathbf{u} = \begin{bmatrix} \mu_0 \\ \mu_0 \end{bmatrix}. \tag{2.404}
$$

Then,

$$
p(\bar{x}|\tau) = \mathcal{N}\left(\bar{x}|\mu_0, \left(N^{-1} + \beta^{-1}\right)\tau^{-1}\right). \tag{2.405}
$$

Then, $\ln p(\bar{x}|\tau)$ except the terms independent of $\mathbf{x}$ can be written as

$$
l_4(\tau) = -\frac{\tau(\bar{x} - \mu_0)^2}{2(N^{-1} + \beta^{-1})}. \tag{2.406}
$$

Then, $\ln p(\tau|\mathbf{x})$ except the terms independent of $\tau$ and $\mathbf{x}$ can be written as

$$l_1(\tau) - l_3(\tau) + l_4(\tau) = (a' - 1)\ln\tau - b'\tau, \tag{2.407}$$

where

$$\begin{aligned} a' &= a + \frac{N}{2}, \\ b' &= b + \frac{1}{2}\sum_{n=1}^{N}(x_n - \bar{x})^2 + \frac{(\bar{x} - \mu_0)^2}{2(N^{-1} + \beta^{-1})}. \end{aligned} \tag{2.408}$$

Then,

$$p(\tau|\mathbf{x}) = \mathrm{Gam}(\tau|a', b'). \tag{2.409}$$

Therefore,

$$p(\mu, \tau|\mathbf{x}) = \mathcal{N}\left(\mu|m, s^2\right)\mathrm{Gam}\left(\tau|a', b'\right). \tag{2.410}$$

## 2.45

Let $\mathbf{x}_1, \cdots, \mathbf{x}_N$ be conditionally independent variables in $D$ dimensions such that

$$\begin{gathered} p(\mathbf{x}_n|\boldsymbol{\Lambda}) = \mathcal{N}\left(\mathbf{x}_n|\boldsymbol{\mu}, \boldsymbol{\Lambda}^{-1}\right), \\ p(\boldsymbol{\Lambda}) = \mathcal{W}(\boldsymbol{\Lambda}|\mathbf{W}, \nu), \end{gathered} \tag{2.411}$$

where

$$\mathcal{W}(\boldsymbol{\Lambda}|\mathbf{W}, \nu) = B(\mathbf{W}, \nu)|\det\boldsymbol{\Lambda}|^{\frac{\nu-D-1}{2}}\exp\left(-\frac{1}{2}\mathrm{tr}\left(\mathbf{W}^{-1}\boldsymbol{\Lambda}\right)\right). \tag{2.412}$$

By Bayes' theorem,

$$p(\boldsymbol{\Lambda}|\mathbf{X})p(\mathbf{X}) = p(\mathbf{X}|\boldsymbol{\Lambda})p(\boldsymbol{\Lambda}). \tag{2.413}$$

The logarithm of the right hand side except the terms independent of $\boldsymbol{\Lambda}$ can be written as

$$\begin{aligned} &-\frac{N}{2}\ln\left|\det\left(\boldsymbol{\Lambda}^{-1}\right)\right| - \frac{1}{2}\sum_{n=1}^{N}(\mathbf{x}_n - \boldsymbol{\mu})^{\intercal}\boldsymbol{\Lambda}(\mathbf{x}_n - \boldsymbol{\mu}) \\ &+\frac{\nu - D - 1}{2}\ln|\det\boldsymbol{\Lambda}| - \frac{1}{2}\mathrm{tr}\left(\mathbf{W}^{-1}\boldsymbol{\Lambda}\right) \\ =&\frac{\nu + N - D - 1}{2}\ln|\det\boldsymbol{\Lambda}| - \frac{1}{2}\mathrm{tr}\left(\left(\mathbf{W}^{-1} + \mathbf{S}\right)\boldsymbol{\Lambda}\right), \end{aligned} \tag{2.414}$$

where

$$\mathbf{S} = \sum_{n=1}^{N} (\mathbf{x}_n - \boldsymbol{\mu})(\mathbf{x}_n - \boldsymbol{\mu})^\intercal. \tag{2.415}$$

Then,

$$p(\boldsymbol{\Lambda}|\mathbf{X}) = \mathcal{W}\left(\boldsymbol{\Lambda}|\left(\mathbf{W}^{-1} + \mathbf{S}\right)^{-1}, \nu + N\right). \tag{2.416}$$

Therefore, $\mathcal{W}$ is a conjugate prior distribution of $\boldsymbol{\Lambda}$.

## 2.46

Let $x$ be a variable such that

$$\begin{aligned} p(x|\tau) &= \mathcal{N}\left(x|\mu, \tau^{-1}\right), \\ p(\tau) &= \mathrm{Gam}(\tau|a, b). \end{aligned} \tag{2.417}$$

By marginalisation,

$$p(x) = \int_0^\infty p(x|\tau)p(\tau)d\tau. \tag{2.418}$$

The right hand side can be written as

$$\begin{aligned} &\int_0^\infty \left(2\pi\tau^{-1}\right)^{-\frac{1}{2}} \exp\left(-\frac{\tau}{2}(x-\mu)^2\right) \frac{b^a}{\Gamma(a)} \tau^{a-1} \exp(-b\tau) d\tau \\ =&(2\pi)^{-\frac{1}{2}} \frac{b^a}{\Gamma(a)} \int_0^\infty \tau^{a-\frac{1}{2}} \exp\left(-\left(b + \frac{(x-\mu)^2}{2}\right)\tau\right) d\tau. \end{aligned} \tag{2.419}$$

By the transformation

$$\tau' = \left(b + \frac{(x-\mu)^2}{2}\right)\tau, \tag{2.420}$$

the integral of the right hand side can be written as

$$\begin{aligned} &\int_0^\infty \left(\frac{\tau'}{b + \frac{(x-\mu)^2}{2}}\right)^{a-\frac{1}{2}} \exp(-\tau') \frac{1}{b + \frac{(x-\mu)^2}{2}} d\tau' \\ =&\Gamma\left(a + \frac{1}{2}\right)\left(b + \frac{(x-\mu)^2}{2}\right)^{-a-\frac{1}{2}}. \end{aligned} \tag{2.421}$$

Then,

$$p(x) = (2\pi)^{-\frac{1}{2}} \frac{\Gamma(a+\frac{1}{2})}{\Gamma(a)} b^a \left(b + \frac{(x-\mu)^2}{2}\right)^{-a-\frac{1}{2}}. \tag{2.422}$$

The right hand side can be written as

$$(2\pi)^{-\frac{1}{2}} \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})} \left(\frac{\nu\sigma^2}{2}\right)^{-\frac{1}{2}} \left(1 + \frac{(x-\mu)^2}{\nu\sigma^2}\right)^{-\frac{\nu+1}{2}}. \tag{2.423}$$

where

$$\begin{aligned} \nu &= 2a, \\ \sigma^2 &= \frac{b}{a}. \end{aligned} \tag{2.424}$$

Therefore,

$$p(x) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})} \left(\pi\nu\sigma^2\right)^{-\frac{1}{2}} \left(1 + \frac{(x-\mu)^2}{\nu\sigma^2}\right)^{-\frac{\nu+1}{2}}. \tag{2.425}$$

## 2.47

We have

$$\mathrm{St}(x|\mu,\sigma^2,\nu) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})} \left(\pi\nu\sigma^2\right)^{-\frac{1}{2}} \left(1 + \frac{(x-\mu)^2}{\nu\sigma^2}\right)^{-\frac{\nu+1}{2}}. \tag{2.426}$$

We have

$$\left(1 + \frac{(x-\mu)^2}{\nu\sigma^2}\right)^{-\frac{\nu+1}{2}} = \left(1 + \frac{1}{y}\right)^{-\frac{(x-\mu)^2}{2\sigma^2}y - \frac{1}{2}}, \tag{2.427}$$

where

$$\frac{1}{y} = \frac{(x-\mu)^2}{\nu\sigma^2}, \tag{2.428}$$

We have

$$\lim_{x\to\infty} \left(1 + \frac{1}{x}\right)^x = e. \tag{2.429}$$

Then,

$$\lim_{y\to\infty} \left(1 + \frac{1}{y}\right)^{-\frac{(x-\mu)^2}{2\sigma^2}y - \frac{1}{2}} = \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right). \tag{2.430}$$

Therefore,

$$\lim_{\nu\to\infty} \mathrm{St}(x|\mu,\sigma^2,\nu) = \mathcal{N}(x|\mu,\sigma^2). \tag{2.431}$$

## 2.48

Let $\mathbf{x}$ be a variable in $D$ dimensions such that

$$\begin{aligned} p(\mathbf{x}|\eta) &= \mathcal{N}\left(\mathbf{x}|\boldsymbol{\mu}, \eta^{-1}\boldsymbol{\Sigma}\right), \\ p(\eta) &= \text{Gam}\left(\eta \mid \frac{\nu}{2}, \frac{\nu}{2}\right). \end{aligned} \tag{2.432}$$

By marginalisation,

$$p(\mathbf{x}) = \int_0^\infty p(\mathbf{x}|\eta)p(\eta)d\eta. \tag{2.433}$$

The logarithm of the integrand can be written as

$$\begin{aligned} &-\frac{D}{2}\ln(2\pi) - \frac{1}{2}\ln\left|\det\left(\eta^{-1}\boldsymbol{\Sigma}\right)\right| - \frac{\eta}{2}(\mathbf{x}-\boldsymbol{\mu})^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) \\ &+\frac{\nu}{2}\ln\frac{\nu}{2} - \ln\Gamma\left(\frac{\nu}{2}\right) + \left(\frac{\nu}{2}-1\right)\ln\eta - \frac{\nu}{2}\eta \\ =&-\frac{D}{2}\ln(2\pi) - \frac{1}{2}\ln|\det\boldsymbol{\Sigma}| + \frac{\nu}{2}\ln\frac{\nu}{2} - \ln\Gamma\left(\frac{\nu}{2}\right) \\ &+\left(\frac{D+\nu}{2}-1\right)\ln\eta - \frac{\nu+(\mathbf{x}-\boldsymbol{\mu})^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}{2}\eta. \end{aligned} \tag{2.434}$$

Then, the integral can be written as

$$(2\pi)^{-\frac{D}{2}}\frac{(\frac{\nu}{2})^{\frac{\nu}{2}}}{\Gamma(\frac{\nu}{2})}|\det\boldsymbol{\Sigma}|^{-\frac{1}{2}}I, \tag{2.435}$$

where

$$I = \int_0^\infty \eta^{\frac{D+\nu}{2}-1}\exp\left(-\frac{\nu+(\mathbf{x}-\boldsymbol{\mu})^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}{2}\eta\right)d\eta. \tag{2.436}$$

By the transformation

$$\eta' = \frac{\nu+(\mathbf{x}-\boldsymbol{\mu})^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}{2}\eta, \tag{2.437}$$

$I$ can be written as

$$\begin{aligned} &\left(\frac{2}{\nu+(\mathbf{x}-\boldsymbol{\mu})^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}\right)^{\frac{D+\nu}{2}}\int_0^\infty \eta'^{\frac{D+\nu}{2}-1}\exp(-\eta')d\eta' \\ =&\Gamma\left(\frac{D+\nu}{2}\right)\left(\frac{2}{\nu+(\mathbf{x}-\boldsymbol{\mu})^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}\right)^{\frac{D+\nu}{2}}. \end{aligned} \tag{2.438}$$

Then, $\ln p(\mathbf{x})$ can be written as

$$\begin{aligned} &-\frac{D}{2}\ln(2\pi)+\frac{\nu}{2}\ln\left(\frac{\nu}{2}\right)-\ln\Gamma\left(\frac{\nu}{2}\right)-\frac{1}{2}\ln|\det\boldsymbol{\Sigma}| \\ &+\ln\Gamma\left(\frac{D+\nu}{2}\right)+\frac{D+\nu}{2}\ln\left(\frac{2}{\nu+(\mathbf{x}-\boldsymbol{\mu})^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}\right). \end{aligned} \tag{2.439}$$

We have

$$\begin{aligned} &\frac{\nu}{2}\ln\left(\frac{\nu}{2}\right)+\frac{D+\nu}{2}\ln\left(\frac{2}{\nu+(\mathbf{x}-\boldsymbol{\mu})^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}\right) \\ =&-\frac{D}{2}\ln\left(\frac{\nu}{2}\right)-\frac{D+\nu}{2}\ln\left(1+\frac{(\mathbf{x}-\boldsymbol{\mu})^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}{\nu}\right). \end{aligned} \tag{2.440}$$

Then, $\ln p(\mathbf{x})$ can be written as

$$\begin{aligned} &-\frac{D}{2}\ln(\pi\nu)-\ln\Gamma\left(\frac{\nu}{2}\right)-\frac{1}{2}\ln|\det\boldsymbol{\Sigma}|+\ln\Gamma\left(\frac{D+\nu}{2}\right) \\ &-\frac{D+\nu}{2}\ln\left(1+\frac{(\mathbf{x}-\boldsymbol{\mu})^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}{\nu}\right). \end{aligned} \tag{2.441}$$

Therefore,

$$p(\mathbf{x})=c\left(1+\frac{(\mathbf{x}-\boldsymbol{\mu})^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}{\nu}\right)^{-\frac{D+\nu}{2}}, \tag{2.442}$$

where

$$c=\frac{\Gamma(\frac{D+\nu}{2})}{\Gamma(\frac{\nu}{2})}(\pi\nu)^{-\frac{D}{2}}|\det\boldsymbol{\Sigma}|^{-\frac{1}{2}}. \tag{2.443}$$

## 2.49

Let $\mathbf{x}$ be a variable such that

$$p(\mathbf{x})=\mathrm{St}(\mathbf{x}|\boldsymbol{\mu},\boldsymbol{\Sigma},\nu). \tag{2.444}$$

We have

$$\mathrm{St}(\mathbf{x}|\boldsymbol{\mu},\boldsymbol{\Sigma},\nu)=\int\mathcal{N}\left(\mathbf{x}|\boldsymbol{\mu},\eta^{-1}\boldsymbol{\Sigma}\right)\mathrm{Gam}\left(\eta\mid\frac{\nu}{2},\frac{\nu}{2}\right)d\eta. \tag{2.445}$$

**(a)**

We have

$$\mathrm{E}\,\mathbf{x} = \int \mathbf{x}\mathrm{St}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}, \nu) d\mathbf{x}. \tag{2.446}$$

The right hand side can be written as

$$\begin{aligned} &\int \mathbf{x} \left( \int \mathcal{N}\left(\mathbf{x}|\boldsymbol{\mu}, \eta^{-1}\boldsymbol{\Sigma}\right) \mathrm{Gam}\left(\eta \mid \frac{\nu}{2}, \frac{\nu}{2}\right) d\eta \right) d\mathbf{x} \\ =&\int \left( \int \mathbf{x}\mathcal{N}\left(\mathbf{x}|\boldsymbol{\mu}, \eta^{-1}\boldsymbol{\Sigma}\right) d\mathbf{x} \right) \mathrm{Gam}\left(\eta \mid \frac{\nu}{2}, \frac{\nu}{2}\right) d\eta. \end{aligned} \tag{2.447}$$

The right hand side can be written as

$$\boldsymbol{\mu} \int \mathrm{Gam}\left(\eta \mid \frac{\nu}{2}, \frac{\nu}{2}\right) d\eta = \boldsymbol{\mu}. \tag{2.448}$$

Therefore,

$$\mathrm{E}\,\mathbf{x} = \boldsymbol{\mu}. \tag{2.449}$$

**(b)**

By (a), we have

$$\mathrm{var}\,\mathbf{x} = \int (\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^{\intercal} \mathrm{St}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}, \nu) d\mathbf{x}. \tag{2.450}$$

The right hand side can be written as

$$\begin{aligned} &\int (\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^{\intercal} \left( \int \mathcal{N}\left(\mathbf{x}|\boldsymbol{\mu}, \eta^{-1}\boldsymbol{\Sigma}\right) \mathrm{Gam}\left(\eta \mid \frac{\nu}{2}, \frac{\nu}{2}\right) d\eta \right) d\mathbf{x} \\ =&\int \left( \int (\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^{\intercal}\mathcal{N}\left(\mathbf{x}|\boldsymbol{\mu}, \eta^{-1}\boldsymbol{\Sigma}\right) d\mathbf{x} \right) \mathrm{Gam}\left(\eta \mid \frac{\nu}{2}, \frac{\nu}{2}\right) d\eta. \end{aligned} \tag{2.451}$$

The right hand side can be written as

$$\begin{aligned} &\left( \int \eta^{-1}\mathrm{Gam}\left(\eta \mid \frac{\nu}{2}, \frac{\nu}{2}\right) d\eta \right) \boldsymbol{\Sigma} \\ =&\frac{(\frac{\nu}{2})^{\frac{\nu}{2}}}{\Gamma(\frac{\nu}{2})} \left( \int \eta^{\frac{\nu}{2}-2} \exp\left(-\frac{\nu}{2}\eta\right) d\eta \right) \boldsymbol{\Sigma}. \end{aligned} \tag{2.452}$$

By the transformation

$$\eta' = \frac{\nu}{2}\eta, \tag{2.453}$$

the integral of the right hand side can be written as

$$\int \left(\frac{2}{\nu}\eta'\right)^{\frac{\nu}{2}-2} \exp(-\eta')\frac{2}{\nu}d\eta' = \left(\frac{2}{\nu}\right)^{\frac{\nu}{2}-1} \Gamma\left(\frac{\nu}{2}-1\right). \tag{2.454}$$

Then,

$$\operatorname{var}\mathbf{x} = \frac{(\frac{\nu}{2})^{\frac{\nu}{2}}}{\Gamma(\frac{\nu}{2})} \left(\frac{2}{\nu}\right)^{\frac{\nu}{2}-1} \Gamma\left(\frac{\nu}{2}-1\right)\mathbf{\Sigma}. \tag{2.455}$$

Therefore,

$$\operatorname{var}\mathbf{x} = \frac{\nu}{\nu-2}\mathbf{\Sigma}. \tag{2.456}$$

**(c)**

Setting the derivative of $p$ with respect to $\mathbf{x}$ to zero gives

$$\mathbf{0} = -\frac{1}{2}\left(\mathbf{\Sigma}^{-1} + \left(\mathbf{\Sigma}^{-1}\right)^{\intercal}\right)(\mathbf{x}-\boldsymbol{\mu})I, \tag{2.457}$$

where

$$I = \int \eta\mathcal{N}\left(\mathbf{x}|\boldsymbol{\mu}, \eta^{-1}\mathbf{\Sigma}\right)\operatorname{Gam}\left(\eta \mid \frac{\nu}{2}, \frac{\nu}{2}\right)d\eta. \tag{2.458}$$

Therefore,

$$\operatorname{mode}\mathbf{x} = \boldsymbol{\mu}. \tag{2.459}$$

## 2.50

We have

$$\operatorname{St}(\mathbf{x}|\boldsymbol{\mu}, \mathbf{\Sigma}, \nu) = c\left(1 + \frac{(\mathbf{x}-\boldsymbol{\mu})^{\intercal}\mathbf{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}{\nu}\right)^{-\frac{D+\nu}{2}}, \tag{2.460}$$

where

$$c = \frac{\Gamma(\frac{D+\nu}{2})}{\Gamma(\frac{\nu}{2})}(\pi\nu)^{-\frac{D}{2}}(\det\mathbf{\Sigma})^{-\frac{1}{2}}. \tag{2.461}$$

We have

$$\left(1+\frac{(\mathbf{x}-\boldsymbol{\mu})^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}{\nu}\right)^{-\frac{D+\nu}{2}} = \left(1+\frac{1}{y}\right)^{-\frac{(\mathbf{x}-\boldsymbol{\mu})^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}{2}y-\frac{D}{2}}, \quad (2.462)$$

where

$$\frac{1}{y} = \frac{(\mathbf{x}-\boldsymbol{\mu})^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}{\nu}. \quad (2.463)$$

We have

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x = e. \quad (2.464)$$

Then,

$$\begin{aligned}&\lim_{y\to\infty}\left(1+\frac{1}{y}\right)^{-\frac{(\mathbf{x}-\boldsymbol{\mu})^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}{2}y-\frac{D}{2}}\\ &= \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right).\end{aligned} \quad (2.465)$$

Therefore,

$$\lim_{\nu\to\infty}\mathrm{St}(\mathbf{x}|\boldsymbol{\mu},\boldsymbol{\Sigma},\nu) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu},\boldsymbol{\Sigma}). \quad (2.466)$$

## 2.51

### (a)

We have

$$\exp(iA)\exp(-iA) = 1. \quad (2.467)$$

The left hand side can be written as

$$(\cos A + i\sin A)(\cos A - i\sin A) = \cos^2 A + \sin^2 A. \quad (2.468)$$

Therefore,

$$\cos^2 A + \sin^2 A = 1. \quad (2.469)$$

### (b)

We have

$$\cos(A-B) = \mathrm{Re}\left(\exp\left(i(A-B)\right)\right). \quad (2.470)$$

The right hand side can be written as

$$\mathrm{Re}\left(\exp(iA)\exp(-iB)\right) = \mathrm{Re}\left((\cos A + i\sin A)(\cos B - i\sin B)\right). \tag{2.471}$$

Therefore,

$$\cos(A - B) = \cos A\cos B + \sin A\sin B. \tag{2.472}$$

**(c)**

We have

$$\sin(A - B) = \mathrm{Im}\left(\exp\left(i(A - B)\right)\right). \tag{2.473}$$

The right hand side can be written as

$$\mathrm{Im}\left(\exp(iA)\exp(-iB)\right) = \left((\cos A + i\sin A)(\cos B - i\sin B)\right). \tag{2.474}$$

Therefore,

$$\sin(A - B) = \sin A\cos B - \cos A\sin B. \tag{2.475}$$

## 2.52

Let

$$f(\theta|\theta_0, m) = \frac{1}{2\pi I_0(m)}\exp\left(m\cos(\theta - \theta_0)\right), \tag{2.476}$$

where

$$I_0(m) = \frac{1}{2\pi}\int_0^{2\pi}\exp(m\cos\theta)d\theta. \tag{2.477}$$

By the Taylor series

$$\cos\alpha = 1 - \frac{1}{2}\alpha^2 + O\left(\alpha^4\right), \tag{2.478}$$

the right hand side can be written as

$$\begin{aligned}&\frac{\exp\left(m\left(1 - \frac{1}{2}(\theta - \theta_0)^2 + O\left((\theta - \theta_0)^4\right)\right)\right)}{\int_0^{2\pi}\exp\left(m\left(1 - \frac{1}{2}\theta^2 + O(\theta^4)\right)\right)d\theta} \\ =&\exp\left(-\frac{m}{2}(\theta - \theta_0)^2\right)\frac{\exp\left(mO\left((\theta - \theta_0)^4\right)\right)}{\int_0^{2\pi}\exp\left(m\left(-\frac{1}{2}\theta^2 + O(\theta^4)\right)\right)d\theta}.\end{aligned} \tag{2.479}$$

Therefore,

$$\lim_{m\to\infty} f(\theta|\theta_0, m) = \mathcal{N}\left(\theta|\theta_0, m^{-1}\right). \tag{2.480}$$

## 2.53

Let

$$\sum_{n=1}^{N} \sin(\theta_n - \theta_0) = 0. \tag{2.481}$$

The left hand side can be written as

$$\sum_{n=1}^{N} (\sin\theta_n \cos\theta_0 - \cos\theta_n \sin\theta_0) = \cos\theta_0 \sum_{n=1}^{N} \sin\theta_n - \sin\theta_0 \sum_{n=1}^{N} \cos\theta_n. \tag{2.482}$$

Therefore,

$$\theta_0 = \arctan\left(\frac{\sum_{n=1}^{N} \sin\theta_n}{\sum_{n=1}^{N} \cos\theta_n}\right). \tag{2.483}$$

## 2.54

Let

$$f(\theta|\theta_0, m) = \frac{1}{2\pi I_0(m)} \exp\left(m\cos(\theta - \theta_0)\right), \tag{2.484}$$

where

$$I_0(m) = \frac{1}{2\pi} \int_0^{2\pi} \exp(m\cos\theta) d\theta. \tag{2.485}$$

Setting the first and seconnd derivatives with respect to $\theta$ to zero gives

$$\begin{aligned} 0 &= -m\sin(\theta - \theta_0) f(\theta|\theta_0, m), \\ 0 &= \left(m^2 \sin^2(\theta - \theta_0) - m\cos(\theta - \theta_0)\right) f(\theta|\theta_0, m). \end{aligned} \tag{2.486}$$

Therefore,

$$\begin{aligned} \operatorname*{argmax}_{\theta} f(\theta|\theta_0, m) &= \theta_0, \\ \operatorname*{argmin}_{\theta} f(\theta|\theta_0, m) &= \theta_0 - \pi\,\mathrm{sgn}(\theta_0 - \pi). \end{aligned} \tag{2.487}$$

## 2.55

### (a)

Let $\theta_1, \cdots, \theta_N$ be independent variables such that

$$p(\theta_n) = \frac{1}{2\pi I_0(m)} \exp\left(m\cos(\theta - \theta_0)\right), \tag{2.488}$$

where

$$I_0(m) = \frac{1}{2\pi} \int_0^{2\pi} \exp(m \cos\theta) d\theta. \tag{2.489}$$

Then,

$$\ln p(\boldsymbol{\theta}) = -\frac{N}{2} \ln\left(2\pi I_0(m)\right) + m \sum_{n=1}^{N} \cos(\theta_n - \theta_0). \tag{2.490}$$

Setting the derivative of $\ln p$ with respect to $\theta_0$ to zero gives

$$0 = m \sum_{n=1}^{N} \sin(\theta_n - \theta_0). \tag{2.491}$$

Therefore, by 2.53, the maximum likelihood solution to $\theta_0$ is given by

$$\theta_0^{\mathrm{ML}} = \arctan\left(\frac{\sum_{n=1}^{N} \sin\theta_n}{\sum_{n=1}^{N} \cos\theta_n}\right). \tag{2.492}$$

**(b)**

Let

$$\begin{aligned} \bar{r}\cos\bar{\theta} &= \frac{1}{N} \sum_{n=1}^{N} \cos\theta_n, \\ \bar{r}\sin\bar{\theta} &= \frac{1}{N} \sum_{n=1}^{N} \sin\theta_n. \end{aligned} \tag{2.493}$$

By (a),

$$\bar{\theta} = \theta_0^{\mathrm{ML}}. \tag{2.494}$$

We have

$$\begin{aligned} &\frac{1}{N} \sum_{n=1}^{N} \cos\left(\theta_n - \theta_0^{\mathrm{ML}}\right) \\ &= \left(\frac{1}{N} \sum_{n=1}^{N} \cos\theta_n\right) \cos\theta_0^{\mathrm{ML}} + \left(\frac{1}{N} \sum_{n=1}^{N} \sin\theta_n\right) \sin\theta_0^{\mathrm{ML}}. \end{aligned} \tag{2.495}$$

The right hand side can be written as

$$\bar{r}\cos^2\bar{\theta} + \bar{r}\sin^2\bar{\theta} = \bar{r}. \tag{2.496}$$

Therefore,

$$\frac{1}{N}\sum_{n=1}^{N}\cos\left(\theta_n - \theta_0^{\mathrm{ML}}\right) = \bar{r}. \tag{2.497}$$

## 2.56

### (a)

Let

$$\mathrm{Beta}(\mu|a,b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\mu^{a-1}(1-\mu)^{b-1}. \tag{2.498}$$

The right hand side can be written as

$$\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\exp\left((a-1)\ln\mu + (b-1)\ln(1-\mu)\right) \tag{2.499}$$

Therefore, the natural parameters are given by

$$\boldsymbol{\eta} = \begin{bmatrix} a-1 \\ b-1 \end{bmatrix}. \tag{2.500}$$

### (b)

Let

$$\mathrm{Gam}(\lambda|a,b) = \frac{b^a}{\Gamma(a)}\lambda^{a-1}\exp(-b\lambda). \tag{2.501}$$

The right hand side can be written as

$$\frac{b^a}{\Gamma(a)}\exp\left((a-1)\ln\lambda - b\lambda\right). \tag{2.502}$$

Therefore, the natural parameters are given by

$$\boldsymbol{\eta} = \begin{bmatrix} a-1 \\ -b \end{bmatrix}. \tag{2.503}$$

**(c)**

Let
$$f(\theta|\theta_0, m) = \frac{1}{2\pi I_0(m)} \exp\left(m\cos(\theta - \theta_0)\right), \tag{2.504}$$
where
$$I_0(m) = \frac{1}{2\pi}\int_0^{2\pi} \exp(m\cos\theta)d\theta, \tag{2.505}$$
the right hand side can be written as
$$\frac{1}{2\pi I_0(m)} \exp(m\cos\theta_0\cos\theta + m\sin\theta_0\sin\theta). \tag{2.506}$$
Therefore, the natural parameters are given by
$$\boldsymbol{\eta} = \begin{bmatrix} m\cos\theta_0 \\ m\sin\theta_0 \end{bmatrix}. \tag{2.507}$$

## 2.57

We have
$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = (2\pi)^{-\frac{D}{2}}(\det\boldsymbol{\Sigma})^{-\frac{1}{2}}\exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right). \tag{2.508}$$
Therefore,
$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = h(\mathbf{x})g(\boldsymbol{\eta})\exp\left(\boldsymbol{\eta}^\intercal\mathbf{u}(\mathbf{x})\right), \tag{2.509}$$
where
$$\begin{aligned} h(\mathbf{x}) &= (2\pi)^{-\frac{D}{2}}, \\ g(\boldsymbol{\eta}) &= (\det(-2\boldsymbol{\eta}_2))^{-\frac{1}{2}}\exp\left(\frac{1}{4}\boldsymbol{\eta}_1^\intercal\boldsymbol{\eta}_2^{-1}\boldsymbol{\eta}_1\right), \\ \boldsymbol{\eta} &= \begin{bmatrix} \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} \\ -\frac{1}{2}\boldsymbol{\Sigma}^{-1} \end{bmatrix}, \mathbf{u}(\mathbf{x}) = \begin{bmatrix} \mathbf{x} \\ \mathbf{x}\mathbf{x}^\intercal \end{bmatrix}. \end{aligned} \tag{2.510}$$

## 2.58

Let $\mathbf{x}$ be a variable such that
$$p(\mathbf{x}|\boldsymbol{\eta}) = h(\mathbf{x})g(\boldsymbol{\eta})\exp\left(\boldsymbol{\eta}^\intercal\mathbf{u}(\mathbf{x})\right). \tag{2.511}$$

**(a)**

Setting the first derivative of

$$\int p(\mathbf{x}|\boldsymbol{\eta})d\mathbf{x} = 1 \tag{2.512}$$

with respect to $\boldsymbol{\eta}$ to zero gives

$$\begin{aligned}\mathbf{0} =& \nabla g(\boldsymbol{\eta}) \int h(\mathbf{x}) \exp\left(\boldsymbol{\eta}^\intercal \mathbf{u}(\mathbf{x})\right) d\mathbf{x} \\ &+ g(\boldsymbol{\eta}) \int \mathbf{u}(\mathbf{x}) h(\mathbf{x}) \exp\left(\boldsymbol{\eta}^\intercal \mathbf{u}(\mathbf{x})\right) d\mathbf{x}.\end{aligned} \tag{2.513}$$

The right hand side can be written as

$$\frac{\nabla g(\boldsymbol{\eta})}{g(\boldsymbol{\eta})} \int p(\mathbf{x}|\boldsymbol{\eta})d\mathbf{x} + \int \mathbf{u}(\mathbf{x}) p(\mathbf{x}|\boldsymbol{\eta})d\mathbf{x} = \frac{\nabla g(\boldsymbol{\eta})}{g(\boldsymbol{\eta})} + \mathrm{E}\,\mathbf{u}(\mathbf{x}). \tag{2.514}$$

Then,

$$\mathrm{E}\,\mathbf{u}(\mathbf{x}) = -\frac{\nabla g(\boldsymbol{\eta})}{g(\boldsymbol{\eta})}. \tag{2.515}$$

Therefore,

$$\mathrm{E}\,\mathbf{u}(\mathbf{x}) = -\nabla \ln g(\boldsymbol{\eta}). \tag{2.516}$$

**(b)**

Setting the second derivative with respect to $\boldsymbol{\eta}$ to zero gives

$$\begin{aligned}\mathbf{O} =& \nabla\nabla g(\boldsymbol{\eta}) \int h(\mathbf{x}) \exp\left(\boldsymbol{\eta}^\intercal \mathbf{u}(\mathbf{x})\right) d\mathbf{x} \\ &+ 2\nabla g(\boldsymbol{\eta}) \int \mathbf{u}(\mathbf{x})^\intercal h(\mathbf{x}) \exp\left(\boldsymbol{\eta}^\intercal \mathbf{u}(\mathbf{x})\right) d\mathbf{x} \\ &+ g(\boldsymbol{\eta}) \int \mathbf{u}(\mathbf{x})\mathbf{u}(\mathbf{x})^\intercal h(\mathbf{x}) \exp\left(\boldsymbol{\eta}^\intercal \mathbf{u}(\mathbf{x})\right) d\mathbf{x}\end{aligned} \tag{2.517}$$

The right hand side can be written as

$$\begin{aligned}&\frac{\nabla\nabla g(\boldsymbol{\eta})}{g(\boldsymbol{\eta})} \int p(\mathbf{x}|\boldsymbol{\eta})d\mathbf{x} + \frac{2\nabla g(\boldsymbol{\eta})}{g(\boldsymbol{\eta})} \int \mathbf{u}(\mathbf{x})^\intercal p(\mathbf{x}|\boldsymbol{\eta})d\mathbf{x} \\ &+ \int \mathbf{u}(\mathbf{x})\mathbf{u}(\mathbf{x})^\intercal p(\mathbf{x}|\boldsymbol{\eta})d\mathbf{x} \\ =&\frac{\nabla\nabla g(\boldsymbol{\eta})}{g(\boldsymbol{\eta})} - 2\,\mathrm{E}\,\mathbf{u}(\mathbf{x})\,\mathrm{E}\,\mathbf{u}(\mathbf{x})^\intercal + \mathrm{E}\left(\mathbf{u}(\mathbf{x})\mathbf{u}(\mathbf{x})^\intercal\right).\end{aligned} \tag{2.518}$$

Then,

$$\mathrm{E}\left(\mathbf{u}(\mathbf{x})\mathbf{u}(\mathbf{x})^\intercal\right) = -\frac{\nabla\nabla g(\boldsymbol{\eta})}{g(\boldsymbol{\eta})} + \frac{2\nabla g(\boldsymbol{\eta})(\nabla g(\boldsymbol{\eta}))^\intercal}{g^2(\boldsymbol{\eta})}. \tag{2.519}$$

We have

$$\operatorname{var}\mathbf{u}(\mathbf{x}) = \mathrm{E}\left(\mathbf{u}(\mathbf{x})\mathbf{u}(\mathbf{x})^\intercal\right) - \mathrm{E}\,\mathbf{u}(\mathbf{x})\,\mathrm{E}\,\mathbf{u}(\mathbf{x})^\intercal. \tag{2.520}$$

Then,

$$\operatorname{var}\mathbf{u}(\mathbf{x}) = -\frac{\nabla\nabla g(\boldsymbol{\eta})}{g(\boldsymbol{\eta})} + \frac{\nabla g(\boldsymbol{\eta})(\nabla g(\boldsymbol{\eta}))^\intercal}{g^2(\boldsymbol{\eta})}. \tag{2.521}$$

Therefore,

$$\operatorname{var}\mathbf{u}(\mathbf{x}) = -\nabla\nabla\ln g(\boldsymbol{\eta}). \tag{2.522}$$

## 2.59

Let

$$p(x|\sigma) = \frac{1}{\sigma} f\left(\frac{x}{\sigma}\right). \tag{2.523}$$

Then

$$\int p(x|\sigma)dx = \frac{1}{\sigma}\int f\left(\frac{x}{\sigma}\right)dx. \tag{2.524}$$

By the transformation

$$x' = \frac{x}{\sigma}, \tag{2.525}$$

the right hand side can be written as

$$\frac{1}{\sigma}\int f(x')\sigma dx' = \int f(x')dx'. \tag{2.526}$$

Therefore, $p(x|\sigma)$ will be normalised if $f(x)$ is normalised.

## 2.60

Let $\mathbf{x}_1, \cdots, \mathbf{x}_N$ be variables such that

$$\mathbf{x}_n \in \mathcal{R}_k \Rightarrow p(\mathbf{x}_n) = \pi_k. \tag{2.527}$$

Let $n_k$ be the number of observations which fall in $\mathcal{R}_k$ and let

$$\Delta_k = \int_{\mathcal{R}_k} d\mathbf{x}_n. \tag{2.528}$$

In order to maximise $\ln p(\mathbf{X})$ under the constraint

$$\int p(\mathbf{x}_n)d\mathbf{x}_n = 1, \tag{2.529}$$

let

$$L = \ln p(\mathbf{X}) + \lambda \left( \int p(\mathbf{x}_n)d\mathbf{x}_n - 1 \right). \tag{2.530}$$

We have

$$\begin{aligned} \ln p(\mathbf{X}) &= \sum_{k=1}^{K} n_k \ln \pi_k, \\ \int p(\mathbf{x}_n)d\mathbf{x}_n &= \sum_{k=1}^{K} \pi_k \Delta_k. \end{aligned} \tag{2.531}$$

Then,

$$L = \sum_{k=1}^{K} n_k \ln \pi_k + \lambda \left( \sum_{k=1}^{K} \pi_k \Delta_k - 1 \right). \tag{2.532}$$

Setting the derivatives of $L$ with respect to $\pi_k$ and $\lambda$ to zero gives

$$\begin{aligned} \frac{n_k}{\pi_k} + \lambda \Delta_k = 0, \\ \sum_{k=1}^{K} \pi_k \Delta_k - 1 = 0. \end{aligned} \tag{2.533}$$

We have

$$\sum_{k=1}^{K} n_k = N. \tag{2.534}$$

Therefore, the maximum likelihood solution to $\pi_k$ is given by

$$\pi_{k\mathrm{ML}} = \frac{n_k}{N\Delta_k}. \tag{2.535}$$

## 2.61 (Incomplete)

Let $\mathbf{x}$ be a variable and $\mathbf{x}_1, \cdots, \mathbf{x}_N$ be observations. Let

$$p(\mathbf{x}) = \frac{K}{NV(\mathbf{x})}, \tag{2.536}$$

where

$$V(\mathbf{x}) = \int_{\|\mathbf{x}'-\mathbf{x}\|\leq\|\mathbf{x}_{(K)}-\mathbf{x}\|} d\mathbf{x}', \tag{2.537}$$

$K$ is a constant and $\mathbf{x}_{(K)}$ is the $K$th nearest observation from the point $\mathbf{x}$.

# 3 Linear Models for Regression

## 3.1

### (a)

We have

$$\tanh a = \frac{\exp(a) - \exp(-a)}{\exp(a) + \exp(-a)}. \tag{3.1}$$

The right hand side can be written as

$$\frac{1 - \exp(-2a)}{1 + \exp(-2a)} = \frac{2}{1 + \exp(-2a)} - 1. \tag{3.2}$$

Therefore,

$$\tanh a = 2\sigma(2a) - 1, \tag{3.3}$$

where

$$\sigma(a) = \frac{1}{1 + \exp(-a)}. \tag{3.4}$$

### (b)

Let

$$y(x_n, \mathbf{w}) = w_0 + \sum_{m=1}^{M} w_j \sigma\left(\frac{x - \mu_j}{s}\right). \tag{3.5}$$

By the result above, the right hand side can be written as

$$\begin{aligned} & w_0 f + \sum_{m=1}^{M} w_m \frac{1 + \tanh\left(\frac{x-\mu_m}{2s}\right)}{2} \\ =& w_0 + \frac{1}{2}\sum_{m=1}^{M} w_m + \frac{1}{2}\sum_{m=1}^{M} w_m \tanh\left(\frac{x - \mu_m}{2s}\right). \end{aligned} \tag{3.6}$$

Therefore, $y(x_n, \mathbf{w})$ is equivalent to

$$y(x_n, \mathbf{u}) = u_0 + \sum_{m=1}^{M} u_m \tanh\left(\frac{x - \mu_m}{2s}\right), \tag{3.7}$$

where

$$
\begin{aligned}
u_0 &= w_0 + \frac{1}{2}\sum_{m=1}^{M} w_m, \\
u_m &= \frac{1}{2} w_m.
\end{aligned}
\tag{3.8}
$$

## 3.2 (Incomplete)

Let $\mathbf{\Phi}$ be an $N \times M$ matarix. Then, for any vector $\mathbf{v}$ in $N$ dimensions,

$$
\mathbf{\Phi}\left(\mathbf{\Phi}^{\intercal}\mathbf{\Phi}\right)^{-1}\mathbf{\Phi}^{\intercal}\mathbf{v} \tag{3.9}
$$

is a projection of $\mathbf{v}$ onto the space spanned by the columns of $\mathbf{\Phi}$?

Additionally, for a vector $\mathbf{t}$ in $N$ dimensions,

$$
\left(\mathbf{\Phi}^{\intercal}\mathbf{\Phi}\right)^{-1}\mathbf{\Phi}^{\intercal}\mathbf{t} \tag{3.10}
$$

is an orthogonal projection of $\mathbf{t}$ onto the space spanned by the columns of $\mathbf{\Phi}$?

## 3.3

Let

$$
E(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N} r_n \left(t_n - \mathbf{w}^{\intercal}\boldsymbol{\phi}_n\right)^2 . \tag{3.11}
$$

The right hand side can be written as

$$
\frac{1}{2}\left\|\mathbf{t}' - \mathbf{\Phi}'\mathbf{w}\right\|^2, \tag{3.12}
$$

where

$$
\mathbf{t}' = \begin{bmatrix} \sqrt{r_1}t_1 \\ \vdots \\ \sqrt{r_N}t_N \end{bmatrix}, \mathbf{\Phi}' = \begin{bmatrix} \sqrt{r_1}\boldsymbol{\phi}(\mathbf{x}_1)^{\intercal} \\ \vdots \\ \sqrt{r_N}\boldsymbol{\phi}(\mathbf{x}_N)^{\intercal} \end{bmatrix}. \tag{3.13}
$$

Setting the derivative of $E(\mathbf{w})$ with respect to $\mathbf{w}$ to zero gives

$$
\mathbf{0} = -\mathbf{\Phi}'^{\intercal}(\mathbf{t}' - \mathbf{\Phi}'\mathbf{w}). \tag{3.14}
$$

Therefore,

$$
\operatorname*{argmin}_{\mathbf{w}} E(\mathbf{w}) = \left(\mathbf{\Phi}'^{\intercal}\mathbf{\Phi}'\right)^{-1}\mathbf{\Phi}'^{\intercal}\mathbf{t}'. \tag{3.15}
$$

## 3.4

Let

$$\tilde{E}(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}(t_n - \tilde{y}_n)^2, \tag{3.16}$$

where

$$\begin{aligned} \tilde{y}_n &= w_0 + \mathbf{w}^\intercal(\mathbf{x}_n + \boldsymbol{\epsilon}_n), \\ p(\boldsymbol{\epsilon}_n) &= \mathcal{N}\left(\boldsymbol{\epsilon}_n|\mathbf{0}, \sigma^2\mathbf{I}\right). \end{aligned} \tag{3.17}$$

We have

$$\mathrm{E}\,\tilde{E}(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}\mathrm{E}\left(t_n - y_n - \mathbf{w}^\intercal\boldsymbol{\epsilon}_n\right)^2, \tag{3.18}$$

where

$$y_n = w_0 + \mathbf{w}^\intercal\mathbf{x}_n. \tag{3.19}$$

The summand of the right hand side can be written as

$$(t_n - y_n)^2 - 2(t_n - y_n)\mathbf{w}^\intercal\,\mathrm{E}\,\boldsymbol{\epsilon}_n + \mathbf{w}^\intercal\left(\mathrm{E}\,\boldsymbol{\epsilon}\boldsymbol{\epsilon}^\intercal\right)\mathbf{w} = (t_n - y_n)^2 + \sigma^2\|\mathbf{w}\|^2. \tag{3.20}$$

Therefore,

$$\mathrm{E}\,\tilde{E}(\mathbf{w}) = E(\mathbf{w}) + \frac{N\sigma^2}{2}\|\mathbf{w}\|^2, \tag{3.21}$$

where

$$E(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}(t_n - y_n)^2. \tag{3.22}$$

## 3.5

Let

$$E(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}\left(t_n - \mathbf{w}^\intercal\boldsymbol{\phi}(\mathbf{x}_n)\right)^2, \tag{3.23}$$

where

$$\sum_{m=1}^{M}|w_m|^q \le \eta. \tag{3.24}$$

In order to minimise $E(\mathbf{w})$, let

$$L(\mathbf{w}, \lambda) = E(\mathbf{w}) + \lambda\left(\sum_{m=1}^{M}|w_m|^q - \eta\right). \tag{3.25}$$

Therefore,

$$\eta = \sum_{m=1}^{M} |w_m^*(\lambda)|^q, \tag{3.26}$$

where

$$\mathbf{w}^*(\lambda) = \operatorname*{argmin}_{\mathbf{w}} L(\mathbf{w}, \lambda). \tag{3.27}$$

## 3.6

Let $\mathbf{t}_1, \cdots, \mathbf{t}_N$ be conditionally independent variables in $D$ dimensions such that

$$p(\mathbf{t}_n|\mathbf{y}_n) = \mathcal{N}\left(\mathbf{t}_n|\mathbf{y}_n, \mathbf{\Sigma}\right), \tag{3.28}$$

where

$$\mathbf{y}_n = \mathbf{W}^\intercal \boldsymbol{\phi}_n. \tag{3.29}$$

Then,

$$\begin{aligned}\ln p(\mathbf{T}|\mathbf{Y}) = &-\frac{ND}{2}\ln(2\pi) - \frac{N}{2}\ln(\det \mathbf{\Sigma}) \\ &-\frac{1}{2}\sum_{n=1}^{N}\left(\mathbf{t}_n - \mathbf{W}^\intercal\boldsymbol{\phi}_n\right)^\intercal \mathbf{\Sigma}^{-1}\left(\mathbf{t}_n - \mathbf{W}^\intercal\boldsymbol{\phi}_n\right).\end{aligned} \tag{3.30}$$

By 3.21, setting the derivatives of $\ln p(\mathbf{T}|\mathbf{Y})$ with respect to $\mathbf{W}$ and $\mathbf{\Sigma}$ to zero gives

$$\begin{aligned}\mathbf{O} &= -\frac{1}{2}\left(\mathbf{\Sigma}^{-1} + \left(\mathbf{\Sigma}^{-1}\right)^\intercal\right)\sum_{n=1}^{N}\left(\mathbf{t}_n - \mathbf{W}^\intercal\boldsymbol{\phi}_n\right)\boldsymbol{\phi}_n^\intercal, \\ \mathbf{O} &= -\frac{N}{2}\left(\mathbf{\Sigma}^{-1}\right)^\intercal + \frac{1}{2}\left(\mathbf{\Sigma}^{-1}\right)^2\sum_{n=1}^{N}\left(\mathbf{t}_n - \mathbf{W}^\intercal\boldsymbol{\phi}_n\right)\left(\mathbf{t}_n - \mathbf{W}^\intercal\boldsymbol{\phi}_n\right)^\intercal.\end{aligned} \tag{3.31}$$

Therefore, the maximum likelihood solutions to $\mathbf{W}$ and $\mathbf{\Sigma}$ are given by

$$\begin{aligned}\mathbf{W}_{\mathrm{ML}} &= \left(\mathbf{\Phi}^\intercal\mathbf{\Phi}\right)^{-1}\mathbf{\Phi}^\intercal\mathbf{t}, \\ \mathbf{\Sigma}_{\mathrm{ML}} &= \frac{1}{N}\sum_{n=1}^{N}\left(\mathbf{t}_n - \mathbf{W}_{\mathrm{ML}}^\intercal\boldsymbol{\phi}_n\right)\left(\mathbf{t}_n - \mathbf{W}_{\mathrm{ML}}^\intercal\boldsymbol{\phi}_n\right)^\intercal.\end{aligned} \tag{3.32}$$

## 3.7

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal\boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0). \end{aligned} \tag{3.33}$$

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{w})p(\mathbf{w}). \tag{3.34}$$

The logarithm of the right hand side except the terms independent of $\mathbf{t}$ and $\mathbf{w}$ can be written as

$$\begin{aligned} &-\frac{\beta}{2}\|\mathbf{t} - \mathbf{\Phi}\mathbf{w}\|^2 - \frac{1}{2}(\mathbf{w} - \mathbf{m}_0)^\intercal\mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{m}_0) \\ =&-\frac{1}{2}\mathbf{v}^\intercal\mathbf{M}\mathbf{v} + \mathbf{v}^\intercal\mathbf{u} - \frac{1}{2}\mathbf{m}_0^\intercal\mathbf{S}_0^{-1}\mathbf{m}_0, \end{aligned} \tag{3.35}$$

where

$$\mathbf{v} = \begin{bmatrix}\mathbf{w} \\ \mathbf{t}\end{bmatrix}, \mathbf{M} = \begin{bmatrix}\mathbf{S}_0^{-1} + \beta\mathbf{\Phi}^\intercal\mathbf{\Phi} & -\beta\mathbf{\Phi}^\intercal \\ -\beta\mathbf{\Phi} & \beta\mathbf{I}\end{bmatrix}, \mathbf{u} = \begin{bmatrix}\mathbf{S}_0^{-1}\mathbf{m}_0 \\ \mathbf{0}\end{bmatrix}. \tag{3.36}$$

Therefore,

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}(\mathbf{w}|\mathbf{m}, \mathbf{S}), \tag{3.37}$$

where

$$\begin{aligned} \mathbf{m} &= \mathbf{S}\left(\mathbf{S}_0^{-1}\mathbf{m}_0 + \beta\mathbf{\Phi}^\intercal\mathbf{t}\right), \\ \mathbf{S} &= \left(\mathbf{S}_0^{-1} + \beta\mathbf{\Phi}^\intercal\mathbf{\Phi}\right)^{-1}. \end{aligned} \tag{3.38}$$

## 3.8

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal\boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0). \end{aligned} \tag{3.39}$$

By 3.7,

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}(\mathbf{w}|\mathbf{m}, \mathbf{S}), \tag{3.40}$$

where

$$\begin{aligned} \mathbf{m} &= \mathbf{S}\left(\mathbf{S}_0^{-1}\mathbf{m}_0 + \beta\mathbf{\Phi}^\intercal\mathbf{t}\right), \\ \mathbf{S} &= \left(\mathbf{S}_0^{-1} + \beta\mathbf{\Phi}^\intercal\mathbf{\Phi}\right)^{-1}. \end{aligned} \tag{3.41}$$

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{t}')p(\mathbf{t}') = p(\mathbf{t}'|\mathbf{w})p(\mathbf{w}), \tag{3.42}$$

where

$$\mathbf{t}' = \begin{bmatrix} \mathbf{t} \\ t_{N+1} \end{bmatrix}. \tag{3.43}$$

By Bayes' theorem, the right hand side can be written as

$$p(t_{N+1}|\mathbf{w})p(\mathbf{t}|\mathbf{w})p(\mathbf{w}) = p(t_{N+1}|\mathbf{w})p(\mathbf{w}|\mathbf{t})p(\mathbf{t}). \tag{3.44}$$

The logarithm of the right hand side except the terms independent of $\mathbf{w}$ can be written as

$$\begin{aligned} & -\frac{\beta}{2}\left(t_{N+1} - \mathbf{w}^\intercal \boldsymbol{\phi}_{N+1}\right)^2 - \frac{1}{2}\left(\mathbf{w} - \mathbf{m}\right)^\intercal \mathbf{S}^{-1}\left(\mathbf{w} - \mathbf{m}\right) \\ =& -\frac{1}{2}\mathbf{v}^\intercal \mathbf{M}\mathbf{v} + \mathbf{v}^\intercal \mathbf{u} - \frac{1}{2}\mathbf{m}^\intercal \mathbf{S}^{-1}\mathbf{m}, \end{aligned} \tag{3.45}$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ t_{N+1} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \mathbf{S}'^{-1} & -\beta\boldsymbol{\phi}_{N+1} \\ -\beta\boldsymbol{\phi}_{N+1}^\intercal & \beta \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \mathbf{S}^{-1}\mathbf{m} \\ 0 \end{bmatrix}. \tag{3.46}$$

Then,

$$p(\mathbf{w}|\mathbf{t}') = \mathcal{N}\left(\mathbf{w}|\mathbf{m}', \mathbf{S}'\right), \tag{3.47}$$

where

$$\begin{aligned} \mathbf{m}' &= \mathbf{S}'\left(\mathbf{S}^{-1}\mathbf{m} + \beta t_{N+1}\boldsymbol{\phi}_{N+1}\right), \\ \mathbf{S}' &= \left(\mathbf{S}^{-1} + \beta\boldsymbol{\phi}_{N+1}\boldsymbol{\phi}_{N+1}^\intercal\right)^{-1}. \end{aligned} \tag{3.48}$$

We have

$$\begin{aligned} \mathbf{S}^{-1}\mathbf{m} + \beta t_{N+1}\boldsymbol{\phi}_{N+1} &= \mathbf{S}_0^{-1}\mathbf{m}_0 + \beta\boldsymbol{\Phi}'^\intercal\mathbf{t}', \\ \mathbf{S}^{-1} + \beta\boldsymbol{\phi}_{N+1}\boldsymbol{\phi}_{N+1}^\intercal &= \mathbf{S}_0^{-1} + \beta\boldsymbol{\Phi}'^\intercal\boldsymbol{\Phi}', \end{aligned} \tag{3.49}$$

where

$$\boldsymbol{\Phi}' = \begin{bmatrix} \boldsymbol{\Phi} \\ \boldsymbol{\phi}_{N+1}^\intercal \end{bmatrix}. \tag{3.50}$$

Therefore,

$$\begin{aligned} \mathbf{m}' &= \mathbf{S}'\left(\mathbf{S}_0^{-1}\mathbf{m}_0 + \beta\boldsymbol{\Phi}'^\intercal\mathbf{t}'\right), \\ \mathbf{S}' &= \left(\mathbf{S}_0^{-1} + \beta\boldsymbol{\Phi}'^\intercal\boldsymbol{\Phi}'\right)^{-1}. \end{aligned} \tag{3.51}$$

## 3.9

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal \boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0). \end{aligned} \tag{3.52}$$

By 3.7,

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}(\mathbf{w}|\mathbf{m}, \mathbf{S}), \tag{3.53}$$

where

$$\begin{aligned} \mathbf{m} &= \mathbf{S}\left(\mathbf{S}_0^{-1}\mathbf{m}_0 + \beta\boldsymbol{\Phi}^\intercal\mathbf{t}\right), \\ \mathbf{S} &= \left(\mathbf{S}_0^{-1} + \beta\boldsymbol{\Phi}^\intercal\boldsymbol{\Phi}\right)^{-1}. \end{aligned} \tag{3.54}$$

Let

$$\mathbf{t}' = \begin{bmatrix} \mathbf{t} \\ t_{N+1} \end{bmatrix}, \boldsymbol{\Phi} = \begin{bmatrix} \boldsymbol{\Phi} \\ \boldsymbol{\phi}_{N+1}^\intercal \end{bmatrix}. \tag{3.55}$$

Then,

$$p(\mathbf{w}|\mathbf{t}') = \mathcal{N}\left(\mathbf{w}|\mathbf{m}', \mathbf{t}'\right), \tag{3.56}$$

where

$$\begin{aligned} \mathbf{m}' &= \mathbf{S}'\left(\mathbf{S}_0^{-1}\mathbf{m}_0 + \beta\boldsymbol{\Phi}'^\intercal\mathbf{t}'\right), \\ \mathbf{S}' &= \left(\mathbf{S}_0^{-1} + \beta\boldsymbol{\Phi}'^\intercal\boldsymbol{\Phi}'\right)^{-1}. \end{aligned} \tag{3.57}$$

## 3.10

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal \boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0\right). \end{aligned} \tag{3.58}$$

By 3.7,

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}\left(\mathbf{w}|\mathbf{m}, \mathbf{S}\right), \tag{3.59}$$

where

$$\begin{aligned} \mathbf{m} &= \mathbf{S}\left(\mathbf{S}_0^{-1}\mathbf{m}_0 + \beta\boldsymbol{\Phi}^\intercal\mathbf{t}\right), \\ \mathbf{S} &= \left(\mathbf{S}_0^{-1} + \beta\boldsymbol{\Phi}^\intercal\boldsymbol{\Phi}\right)^{-1}. \end{aligned} \tag{3.60}$$

By marginalisation,

$$p(t_{N+1}|\mathbf{t}) = \int p(t_{N+1}|\mathbf{w})p(\mathbf{w}|\mathbf{t})d\mathbf{w}. \tag{3.61}$$

The logarithm of the integrand of the right hand side except the terms independent of $t_{N+1}$ and $\mathbf{w}$ can be written as

$$\begin{aligned}&-\frac{\beta}{2}\left(t_{N+1}-\mathbf{w}^\intercal\boldsymbol{\phi}_{N+1}\right)^2-\frac{1}{2}(\mathbf{w}-\mathbf{m})^\intercal\mathbf{S}^{-1}(\mathbf{w}-\mathbf{m})\\=&-\frac{1}{2}\mathbf{v}^\intercal\mathbf{M}\mathbf{v}+\mathbf{v}^\intercal\mathbf{u}-\frac{1}{2}\mathbf{m}^\intercal\mathbf{S}^{-1}\mathbf{m},\end{aligned}\tag{3.62}$$

where

$$\mathbf{v}=\begin{bmatrix}\mathbf{w}\\t_{N+1}\end{bmatrix},\mathbf{M}=\begin{bmatrix}\mathbf{S}^{-1}+\beta\boldsymbol{\phi}_{N+1}\boldsymbol{\phi}_{N+1}^\intercal & -\beta\boldsymbol{\phi}_{N+1}\\-\beta\boldsymbol{\phi}_{N+1}^\intercal & \beta\end{bmatrix},\mathbf{u}=\begin{bmatrix}\mathbf{S}^{-1}\mathbf{m}\\0\end{bmatrix}.\tag{3.63}$$

By 2.24,

$$\mathbf{M}^{-1}=\begin{bmatrix}\mathbf{S} & \mathbf{S}\boldsymbol{\phi}_{N+1}\\\boldsymbol{\phi}_{N+1}^\intercal\mathbf{S} & \beta^{-1}+\boldsymbol{\phi}_{N+1}^\intercal\mathbf{S}\boldsymbol{\phi}_{N+1}\end{bmatrix}.\tag{3.64}$$

Then,

$$\mathbf{M}^{-1}\mathbf{u}=\begin{bmatrix}\mathbf{m}\\\mathbf{m}^\intercal\boldsymbol{\phi}_{N+1}\end{bmatrix}.\tag{3.65}$$

Therefore,

$$p(t_{N+1}|\mathbf{t})=\mathcal{N}\left(t_{N+1}|\mathbf{m}^\intercal\boldsymbol{\phi}_{N+1},\sigma^2(\boldsymbol{\phi}_{N+1})\right),\tag{3.66}$$

where

$$\sigma^2(\boldsymbol{\phi})=\beta^{-1}+\boldsymbol{\phi}^\intercal\mathbf{S}\boldsymbol{\phi}.\tag{3.67}$$

## 3.11

Let $t_1,\cdots,t_N$ be conditionally independent variables such that

$$\begin{aligned}p(t_n|\mathbf{w})&=\mathcal{N}\left(t_n|\mathbf{w}^\intercal\boldsymbol{\phi}_n,\beta^{-1}\right),\\p(\mathbf{w})&=\mathcal{N}\left(\mathbf{w}|\mathbf{m}_0,\mathbf{S}_0\right).\end{aligned}\tag{3.68}$$

By 3.7,

$$p(\mathbf{w}|\mathbf{t})=\mathcal{N}\left(\mathbf{w}|\mathbf{m},\mathbf{S}\right),\tag{3.69}$$

where

$$\begin{aligned}\mathbf{m}&=\mathbf{S}\left(\mathbf{S}_0^{-1}\mathbf{m}_0+\beta\boldsymbol{\Phi}^\intercal\mathbf{t}\right),\\\mathbf{S}&=\left(\mathbf{S}_0^{-1}+\beta\boldsymbol{\Phi}^\intercal\boldsymbol{\Phi}\right)^{-1}.\end{aligned}\tag{3.70}$$

By 3.10,

$$p(t_{N+1}|\mathbf{t})=\mathcal{N}\left(t_{N+1}|\mathbf{m}^\intercal\boldsymbol{\phi}_{N+1},\sigma^2(\boldsymbol{\phi}_{N+1})\right),\tag{3.71}$$

where

$$\sigma^2(\boldsymbol{\phi}) = \beta^{-1} + \boldsymbol{\phi}^\intercal \mathbf{S} \boldsymbol{\phi}. \tag{3.72}$$

Then,

$$p(t_{N+2}|\mathbf{t}') = \mathcal{N}\left(t_{N+2}|\mathbf{m}'^\intercal \boldsymbol{\phi}_{N+2}, \sigma'^2(\boldsymbol{\phi}_{N+2})\right), \tag{3.73}$$

where

$$\begin{aligned} \mathbf{m}' &= \mathbf{S}'\left(\mathbf{S}_0^{-1}\mathbf{m}_0 + \beta \boldsymbol{\Phi}'^\intercal \mathbf{t}'\right), \\ \mathbf{S}' &= \left(\mathbf{S}_0^{-1} + \beta \boldsymbol{\Phi}'^\intercal \boldsymbol{\Phi}'\right)^{-1}, \\ \sigma'^2(\boldsymbol{\phi}) &= \beta^{-1} + \boldsymbol{\phi}^\intercal \mathbf{S}' \boldsymbol{\phi}, \end{aligned} \tag{3.74}$$

and

$$\mathbf{t}' = \begin{bmatrix} \mathbf{t} \\ t_{N+1} \end{bmatrix}, \boldsymbol{\Phi} = \begin{bmatrix} \boldsymbol{\Phi} \\ \boldsymbol{\phi}_{N+1}^\intercal \end{bmatrix}. \tag{3.75}$$

We have

$$\sigma^2(\boldsymbol{\phi}) - \sigma'^2(\boldsymbol{\phi}) = \boldsymbol{\phi}^\intercal \left(\mathbf{S} - \mathbf{S}'\right) \boldsymbol{\phi}, \tag{3.76}$$

and

$$\mathbf{S}' = \left(\mathbf{S}^{-1} + \beta \boldsymbol{\phi}_{N+1} \boldsymbol{\phi}_{N+1}^\intercal\right)^{-1}. \tag{3.77}$$

By 2.26,

$$\mathbf{S}' = \mathbf{S} - c\beta^2 \mathbf{S} \boldsymbol{\phi}_{N+1} \boldsymbol{\phi}_{N+1}^\intercal \mathbf{S}, \tag{3.78}$$

where

$$c = \left(\beta + \beta^2 \boldsymbol{\phi}_{N+1}^\intercal \mathbf{S} \boldsymbol{\phi}_{N+1}\right)^{-1}. \tag{3.79}$$

Then,

$$\sigma^2(\boldsymbol{\phi}) - \sigma'^2(\boldsymbol{\phi}) = c\beta^2 \left(\boldsymbol{\phi}^\intercal \mathbf{S} \boldsymbol{\phi}_{N+1}\right)^2. \tag{3.80}$$

Therefore,

$$\sigma^2(\boldsymbol{\phi}) \geq \sigma'^2(\boldsymbol{\phi}). \tag{3.81}$$

## 3.12

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}, \beta) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal \boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}|\beta) &= \mathcal{N}\left(\mathbf{w}|\mathbf{m}_0, \beta^{-1}\mathbf{S}_0\right), \\ p(\beta) &= \mathrm{Gam}(\beta|a_0, b_0). \end{aligned} \tag{3.82}$$

where $\mathbf{w}$ is a vector in $M$ dimensions.

**(a)**

By Bayes' theorem,

$$p(\mathbf{w}|\beta, \mathbf{t})p(\mathbf{t}|\beta) = p(\mathbf{t}|\mathbf{w}, \beta)p(\mathbf{w}|\beta). \tag{3.83}$$

The logarithm of the right hand side except the terms independent of $\mathbf{w}$ can be written as

$$\begin{aligned} & -\frac{\beta}{2}\|\mathbf{t} - \boldsymbol{\Phi}\mathbf{w}\|^2 - \frac{\beta}{2}(\mathbf{w} - \mathbf{m}_0)^\intercal \mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{m}_0) \\ = & -\frac{1}{2}\mathbf{v}^\intercal \mathbf{M}\mathbf{v} + \mathbf{v}^\intercal \mathbf{u} - \frac{\beta}{2}\mathbf{m}_0^\intercal \mathbf{S}_0^{-1}\mathbf{m}_0, \end{aligned} \tag{3.84}$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ \mathbf{t} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \beta\left(\mathbf{S}_0^{-1} + \boldsymbol{\Phi}^\intercal\boldsymbol{\Phi}\right) & -\beta\boldsymbol{\Phi}^\intercal \\ -\beta\boldsymbol{\Phi} & \beta\mathbf{I} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \beta\mathbf{S}_0^{-1}\mathbf{m}_0 \\ \mathbf{0} \end{bmatrix}. \tag{3.85}$$

Therefore,

$$p(\mathbf{w}|\beta, \mathbf{t}) = \mathcal{N}\left(\mathbf{w}|\mathbf{m}, \beta^{-1}\mathbf{S}\right), \tag{3.86}$$

where

$$\begin{aligned} \mathbf{m} &= \mathbf{S}\left(\mathbf{S}_0^{-1}\mathbf{m}_0 + \boldsymbol{\Phi}^\intercal\mathbf{t}\right), \\ \mathbf{S} &= \left(\mathbf{S}_0^{-1} + \boldsymbol{\Phi}^\intercal\boldsymbol{\Phi}\right)^{-1}. \end{aligned} \tag{3.87}$$

**(b)**

By Bayes' theorem,

$$p(\beta|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\beta)p(\beta). \tag{3.88}$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix} \beta^{-1}\mathbf{S}_0 & \beta^{-1}\mathbf{S}_0\boldsymbol{\Phi}^\intercal \\ \beta^{-1}\boldsymbol{\Phi}\mathbf{S}_0 & \beta^{-1}\left(\mathbf{I} + \boldsymbol{\Phi}\mathbf{S}_0\boldsymbol{\Phi}^\intercal\right) \end{bmatrix}, \tag{3.89}$$

so that

$$\mathbf{M}^{-1}\mathbf{u} = \begin{bmatrix} \mathbf{m}_0 \\ \boldsymbol{\Phi}\mathbf{m}_0 \end{bmatrix}. \tag{3.90}$$

Then,

$$p(\mathbf{t}|\beta) = \mathcal{N}\left(\mathbf{t}|\boldsymbol{\Phi}\mathbf{m}_0, \beta^{-1}\left(\mathbf{I} + \boldsymbol{\Phi}\mathbf{S}_0\boldsymbol{\Phi}^\intercal\right)\right). \tag{3.91}$$

Then, the logarithm of $p(\mathbf{t}|\beta)p(\beta)$ except the terms independent of $\beta$ can be written as

$$\begin{aligned}
&-\frac{N}{2}\ln\beta^{-1} - \frac{\beta}{2}(\mathbf{t} - \mathbf{\Phi}\mathbf{m}_0)^\intercal \left(\mathbf{I} + \mathbf{\Phi}\mathbf{S}_0\mathbf{\Phi}^\intercal\right)^{-1} (\mathbf{t} - \mathbf{\Phi}\mathbf{m}_0) \\
&+ (a_0 - 1)\ln\beta - b_0\beta \\
=&(a - 1)\ln\beta - b\beta,
\end{aligned} \tag{3.92}$$

where

$$\begin{aligned}
a &= a_0 + \frac{N}{2}, \\
b &= b_0 + \frac{1}{2}(\mathbf{t} - \mathbf{\Phi}\mathbf{m}_0)^\intercal \left(\mathbf{I} + \mathbf{\Phi}\mathbf{S}_0\mathbf{\Phi}^\intercal\right)^{-1} (\mathbf{t} - \mathbf{\Phi}\mathbf{m}_0).
\end{aligned} \tag{3.93}$$

Therefore,

$$p(\beta|\mathbf{t}) = \mathrm{Gam}(\beta|a, b). \tag{3.94}$$

**(c)**

By Bayes' theorem,

$$p(\mathbf{w}|\beta, \mathbf{t})p(\beta|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{w}, \beta)p(\mathbf{w}|\beta)p(\beta). \tag{3.95}$$

Then, by (a) and (b),

$$p(\mathbf{t}) = \frac{\mathcal{N}\left(\mathbf{t}|\mathbf{\Phi}\mathbf{w}, \beta^{-1}\mathbf{I}\right)\mathcal{N}\left(\mathbf{w}|\mathbf{m}_0, \beta^{-1}\mathbf{S}_0\right)\mathrm{Gam}(\beta|a_0, b_0)}{\mathcal{N}\left(\mathbf{w}|\mathbf{m}, \beta^{-1}\mathbf{S}\right)\mathrm{Gam}(\beta|a, b)}. \tag{3.96}$$

The logarithm of the right hand side can be written as

$$\begin{aligned}
&-\frac{N}{2}\ln(2\pi) - \frac{N}{2}\ln\beta^{-1} - \frac{\beta}{2}\|\mathbf{t} - \mathbf{\Phi}\mathbf{w}\|^2 \\
&-\frac{M}{2}\ln(2\pi) - \frac{M}{2}\ln\beta^{-1} - \frac{1}{2}\det\mathbf{S}_0 - \frac{\beta}{2}(\mathbf{w} - \mathbf{m}_0)^\intercal\mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{m}_0) \\
&+ a_0\ln b_0 - \ln\Gamma(a_0) + (a_0 - 1)\ln\beta - b_0\beta \\
&+ \frac{M}{2}\ln(2\pi) + \frac{M}{2}\ln\beta^{-1} + \frac{1}{2}\det\mathbf{S} + \frac{\beta}{2}(\mathbf{w} - \mathbf{m})^\intercal\mathbf{S}^{-1}(\mathbf{w} - \mathbf{m}) \\
&- a\ln b + \ln\Gamma(a) - (a - 1)\ln\beta + b\beta \\
=& -\frac{N}{2}\ln(2\pi) - \frac{1}{2}\det\mathbf{S}_0 + a_0\ln b_0 - \ln\Gamma(a_0) + \frac{1}{2}\det\mathbf{S} \\
&- a\ln b + \ln\Gamma(a) + a'\ln\beta - b'\beta,
\end{aligned} \tag{3.97}$$

where

$$
\begin{aligned}
a' =& a_0 + \frac{N}{2} - a, \\
b' =& b_0 + \frac{1}{2}\|\mathbf{t} - \mathbf{\Phi}\mathbf{w}\|^2 + \frac{1}{2}(\mathbf{w} - \mathbf{m}_0)^\intercal \mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{m}_0) \\
&- b - \frac{1}{2}(\mathbf{w} - \mathbf{m})^\intercal \mathbf{S}^{-1}(\mathbf{w} - \mathbf{m}).
\end{aligned}
\tag{3.98}
$$

By (a), $a'$ and $b'$ can be written as zero. Then, $p(\mathbf{t})$ except the terms independent of $\mathbf{t}$ can be written as

$$
b^{-a} = \left(1 + (\mathbf{t} - \boldsymbol{\mu})^\intercal (\nu \mathbf{\Sigma})^{-1} (\mathbf{t} - \boldsymbol{\mu})\right)^{-\frac{\nu+N}{2}} b_0^{-a}, \tag{3.99}
$$

where

$$
\begin{aligned}
\boldsymbol{\mu} &= \mathbf{\Phi}\mathbf{m}_0, \\
\mathbf{\Sigma} &= \frac{b_0}{a_0}\left(\mathbf{I} + \mathbf{\Phi}\mathbf{S}_0\mathbf{\Phi}^\intercal\right), \\
\nu &= 2a_0.
\end{aligned}
\tag{3.100}
$$

Therefore, by 2.48,

$$
p(\mathbf{t}) = \mathrm{St}(\mathbf{t}|\boldsymbol{\mu}, \mathbf{\Sigma}, \nu). \tag{3.101}
$$

## 3.13

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$
\begin{aligned}
p(t_n|\mathbf{w}, \beta) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal \boldsymbol{\phi}_n, \beta^{-1}\right), \\
p(\mathbf{w}|\beta) &= \mathcal{N}\left(\mathbf{w}|\mathbf{m}_0, \beta^{-1}\mathbf{S}_0\right), \\
p(\beta) &= \mathrm{Gam}(\beta|a_0, b_0).
\end{aligned}
\tag{3.102}
$$

where $\mathbf{w}$ is a vector in $M$ dimensions. By marginalisation,

$$
\begin{aligned}
p(t_{N+1}|\mathbf{t}) &= \int_0^\infty p(t_{N+1}|\beta, \mathbf{t}) p(\beta|\mathbf{t}) d\beta, \\
p(t_{N+1}|\beta, \mathbf{t}) &= \int p(t_{N+1}|\mathbf{w}, \beta) p(\mathbf{w}|\beta, \mathbf{t}) d\mathbf{w}.
\end{aligned}
\tag{3.103}
$$

By 3.12(a),

$$
p(\mathbf{w}, \beta|\mathbf{t}) = \mathcal{N}\left(\mathbf{w}|\mathbf{m}, \beta^{-1}\mathbf{S}\right), \tag{3.104}
$$

where

$$\begin{aligned}\mathbf{m} &= \mathbf{S}\left(\mathbf{S}_0^{-1}\mathbf{m}_0 + \mathbf{\Phi}^\intercal\mathbf{t}\right), \\ \mathbf{S} &= \left(\mathbf{S}_0^{-1} + \mathbf{\Phi}^\intercal\mathbf{\Phi}\right)^{-1}.\end{aligned} \tag{3.105}$$

Then, the logarithm of $p(t_{N+1}|\mathbf{w},\beta)p(\mathbf{w}|\beta,\mathbf{t})$ except the terms independent of $\mathbf{w}$ can be written as

$$\begin{aligned}&-\frac{\beta}{2}\left(t_{N+1} - \mathbf{w}^\intercal\boldsymbol{\phi}_{N+1}\right)^2 - \frac{\beta}{2}(\mathbf{w}-\mathbf{m})^\intercal\mathbf{S}^{-1}(\mathbf{w}-\mathbf{m}) \\ =&-\frac{\beta}{2}\mathbf{v}^\intercal\mathbf{M}\mathbf{v} + \beta\mathbf{v}^\intercal\mathbf{u} - \frac{\beta}{2}\mathbf{m}^\intercal\mathbf{S}^{-1}\mathbf{m},\end{aligned} \tag{3.106}$$

where

$$\mathbf{v} = \begin{bmatrix}\mathbf{w} \\ t_{N+1}\end{bmatrix}, \mathbf{M} = \begin{bmatrix}\mathbf{S}^{-1} + \boldsymbol{\phi}_{N+1}\boldsymbol{\phi}_{N+1}^\intercal & -\boldsymbol{\phi}_{N+1} \\ -\boldsymbol{\phi}_{N+1}^\intercal & 1\end{bmatrix}, \mathbf{u} = \begin{bmatrix}\mathbf{S}^{-1}\mathbf{m} \\ 0\end{bmatrix}. \tag{3.107}$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix}\mathbf{S} & \mathbf{S}\boldsymbol{\phi}_{N+1} \\ \boldsymbol{\phi}_{N+1}^\intercal\mathbf{S} & 1 + \boldsymbol{\phi}_{N+1}^\intercal\mathbf{S}\boldsymbol{\phi}_{N+1}\end{bmatrix}, \tag{3.108}$$

so that

$$\mathbf{M}^{-1}\mathbf{u} = \begin{bmatrix}\mathbf{m} \\ \mathbf{m}^\intercal\boldsymbol{\phi}_{N+1}\end{bmatrix}. \tag{3.109}$$

Then,

$$p(t_{N+1}|\beta,\mathbf{t}) = \mathcal{N}\left(t_{N+1}|\mathbf{m}^\intercal\boldsymbol{\phi}_{N+1}, \beta^{-1}\left(1 + \boldsymbol{\phi}_{N+1}^\intercal\mathbf{S}\boldsymbol{\phi}_{N+1}\right)\right). \tag{3.110}$$

By 3.12(b),

$$p(\beta|\mathbf{t}) = \mathrm{Gam}(\beta|a,b), \tag{3.111}$$

where

$$\begin{aligned}a &= a_0 + \frac{N}{2}, \\ b &= b_0 + \frac{1}{2}(\mathbf{t}-\mathbf{\Phi}\mathbf{m}_0)^\intercal\left(\mathbf{I} + \mathbf{\Phi}\mathbf{S}_0\mathbf{\Phi}^\intercal\right)^{-1}(\mathbf{t}-\mathbf{\Phi}\mathbf{m}_0).\end{aligned} \tag{3.112}$$

Then, the logarithm of $p(t_{N+1}|\beta,\mathbf{t})p(\beta|\mathbf{t})$ except the terms independent of $\beta$ can be written as

$$\begin{aligned}&-\frac{1}{2}\ln\beta^{-1} - \frac{\left(t_{N+1} - \mathbf{m}^\intercal\boldsymbol{\phi}_{N+1}\right)^2}{2\left(1 + \boldsymbol{\phi}_{N+1}^\intercal\mathbf{S}\boldsymbol{\phi}_{N+1}\right)}\beta + (a-1)\ln\beta - b\beta \\ =&\,(a'-1)\ln\beta - b'\beta,\end{aligned} \tag{3.113}$$

where

$$\begin{aligned} a' &= a + \frac{1}{2}, \\ b' &= b + \frac{\left(t_{N+1} - \mathbf{m}^\intercal \boldsymbol{\phi}_{N+1}\right)^2}{2\left(1 + \boldsymbol{\phi}_{N+1}^\intercal \mathbf{S} \boldsymbol{\phi}_{N+1}\right)}. \end{aligned} \tag{3.114}$$

We have

$$\int_0^\infty \beta^{a'-1} \exp(-b'\beta) d\beta = \Gamma(a') b'^{-a'}. \tag{3.115}$$

Then, $p(t_{N+1}|\mathbf{t})$ except the terms independent of $t_{N+1}$ can be written as

$$b'^{-a'} = \left(1 + \frac{\left(t_{N+1} - \mu\right)^2}{\nu\sigma^2}\right)^{-\frac{\nu+1}{2}} b^{-a'}, \tag{3.116}$$

where

$$\begin{aligned} \mu &= \mathbf{m}^\intercal \boldsymbol{\phi}_{N+1}, \\ \sigma^2 &= \frac{b}{a}\left(1 + \boldsymbol{\phi}_{N+1}^\intercal \mathbf{S} \boldsymbol{\phi}_{N+1}\right), \\ \nu &= 2a. \end{aligned} \tag{3.117}$$

Thererfore, by 2.46,

$$p(t_{N+1}|\mathbf{t}) = \mathrm{St}(t_{N+1}|\mu, \sigma^2, \nu). \tag{3.118}$$

## 3.14 (Incomplete)

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal \boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I}\right), \end{aligned} \tag{3.119}$$

where $\mathbf{w}$ is a vector in $M$ dimensions.

### (a)

By 3.7,

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}\left(\mathbf{w}|\mathbf{m}, \mathbf{S}\right), \tag{3.120}$$

where

$$\begin{aligned} \mathbf{m} &= \beta \mathbf{S} \boldsymbol{\Phi}^\intercal \mathbf{t}, \\ \mathbf{S} &= \left(\alpha \mathbf{I} + \beta \boldsymbol{\Phi}^\intercal \boldsymbol{\Phi}\right)^{-1}. \end{aligned} \tag{3.121}$$

Let

$$y(\boldsymbol{\phi}, \mathbf{w}) = \mathbf{w}^\intercal \boldsymbol{\phi}. \tag{3.122}$$

Then,

$$y(\boldsymbol{\phi}, \mathbf{m}) = \beta \boldsymbol{\phi}^\intercal \mathbf{S} \boldsymbol{\Phi}^\intercal \mathbf{t}. \tag{3.123}$$

The right hand side can be written as

$$\sum_{n=1}^{N} k(\boldsymbol{\phi}, \boldsymbol{\phi}_n) t_n, \tag{3.124}$$

where

$$k(\boldsymbol{\phi}, \boldsymbol{\phi}') = \beta \boldsymbol{\phi}^\intercal \mathbf{S} \boldsymbol{\phi}'. \tag{3.125}$$

Let us suppose that $\phi_1, \cdots, \phi_M$ are linearly independent, $N > M$ and

$$\phi_1 = 1. \tag{3.126}$$

Then, we can construct a new basis set $\psi_1, \cdots, \psi_M$ such that

$$\boldsymbol{\Psi}^\intercal \boldsymbol{\Psi} = \mathbf{I}? \tag{3.127}$$

where

$$\psi_1 = 1. \tag{3.128}$$

**(b)**

## 3.15

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n | \mathbf{w}^\intercal \boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w} | \mathbf{0}, \alpha^{-1} \mathbf{I}\right), \end{aligned} \tag{3.129}$$

where $\mathbf{w}$ is a vector in $M$ dimensions. By 3.19,

$$\begin{aligned} \ln p(\mathbf{t}) =& \frac{M}{2} \ln \alpha + \frac{N}{2} \ln \beta - \frac{N}{2} \ln(2\pi) + \frac{1}{2} \ln(\det \mathbf{S}) \\ & - E(\mathbf{m}, \alpha, \beta), \end{aligned} \tag{3.130}$$

where

$$
\begin{aligned}
E(\mathbf{w}, \alpha, \beta) &= \frac{\beta}{2}\|\mathbf{t} - \mathbf{\Phi}\mathbf{w}\|^2 + \frac{\alpha}{2}\|\mathbf{w}\|^2, \\
\mathbf{m} &= \beta\mathbf{S}\mathbf{\Phi}^\intercal\mathbf{t}, \\
\mathbf{S}^{-1} &= \alpha\mathbf{I} + \beta\mathbf{\Phi}^\intercal\mathbf{\Phi}.
\end{aligned}
\tag{3.131}
$$

By 3.20, the maximum likelihood solutions to $\alpha$ and $\beta$ are given by

$$
\begin{aligned}
\alpha_{\mathrm{ML}} &= \frac{\gamma}{\|\mathbf{m}\|^2} \\
\beta_{\mathrm{ML}} &= \frac{N - \gamma}{\|\mathbf{t} - \mathbf{\Phi}\mathbf{m}\|^2}
\end{aligned}
\tag{3.132}
$$

where

$$
\gamma = \sum_{m=1}^{M} \frac{\beta\lambda_m}{\alpha + \beta\lambda_m}, \tag{3.133}
$$

and $\lambda_1, \cdots, \lambda_M$ are the eigenvalues of $\mathbf{\Phi}^\intercal\mathbf{\Phi}$. Therefore,

$$
E(\mathbf{m}, \alpha_{\mathrm{ML}}, \beta_{\mathrm{ML}}) = \frac{N}{2}. \tag{3.134}
$$

## 3.16

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$
\begin{aligned}
p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal\boldsymbol{\phi}_n, \beta^{-1}\right), \\
p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I}\right),
\end{aligned}
\tag{3.135}
$$

where $\mathbf{w}$ is a vector in $M$ dimensions. By marginalisation,

$$
p(\mathbf{t}) = \int p(\mathbf{t}|\mathbf{w})p(\mathbf{w})d\mathbf{w}. \tag{3.136}
$$

The logarithm of the integrand of the right hand side except the terms independent of $\mathbf{w}$ can be written as

$$
-\frac{\beta}{2}\|\mathbf{t} - \mathbf{\Phi}\mathbf{w}\|^2 - \frac{\alpha}{2}\mathbf{w}^\intercal\mathbf{w} = -\frac{1}{2}\mathbf{v}^\intercal\mathbf{M}\mathbf{v}, \tag{3.137}
$$

where

$$
\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ \mathbf{t} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \alpha\mathbf{I} + \beta\mathbf{\Phi}^\intercal\mathbf{\Phi} & -\beta\mathbf{\Phi}^\intercal \\ -\beta\mathbf{\Phi} & \beta\mathbf{I} \end{bmatrix}. \tag{3.138}
$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix} \alpha^{-1}\mathbf{I} & \alpha^{-1}\mathbf{\Phi}^\intercal \\ \alpha^{-1}\mathbf{\Phi} & \alpha^{-1}\mathbf{\Phi}\mathbf{\Phi}^\intercal + \beta^{-1}\mathbf{I} \end{bmatrix}. \tag{3.139}$$

Therefore,

$$p(\mathbf{t}) = \mathcal{N}\left(\mathbf{t}|\mathbf{0}, \alpha^{-1}\mathbf{\Phi}\mathbf{\Phi}^\intercal + \beta^{-1}\mathbf{I}\right). \tag{3.140}$$

## 3.17

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal\boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I}), \end{aligned} \tag{3.141}$$

where $\mathbf{w}$ is a vector in $M$ dimensions. By marginalisation,

$$p(\mathbf{t}) = \int p(\mathbf{t}|\mathbf{w})p(\mathbf{w})d\mathbf{w}. \tag{3.142}$$

The logarithm of the integrand of the right hand side can be written as

$$\begin{aligned} &-\frac{N}{2}\ln\left(2\pi\beta^{-1}\right) - \frac{\beta}{2}\|\mathbf{t} - \mathbf{\Phi}\mathbf{w}\|^2 \\ &-\frac{M}{2}\ln(2\pi) - \frac{1}{2}\ln\left(\det\left(\alpha^{-1}\mathbf{I}\right)\right) - \frac{\alpha}{2}\|\mathbf{w}\|^2. \end{aligned} \tag{3.143}$$

Therefore,

$$p(\mathbf{t}) = \left(\frac{\beta}{2\pi}\right)^{\frac{N}{2}}\left(\frac{\alpha}{2\pi}\right)^{\frac{M}{2}}\int \exp\left(-E(\mathbf{w})\right)d\mathbf{w}, \tag{3.144}$$

where

$$E(\mathbf{w}) = \frac{\beta}{2}\|\mathbf{t} - \mathbf{\Phi}\mathbf{w}\|^2 + \frac{\alpha}{2}\|\mathbf{w}\|^2. \tag{3.145}$$

## 3.18

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal\boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I}\right), \end{aligned} \tag{3.146}$$

where $\mathbf{w}$ is a vector in $M$ dimensions. By 3.17,

$$p(\mathbf{t}) = \left(\frac{\beta}{2\pi}\right)^{\frac{N}{2}} \left(\frac{\alpha}{2\pi}\right)^{\frac{M}{2}} \int \exp\left(-E(\mathbf{w})\right) d\mathbf{w}, \tag{3.147}$$

where

$$E(\mathbf{w}) = \frac{\beta}{2}\|\mathbf{t} - \mathbf{\Phi}\mathbf{w}\|^2 + \frac{\alpha}{2}\|\mathbf{w}\|^2. \tag{3.148}$$

We have

$$\nabla E(\mathbf{w}) = -\beta\mathbf{\Phi}^\intercal(\mathbf{t} - \mathbf{\Phi}\mathbf{w}) + \alpha\mathbf{w}. \tag{3.149}$$

Then,

$$\begin{aligned} \nabla E(\mathbf{w}) &= \mathbf{S}^{-1}(\mathbf{w} - \mathbf{m}), \\ \nabla\nabla E(\mathbf{w}) &= \mathbf{S}^{-1}. \end{aligned} \tag{3.150}$$

where

$$\begin{aligned} \mathbf{m} &= \beta\mathbf{S}\mathbf{\Phi}^\intercal\mathbf{t}, \\ \mathbf{S} &= \left(\beta\mathbf{\Phi}^\intercal\mathbf{\Phi} + \alpha\mathbf{I}\right)^{-1}. \end{aligned} \tag{3.151}$$

Then,

$$E(\mathbf{w}) = E(\mathbf{m}) + \frac{1}{2}(\mathbf{w} - \mathbf{m})^\intercal\mathbf{S}^{-1}(\mathbf{w} - \mathbf{m}). \tag{3.152}$$

## 3.19

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal\boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I}\right), \end{aligned} \tag{3.153}$$

where $\mathbf{w}$ is a vector in $M$ dimensions. By 3.17,

$$p(\mathbf{t}) = \left(\frac{\beta}{2\pi}\right)^{\frac{N}{2}} \left(\frac{\alpha}{2\pi}\right)^{\frac{M}{2}} \int \exp\left(-E(\mathbf{w})\right) d\mathbf{w}, \tag{3.154}$$

where

$$E(\mathbf{w}) = \frac{\beta}{2}\|\mathbf{t} - \mathbf{\Phi}\mathbf{w}\|^2 + \frac{\alpha}{2}\|\mathbf{w}\|^2. \tag{3.155}$$

By 3.18,

$$E(\mathbf{w}) = E(\mathbf{m}) + \frac{1}{2}(\mathbf{w} - \mathbf{m})^\intercal\mathbf{S}^{-1}(\mathbf{w} - \mathbf{m}), \tag{3.156}$$

where

$$\begin{aligned}\mathbf{m} &= \beta\mathbf{S}\mathbf{\Phi}^\intercal\mathbf{t},\\ \mathbf{S} &= \left(\beta\mathbf{\Phi}^\intercal\mathbf{\Phi}+\alpha\mathbf{I}\right)^{-1}.\end{aligned} \tag{3.157}$$

Then, the integral can be written as

$$\begin{aligned}&\exp\left(-E(\mathbf{m})\right)\int\exp\left(-\frac{1}{2}(\mathbf{w}-\mathbf{m})^\intercal\mathbf{S}^{-1}(\mathbf{w}-\mathbf{m})\right)d\mathbf{w}\\ =&(2\pi)^{\frac{M}{2}}(\det\mathbf{S})^{\frac{1}{2}}\exp\left(-E(\mathbf{m})\right).\end{aligned} \tag{3.158}$$

Therefore,

$$\ln p(\mathbf{t}) = \frac{M}{2}\ln\alpha + \frac{N}{2}\ln\beta - \frac{N}{2}\ln(2\pi) + \frac{1}{2}\ln(\det\mathbf{S}) - E(\mathbf{m}). \tag{3.159}$$

## 3.20

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned}p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal\boldsymbol{\phi}_n, \beta^{-1}\right),\\ p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I}\right),\end{aligned} \tag{3.160}$$

where $\mathbf{w}$ is a vector in $M$ dimensions. By 3.19,

$$\ln p(\mathbf{t}) = \frac{M}{2}\ln\alpha + \frac{N}{2}\ln\beta - \frac{N}{2}\ln(2\pi) + \frac{1}{2}\ln(\det\mathbf{S}) - E(\mathbf{m}), \tag{3.161}$$

where

$$E(\mathbf{w}) = \frac{\beta}{2}\|\mathbf{t}-\mathbf{\Phi}\mathbf{w}\|^2 + \frac{\alpha}{2}\|\mathbf{w}\|^2, \tag{3.162}$$

and

$$\begin{aligned}\mathbf{m} &= \beta\mathbf{S}\mathbf{\Phi}^\intercal\mathbf{t},\\ \mathbf{S} &= \left(\beta\mathbf{\Phi}^\intercal\mathbf{\Phi}+\alpha\mathbf{I}\right)^{-1}.\end{aligned} \tag{3.163}$$

By 3.21, setting the derivatives of $\ln p(\mathbf{t})$ with respect to $\alpha$ and $\beta$ to zero gives

$$\begin{aligned}0 &= \frac{M}{2\alpha} + \frac{1}{2}\operatorname{tr}\left(\mathbf{S}^{-1}\frac{\partial\mathbf{S}}{\partial\alpha}\right) - \frac{\partial E(\mathbf{m})}{\partial\alpha},\\ 0 &= \frac{N}{2\beta} + \frac{1}{2}\operatorname{tr}\left(\mathbf{S}^{-1}\frac{\partial\mathbf{S}}{\partial\beta}\right) - \frac{\partial E(\mathbf{m})}{\partial\beta}.\end{aligned} \tag{3.164}$$

We have

$$\mathbf{O} = \frac{\partial \mathbf{S}^{-1}\mathbf{S}}{\partial \alpha},$$
$$\mathbf{O} = \frac{\partial \mathbf{S}^{-1}\mathbf{S}}{\partial \beta}. \tag{3.165}$$

The right hand sides can be written as

$$\mathbf{S}^{-1}\frac{\partial \mathbf{S}}{\partial \alpha} + \frac{\partial \mathbf{S}^{-1}}{\partial \alpha}\mathbf{S} = \mathbf{S}^{-1}\frac{\partial \mathbf{S}}{\partial \alpha} + \mathbf{S},$$
$$\mathbf{S}^{-1}\frac{\partial \mathbf{S}}{\partial \beta} + \frac{\partial \mathbf{S}^{-1}}{\partial \beta}\mathbf{S} = \mathbf{S}^{-1}\frac{\partial \mathbf{S}}{\partial \beta} + \mathbf{\Phi}^\intercal\mathbf{\Phi}\mathbf{S}. \tag{3.166}$$

Then,

$$\mathbf{S}^{-1}\frac{\partial \mathbf{S}}{\partial \alpha} = -\mathbf{S},$$
$$\mathbf{S}^{-1}\frac{\partial \mathbf{S}}{\partial \beta} = -\mathbf{\Phi}^\intercal\mathbf{\Phi}\mathbf{S}. \tag{3.167}$$

We have

$$\operatorname{tr}\mathbf{S} = \sum_{m=1}^{M}(\beta\lambda_m + \alpha)^{-1},$$
$$\operatorname{tr}(\mathbf{\Phi}^\intercal\mathbf{\Phi}\mathbf{S}) = \sum_{m=1}^{M}\lambda_m(\beta\lambda_m + \alpha)^{-1}, \tag{3.168}$$

where $\lambda_1, \cdots, \lambda_M$ are the eigenvalues of $\mathbf{\Phi}^\intercal\mathbf{\Phi}$. The right hand sides can be written as $\alpha^{-1}(M-\gamma)$ and $\beta^{-1}\gamma$, where

$$\gamma = \sum_{m=1}^{M}\frac{\beta\lambda_m}{\alpha + \beta\lambda_m}. \tag{3.169}$$

We have

$$\frac{\partial E(\mathbf{m})}{\partial \alpha} = \nabla E(\mathbf{m})\frac{\partial \mathbf{m}}{\partial \alpha} + \frac{1}{2}\|\mathbf{m}\|^2,$$
$$\frac{\partial E(\mathbf{m})}{\partial \beta} = \nabla E(\mathbf{m})\frac{\partial \mathbf{m}}{\partial \beta} + \frac{1}{2}\|\mathbf{t} - \mathbf{\Phi}\mathbf{m}\|^2. \tag{3.170}$$

We have

$$\nabla E(\mathbf{m}) = \mathbf{0}. \tag{3.171}$$

Then,

$$\begin{aligned} 0 &= \frac{M}{2\alpha} - \frac{M-\gamma}{2\alpha} - \frac{1}{2}\|\mathbf{m}\|^2, \\ 0 &= \frac{N}{2\beta} - \frac{\gamma}{2\beta} - \frac{1}{2}\|\mathbf{t} - \mathbf{\Phi}\mathbf{m}\|^2. \end{aligned} \tag{3.172}$$

Therefore, the maximum likelihood solutions to $\alpha$ and $\beta$ are given by

$$\begin{aligned} \alpha_{\mathrm{ML}} &= \frac{\gamma}{\|\mathbf{m}\|^2}, \\ \beta_{\mathrm{ML}} &= \frac{N-\gamma}{\|\mathbf{t} - \mathbf{\Phi}\mathbf{m}\|^2}. \end{aligned} \tag{3.173}$$

## 3.21

Let $\mathbf{\Sigma}$ be a $M \times M$ real symmetric matrix such that

$$\mathbf{\Sigma}\mathbf{u}_m = \lambda_m \mathbf{u}_m, \tag{3.174}$$

where $\mathbf{u}_1, \cdots, \mathbf{u}_M$ are unit vectors. Let

$$\begin{aligned} \mathbf{\Lambda} &= \mathrm{diag}(\lambda_1, \cdots, \lambda_M), \\ \mathbf{U} &= [\mathbf{u}_1 \cdots \mathbf{u}_M]. \end{aligned} \tag{3.175}$$

By 2.19,

$$\begin{aligned} \mathbf{\Sigma} &= \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\intercal, \\ \mathbf{U}^\intercal\mathbf{U} &= \mathbf{I}. \end{aligned} \tag{3.176}$$

Then,

$$\det \mathbf{\Sigma} = \prod_{m=1}^{M} \lambda_m, \tag{3.177}$$

so that

$$\ln(\det \mathbf{\Sigma}) = \sum_{m=1}^{M} \ln \lambda_m. \tag{3.178}$$

Then,

$$\frac{\partial}{\partial \alpha} \ln(\det \mathbf{\Sigma}) = \sum_{m=1}^{M} \frac{\partial \lambda_m}{\partial \alpha} \frac{1}{\lambda_m}. \tag{3.179}$$

Then,

$$\frac{\partial}{\partial \alpha} \ln(\det \boldsymbol{\Sigma}) = \operatorname{tr}\left(\boldsymbol{\Lambda}^{-1} \frac{\partial \boldsymbol{\Lambda}}{\partial \alpha}\right). \tag{3.180}$$

The right hand side can be written as

$$\operatorname{tr}\left(\mathbf{U}\boldsymbol{\Lambda}^{-1}\mathbf{U}^{\intercal} \frac{\partial \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^{\intercal}}{\partial \alpha}\right) = \operatorname{tr}\left(\boldsymbol{\Sigma}^{-1} \frac{\partial \boldsymbol{\Sigma}}{\partial \alpha}\right). \tag{3.181}$$

Therefore,

$$\frac{\partial}{\partial \alpha} \ln(\det \boldsymbol{\Sigma}) = \operatorname{tr}\left(\boldsymbol{\Sigma}^{-1} \frac{\partial \boldsymbol{\Sigma}}{\partial \alpha}\right). \tag{3.182}$$

## 3.22

Refer to 3.20.

## 3.23

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}, \beta) &= \mathcal{N}\left(t_n|\mathbf{w}^{\intercal}\boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}|\beta) &= \mathcal{N}\left(\mathbf{w}|\mathbf{m}_0, \beta^{-1}\mathbf{S}_0\right), \\ p(\beta) &= \operatorname{Gam}(\beta|a_0, b_0). \end{aligned} \tag{3.183}$$

where $\mathbf{w}$ is a vector in $M$ dimensions. By marginalisation,

$$\begin{aligned} p(\mathbf{t}) &= \int_0^\infty p(\mathbf{t}|\beta)p(\beta)d\beta, \\ p(\mathbf{t}|\beta) &= \int p(\mathbf{t}|\mathbf{w}, \beta)p(\mathbf{w}|\beta)d\mathbf{w}. \end{aligned} \tag{3.184}$$

The logarithm of $p(\mathbf{t}|\mathbf{w}, \beta)p(\mathbf{w}|\beta)$ except the terms independent of $\mathbf{w}$ can be written as

$$\begin{aligned} &-\frac{\beta}{2}\|\mathbf{t} - \boldsymbol{\Phi}\mathbf{w}\|^2 - \frac{\beta}{2}(\mathbf{w} - \mathbf{m}_0)^{\intercal}\mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{m}_0) \\ =& -\frac{1}{2}\mathbf{v}^{\intercal}\mathbf{M}\mathbf{v} + \mathbf{v}^{\intercal}\mathbf{u} - \frac{\beta}{2}\mathbf{m}_0^{\intercal}\mathbf{S}_0^{-1}\mathbf{m}_0, \end{aligned} \tag{3.185}$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ \mathbf{t} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \beta\left(\mathbf{S}_0^{-1} + \boldsymbol{\Phi}^{\intercal}\boldsymbol{\Phi}\right) & -\beta\boldsymbol{\Phi}^{\intercal} \\ -\beta\boldsymbol{\Phi} & \beta\mathbf{I} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \beta\mathbf{S}_0^{-1}\mathbf{m}_0 \\ \mathbf{0} \end{bmatrix}. \tag{3.186}$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix} \beta^{-1}\mathbf{S}_0 & \beta^{-1}\mathbf{S}_0\boldsymbol{\Phi}^\intercal \\ \beta^{-1}\boldsymbol{\Phi}\mathbf{S}_0 & \beta^{-1}\left(\mathbf{I}+\boldsymbol{\Phi}\mathbf{S}_0\boldsymbol{\Phi}^\intercal\right) \end{bmatrix}, \tag{3.187}$$

so that

$$\mathbf{M}^{-1}\mathbf{u} = \begin{bmatrix} \mathbf{m}_0 \\ \boldsymbol{\Phi}\mathbf{m}_0 \end{bmatrix}. \tag{3.188}$$

Then,

$$p(\mathbf{t}|\beta) = \mathcal{N}\left(\mathbf{t}|\boldsymbol{\Phi}\mathbf{m}_0, \beta^{-1}\left(\mathbf{I}+\boldsymbol{\Phi}\mathbf{S}_0\boldsymbol{\Phi}^\intercal\right)\right). \tag{3.189}$$

Then, the logarithm of $p(\mathbf{t}|\beta)p(\beta)$ excpet the terms independent of $\beta$ can be written as

$$\begin{aligned} & -\frac{N}{2}\ln\beta^{-1} - \frac{\beta}{2}(\mathbf{t}-\boldsymbol{\Phi}\mathbf{m}_0)^\intercal\left(\mathbf{I}+\boldsymbol{\Phi}\mathbf{S}_0\boldsymbol{\Phi}^\intercal\right)^{-1}(\mathbf{t}-\boldsymbol{\Phi}\mathbf{m}_0) \\ & + (a_0-1)\ln\beta - b_0\beta \\ =& (a-1)\ln\beta - b\beta, \end{aligned} \tag{3.190}$$

where

$$\begin{aligned} a &= a_0 + \frac{N}{2}, \\ b &= b_0 + \frac{1}{2}(\mathbf{t}-\boldsymbol{\Phi}\mathbf{m}_0)\left(\mathbf{I}+\boldsymbol{\Phi}\mathbf{S}_0\boldsymbol{\Phi}^\intercal\right)^{-1}(\mathbf{t}-\boldsymbol{\Phi}\mathbf{m}_0)^\intercal. \end{aligned} \tag{3.191}$$

We have

$$\int_0^\infty \beta^{a_0-1}\exp(-b\beta)d\beta = b^{-a}\Gamma(a). \tag{3.192}$$

We have

$$b^{-a} = \left(1+(\mathbf{t}-\boldsymbol{\mu})^\intercal(\nu\boldsymbol{\Sigma})^{-1}(\mathbf{t}-\boldsymbol{\mu})\right)^{-\frac{\nu+N}{2}} b_0^{-a}, \tag{3.193}$$

where

$$\begin{aligned} \boldsymbol{\mu} &= \boldsymbol{\Phi}\mathbf{m}_0, \\ \boldsymbol{\Sigma} &= \frac{b_0}{a_0}\left(\mathbf{I}+\boldsymbol{\Phi}\mathbf{S}_0\boldsymbol{\Phi}^\intercal\right), \\ \nu &= 2a_0. \end{aligned} \tag{3.194}$$

Therefore, by 2.48,

$$p(\mathbf{t}) = \mathrm{St}(\mathbf{t}|\boldsymbol{\mu}, \boldsymbol{\Lambda}, \nu), \tag{3.195}$$

## 3.24

Refer to 3.12.

# 4 Linear Models for Classification

## 4.1

Let $x_1, \cdots, x_M$ and $y_1, \cdots, y_N$ be two sets of data points. Then, the corresponding convex hulls are defined as the sets of all points $\mathbf{x}$ and $\mathbf{y}$ such that

$$\begin{aligned} \mathbf{x} &= \sum_{m=1}^{M} \alpha_m \mathbf{x}_m, \\ \mathbf{y} &= \sum_{n=1}^{N} \beta_n \mathbf{y}_n, \end{aligned} \tag{4.1}$$

where

$$\begin{gathered} \sum_{m=1}^{M} \alpha_m = \sum_{n=1}^{N} \beta_n = 1, \\ \alpha_m \geq 0, \beta_n \geq 0. \end{gathered} \tag{4.2}$$

Let us assume that $\alpha_1, \cdots, \alpha_M$ and $\beta_1, \cdots, \beta_N$ below are subject to the constraints above.

### (a)

If the convex hulls intersect, then there exist $\alpha_1, \cdots, \alpha_M$ and $\beta_1, \cdots, \beta_N$ such that

$$\sum_{m=1}^{M} \alpha_m \mathbf{x}_m = \sum_{n=1}^{N} \beta_n \mathbf{y}_n. \tag{4.3}$$

Then,

$$\sum_{m=1}^{M} \alpha_m \left( \hat{\mathbf{w}}^\intercal \mathbf{x}_m + w_0 \right) = \hat{\mathbf{w}}^\intercal \sum_{m=1}^{M} \alpha_m \mathbf{x}_m + w_0 \sum_{m} \alpha_m, \tag{4.4}$$

for any $\hat{\mathbf{w}}$ and $w_0$. The right hand side can be written as

$$\hat{\mathbf{w}}^\intercal \sum_{n=1}^{N} \beta_n \mathbf{y}_n + w_0 \sum_{n=1}^{N} \beta_n = \sum_{n=1}^{N} \beta_n \left( \hat{\mathbf{w}}^\intercal \mathbf{y}_n + w_0 \right). \tag{4.5}$$

Therefore, there do not exist $\hat{\mathbf{w}}$ and $w_0$ such that

$$\begin{aligned} \hat{\mathbf{w}}^\intercal \mathbf{x}_m + w_0 > 0, \\ \hat{\mathbf{w}}^\intercal \mathbf{y}_n + w_0 < 0. \end{aligned} \tag{4.6}$$

**(b)**

If there exist $\hat{\mathbf{w}}$ and $w_0$ such that

$$\begin{aligned} \hat{\mathbf{w}}^\intercal \mathbf{x}_m + w_0 > 0, \\ \hat{\mathbf{w}}^\intercal \mathbf{y}_n + w_0 < 0, \end{aligned} \tag{4.7}$$

then

$$\begin{aligned} \sum_{m=1}^{M} \alpha_m \left(\hat{\mathbf{w}}^\intercal \mathbf{x}_m + w_0\right) > 0, \\ \sum_{n=1}^{N} \beta_n \left(\hat{\mathbf{w}}^\intercal \mathbf{y}_n + w_0\right) < 0. \end{aligned} \tag{4.8}$$

The left hand sides can be written as

$$\begin{aligned} \hat{\mathbf{w}}^\intercal \sum_{m=1}^{M} \alpha_m \mathbf{x}_m + w_0 \sum_{m=1}^{M} \alpha_m = \hat{\mathbf{w}}^\intercal \sum_{m=1}^{M} \alpha_m \mathbf{x}_m + w_0, \\ \hat{\mathbf{w}}^\intercal \sum_{n=1}^{N} \beta_n \mathbf{y}_n + w_0 \sum_{n=1}^{N} \beta_n = \hat{\mathbf{w}}^\intercal \sum_{n=1}^{N} \beta_n \mathbf{y}_n + w_0. \end{aligned} \tag{4.9}$$

Then, there do not exist $\alpha_1, \cdots, \alpha_M$ and $\beta_1, \cdots, \beta_N$ such that

$$\sum_{m=1}^{M} \alpha_m \mathbf{x}_m = \sum_{n=1}^{N} \beta_n \mathbf{y}_n. \tag{4.10}$$

Therefore, the convex hulls do not intersect.

## 4.2 (Incomplete)

Let $\mathbf{x}_1, \cdots, \mathbf{x}_N$ and $\mathbf{w}_1, \cdots, \mathbf{w}_K$ be variables in $M$ dimensions and $\mathbf{t}_1, \cdots, \mathbf{t}_N$ be variables in $K$ dimensions. Let

$$E(\tilde{\mathbf{W}}) = \frac{1}{2} \operatorname{tr}\left((\tilde{\mathbf{X}}\tilde{\mathbf{W}} - \mathbf{T})^\intercal (\tilde{\mathbf{X}}\tilde{\mathbf{W}} - \mathbf{T})\right), \tag{4.11}$$

where

$$\tilde{\mathbf{X}} = \begin{bmatrix} 1 & \mathbf{x}_1^\intercal \\ \vdots & \vdots \\ 1 & \mathbf{x}_N^\intercal \end{bmatrix}, \tilde{\mathbf{W}} = \begin{bmatrix} w_{10} & \cdots & w_{K0} \\ \mathbf{w}_1 & \cdots & \mathbf{w}_K \end{bmatrix}, \mathbf{T} = \begin{bmatrix} \mathbf{t}_1^\intercal \\ \vdots \\ \mathbf{t}_N^\intercal \end{bmatrix}. \tag{4.12}$$

Setting the derivative of $E(\tilde{\mathbf{W}})$ with respect to $\tilde{\mathbf{W}}$ to zero gives

$$\mathbf{O} = \tilde{\mathbf{X}}^\intercal(\tilde{\mathbf{X}}\tilde{\mathbf{W}} - \mathbf{T}). \tag{4.13}$$

Then,

$$\operatorname*{argmin}_{\tilde{\mathbf{W}}} E(\tilde{\mathbf{W}}) = \tilde{\mathbf{W}}^*, \tag{4.14}$$

where

$$\tilde{\mathbf{W}}^* = \left(\tilde{\mathbf{X}}^\intercal\tilde{\mathbf{X}}\right)^{-1}\tilde{\mathbf{X}}^\intercal\mathbf{T}. \tag{4.15}$$

Then,

$$(\tilde{\mathbf{W}}^*)^\intercal\tilde{\mathbf{x}} - \mathbf{t}_n = \mathbf{T}^\intercal\tilde{\mathbf{X}}\left(\tilde{\mathbf{X}}^\intercal\tilde{\mathbf{X}}\right)^{-1}\tilde{\mathbf{x}} - \mathbf{t}_n, \tag{4.16}$$

where $\tilde{\mathbf{x}}$ is a vector in $M+1$ dimensions whose first element is 1. The right hand side can be written as

$$\mathbf{T}^\intercal\left(\tilde{\mathbf{X}}\left(\tilde{\mathbf{X}}^\intercal\tilde{\mathbf{X}}\right)^{-1}\tilde{\mathbf{x}} - \mathbf{v}_n\right) = \mathbf{0}? \tag{4.17}$$

where $\mathbf{v}_n$ is a vector in $N$ dimensions whose $n$th element is 1 and other elements are zero. Then,

$$(\tilde{\mathbf{W}}^*)^\intercal\tilde{\mathbf{x}} - \mathbf{t}_n = \mathbf{0}. \tag{4.18}$$

Therefore, if

$$\mathbf{a}^\intercal\mathbf{t}_n + b = 0, \tag{4.19}$$

then

$$\mathbf{a}^\intercal(\tilde{\mathbf{W}}^*)^\intercal\tilde{\mathbf{x}} + b = 0. \tag{4.20}$$

## 4.3 (Incomplete)

## 4.4

Let $\mathbf{x}_1, \cdots, \mathbf{x}_N$ be variables and let

$$\mathbf{m}_k = \frac{1}{N_k}\sum_{n\in\mathcal{C}_k}\mathbf{x}_n, \tag{4.21}$$

where

$$N_k = \sum_{n\in\mathcal{C}_k} 1. \tag{4.22}$$

In order to maximise $\mathbf{w}^\intercal(\mathbf{m}_2 - \mathbf{m}_1)$ under the constraint

$$\|\mathbf{w}\|^2 = 1, \tag{4.23}$$

let

$$L = \mathbf{w}^\intercal(\mathbf{m}_2 - \mathbf{m}_1) + \lambda\left(\|\mathbf{w}\|^2 - 1\right). \tag{4.24}$$

Setting the derivatives of $L$ with respect to $\mathbf{w}$ and $\lambda$ to zero gives

$$\begin{aligned} \mathbf{m}_2 - \mathbf{m}_1 + 2\lambda\mathbf{w} &= \mathbf{0}, \\ \|\mathbf{w}\|^2 - 1 &= 0. \end{aligned} \tag{4.25}$$

Therefore, the optimal solution to $\mathbf{w}$ satisfy

$$\mathbf{w}^* = c(\mathbf{m}_2 - \mathbf{m}_1), \tag{4.26}$$

where $c$ is a constant.

## 4.5

Let $\mathbf{x}_1, \cdots, \mathbf{x}_N$ be variables and let

$$\mathbf{m}_k = \frac{1}{N_k}\sum_{n\in\mathcal{C}_k}\mathbf{x}_n, \tag{4.27}$$

where

$$N_k = \sum_{n\in\mathcal{C}_k} 1. \tag{4.28}$$

Let

$$J(\mathbf{w}) = \frac{(m_2 - m_1)^2}{s_1^2 + s_2^2}, \tag{4.29}$$

where

$$\begin{aligned} s_k^2 &= \sum_{n\in\mathcal{C}_k}(y_n - m_k)^2, \\ y_n &= \mathbf{w}^\intercal\mathbf{x}_n, \\ m_k &= \mathbf{w}^\intercal\mathbf{m}_k. \end{aligned} \tag{4.30}$$

Then, $J(\mathbf{w})$ can be written as

$$\frac{\left(\mathbf{w}^\intercal(\mathbf{m}_2 - \mathbf{m}_1)\right)^2}{\sum_{n\in\mathcal{C}_1}\left(\mathbf{w}^\intercal(\mathbf{x}_n - \mathbf{m}_1)\right)^2 + \sum_{n\in\mathcal{C}_2}\left(\mathbf{w}^\intercal(\mathbf{x}_n - \mathbf{m}_2)\right)^2} = \frac{\mathbf{w}^\intercal\mathbf{S}_\mathrm{B}\mathbf{w}}{\mathbf{w}^\intercal\mathbf{S}_\mathrm{W}\mathbf{w}}, \tag{4.31}$$

where

$$\mathbf{S}_{\mathrm{B}} = (\mathbf{m}_2 - \mathbf{m}_1)(\mathbf{m}_2 - \mathbf{m}_1)^{\intercal},$$
$$\mathbf{S}_{\mathrm{W}} = \sum_{k=1,2} \sum_{n \in \mathcal{C}_k} (\mathbf{x}_n - \mathbf{m}_k)(\mathbf{x}_n - \mathbf{m}_k)^{\intercal}. \tag{4.32}$$

## 4.6

Let $\mathbf{x}_1, \cdots, \mathbf{x}_N$ be variables and let

$$\mathbf{m}_k = \frac{1}{N_k} \sum_{n \in \mathcal{C}_k} \mathbf{x}_n, \tag{4.33}$$

where $k \in \{1, 2\}$ and

$$N_k = \sum_{n \in \mathcal{C}_k} 1. \tag{4.34}$$

Let

$$E = \frac{1}{2} \sum_{n=1}^{N} \left( \mathbf{w}^{\intercal} \mathbf{x}_n + w_0 - t_n \right)^2, \tag{4.35}$$

where

$$t_n = \begin{cases} \frac{N}{N_1}, & n \in \mathcal{C}_1, \\ -\frac{N}{N_2}, & n \in \mathcal{C}_2. \end{cases} \tag{4.36}$$

Setting the derivative of $E$ with respect to $\mathbf{w}$ and $w_0$ gives

$$0 = \sum_{n=1}^{N} \left( \mathbf{w}^{\intercal} \mathbf{x}_n + w_0 - t_n \right),$$
$$\mathbf{0} = \sum_{n=1}^{N} \left( \mathbf{w}^{\intercal} \mathbf{x}_n + w_0 - t_n \right) \mathbf{x}_n. \tag{4.37}$$

The right hand side of the first equation can be written as

$$\mathbf{w}^{\intercal} \sum_{n=1}^{N} \mathbf{x}_n + N w_0 - \sum_{n=1}^{N} t_n = N \left( \mathbf{w}^{\intercal} \mathbf{m} + w_0 \right), \tag{4.38}$$

where

$$\mathbf{m} = \frac{1}{N} \sum_{n=1}^{N} \mathbf{x}_n. \tag{4.39}$$

Then,

$$w_0 = -\mathbf{w}^\intercal \mathbf{m}. \tag{4.40}$$

Then, the right hand side of the second equation above can be written as

$$\sum_{n=1}^{N} \left(\mathbf{w}^\intercal \left(\mathbf{x}_n - \mathbf{m}\right) - t_n\right) \mathbf{x}_n = \mathbf{v}_1 + \mathbf{v}_2, \tag{4.41}$$

where

$$\begin{aligned} \mathbf{v}_1 &= \sum_{n \in \mathcal{C}_1} \left(\mathbf{w}^\intercal (\mathbf{x}_n - \mathbf{m}) - \frac{N}{N_1}\right) \mathbf{x}_n, \\ \mathbf{v}_2 &= \sum_{n \in \mathcal{C}_2} \left(\mathbf{w}^\intercal (\mathbf{x}_n - \mathbf{m}) + \frac{N}{N_2}\right) \mathbf{x}_n. \end{aligned} \tag{4.42}$$

We have

$$\begin{aligned} \mathbf{m} &= \frac{N_1}{N}\mathbf{m}_1 + \frac{N_2}{N}\mathbf{m}_2, \\ \sum_{n \in \mathcal{C}_1} (\mathbf{x}_n - \mathbf{m}_1) &= \mathbf{0}, \\ \sum_{n \in \mathcal{C}_2} (\mathbf{x}_n - \mathbf{m}_2) &= \mathbf{0}. \end{aligned} \tag{4.43}$$

Then, $\mathbf{v}_1$ and $\mathbf{v}_2$ can be written as

$$\begin{aligned} &\sum_{n \in \mathcal{C}_1} \left(\mathbf{w}^\intercal \left(\mathbf{x}_n - \mathbf{m}_1 + \frac{N_2}{N}(\mathbf{m}_1 - \mathbf{m}_2)\right) - \frac{N}{N_1}\right) (\mathbf{x}_n - \mathbf{m}_1 + \mathbf{m}_1) \\ &= \left(\sum_{n \in \mathcal{C}_1} (\mathbf{x}_n - \mathbf{m}_1)(\mathbf{x}_n - \mathbf{m}_1)^\intercal\right) \mathbf{w} + \frac{N_1 N_2}{N} (\mathbf{m}_1 - \mathbf{m}_2)\mathbf{m}_1^\intercal \mathbf{w} - N\mathbf{m}_1, \end{aligned} \tag{4.44}$$

and

$$\begin{aligned} &\sum_{n \in \mathcal{C}_2} \left(\mathbf{w}^\intercal \left(\mathbf{x}_n - \mathbf{m}_2 + \frac{N_1}{N}(\mathbf{m}_2 - \mathbf{m}_1)\right) + \frac{N}{N_2}\right) (\mathbf{x}_n - \mathbf{m}_2 + \mathbf{m}_2) \\ &= \left(\sum_{n \in \mathcal{C}_2} (\mathbf{x}_n - \mathbf{m}_2)(\mathbf{x}_n - \mathbf{m}_2)^\intercal\right) \mathbf{w} + \frac{N_1 N_2}{N} (\mathbf{m}_2 - \mathbf{m}_1)\mathbf{m}_2^\intercal \mathbf{w} + N\mathbf{m}_2. \end{aligned} \tag{4.45}$$

Then,

$$\begin{aligned}\mathbf{0} = &\left(\sum_{k=1,2}\sum_{n\in\mathcal{C}_k}(\mathbf{x}_n - \mathbf{m}_k)(\mathbf{x}_n - \mathbf{m}_k)^\intercal\right)\mathbf{w} \\ &+ \frac{N_1 N_2}{N}(\mathbf{m}_1 - \mathbf{m}_2)(\mathbf{m}_1 - \mathbf{m}_2)^\intercal \mathbf{w} - N(\mathbf{m}_1 - \mathbf{m}_2).\end{aligned} \tag{4.46}$$

Therefore, the optimal solution to $\mathbf{w}$ satisfies

$$\left(\mathbf{S}_\mathrm{W} + \frac{N_1 N_2}{N}\mathbf{S}_\mathrm{B}\right)\mathbf{w}^* = N(\mathbf{m}_1 - \mathbf{m}_2). \tag{4.47}$$

where

$$\begin{aligned}\mathbf{S}_\mathrm{W} &= \sum_{k=1,2}\sum_{n\in\mathcal{C}_k}(\mathbf{x}_n - \mathbf{m}_k)(\mathbf{x}_n - \mathbf{m}_k)^\intercal, \\ \mathbf{S}_\mathrm{B} &= (\mathbf{m}_2 - \mathbf{m}_1)(\mathbf{m}_2 - \mathbf{m}_1)^\intercal.\end{aligned} \tag{4.48}$$

## 4.7

Let

$$\sigma(a) = \frac{1}{1+\exp(-a)}. \tag{4.49}$$

### (a)

We have

$$\sigma(-a) = \frac{1}{1+\exp(a)}. \tag{4.50}$$

The right hand side can be written as

$$1 - \frac{\exp(a)}{1+\exp(a)} = 1 - \frac{1}{1+\exp(-a)}. \tag{4.51}$$

Therefore,

$$\sigma(-a) = 1 - \sigma(a). \tag{4.52}$$

**(b)**

We have

$$\exp(-a) = \frac{1}{\sigma(a)} - 1. \tag{4.53}$$

Then,

$$a = -\ln\left(\frac{1}{\sigma(a)} - 1\right). \tag{4.54}$$

Therefore,

$$\sigma^{-1}(y) = \ln\left(\frac{y}{1-y}\right). \tag{4.55}$$

## 4.8

Let $\mathbf{x}$ be a variable in $D$ dimensions such that

$$p(\mathbf{x}|\mathcal{C}_k) = \mathcal{N}\left(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}\right), \tag{4.56}$$

where

$$p(\mathcal{C}_1) + p(\mathcal{C}_2) = 1. \tag{4.57}$$

By Bayes' theorem,

$$p(\mathcal{C}_1|\mathbf{x}) = \frac{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1)}{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1) + p(\mathbf{x}|\mathcal{C}_2)p(\mathcal{C}_2)}. \tag{4.58}$$

The right hand side can be written as

$$\sigma(a) = \frac{1}{1+\exp(-a)}, \tag{4.59}$$

where

$$a = \ln\left(\frac{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1)}{p(\mathbf{x}|\mathcal{C}_2)p(\mathcal{C}_2)}\right). \tag{4.60}$$

Substituting the expressions above of $p(\mathbf{x}|\mathcal{C}_k)$, we have

$$\begin{aligned} a = &-\frac{D}{2}\ln(2\pi) - \frac{1}{2}\ln(\det\Sigma) - \frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_1)^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}_1) + \ln p(\mathcal{C}_1) \\ &+ \frac{D}{2}\ln(2\pi) + \frac{1}{2}\ln(\det\Sigma) + \frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_2)^\intercal\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}_2) - \ln p(\mathcal{C}_2). \end{aligned} \tag{4.61}$$

Therefore,

$$p(\mathcal{C}_1|\mathbf{x}) = \sigma\left(\mathbf{w}^\intercal\mathbf{x} + w_0\right), \tag{4.62}$$

where

$$\begin{aligned}\mathbf{w} &= \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2),\\ w_0 &= -\frac{1}{2}\boldsymbol{\mu}_1^\intercal\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_1 + \frac{1}{2}\boldsymbol{\mu}_2^\intercal\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_2 + \ln p(\mathcal{C}_1) - \ln p(\mathcal{C}_2).\end{aligned} \tag{4.63}$$

## 4.9

Let $\mathbf{t}_1, \cdots, \mathbf{t}_N$ be independent variables from the standard basis in $K$ dimensions such that

$$p(\mathbf{t}_n, \boldsymbol{\phi}_n) = \prod_{k=1}^{K} y_{nk}^{t_{nk}}, \tag{4.64}$$

where

$$\begin{aligned}y_{nk} &= p(\boldsymbol{\phi}_n, \mathcal{C}_k),\\ p(\mathcal{C}_k) &= \pi_k,\end{aligned} \tag{4.65}$$

By Bayes' theorem,

$$y_{nk} = \pi_k p(\boldsymbol{\phi}_n|\mathcal{C}_k). \tag{4.66}$$

Then,

$$\ln p(\mathbf{T}, \boldsymbol{\Phi}) = \sum_{n=1}^{N}\sum_{k=1}^{K} t_{nk}\left(\ln \pi_k + \ln p(\boldsymbol{\phi}_n|\mathcal{C}_k)\right). \tag{4.67}$$

In order to maximise $\ln p(\mathbf{T}, \boldsymbol{\Phi})$ under the constraint

$$\sum_{k=1}^{K} \pi_k = 1, \tag{4.68}$$

let

$$L = \ln p(\mathbf{T}, \boldsymbol{\Phi}) + \lambda\left(\sum_{k=1}^{K}\pi_k - 1\right). \tag{4.69}$$

Setting the derivatives of $L$ with respect to $\pi_k$ and $\lambda$ to zero gives

$$\begin{aligned}0 &= \frac{N_k}{\pi_k} + \lambda,\\ 0 &= \sum_{k=1}^{K}\pi_k - 1,\end{aligned} \tag{4.70}$$

where

$$N_k = \sum_{n=1}^{N} t_{nk}. \tag{4.71}$$

Then,

$$\lambda = -\sum_{k=1}^{K} N_k. \tag{4.72}$$

The right hand side can be written as $-N$. Therefore, the maximum likelihood solution to $\pi_k$ is given by

$$\pi_{k\mathrm{ML}} = \frac{N_k}{N}. \tag{4.73}$$

## 4.10

Let $\mathbf{t}_1, \cdots, \mathbf{t}_N$ be independent variables from the standard basis in $K$ dimensions such that

$$p(\mathbf{t}_n, \boldsymbol{\phi}_n) = \prod_{k=1}^{K} y_{nk}^{t_{nk}}, \tag{4.74}$$

where

$$\begin{aligned} y_{nk} &= p(\boldsymbol{\phi}_n, \mathcal{C}_k), \\ p(\boldsymbol{\phi}_n | \mathcal{C}_k) &= \mathcal{N}\left(\boldsymbol{\phi}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}\right), \\ p(\mathcal{C}_k) &= \pi_k. \end{aligned} \tag{4.75}$$

By Bayes' theorem,

$$y_{nk} = \pi_k \mathcal{N}\left(\boldsymbol{\phi}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}\right). \tag{4.76}$$

Then,

$$\ln p(\mathbf{T}, \boldsymbol{\Phi}) = \sum_{n=1}^{N} \sum_{k=1}^{K} t_{nk} \left( \ln \mathcal{N}\left(\boldsymbol{\phi}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}\right) + \ln \pi_k \right). \tag{4.77}$$

The right hand side except the terms independent of $\boldsymbol{\mu}_k$ and $\boldsymbol{\Sigma}$ can be written as

$$l = \sum_{n=1}^{N} \sum_{k=1}^{K} t_{nk} \left( -\frac{1}{2} \ln(\det \boldsymbol{\Sigma}) - \frac{1}{2} (\boldsymbol{\phi}_n - \boldsymbol{\mu}_k)^{\intercal} \boldsymbol{\Sigma}^{-1} (\boldsymbol{\phi}_n - \boldsymbol{\mu}_k) \right). \tag{4.78}$$

By 3.21, setting the derivatives of $l$ with respect to $\boldsymbol{\mu}_k$ and $\boldsymbol{\Sigma}$ to zero gives

$$
\begin{aligned}
\mathbf{0} &= \frac{1}{2}\sum_{n=1}^{N} t_{nk}\left(\boldsymbol{\Sigma}^{-1} + \left(\boldsymbol{\Sigma}^{-1}\right)^{\mathsf{T}}\right)(\boldsymbol{\phi}_n - \boldsymbol{\mu}_k), \\
\mathbf{O} &= -\frac{1}{2}\sum_{n=1}^{N}\sum_{k=1}^{K} t_{nk}\left(\left(\boldsymbol{\Sigma}^{-1}\right)^{\mathsf{T}} - \left(\boldsymbol{\Sigma}^{-1}\right)^{2}(\boldsymbol{\phi}_n - \boldsymbol{\mu}_k)(\boldsymbol{\phi}_n - \boldsymbol{\mu}_k)^{\mathsf{T}}\right).
\end{aligned}
\tag{4.79}
$$

Therefore, the maximum likelihood solutions to $\boldsymbol{\mu}_k$ and $\boldsymbol{\Sigma}$ are given by

$$
\begin{aligned}
\boldsymbol{\mu}_{k\mathrm{ML}} &= \frac{1}{N_k}\sum_{n=1}^{N} t_{nk}\boldsymbol{\phi}_n, \\
\boldsymbol{\Sigma}_{\mathrm{ML}} &= \frac{1}{N}\sum_{k=1}^{K} N_k \mathbf{S}_k,
\end{aligned}
\tag{4.80}
$$

where

$$
\begin{aligned}
N_k &= \sum_{n=1}^{N} t_{nk}, \\
\mathbf{S}_k &= \frac{1}{N_k}\sum_{n=1}^{N} t_{nk}(\boldsymbol{\phi}_n - \boldsymbol{\mu}_k)(\boldsymbol{\phi}_n - \boldsymbol{\mu}_k)^{\mathsf{T}}.
\end{aligned}
\tag{4.81}
$$

## 4.11

Let $\boldsymbol{\phi}$ be a variable in $M$ dimensions each of whose component is a binary code with length $L$ such that

$$
p(\boldsymbol{\phi}|\mathcal{C}_k) = \prod_{m=1}^{M}\prod_{l=1}^{L} \mu_{kml}^{\phi_{ml}}. \tag{4.82}
$$

By Bayes' theorem,

$$
p(\mathcal{C}_k|\boldsymbol{\phi}) = \frac{p(\boldsymbol{\phi}|\mathcal{C}_k)p(\mathcal{C}_k)}{\sum_{k'=1}^{K} p(\boldsymbol{\phi}|\mathcal{C}_{k'})p(\mathcal{C}_{k'})}. \tag{4.83}
$$

Therefore,

$$
p(\mathcal{C}_k|\boldsymbol{\phi}) = \frac{\exp\left(a_k(\boldsymbol{\phi})\right)}{\sum_{k'=1}^{K}\exp\left(a_{k'}(\boldsymbol{\phi})\right)}, \tag{4.84}
$$

where

$$
a_k(\boldsymbol{\phi}) = \left(\sum_{m=1}^{M}\sum_{l=1}^{L} \phi_{ml}\ln\mu_{kml}\right) + \ln p(\mathcal{C}_k). \tag{4.85}
$$

## 4.12

Let

$$\sigma(a) = \frac{1}{1+\exp(-a)}. \tag{4.86}$$

Then,

$$\frac{d\sigma(a)}{da} = \frac{\exp(-a)}{(1+\exp(-a))^2}. \tag{4.87}$$

The right hand side can be written as

$$\frac{1}{1+\exp(-a)} - \frac{1}{(1+\exp(-a))^2} = \sigma(a) - (\sigma(a))^2. \tag{4.88}$$

Therefore,

$$\frac{d\sigma(a)}{da} = \sigma(a)\left(1-\sigma(a)\right). \tag{4.89}$$

## 4.13

Let $t_1, \cdots, t_N$ be conditionally independent binary variables such that

$$p(t_n|\mathbf{w}) = y_n^{t_n}(1-y_n)^{1-t_n}, \tag{4.90}$$

where

$$\begin{aligned} y_n &= \sigma\left(\mathbf{w}^\intercal \boldsymbol{\phi}_n\right), \\ \sigma(a) &= \frac{1}{1+\exp(-a)}. \end{aligned} \tag{4.91}$$

Let

$$E(\mathbf{w}) = -\ln p(\mathbf{t}|\mathbf{w}). \tag{4.92}$$

The right hand side can be written as

$$-\sum_{n=1}^{N}\left(t_n \ln y_n + (1-t_n)\ln(1-y_n)\right). \tag{4.93}$$

By 4.12,

$$\nabla E(\mathbf{w}) = -\sum_{n=1}^{N}\left(\frac{t_n}{y_n} y_n(1-y_n)\boldsymbol{\phi}_n - \frac{1-t_n}{1-y_n} y_n(1-y_n)\boldsymbol{\phi}_n\right). \tag{4.94}$$

The right hand side can be written as

$$-\sum_{n=1}^{N}(t_n(1-y_n)\boldsymbol{\phi}_n-(1-t_n)y_n\boldsymbol{\phi}_n)=\sum_{n=1}^{N}(y_n-t_n)\boldsymbol{\phi}_n. \tag{4.95}$$

Therefore,

$$\nabla E(\mathbf{w})=\sum_{n=1}^{N}(y_n-t_n)\boldsymbol{\phi}_n. \tag{4.96}$$

## 4.14

Let $t_1,\cdots,t_N$ be conditionally independent binary variables such that

$$p(t_n|\mathbf{w})=y_n^{t_n}(1-y_n)^{1-t_n}, \tag{4.97}$$

where

$$\begin{aligned} y_n &= \sigma\left(\mathbf{w}^\intercal\boldsymbol{\phi}_n\right), \\ \sigma(a) &= \frac{1}{1+\exp(-a)}. \end{aligned} \tag{4.98}$$

Let

$$E(\mathbf{w})=-\ln p(\mathbf{t}|\mathbf{w}). \tag{4.99}$$

By 4.13, setting the derivative of $E(\mathbf{w})$ with respect to $\mathbf{w}$ to zero gives

$$\mathbf{0}=\sum_{n=1}^{N}\left(y_n-t_n\right)\boldsymbol{\phi}_n. \tag{4.100}$$

If $\boldsymbol{\phi}_1,\cdots,\boldsymbol{\phi}_N$ are linearly independent, then

$$y_n=t_n. \tag{4.101}$$

Then,

$$\sigma\left(\mathbf{w}^\intercal\boldsymbol{\phi}_n\right)=\begin{cases}1, & t_n=1,\\ 0, & \text{otherwise.}\end{cases} \tag{4.102}$$

Threrefore,

$$\mathbf{w}^\intercal\boldsymbol{\phi}_n=\begin{cases}\infty, & t_n=1,\\ -\infty, & \text{otherwise.}\end{cases} \tag{4.103}$$

## 4.15

Let $t_1, \cdots, t_N$ be conditionally independent binary variables such that

$$p(t_n|\mathbf{w}) = y_n^{t_n}(1-y_n)^{1-t_n}, \tag{4.104}$$

where

$$y_n = \sigma\left(\mathbf{w}^\intercal \boldsymbol{\phi}_n\right), \quad \sigma(a) = \frac{1}{1+\exp(-a)}. \tag{4.105}$$

Let

$$E(\mathbf{w}) = -\ln p(\mathbf{t}|\mathbf{w}). \tag{4.106}$$

By 4.13,

$$\nabla E(\mathbf{w}) = \sum_{n=1}^{N}(y_n - t_n)\boldsymbol{\phi}_n. \tag{4.107}$$

By 4.12,

$$\nabla\nabla E(\mathbf{w}) = \sum_{n=1}^{N} y_n(1-y_n)\boldsymbol{\phi}_n\boldsymbol{\phi}_n^\intercal. \tag{4.108}$$

The right hand side can be written as

$$\mathbf{H} = \boldsymbol{\Phi}^\intercal\mathbf{R}\boldsymbol{\Phi}, \tag{4.109}$$

where

$$R_{nn'} = y_n(1-y_n)I_{nn'}, \tag{4.110}$$

Then,

$$\mathbf{u}^\intercal\mathbf{H}\mathbf{u} = \mathbf{v}^\intercal\mathbf{R}\mathbf{v}, \tag{4.111}$$

where

$$\mathbf{v} = \boldsymbol{\Phi}\mathbf{u}. \tag{4.112}$$

The right hand side can be written as

$$\sum_{n=1}^{N}\sum_{n'=1}^{N} v_n y_n(1-y_n)I_{nn'}v_{n'} = \sum_{n=1}^{N} y_n(1-y_n)v_n^2. \tag{4.113}$$

We have

$$y_n(1-y_n) > 0. \tag{4.114}$$

Then,

$$\mathbf{u}^\intercal\mathbf{H}\mathbf{u} > 0, \tag{4.115}$$

so that $\mathbf{H}$ is positive definite. Therefore, $E$ is a convex function of $\mathbf{w}$ and it has a unique minimum.

## 4.16

Let $t_1, \cdots, t_N$ be independent binary variables such that

$$p(t_n = 1) = \pi_n. \tag{4.116}$$

Then,

$$p(t_n) = \pi_n^{t_n}(1-\pi_n)^{1-t_n}, \tag{4.117}$$

so that

$$p(\mathbf{t}) = \prod_{n=1}^{N} \pi_n^{t_n}(1-\pi_n)^{1-t_n}. \tag{4.118}$$

Therefore,

$$-\ln p(\mathbf{t}) = -\sum_{n=1}^{N} \left(t_n \ln \pi_n + (1-t_n)\ln(1-\pi_n)\right). \tag{4.119}$$

## 4.17

Let

$$y_k = \frac{\exp(a_k)}{\sum_{k'=1}^{K}\exp(a_{k'})}. \tag{4.120}$$

Then,

$$\frac{\partial y_k}{\partial a_k} = \frac{\exp(a_k)}{\sum_{k'=1}^{K}\exp(a_{k'})} - \frac{\exp(2a_k)}{\left(\sum_{k'=1}^{K}\exp(a_{k'})\right)^2}. \tag{4.121}$$

The right hand side can be written as $y_k(1-y_k)$. If $k \neq k'$, then

$$\frac{\partial y_k}{\partial a_{k'}} = -\frac{\exp(a_k + a_{k'})}{\left(\sum_{k''=1}^{K}\exp(a_{k''})\right)^2}. \tag{4.122}$$

The right hand side can be written as $-y_k y_{k'}$. Therefore,

$$\frac{\partial y_k}{\partial a_{k'}} = y_k(I_{kk'} - y_{k'}). \tag{4.123}$$

## 4.18

Let $\mathbf{t}_1, \cdots, \mathbf{t}_N$ be conditionally independent variables from the standard basis in $K$ dimensions such that

$$p(\mathbf{t}_n|\mathbf{W}) = \prod_{k=1}^{K} y_{nk}^{t_{nk}}, \tag{4.124}$$

where

$$\begin{aligned} y_{nk} &= \frac{\exp(a_{nk})}{\sum_{k'=1}^{K} \exp(a_{nk'})}, \\ a_{nk} &= \mathbf{w}_k^{\mathsf{T}} \boldsymbol{\phi}_n. \end{aligned} \tag{4.125}$$

Let

$$E(\mathbf{W}) = -\ln p(\mathbf{T}|\mathbf{W}). \tag{4.126}$$

The right hand side can be written as

$$-\sum_{n=1}^{N} \sum_{k=1}^{K} t_{nk} \ln y_{nk}. \tag{4.127}$$

By 4.17,

$$\nabla_{\mathbf{w}_{k'}} E(\mathbf{W}) = -\sum_{n=1}^{N} \sum_{k=1}^{K} (I_{kk'} - y_{nk'}) t_{nk} \boldsymbol{\phi}_n. \tag{4.128}$$

The right hand side can be written as

$$-\sum_{n=1}^{N} \left( \sum_{k=1}^{K} (I_{kk'} - y_{nk'}) t_{nk} \right) \boldsymbol{\phi}_n = -\sum_{n=1}^{N} (t_{nk'} - y_{nk'}) \boldsymbol{\phi}_n. \tag{4.129}$$

Therefore,

$$\nabla_{\mathbf{w}_k} E(\mathbf{W}) = \sum_{n=1}^{N} (y_{nk} - t_{nk}) \, \boldsymbol{\phi}_n. \tag{4.130}$$

## 4.19

Let $t_1, \cdots, t_N$ be conditionally independent binary variables such that

$$p(t_n = 1|\boldsymbol{\phi}_n) = \Phi(a_n), \tag{4.131}$$

where

$$\begin{aligned} \Phi(a) &= \int_{-\infty}^{a} \mathcal{N}(\theta|0,1) d\theta, \\ a_n &= \mathbf{w}^{\mathsf{T}} \boldsymbol{\phi}_n. \end{aligned} \tag{4.132}$$

**(a)**

We have

$$p(t_n|\boldsymbol{\phi}_n) = (\Phi(a_n))^{t_n} (1 - \Phi(a_n))^{1-t_n}. \tag{4.133}$$

Let

$$E(\mathbf{w}) = -\ln p(\mathbf{t}|\boldsymbol{\Phi}). \tag{4.134}$$

The right hand side can be written as

$$-\sum_{n=1}^{N} (t_n \ln \Phi(a_n) + (1 - t_n) \ln (1 - \Phi(a_n))). \tag{4.135}$$

Then,

$$\nabla E(\mathbf{w}) = -\sum_{n=1}^{N} \left( t_n \frac{\mathcal{N}(a_n|0,1)}{\Phi(a_n)} - (1 - t_n) \frac{\mathcal{N}(a_n|0,1)}{1 - \Phi(a_n)} \right) \boldsymbol{\phi}_n. \tag{4.136}$$

The right hand side can be written as

$$\begin{aligned} &-\sum_{n=1}^{N} \left( \frac{t_n}{\Phi(a_n)} - \frac{1 - t_n}{1 - \Phi(a_n)} \right) \mathcal{N}(a_n|0,1) \boldsymbol{\phi}_n \\ =&\sum_{n=1}^{N} \frac{\mathcal{N}(a_n|0,1)}{\Phi(a_n)(1 - \Phi(a_n))} (\Phi(a_n) - t_n) \boldsymbol{\phi}_n. \end{aligned} \tag{4.137}$$

Therefore,

$$\nabla E(\mathbf{w}) = \sum_{n=1}^{N} \frac{\mathcal{N}(a_n|0,1)}{\Phi(a_n)(1 - \Phi(a_n))} (\Phi(a_n) - t_n) \boldsymbol{\phi}_n. \tag{4.138}$$

**(b)**

We have

$$
\begin{aligned}
\nabla\nabla E(\mathbf{w}) =& \sum_{n=1}^{N} \frac{-a_n \mathcal{N}(a_n|0,1)}{\Phi(a_n)\left(1-\Phi(a_n)\right)} \left(\Phi(a_n) - t_n\right) \boldsymbol{\phi}_n \boldsymbol{\phi}_n^{\intercal} \\
&- \sum_{n=1}^{N} \frac{\left(\mathcal{N}(a_n|0,1)\right)^2}{\left(\Phi(a_n)\right)^2\left(1-\Phi(a_n)\right)} \left(\Phi(a_n) - t_n\right) \boldsymbol{\phi}_n \boldsymbol{\phi}_n^{\intercal} \\
&+ \sum_{n=1}^{N} \frac{\left(\mathcal{N}(a_n|0,1)\right)^2}{\Phi(a_n)\left(1-\Phi(a_n)\right)^2} \left(\Phi(a_n) - t_n\right) \boldsymbol{\phi}_n \boldsymbol{\phi}_n^{\intercal} \\
&+ \sum_{n=1}^{N} \frac{\left(\mathcal{N}(a_n|0,1)\right)^2}{\Phi(a_n)\left(1-\Phi(a_n)\right)} \boldsymbol{\phi}_n \boldsymbol{\phi}_n^{\intercal}.
\end{aligned}
\tag{4.139}
$$

Therefore,

$$
\nabla\nabla E(\mathbf{w}) = \sum_{n=1}^{N} b_n \boldsymbol{\phi}_n \boldsymbol{\phi}_n^{\intercal}, \tag{4.140}
$$

where

$$
\begin{aligned}
b_n =& \left(\frac{\mathcal{N}(a_n|0,1)}{\Phi(a_n)\left(1-\Phi(a_n)\right)}\right)^2 \left(\left(\Phi(a_n)\right)^2 - 2t_n\Phi(a_n) + t_n\right) \\
&- \frac{\mathcal{N}(a_n|0,1)}{\Phi(a_n)\left(1-\Phi(a_n)\right)} a_n \left(\Phi(a_n) - t_n\right).
\end{aligned}
\tag{4.141}
$$

## 4.20

Let $\mathbf{t}_1, \cdots, \mathbf{t}_N$ be conditionally independent variables from the standard basis in $K$ dimensions such that

$$
p(\mathbf{t}_n|\mathbf{W}) = \prod_{k=1}^{K} y_{nk}^{t_{nk}}, \tag{4.142}
$$

where

$$
\begin{aligned}
y_{nk} &= \frac{\exp(a_{nk})}{\sum_{k'=1}^{K} \exp(a_{nk'})}, \\
a_{nk} &= \mathbf{w}_k^{\intercal} \boldsymbol{\phi}_n.
\end{aligned}
\tag{4.143}
$$

Let

$$E(\mathbf{W}) = -\ln p(\mathbf{T}|\mathbf{W}). \tag{4.144}$$

By 4.18,

$$\nabla_{\mathbf{w}_k} E(\mathbf{W}) = \sum_{n=1}^{N} (y_{nk} - t_{nk})\, \boldsymbol{\phi}_n. \tag{4.145}$$

By 4.17,

$$\nabla_{\mathbf{w}_k} \nabla_{\mathbf{w}_{k'}} E(\mathbf{W}) = \sum_{n=1}^{N} y_{nk}(I_{kk'} - y_{nk'}) \boldsymbol{\phi}_n \boldsymbol{\phi}_n^{\mathsf{T}}. \tag{4.146}$$

The right hand side can be written as

$$\mathbf{H}_{kk'} = \boldsymbol{\Phi}^{\mathsf{T}} \mathbf{R}_{kk'} \boldsymbol{\Phi}, \tag{4.147}$$

where

$$R_{kk'nn'} = y_{nk}(I_{kk'} - y_{nk'}) I_{nn'}. \tag{4.148}$$

Let

$$\mathbf{H} = \begin{bmatrix} \mathbf{H}_{11} & \cdots & \mathbf{H}_{1K} \\ \vdots & \ddots & \vdots \\ \mathbf{H}_{K1} & \cdots & \mathbf{H}_{KK} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \mathbf{u}_1 \\ \vdots \\ \mathbf{u}_K \end{bmatrix}, \tag{4.149}$$

where $\mathbf{u}_1, \cdots, \mathbf{u}_K$ are vectors in the same dimension as $\mathbf{w}$. Then,

$$\mathbf{u}^{\mathsf{T}} \mathbf{H} \mathbf{u} = \sum_{k=1}^{K} \sum_{k'=1}^{K} \mathbf{v}_k^{\mathsf{T}} \mathbf{R}_{kk'} \mathbf{v}_{k'}, \tag{4.150}$$

where

$$\mathbf{v}_k = \boldsymbol{\Phi} \mathbf{u}_k. \tag{4.151}$$

The right hand side can be written as

$$\begin{aligned} &\sum_{k=1}^{K} \sum_{k'=1}^{K} \sum_{n=1}^{N} \sum_{n'=1}^{N} v_{kn} y_{nk} (I_{kk'} - y_{nk'}) I_{nn'} v_{k'n'} \\ =&\sum_{k=1}^{K} \sum_{k'=1}^{K} \sum_{n=1}^{N} v_{kn} y_{nk} (I_{kk'} - y_{nk'}) v_{k'n}. \end{aligned} \tag{4.152}$$

The right hand side can be written as

$$\sum_{n=1}^{N} \sum_{k=1}^{K} y_{nk} v_{kn} \left( v_{kn} - \sum_{k'=1}^{K} y_{nk'} v_{k'n} \right). \tag{4.153}$$

By Jensen's inequality,

$$\begin{aligned}
&\sum_{k=1}^{K} y_{nk} v_{kn} \left( v_{kn} - \sum_{k'=1}^{K} y_{nk'} v_{k'n} \right) \\
&\geq \left( \sum_{k=1}^{K} y_{nk} v_{kn} \right) \left( \sum_{k=1}^{K} y_{nk} v_{kn} - \sum_{k'=1}^{K} y_{nk'} v_{k'n} \right),
\end{aligned} \tag{4.154}$$

where the right hand side is zero. Then,

$$\mathbf{u}^{\intercal}\mathbf{H}\mathbf{u} \geq 0. \tag{4.155}$$

Therefore, $\mathbf{H}$ is positive semidifinite.

## 4.21

Let

$$\Phi(a) = \int_{-\infty}^{a} \mathcal{N}(\theta|0,1) d\theta. \tag{4.156}$$

The right hand side can be written as

$$\int_{-\infty}^{0} \mathcal{N}(\theta|0,1) d\theta + \int_{0}^{a} \mathcal{N}(\theta|0,1) d\theta = \frac{1}{2} + \frac{1}{\sqrt{2\pi}} \int_{0}^{a} \exp\left(-\frac{\theta^2}{2}\right) d\theta. \tag{4.157}$$

The second term of the right hand side can be written as

$$\frac{1}{\sqrt{2\pi}} \int_{0}^{\frac{a}{\sqrt{2}}} \exp\left(-t^2\right) \sqrt{2} dt = \frac{1}{2} \mathrm{erf}\left(\frac{a}{\sqrt{2}}\right), \tag{4.158}$$

where

$$\mathrm{erf}(a) = \frac{2}{\sqrt{\pi}} \int_{0}^{a} \exp\left(-t^2\right) dt. \tag{4.159}$$

Therefore,

$$\Phi(a) = \frac{1}{2}\left(1 + \mathrm{erf}\left(\frac{a}{\sqrt{2}}\right)\right). \tag{4.160}$$

## 4.22

Let $\mathcal{D}$ be a set of conditionally independent variables dependent on $\boldsymbol{\theta}$ in $M$ dimensions. By marginalisation,

$$p(\mathcal{D}) = \int p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})d\boldsymbol{\theta}. \tag{4.161}$$

Let the logarithm of the integrand be $-E(\boldsymbol{\theta})$. We have the Taylor series

$$\begin{aligned} E(\boldsymbol{\theta}) =& E(\boldsymbol{\theta}_{\text{MAP}}) + \frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{MAP}})^\intercal \mathbf{H}_{\text{MAP}}(\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{MAP}}) \\ &+ O\left(\|\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{MAP}}\|^3\right), \end{aligned} \tag{4.162}$$

where $\boldsymbol{\theta}_{\text{MAP}}$ is a stationary point of $E$ and

$$\mathbf{H}_{\text{MAP}} = \nabla\nabla E(\boldsymbol{\theta}_{\text{MAP}}). \tag{4.163}$$

Then,

$$\int p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})d\boldsymbol{\theta} \simeq \exp\left(-E(\boldsymbol{\theta}_{\text{MAP}})\right) I, \tag{4.164}$$

where

$$I = \int \exp\left(-\frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{MAP}})^\intercal \mathbf{H}_{\text{MAP}}(\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{MAP}})\right) d\boldsymbol{\theta}. \tag{4.165}$$

Therefore,

$$p(\mathcal{D}) \simeq (2\pi)^{\frac{M}{2}} (\det \mathbf{H}_{\text{MAP}})^{-\frac{1}{2}} \exp\left(-E(\boldsymbol{\theta}_{\text{MAP}})\right), \tag{4.166}$$

so that

$$\ln p(\mathcal{D}) \simeq \frac{M}{2}\ln(2\pi) - \frac{1}{2}\ln(\det \mathbf{H}_{\text{MAP}}) - E(\boldsymbol{\theta}_{\text{MAP}}). \tag{4.167}$$

## 4.23

Let $\mathcal{D}$ be a set of $t_1, \cdots, t_N$ which are conditionally independent and identically distributed variables dependent on $\boldsymbol{\theta}$ in $M$ dimensions such that

$$p(\boldsymbol{\theta}) = \mathcal{N}\left(\boldsymbol{\theta}|\mathbf{m}_0, \mathbf{S}_0\right). \tag{4.168}$$

By 4.22,

$$\ln p(\mathcal{D}) \simeq \frac{M}{2}\ln(2\pi) - \frac{1}{2}\ln\left(\det \mathbf{H}_{\text{MAP}}\right) - E(\boldsymbol{\theta}_{\text{MAP}}), \tag{4.169}$$

where

$$E(\boldsymbol{\theta}) = -\ln p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta}), \tag{4.170}$$

$\boldsymbol{\theta}_{\text{MAP}}$ is a stationary point of $E$ and

$$\mathbf{H}_{\text{MAP}} = \nabla\nabla E(\boldsymbol{\theta}_{\text{MAP}}). \tag{4.171}$$

We have

$$\mathbf{H}_{\text{MAP}} = \mathbf{H}_{\mathcal{D}\text{MAP}} + \mathbf{S}_0^{-1}, \tag{4.172}$$

where

$$\mathbf{H}_{\mathcal{D}\text{MAP}} = -\nabla\nabla \ln p(\mathcal{D}|\boldsymbol{\theta}_{\text{MAP}}). \tag{4.173}$$

The right hand side of the approximation can be written as

$$\begin{aligned}&\frac{M}{2}\ln(2\pi) - \frac{1}{2}\ln\left(\det\left(\mathbf{H}_{\mathcal{D}\text{MAP}} + \mathbf{S}_0^{-1}\right)\right)\\ &+ \ln p(\mathcal{D}|\boldsymbol{\theta}_{\text{MAP}}) + \ln p(\boldsymbol{\theta}_{\text{MAP}}).\end{aligned} \tag{4.174}$$

We have

$$\begin{aligned}\ln p(\boldsymbol{\theta}_{\text{MAP}}) = &-\frac{M}{2}\ln(2\pi) - \frac{1}{2}\ln(\det \mathbf{S}_0)\\ &-\frac{1}{2}(\boldsymbol{\theta}_{\text{MAP}} - \mathbf{m}_0)^\intercal \mathbf{S}_0^{-1}(\boldsymbol{\theta}_{\text{MAP}} - \mathbf{m}_0).\end{aligned} \tag{4.175}$$

Then,

$$\begin{aligned}\ln p(\mathcal{D}) \simeq &-\frac{1}{2}\ln\left(\det\left(\mathbf{H}_{\mathcal{D}\text{MAP}}\mathbf{S}_0 + \mathbf{I}\right)\right) + \ln p(\mathcal{D}|\boldsymbol{\theta}_{\text{MAP}})\\ &-\frac{1}{2}(\boldsymbol{\theta}_{\text{MAP}} - \mathbf{m}_0)^\intercal \mathbf{S}_0^{-1}(\boldsymbol{\theta}_{\text{MAP}} - \mathbf{m}_0).\end{aligned} \tag{4.176}$$

Then, for a large $\mathbf{S}_0$,

$$\ln p(\mathcal{D}) \simeq \ln p(\mathcal{D}|\boldsymbol{\theta}_{\text{MAP}}) - \frac{1}{2}\ln\left(\det\left(\mathbf{H}_{\mathcal{D}\text{MAP}}\mathbf{S}_0\right)\right). \tag{4.177}$$

We have

$$\mathbf{H}_{\mathcal{D}\text{MAP}} = N\bar{\mathbf{H}}_{\text{MAP}}, \tag{4.178}$$

where

$$\begin{aligned}\bar{\mathbf{H}}_{\text{MAP}} &= \frac{1}{N}\sum_{n=1}^{N}\mathbf{H}_{n\text{MAP}},\\ \mathbf{H}_{n\text{MAP}} &= -\nabla\nabla \ln p(t_n|\boldsymbol{\theta}_{\text{MAP}}).\end{aligned} \tag{4.179}$$

Then,

$$\det\left(\mathbf{H}_{\mathcal{D}\mathrm{MAP}}\mathbf{S}_0\right) = N^M \det\left(\bar{\mathbf{H}}_{\mathrm{MAP}}\mathbf{S}_0\right), \tag{4.180}$$

so that

$$\ln\left(\det\left(\mathbf{H}_{\mathcal{D}\mathrm{MAP}}\mathbf{S}_0\right)\right) \simeq M \ln N. \tag{4.181}$$

Therefore,

$$\ln p(\mathcal{D}) \simeq \ln p(\mathcal{D}|\boldsymbol{\theta}_{\mathrm{MAP}}) - \frac{M}{2}\ln N. \tag{4.182}$$

## 4.24

Let $t_1, \cdots, t_N$ be conditionally independent binary variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= y_n^{t_n}(1-y_n)^{1-t_n}, \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0), \end{aligned} \tag{4.183}$$

where

$$\begin{aligned} y_n &= \sigma\left(\mathbf{w}^\intercal \boldsymbol{\phi}_n\right), \\ \sigma(a) &= \frac{1}{1+\exp(-a)}. \end{aligned} \tag{4.184}$$

By marginalisation,

$$p(t_{N+1}|\mathbf{t}) = \int p(t_{N+1}|\mathbf{w})p(\mathbf{w}|\mathbf{t})d\mathbf{w}. \tag{4.185}$$

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{w})p(\mathbf{w}). \tag{4.186}$$

The logarithm of the right hand side except the terms independent of $\mathbf{w}$ can be written as $-E(\mathbf{w})$ where

$$\begin{aligned} E(\mathbf{w}) = &\sum_{n=1}^{N}\left(t_n \ln y_n + (1-t_n)\ln(1-y_n)\right) \\ &- \frac{1}{2}(\mathbf{w}-\mathbf{m}_0)^\intercal \mathbf{S}_0^{-1}(\mathbf{w}-\mathbf{m}_0). \end{aligned} \tag{4.187}$$

Let $\mathbf{w}_{\mathrm{MAP}}$ be a stationary point of $E$. Then, we have the Taylor series

$$\begin{aligned} E(\mathbf{w}) = &E(\mathbf{w}_{\mathrm{MAP}}) + \frac{1}{2}(\mathbf{w}-\mathbf{w}_{\mathrm{MAP}})^\intercal \mathbf{H}_{\mathrm{MAP}}(\mathbf{w}-\mathbf{w}_{\mathrm{MAP}}) \\ &+ O(\|\mathbf{w}-\mathbf{w}_{\mathrm{MAP}}\|^3), \end{aligned} \tag{4.188}$$

where

$$\mathbf{H}_{\mathrm{MAP}} = \nabla\nabla E(\mathbf{w}_{\mathrm{MAP}}). \tag{4.189}$$

Then,

$$p(\mathbf{w}|\mathbf{t}) \simeq \mathcal{N}\left(\mathbf{w}|\mathbf{w}_{\mathrm{MAP}}, \mathbf{H}_{\mathrm{MAP}}^{-1}\right). \tag{4.190}$$

Then,

$$\begin{aligned} &\int p(t_{N+1}|\mathbf{w})p(\mathbf{w}|\mathbf{t})d\mathbf{w} \\ \simeq &\int \sigma\left(a_{N+1}\right)^{t_{N+1}} \left(1-\sigma(a_{N+1})\right)^{1-t_{N+1}} p(a_{N+1})da_{N+1}, \end{aligned} \tag{4.191}$$

where

$$p(a_{N+1}) = \int \delta\left(a_{N+1} - \mathbf{w}^{\intercal}\boldsymbol{\phi}_{N+1}\right)\mathcal{N}\left(\mathbf{w}|\mathbf{w}_{\mathrm{MAP}}, \mathbf{H}_{\mathrm{MAP}}^{-1}\right)d\mathbf{w}, \tag{4.192}$$

and $\delta$ is the Dirac delta function. Since $t_{N+1}$ is a binary variable, the right hand side of the approximation can be written as

$$\upsilon_{N+1}^{t_{N+1}}(1-\upsilon_{N+1})^{1-t_{N+1}}, \tag{4.193}$$

where

$$\upsilon_{N+1} = \int \sigma(a_{N+1})p(a_{N+1})da_{N+1}. \tag{4.194}$$

We have

$$\mathrm{E}\, a_{N+1} = \int a_{N+1}p(a_{N+1})da_{N+1}. \tag{4.195}$$

The right hand side can be written as

$$\int \mathbf{w}^{\intercal}\boldsymbol{\phi}_{N+1}\mathcal{N}\left(\mathbf{w}|\mathbf{w}_{\mathrm{MAP}}, \mathbf{H}_{\mathrm{MAP}}^{-1}\right)d\mathbf{w} = \mathbf{w}_{\mathrm{MAP}}^{\intercal}\boldsymbol{\phi}_{N+1}. \tag{4.196}$$

Then,

$$\mathrm{var}\, a_{N+1} = \int \left(a_{N+1} - \mathbf{w}_{\mathrm{MAP}}^{\intercal}\boldsymbol{\phi}_{N+1}\right)^2 p(a_{N+1})da_{N+1}. \tag{4.197}$$

The right hand side can be written as

$$\int \left(\left(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}}\right)^{\intercal}\boldsymbol{\phi}_{N+1}\right)^2 p(a_{N+1})da_{N+1} = \boldsymbol{\phi}_{N+1}^{\intercal}\mathbf{H}_{\mathrm{MAP}}^{-1}\boldsymbol{\phi}_{N+1}. \tag{4.198}$$

Then,

$$p(a_{N+1}) = \mathcal{N}\left(a_{N+1}|\mathbf{w}_{\text{MAP}}^{\intercal}\boldsymbol{\phi}_{N+1}, \boldsymbol{\phi}_{N+1}^{\intercal}\mathbf{H}_{\text{MAP}}^{-1}\boldsymbol{\phi}_{N+1}\right). \tag{4.199}$$

Therefore,

$$p(t_{N+1}|\mathbf{t}) = \upsilon_{N+1}^{t_{N+1}}(1-\upsilon_{N+1})^{1-t_{N+1}}, \tag{4.200}$$

where

$$\upsilon_{N+1} = \int \sigma(a)\mathcal{N}\left(a|\mathbf{w}_{\text{MAP}}^{\intercal}\boldsymbol{\phi}_{N+1}, \boldsymbol{\phi}_{N+1}^{\intercal}\mathbf{H}_{\text{MAP}}^{-1}\boldsymbol{\phi}_{N+1}\right) da. \tag{4.201}$$

## 4.25

Let $\lambda$ be a constant such that

$$\left.\frac{d\sigma(a)}{da}\right|_{a=0} = \left.\frac{d\Phi(\lambda a)}{da}\right|_{a=0}, \tag{4.202}$$

where

$$\begin{aligned} \sigma(a) &= \frac{1}{1+\exp(-a)}, \\ \Phi(a) &= \int_{-\infty}^{a} \mathcal{N}(\theta|0,1)d\theta. \end{aligned} \tag{4.203}$$

By 4.12,

$$\frac{d\sigma(a)}{da} = \sigma(a)\left(1-\sigma(a)\right). \tag{4.204}$$

We have

$$\frac{d\Phi(\lambda a)}{da} = \lambda\mathcal{N}(\lambda a|0,1). \tag{4.205}$$

Then,

$$\frac{1}{4} = \lambda(2\pi)^{-\frac{1}{2}}. \tag{4.206}$$

Therefore,

$$\lambda^2 = \frac{\pi}{8}. \tag{4.207}$$

## 4.26

Let

$$I(\mu) = \int \Phi(\lambda a)\mathcal{N}(a|\mu, \sigma^2)da, \tag{4.208}$$

where

$$\Phi(a) = \int_{-\infty}^{a} \mathcal{N}(\theta|0, 1)d\theta. \tag{4.209}$$

By the transformation

$$z = \frac{\theta - \mu}{\sigma}, \tag{4.210}$$

the right hand side can be written as

$$\begin{aligned} &\int \Phi\left(\lambda(\mu + \sigma z)\right)\mathcal{N}(\mu + \sigma z|\mu, \sigma^2)\sigma dz \\ =&\int \Phi\left(\lambda(\mu + \sigma z)\right)\mathcal{N}(z|0, 1)dz. \end{aligned} \tag{4.211}$$

Then,

$$\frac{\partial}{\partial\mu}I(\mu) = \lambda\int \mathcal{N}\left(\lambda(\mu + \sigma z)|0, 1\right)\mathcal{N}(z|0, 1)dz. \tag{4.212}$$

The logarithm of the integrand of the right hand side can be written as

$$\begin{aligned} &-\frac{1}{2}\ln(2\pi) - \frac{\lambda^2(\mu + \sigma z)^2}{2} - \frac{1}{2}\ln(2\pi) - \frac{z^2}{2} \\ =&-\ln(2\pi) - \frac{1 + \sigma^2\lambda^2}{2}\left(z + \frac{\mu\sigma\lambda^2}{1 + \sigma^2\lambda^2}\right)^2 + \frac{\mu^2\sigma^2\lambda^4}{2\left(1 + \sigma^2\lambda^2\right)} - \frac{\mu^2\lambda^2}{2}. \end{aligned} \tag{4.213}$$

The right hand side can be written as

$$\begin{aligned} &-\ln(2\pi) - \frac{1 + \sigma^2\lambda^2}{2}\left(z + \frac{\mu\sigma\lambda^2}{1 + \sigma^2\lambda^2}\right)^2 - \frac{\mu^2\lambda^2}{2\left(1 + \sigma^2\lambda^2\right)} \\ =&-\frac{1}{2}\ln(2\pi) - \frac{1}{2}\ln\left(1 + \sigma^2\lambda^2\right)^{-1} - \frac{1 + \sigma^2\lambda^2}{2}\left(z + \frac{\mu\sigma\lambda^2}{1 + \sigma^2\lambda^2}\right)^2 \\ &-\ln\lambda - \frac{1}{2}\ln(2\pi) - \frac{1}{2}\ln\left(\lambda^{-2} + \sigma^2\right) - \frac{\mu^2}{2\left(\lambda^{-2} + \sigma^2\right)}. \end{aligned} \tag{4.214}$$

Then, the integral can be written as

$$\begin{aligned}&\frac{1}{\lambda}\mathcal{N}\left(\mu|\,0,\lambda^{-2}+\sigma^2\right)\int\mathcal{N}\left(z|-\frac{\mu\sigma\lambda^2}{1+\sigma^2\lambda^2},\left(1+\sigma^2\lambda^2\right)^{-1}\right)dz\\=&\frac{1}{\lambda}\mathcal{N}\left(\mu|0,\lambda^{-2}+\sigma^2\right).\end{aligned}\tag{4.215}$$

Then,

$$\frac{\partial}{\partial\mu}I(\mu)=\mathcal{N}\left(\mu|0,\lambda^{-2}+\sigma^2\right),\tag{4.216}$$

so that

$$I(\mu)=\int_{-\infty}^{\mu}\mathcal{N}\left(m|0,\lambda^{-2}+\sigma^2\right)dm.\tag{4.217}$$

By the transformation

$$m'=\left(\lambda^{-2}+\sigma^2\right)^{-\frac{1}{2}}m,\tag{4.218}$$

the right hand side can be written as

$$\begin{aligned}&\int_{-\infty}^{\left(\lambda^{-2}+\sigma^2\right)^{-\frac{1}{2}}\mu}\left(\lambda^{-2}+\sigma^2\right)^{-\frac{1}{2}}\mathcal{N}\left(m'|0,1\right)\left(\lambda^{-2}+\sigma^2\right)^{\frac{1}{2}}dm'\\=&\int_{-\infty}^{\left(\lambda^{-2}+\sigma^2\right)^{-\frac{1}{2}}\mu}\mathcal{N}\left(m'|0,1\right)dm'.\end{aligned}\tag{4.219}$$

Therefore,

$$I(\mu)=\Phi\left(\left(\lambda^{-2}+\sigma^2\right)^{-\frac{1}{2}}\mu\right).\tag{4.220}$$

# 5 Neural Networks

## 5.1

Let

$$y_k(\mathbf{x}, \mathbf{w}) = \sigma\left(\sum_{m=1}^{M} w_{km}^{(2)} \sigma\left(\sum_{d=1}^{D} w_{md}^{(1)} x_d + w_{m0}^{(1)}\right) + w_{k0}^{(2)}\right), \tag{5.1}$$

where

$$\sigma(a) = \frac{1}{1 + \exp(-a)}. \tag{5.2}$$

We have

$$\sigma(a) = \frac{\exp\left(\frac{a}{2}\right)}{\exp\left(\frac{a}{2}\right) + \exp\left(-\frac{a}{2}\right)}. \tag{5.3}$$

The right hand side can be written as

$$\frac{\exp\left(\frac{a}{2}\right) - \exp\left(-\frac{a}{2}\right)}{\exp\left(\frac{a}{2}\right) + \exp\left(-\frac{a}{2}\right)} + \frac{\exp\left(-\frac{a}{2}\right)}{\exp\left(\frac{a}{2}\right) + \exp\left(-\frac{a}{2}\right)} = \tanh\left(\frac{a}{2}\right) + 1 - \sigma(a). \tag{5.4}$$

Then,

$$\sigma(a) = \frac{1}{2}\left(1 + \tanh\left(\frac{a}{2}\right)\right). \tag{5.5}$$

Then, the argument of the expression of $y_k(\mathbf{x}, \mathbf{w})$ can be written as

$$\begin{aligned} &\sum_{m=1}^{M} w_{km}^{(2)} \left(\frac{1}{2}\left(1 + \tanh\left(\frac{1}{2}\left(\sum_{d=1}^{D} w_{md}^{(1)} x_d + w_{m0}^{(1)}\right)\right)\right)\right) + w_{k0}^{(2)} \\ =&\frac{1}{2}\sum_{m=1}^{M} w_{km}^{(2)} \tanh\left(\frac{1}{2}\sum_{d=1}^{D} w_{md}^{(1)} x_d + \frac{1}{2} w_{m0}^{(1)}\right) + \frac{1}{2}\sum_{m=1}^{M} w_{km}^{(2)} + w_{k0}^{(2)}. \end{aligned} \tag{5.6}$$

Therefore,

$$\begin{aligned} &y_k(\mathbf{x}, \mathbf{w}) \\ =&\sigma\left(\frac{1}{2}\sum_{m=1}^{M} w_{km}^{(2)} \tanh\left(\frac{1}{2}\sum_{d=1}^{D} w_{md}^{(1)} x_d + \frac{1}{2} w_{m0}^{(1)}\right) + \frac{1}{2}\sum_{m=1}^{M} w_{km}^{(2)} + w_{k0}^{(2)}\right). \end{aligned} \tag{5.7}$$

## 5.2

Let $\mathbf{t}_1, \cdots, \mathbf{t}_N$ be conditionally independent variables such that

$$p(\mathbf{t}_n|\mathbf{w}) = \mathcal{N}\left(\mathbf{t}_n|\mathbf{y}_n, \beta^{-1}\mathbf{I}\right), \tag{5.8}$$

where

$$\mathbf{y}_n = \mathbf{y}(\mathbf{x}_n, \mathbf{w}). \tag{5.9}$$

Then, the logarithm of $p(\mathbf{T}|\mathbf{w})$ except the terms independent of $\mathbf{w}$ can be written as

$$-\frac{\beta}{2}\sum_{n=1}^{N}\|\mathbf{t}_n - \mathbf{y}(\mathbf{x}_n, \mathbf{w})\|^2. \tag{5.10}$$

Therefore, maximising $p(\mathbf{T}|\mathbf{w})$ is equivalent to minimising $E(\mathbf{w})$, where

$$E(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}\|\mathbf{y}(\mathbf{x}_n, \mathbf{w}) - \mathbf{t}_n\|^2. \tag{5.11}$$

## 5.3

Let $\mathbf{t}_1, \cdots, \mathbf{t}_N$ be conditionally independent variables such that

$$p(\mathbf{t}_n|\mathbf{w}) = \mathcal{N}\left(\mathbf{t}_n|\mathbf{y}_n, \boldsymbol{\Sigma}\right), \tag{5.12}$$

where

$$\mathbf{y}_n = \mathbf{y}(\mathbf{x}_n, \mathbf{w}). \tag{5.13}$$

### (a)

The logarithm of $\ln p(\mathbf{T}|\mathbf{w})$ except the terms independent of $\boldsymbol{\Sigma}$ can be written as

$$l = -\frac{N}{2}\ln(\det\boldsymbol{\Sigma}) - \frac{1}{2}\sum_{n=1}^{N}\left(\mathbf{t}_n - \mathbf{y}_n\right)^\intercal \boldsymbol{\Sigma}^{-1}\left(\mathbf{t}_n - \mathbf{y}_n\right). \tag{5.14}$$

By 3.21, setting the derivative of $l$ with respect to $\boldsymbol{\Sigma}$ to zero gives

$$\mathbf{O} = -\frac{N}{2}\boldsymbol{\Sigma}^{-1} + \frac{1}{2}\left(\boldsymbol{\Sigma}^{-1}\right)^2\sum_{n=1}^{N}\left(\mathbf{t}_n - \mathbf{y}_n\right)\left(\mathbf{t}_n - \mathbf{y}_n\right)^\intercal. \tag{5.15}$$

Therefore, the maximum likelihood solution to $\boldsymbol{\Sigma}$ is given by

$$\boldsymbol{\Sigma}_{\mathrm{ML}} = \frac{1}{N}\sum_{n=1}^{N}\left(\mathbf{t}_n - \mathbf{y}_n\right)\left(\mathbf{t}_n - \mathbf{y}_n\right)^\intercal. \tag{5.16}$$

**(b)**

If $\mathbf{\Sigma}$ is fixed and known, then the maximum likelihood solution to $\mathbf{w}$ is given by minimising $E(\mathbf{w})$ where

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^{N} (\mathbf{t}_n - \mathbf{y}_n)^\intercal \, \mathbf{\Sigma}^{-1} \, (\mathbf{t}_n - \mathbf{y}_n) \,. \tag{5.17}$$

## 5.4

Let $t_1, \cdots, t_N$ be conditionally independent binary variables such that

$$p(t_n = 1|\mathbf{w}) = (1-\epsilon)y_n + \epsilon(1-y_n), \tag{5.18}$$

where

$$y_n = y(\mathbf{x}_n, \mathbf{w}). \tag{5.19}$$

Then,

$$p(t_n|\mathbf{w}) = \left((1-\epsilon)y_n + \epsilon\,(1-y_n)\right)^{t_n} \left(\epsilon y_n + (1-\epsilon)\,(1-y_n)\right)^{1-t_n}. \tag{5.20}$$

Therefore,

$$\begin{aligned} -\ln p(\mathbf{t}|\mathbf{w}) = &-\sum_{n=1}^{N} t_n \ln\left((1-\epsilon)y_n + \epsilon(1-y_n)\right) \\ &-\sum_{n=1}^{N} (1-t_n) \ln\left(\epsilon y_n + (1-\epsilon)(1-y_n)\right). \end{aligned} \tag{5.21}$$

## 5.5

Let $\mathbf{t}_1, \cdots, \mathbf{t}_N$ be conditionally independent variables from the standard basis in $K$ dimensions such that

$$p(\mathbf{t}_n|\mathbf{w}) = \prod_{k=1}^{K} y_{nk}^{t_{nk}}, \tag{5.22}$$

where

$$y_{nk} = y_k(\mathbf{x}_n, \mathbf{w}). \tag{5.23}$$

Therefore,

$$\ln p(\mathbf{T}|\mathbf{w}) = \sum_{n=1}^{N} \sum_{k=1}^{K} t_{nk} \ln y_{nk}. \tag{5.24}$$

## 5.6

Let $t_1, \cdots, t_N$ be conditionally independent binary variables such that

$$p(t_n|\mathbf{w}) = y_n^{t_n}(1 - y_n)^{1-t_n}, \tag{5.25}$$

where

$$\begin{aligned} y_n &= \sigma(a_n), \\ \sigma(a) &= \frac{1}{1 + \exp(-a)}, \\ a_n &= a(\mathbf{x}_n, \mathbf{w}). \end{aligned} \tag{5.26}$$

Let

$$E(\mathbf{w}) = -\ln p(\mathbf{t}|\mathbf{w}). \tag{5.27}$$

Then,

$$E(\mathbf{w}) = -\sum_{n=1}^{N} \left(t_n \ln y_n + (1 - t_n) \ln(1 - y_n)\right). \tag{5.28}$$

Then,

$$\frac{\partial E(\mathbf{w})}{\partial a_n} = -y_n(1 - y_n) \left(\frac{t_n}{y_n} - \frac{1 - t_n}{1 - y_n}\right). \tag{5.29}$$

Therefore,

$$\frac{\partial E(\mathbf{w})}{\partial a_n} = y_n - t_n. \tag{5.30}$$

## 5.7

Let $\mathbf{t}_1, \cdots, \mathbf{t}_N$ be conditionally independent variables from the standard basis in $K$ dimenstions such that

$$p(\mathbf{t}_n|\mathbf{w}) = \prod_{k=1}^{K} y_{nk}^{t_{nk}}, \tag{5.31}$$

where

$$\begin{aligned} y_{nk} &= y_k(\mathbf{a}_n), \\ y_k(\mathbf{a}) &= \frac{\exp(a_k)}{\sum_{k'=1}^{K} \exp(a_{k'})}, \\ \mathbf{a}_n &= \mathbf{a}(\mathbf{x}_n, \mathbf{w}). \end{aligned} \tag{5.32}$$

Let

$$E(\mathbf{w}) = -\ln p(\mathbf{T}|\mathbf{w}). \tag{5.33}$$

Then,

$$E(\mathbf{w}) = -\sum_{n=1}^{N}\sum_{k=1}^{K} t_{nk} \ln y_{nk}. \tag{5.34}$$

Then,

$$\frac{\partial E(\mathbf{w})}{\partial a_{nk'}} = -\sum_{k=1}^{K} \frac{t_{nk}}{y_{nk}} \frac{\partial y_{nk}}{\partial a_{nk'}}. \tag{5.35}$$

By 4.17,

$$\frac{\partial y_{nk}}{\partial a_{nk'}} = y_{nk}(I_{kk'} - y_{nk'}). \tag{5.36}$$

Then,

$$\frac{\partial E(\mathbf{w})}{\partial a_{nk'}} = \sum_{k=1}^{K} t_{nk}(y_{nk'} - I_{kk'}). \tag{5.37}$$

The right hand side can be written as

$$y_{nk'} \sum_{k=1}^{K} t_{nk} - t_{nk'} = y_{nk'} - t_{nk'}. \tag{5.38}$$

Therefore,

$$\frac{\partial E(\mathbf{w})}{\partial a_{nk}} = y_{nk} - t_{nk}. \tag{5.39}$$

## 5.8

We have

$$\tanh a = \frac{\exp(a) - \exp(-a)}{\exp(a) + \exp(-a)}. \tag{5.40}$$

Then,

$$\frac{d}{da} \tanh a = 1 - \left(\frac{\exp(a) - \exp(-a)}{\exp(a) + \exp(-a)}\right)^2. \tag{5.41}$$

Therefore,

$$\frac{d}{da} \tanh a = 1 - (\tanh a)^2. \tag{5.42}$$

## 5.9

Let $t_1, \cdots, t_N$ be conditionally independent variables from $\{-1, 1\}$ such that

$$p(t_n|\mathbf{w}) = \left(\frac{1+y_n}{2}\right)^{\frac{1+t_n}{2}} \left(\frac{1-y_n}{2}\right)^{\frac{1-t_n}{2}}, \tag{5.43}$$

where

$$\begin{aligned} y_n &= y(\mathbf{x}_n, \mathbf{w}), \\ -1 \leq y(&\mathbf{x}_n, \mathbf{w}) \leq 1. \end{aligned} \tag{5.44}$$

Let

$$E(\mathbf{w}) = -\ln p(\mathbf{t}|\mathbf{w}). \tag{5.45}$$

Then,

$$E(\mathbf{w}) = -\sum_{n=1}^{N} \left( \frac{1+t_n}{2} \ln \frac{1+y_n}{2} + \frac{1-t_n}{2} \ln \frac{1-y_n}{2} \right). \tag{5.46}$$

The appropriate choice of $y$ is tanh.

## 5.10

Let

$$\mathbf{H} = \nabla\nabla E(\mathbf{w}), \tag{5.47}$$

where $E$ is a real function of real vectors. Then, $\mathbf{H}$ is a real symmetric matrix. Therefore, by 2.20, $\mathbf{H}$ is positive definite if and only if all of its eigenvalues are positive.

## 5.11

Let $\mathbf{w}$ be a real vector in $M$ dimensions. We have the Taylor series

$$E(\mathbf{w}) = E(\mathbf{w}^*) + \frac{1}{2} (\mathbf{w} - \mathbf{w}^*)^\intercal \mathbf{H}^* (\mathbf{w} - \mathbf{w}^*) + O\left(\|\mathbf{w} - \mathbf{w}^*\|^3\right), \tag{5.48}$$

where $\mathbf{w}^*$ is a stationary point of $E$ and

$$\mathbf{H}^* = \left.\nabla\nabla E(\mathbf{w})\right|_{\mathbf{w}=\mathbf{w}^*}. \tag{5.49}$$

Let $\mathbf{u}_1, \cdots, \mathbf{u}_M$ be eigenvectors of $\mathbf{H}^*$ such that

$$\mathbf{H}^* \mathbf{u}_m = \lambda_m \mathbf{u}_m. \tag{5.50}$$

Then, there exists $\alpha_1, \cdots, \alpha_M$ such that

$$\mathbf{w} - \mathbf{w}^* = \sum_{m=1}^{M} \alpha_m \mathbf{u}_m. \tag{5.51}$$

By 2.19,

$$\mathbf{u}_m^\intercal \mathbf{u}_{m'} = I_{mm'}. \tag{5.52}$$

Then,

$$E(\mathbf{w}) \simeq E\left(\mathbf{w}^*\right) + \frac{1}{2} \sum_{m=1}^{M} \lambda_m \alpha_m^2. \tag{5.53}$$

Therefore, the contours of constant $E$ are ellipses whose axes are aligned with $\mathbf{u}_1, \cdots, \mathbf{u}_M$ with lengths which are proportional to $\lambda_1^{-\frac{1}{2}}, \cdots, \lambda_M^{-\frac{1}{2}}$.

## 5.12

Let $\mathbf{w}$ be a real vector. We have the Taylor series

$$E(\mathbf{w}) = E\left(\mathbf{w}^*\right) + \frac{1}{2}\left(\mathbf{w} - \mathbf{w}^*\right)^\intercal \mathbf{H}^* \left(\mathbf{w} - \mathbf{w}^*\right) + O\left(\|\mathbf{w} - \mathbf{w}^*\|^3\right), \tag{5.54}$$

where $\mathbf{w}^*$ is a stationary point of $E$ and

$$\mathbf{H}^* = \nabla\nabla E(\mathbf{w})\big|_{\mathbf{w}=\mathbf{w}^*}. \tag{5.55}$$

If $\mathbf{H}^*$ is positive definite, then the second term of the right hand will be non-negative. Then, $\mathbf{w}^*$ is a local minimum of $E$. On the other hand, if $\mathbf{w}^*$ is a local minimum of $E$, then the second term of the right hand side will be non-negative. Then, $\mathbf{H}^*$ is positive definite. Therefore, the necessary and sufficient condition for $\mathbf{w}^*$ to be a local minimum is that $\mathbf{H}^*$ is positive definite.

## 5.13

Let $\mathbf{w}$ be a vector in $M$ dimensions. We have the Taylor series

$$\begin{aligned} E(\mathbf{w}) =& E\left(\hat{\mathbf{w}}\right) + \hat{\mathbf{g}}^{\intercal}\left(\mathbf{w} - \hat{\mathbf{w}}\right) + \frac{1}{2}\left(\mathbf{w} - \hat{\mathbf{w}}\right)^{\intercal}\hat{\mathbf{H}}\left(\mathbf{w} - \hat{\mathbf{w}}\right) \\ &+ O\left(\|\mathbf{w} - \hat{\mathbf{w}}\|^{3}\right), \end{aligned} \tag{5.56}$$

where

$$\begin{aligned} \hat{\mathbf{g}} &= \nabla E(\hat{\mathbf{w}}), \\ \hat{\mathbf{H}} &= \nabla\nabla E(\hat{\mathbf{w}}). \end{aligned} \tag{5.57}$$

Since $\hat{\mathbf{g}}$ is a vector in $M$ dimensions and $\hat{\mathbf{H}}$ is a $M \times M$ symmetric matrix, the number of independent elements of the right hand side is

$$M + \frac{M(M+1)}{2} = \frac{M(M+3)}{2}. \tag{5.58}$$

## 5.14

Let $w$ be a variable. We have the Taylor series

$$\begin{aligned} E_n(w_{mm'} + \epsilon) &= E_n(w_{mm'}) + \left.\frac{\partial E_n}{\partial w}\right|_{w=w_{mm'}} \epsilon + O\left(\epsilon^2\right), \\ E_n(w_{mm'} - \epsilon) &= E_n(w_{mm'}) - \left.\frac{\partial E_n}{\partial w}\right|_{w=w_{mm'}} \epsilon + O\left(\epsilon^2\right). \end{aligned} \tag{5.59}$$

Therefore,

$$\left.\frac{\partial E_n}{\partial w}\right|_{w=w_{mm'}} = \frac{E_n(w_{mm'} + \epsilon) - E_n(w_{mm'} - \epsilon)}{2\epsilon} + O\left(\epsilon^2\right). \tag{5.60}$$

## 5.15 (Incomplete)

Let

$$J_{km} = \frac{\partial y_k}{\partial x_m}. \tag{5.61}$$

The right hand side can be written as

$$\sum_{m'=1}^{M} W_{mm'} \frac{\partial y_k}{\partial a_{m'}}, \tag{5.62}$$

where

$$W_{mm'} = \frac{\partial a_{m'}}{\partial x_m}. \tag{5.63}$$

We have

$$\frac{\partial y_k}{\partial a_{m'}} = \sum_{m''=1}^{M} \frac{\partial y_k}{\partial a_{m''}} \frac{\partial a_{m''}}{\partial a_{m'}}. \tag{5.64}$$

## 5.16

Let $\mathbf{y}_1, \cdots, \mathbf{y}_N$ be vectors dependent on $\mathbf{w}$. Let

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^{N} \|\mathbf{y}_n - \mathbf{t}_n\|^2. \tag{5.65}$$

Then,

$$\nabla E(\mathbf{w}) = \sum_{n=1}^{N} (\nabla \mathbf{y}_n)^\intercal (\mathbf{y}_n - \mathbf{t}_n). \tag{5.66}$$

Therefore,

$$\nabla\nabla E(\mathbf{w}) = \sum_{n=1}^{N} (\nabla \operatorname{vec}(\nabla \mathbf{y}_n)^\intercal)^\intercal (\mathbf{y}_n - \mathbf{t}_n) + \sum_{n=1}^{N} (\nabla \mathbf{y}_n)^\intercal (\nabla \mathbf{y}_n). \tag{5.67}$$

## 5.17

Let $y$ be a scalar dependent on $\mathbf{x}$ and $\mathbf{w}$. Let

$$E(\mathbf{w}) = \frac{1}{2} \iint (y - t)^2 p(\mathbf{x}, t) d\mathbf{x} dt. \tag{5.68}$$

Then,

$$\nabla E(\mathbf{w}) = \iint (y - t) p(\mathbf{x}, t) \nabla y d\mathbf{x} dt. \tag{5.69}$$

The right hand side can be written as

$$\begin{aligned} &\int y \nabla y \left( \int p(\mathbf{x}, t) dt \right) d\mathbf{x} - \int \nabla y \left( \int t p(t|\mathbf{x}) dt \right) p(\mathbf{x}) d\mathbf{x} \\ &= \int y \nabla y p(\mathbf{x}) d\mathbf{x} - \int \nabla y \, \mathrm{E}(t|\mathbf{x}) p(\mathbf{x}) d\mathbf{x}. \end{aligned} \tag{5.70}$$

Then,

$$\begin{aligned}\nabla\nabla E(\mathbf{w}) = \int \nabla y\,(\nabla y)^\intercal\, p(\mathbf{x})d\mathbf{x} + \int y\nabla\nabla y p(\mathbf{x})d\mathbf{x} \\ - \int \nabla\nabla y\,\mathrm{E}(t|\mathbf{x})p(\mathbf{x})d\mathbf{x}.\end{aligned} \tag{5.71}$$

The second and the third terms of the right hand side can be written as

$$\mathrm{E}\left(y\nabla\nabla y\right) - \mathrm{E}\left(\nabla\nabla y\,\mathrm{E}(t|\mathbf{x})\right) = \mathrm{E}\left(\left(y - \mathrm{E}(t|\mathbf{x})\right)\nabla\nabla y\right). \tag{5.72}$$

Therefore, if

$$y = \mathrm{E}(t|\mathbf{x}), \tag{5.73}$$

then

$$\nabla\nabla E(\mathbf{w}) = \int \nabla y\,(\nabla y)^\intercal\, p(\mathbf{x})d\mathbf{x}. \tag{5.74}$$

## 5.18

Let $y_1, \cdots, y_N$ be scalars such that

$$\begin{aligned} y_n &= \mathbf{w}_n^{(2)\intercal}\mathbf{z} + \mathbf{w}_n^{(0)\intercal}\mathbf{x}, \\ z_m &= \tanh\left(\mathbf{w}_m^{(1)\intercal}\mathbf{x}\right). \end{aligned} \tag{5.75}$$

Let

$$E = \frac{1}{2}\sum_{n=1}^{N}(y_n - t_n)^2. \tag{5.76}$$

Then,

$$\begin{aligned} \frac{\partial E}{\partial \mathbf{w}_n^{(0)}} &= \frac{\partial E}{\partial y_n}\frac{\partial y_n}{\partial \mathbf{w}_n^{(0)}}, \\ \frac{\partial E}{\partial \mathbf{w}_m^{(1)}} &= \frac{\partial E}{\partial y_n}\frac{\partial y_n}{\partial \mathbf{z}}\frac{\partial \mathbf{z}}{\partial \mathbf{w}_m^{(1)}}, \\ \frac{\partial E}{\partial \mathbf{w}_n^{(2)}} &= \frac{\partial E}{\partial y_n}\frac{\partial y_n}{\partial \mathbf{w}_n^{(2)}}. \end{aligned} \tag{5.77}$$

Therefore,

$$\begin{aligned} \frac{\partial E}{\partial \mathbf{w}_n^{(0)}} &= (y_n - t_n)\mathbf{x}, \\ \frac{\partial E}{\partial \mathbf{w}_m^{(1)}} &= (y_n - t_n)\mathbf{A}\mathbf{w}_n^{(2)}, \\ \frac{\partial E}{\partial \mathbf{w}_n^{(2)}} &= (y_n - t_n)\mathbf{z}, \end{aligned} \tag{5.78}$$

where

$$A_{mm'} = \left(1 - z_m^2\right) x_{m'}. \tag{5.79}$$

## 5.19

Let $t_1, \cdots, t_N$ be conditionally independent binary variables such that

$$p(t_n|\mathbf{w}) = y_n^{t_n}(1 - y_n)^{1-t_n}, \tag{5.80}$$

where

$$\begin{aligned} y_n &= \sigma(a_n), \\ \sigma(a) &= \frac{1}{1 + \exp(-a)}, \\ a_n &= a(\mathbf{x}_n, \mathbf{w}). \end{aligned} \tag{5.81}$$

Let

$$E(\mathbf{w}) = -\ln p(\mathbf{t}|\mathbf{w}). \tag{5.82}$$

Then,

$$E(\mathbf{w}) = -\sum_{n=1}^{N} \left(t_n \ln y_n + (1 - t_n)\ln(1 - y_n)\right). \tag{5.83}$$

Then,

$$\nabla E(\mathbf{w}) = -\sum_{n=1}^{N} \left(\frac{t_n}{y_n} - \frac{1 - t_n}{1 - y_n}\right) \nabla y_n. \tag{5.84}$$

We have

$$\nabla y_n = \frac{\partial y_n}{\partial a_n} \nabla a_n. \tag{5.85}$$

Then,

$$\nabla y_n = y_n(1 - y_n)\nabla a_n. \tag{5.86}$$

Then,

$$\nabla E(\mathbf{w}) = -\sum_{n=1}^{N} \left(t_n(1 - y_n) - (1 - t_n)y_n\right) \nabla a_n, \tag{5.87}$$

so that

$$\nabla E(\mathbf{w}) = \sum_{n=1}^{N} (y_n - t_n)\nabla a_n. \tag{5.88}$$

Then,

$$\nabla\nabla E(\mathbf{w}) = \sum_{n=1}^{N} \nabla a_n \, (\nabla y_n)^\intercal + \sum_{n=1}^{N} (y_n - t_n) \nabla\nabla a_n. \tag{5.89}$$

Therefore,

$$\nabla\nabla E(\mathbf{w}) = \sum_{n=1}^{N} y_n(1 - y_n) \nabla a_n (\nabla a_n)^\intercal + \sum_{n=1}^{N} (y_n - t_n) \nabla\nabla a_n. \tag{5.90}$$

## 5.20

Let $\mathbf{t}_1, \cdots, \mathbf{t}_N$ be conditionally independent variables from the standard basis in $K$ dimensions such that

$$p(\mathbf{t}_n|\mathbf{w}) = \prod_{k=1}^{K} y_{nk}^{t_{nk}}, \tag{5.91}$$

where

$$\begin{aligned} y_{nk} &= y_k(\mathbf{a}_n), \\ y_k(\mathbf{a}) &= \frac{\exp(a_k)}{\sum_{k'=1}^{K} \exp(a_{k'})}, \\ \mathbf{a}_n &= \mathbf{a}(\mathbf{x}_n, \mathbf{w}). \end{aligned} \tag{5.92}$$

Let

$$E(\mathbf{w}) = -\ln p(\mathbf{T}|\mathbf{w}). \tag{5.93}$$

Then,

$$E(\mathbf{w}) = -\sum_{n=1}^{N} \sum_{k=1}^{K} t_{nk} \ln y_{nk}. \tag{5.94}$$

Then,

$$\nabla E(\mathbf{w}) = -\sum_{n=1}^{N} \sum_{k=1}^{K} \frac{t_{nk}}{y_{nk}} \nabla y_{nk}. \tag{5.95}$$

We have

$$\nabla y_{nk} = \sum_{k'=1}^{K} \frac{\partial y_{nk}}{\partial a_{nk'}} \nabla a_{nk'}. \tag{5.96}$$

By 4.17,

$$\frac{\partial y_{nk}}{\partial a_{nk'}} = y_{nk}(I_{kk'} - y_{nk'}). \tag{5.97}$$

Then,

$$\nabla y_{nk} = y_{nk} \sum_{k'=1}^{K} (I_{kk'} - y_{nk'}) \nabla a_{nk'}. \tag{5.98}$$

Then,

$$\nabla E(\mathbf{w}) = -\sum_{n=1}^{N} \sum_{k=1}^{K} t_{nk} \sum_{k'=1}^{K} (I_{kk'} - y_{nk'}) \nabla a_{nk}, \tag{5.99}$$

so that

$$\nabla E(\mathbf{w}) = \sum_{n=1}^{N} \sum_{k=1}^{K} (y_{nk} - t_{nk}) \nabla a_{nk}. \tag{5.100}$$

Then,

$$\nabla\nabla E(\mathbf{w}) = \sum_{n=1}^{N} \sum_{k=1}^{K} \nabla a_{nk} (\nabla y_{nk})^{\intercal} + \sum_{n=1}^{N} \sum_{k=1}^{K} (y_{nk} - t_{nk}) \nabla\nabla a_{nk}. \tag{5.101}$$

Therefore,

$$\begin{aligned}\nabla\nabla E(\mathbf{w}) =& \sum_{n=1}^{N} \sum_{k=1}^{K} y_{nk} \nabla a_{nk} \sum_{k'=1}^{K} (I_{kk'} - y_{nk'}) (\nabla a_{nk'})^{\intercal} \\ &+ \sum_{n=1}^{N} \sum_{k=1}^{K} (y_{nk} - t_{nk}) \nabla\nabla a_{nk}.\end{aligned} \tag{5.102}$$

## 5.21

Let $\mathbf{y}_1, \cdots, \mathbf{y}_N$ be vectors dependent on $\mathbf{w}$. Let

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^{N} \|\mathbf{y}_n - \mathbf{t}_n\|^2. \tag{5.103}$$

By 5.16,

$$\nabla\nabla E(\mathbf{w}) = \sum_{n=1}^{N} (\nabla \operatorname{vec} (\nabla \mathbf{y}_n)^{\intercal})^{\intercal} (\mathbf{y}_n - \mathbf{t}_n) + \mathbf{S}, \tag{5.104}$$

where

$$\mathbf{S} = \sum_{n=1}^{N} (\nabla \mathbf{y}_n)^{\intercal} (\nabla \mathbf{y}_n). \tag{5.105}$$

Let

$$\mathbf{S}' = \sum_{n=1}^{N+1} (\nabla \mathbf{y}_n)^{\intercal} (\nabla \mathbf{y}_n) . \tag{5.106}$$

Then,

$$\mathbf{S}' = \mathbf{S} + \mathbf{B}_{N+1}^{\intercal} \mathbf{B}_{N+1}, \tag{5.107}$$

where

$$\mathbf{B}_{N+1} = \nabla \mathbf{y}_{N+1}. \tag{5.108}$$

Therefore, by 2.26,

$$\mathbf{S}'^{-1} = \mathbf{S}^{-1} - \mathbf{S}^{-1} \mathbf{B}_{N+1}^{\intercal} \mathbf{M} \mathbf{B}_{N+1} \mathbf{S}^{-1}, \tag{5.109}$$

where

$$\mathbf{M} = \left( \mathbf{I} + \mathbf{B}_{N+1} \mathbf{S}^{-1} \mathbf{B}_{N+1}^{\intercal} \right)^{-1} . \tag{5.110}$$

## 5.22

Let $a_1, \cdots, a_N$ be variables such that

$$\begin{aligned} a_n &= \mathbf{w}_n^{(2)\intercal} \mathbf{z}, \\ z_m &= h(b_m), \\ b_m &= \mathbf{w}_m^{(1)\intercal} \mathbf{x}. \end{aligned} \tag{5.111}$$

Let $E$ be a function of $a_1, \cdots, a_N$. Then,

$$\begin{aligned} \frac{\partial E}{\partial \mathbf{w}_m^{(1)}} &= \sum_{n=1}^{N} \frac{\partial E}{\partial a_n} \frac{\partial a_n}{\partial z_m} \frac{\partial z_m}{\partial b_m} \frac{\partial b_m}{\partial \mathbf{w}_m^{(1)}}, \\ \frac{\partial E}{\partial \mathbf{w}_n^{(2)}} &= \frac{\partial E}{\partial a_n} \frac{\partial a_n}{\partial \mathbf{w}_n^{(2)}}. \end{aligned} \tag{5.112}$$

Then,

$$\begin{aligned} \frac{\partial E}{\partial \mathbf{w}_m^{(1)}} &= h'(b_m) \mathbf{x} \sum_{n=1}^{N} w_{nm}^{(2)} \frac{\partial E}{\partial a_n}, \\ \frac{\partial E}{\partial \mathbf{w}_n^{(2)}} &= \frac{\partial E}{\partial a_n} \mathbf{z}. \end{aligned} \tag{5.113}$$

Therefore,

$$
\begin{aligned}
\frac{\partial^2 E}{\partial \mathbf{w}_m^{(1)} \partial \mathbf{w}_{m'}^{(1)}} =& C_{mm'}^{(11)} \mathbf{x}\mathbf{x}^\intercal, \\
\frac{\partial^2 E}{\partial \mathbf{w}_m^{(1)} \partial \mathbf{w}_n^{(2)}} =& C_{mn}^{(12)} \mathbf{x}\mathbf{z}^\intercal + h'(b_m) \frac{\partial E}{\partial a_n} \mathbf{x}\mathbf{v}^\intercal, \\
\frac{\partial^2 E}{\partial \mathbf{w}_n^{(2)} \partial \mathbf{w}_{n'}^{(2)}} =& C_{nn'}^{(22)} \mathbf{z}\mathbf{z}^\intercal,
\end{aligned}
\tag{5.114}
$$

where $\mathbf{v}$ is a vector from the standard basis in $M$ dimensions such that

$$
v_m = 1, \tag{5.115}
$$

and

$$
\begin{aligned}
C_{mm'}^{(11)} =& h''(b_m) I_{mm'} \sum_{n=1}^{N} w_{nm}^{(2)} \frac{\partial E}{\partial a_n} \\
&+ h'(b_m) h'(b_{m'}) \sum_{n=1}^{N} w_{nm}^{(2)} \sum_{n'=1}^{N} w_{n'm'}^{(2)} \frac{\partial^2 E}{\partial y_n \partial a_{n'}}, \\
C_{mn}^{(12)} =& h'(b_m) \sum_{n'=1}^{N} w_{n'm}^{(2)} \frac{\partial^2 E}{\partial a_n \partial a_{n'}}, \\
C_{nn'}^{(22)} =& \frac{\partial^2 E}{\partial a_n \partial a_{n'}}.
\end{aligned}
\tag{5.116}
$$

## 5.23

Let $a_1, \cdots, a_N$ be variables such that

$$
\begin{aligned}
a_n &= \mathbf{w}_n^{(2)\intercal} \mathbf{z} + \mathbf{w}_n^{(0)\intercal} \mathbf{x}, \\
z_m &= h(b_m), \\
b_m &= \mathbf{w}_m^{(1)\intercal} \mathbf{x}.
\end{aligned}
\tag{5.117}
$$

Let $E$ be a function of $a_1, \cdots, a_N$. Then,

$$
\begin{aligned}
\frac{\partial E}{\partial \mathbf{w}_n^{(0)}} &= \frac{\partial E}{\partial a_n} \frac{\partial a_n}{\partial \mathbf{w}_n^{(0)}}, \\
\frac{\partial E}{\partial \mathbf{w}_m^{(1)}} &= \sum_{n=1}^{N} \frac{\partial E}{\partial a_n} \frac{\partial a_n}{\partial z_m} \frac{\partial z_m}{\partial b_m} \frac{\partial b_m}{\partial \mathbf{w}_m^{(1)}}, \\
\frac{\partial E}{\partial \mathbf{w}_n^{(2)}} &= \frac{\partial E}{\partial a_n} \frac{\partial a_n}{\partial \mathbf{w}_n^{(2)}}.
\end{aligned}
\tag{5.118}
$$

Then,

$$\begin{aligned}\frac{\partial E}{\partial \mathbf{w}_n^{(0)}} &= \frac{\partial E}{\partial a_n}\mathbf{x},\\ \frac{\partial E}{\partial \mathbf{w}_m^{(1)}} &= h'(b_m)\mathbf{x}\sum_{n=1}^{N} w_{nm}^{(2)}\frac{\partial E}{\partial a_n},\\ \frac{\partial E}{\partial \mathbf{w}_n^{(2)}} &= \frac{\partial E}{\partial a_n}\mathbf{z}.\end{aligned} \tag{5.119}$$

Therefore,

$$\begin{aligned}\frac{\partial^2 E}{\partial \mathbf{w}_n^{(0)}\partial \mathbf{w}_{n'}^{(0)}} &= C_{nn'}^{(22)}\mathbf{x}\mathbf{x}^{\intercal},\\ \frac{\partial^2 E}{\partial \mathbf{w}_n^{(0)}\partial \mathbf{w}_m^{(1)}} &= C_{nm}^{(12)}\mathbf{x}\mathbf{x}^{\intercal},\\ \frac{\partial^2 E}{\partial \mathbf{w}_n^{(0)}\partial \mathbf{w}_{n'}^{(2)}} &= C_{nn'}^{(22)}\mathbf{x}\mathbf{z}^{\intercal},\\ \frac{\partial^2 E}{\partial \mathbf{w}_m^{(1)}\partial \mathbf{w}_{m'}^{(1)}} &= C_{mm'}^{(11)}\mathbf{x}\mathbf{x}^{\intercal},\\ \frac{\partial^2 E}{\partial \mathbf{w}_m^{(1)}\partial \mathbf{w}_n^{(2)}} &= C_{mn}^{(12)}\mathbf{x}\mathbf{z}^{\intercal} + h'(b_m)\frac{\partial E}{\partial a_n}\mathbf{x}\mathbf{v}^{\intercal},\\ \frac{\partial^2 E}{\partial \mathbf{w}_n^{(2)}\partial \mathbf{w}_{n'}^{(2)}} &= C_{nn'}^{(22)}\mathbf{z}\mathbf{z}^{\intercal},\end{aligned} \tag{5.120}$$

where $\mathbf{v}$ is a vector from the standard basis in $M$ dimensions such that

$$v_m = 1, \tag{5.121}$$

and

$$\begin{aligned}C_{mm'}^{(11)} =& h''(b_m)I_{mm'}\sum_{n=1}^{N} w_{nm}^{(2)}\frac{\partial E}{\partial a_n}\\ &+ h'(b_m)h'(b_{m'})\sum_{n=1}^{N} w_{nm}^{(2)}\sum_{n'=1}^{N} w_{n'm'}^{(2)}\frac{\partial^2 E}{\partial a_n a_{n'}},\\ C_{nm}^{(12)} =& h'(b_m)\sum_{n'=1}^{N} w_{n'm}^{(2)}\frac{\partial^2 E}{\partial a_n \partial a_n'},\\ C_{nn'}^{(22)} =& \frac{\partial^2 E}{\partial a_n \partial a_{n'}}.\end{aligned} \tag{5.122}$$

## 5.24

Let $y_1, \cdots, y_N$ be variables such that

$$\begin{aligned} y_n &= \mathbf{w}_n^\intercal \mathbf{z} + w_{n0}, \\ z_m &= h\left(\mathbf{w}_m^\intercal \mathbf{x} + w_{m0}\right). \end{aligned} \tag{5.123}$$

### (a)

Let

$$\tilde{\mathbf{x}} = a\mathbf{x} + b\mathbf{v}, \tag{5.124}$$

where

$$\mathbf{v} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}.$$

Then,

$$z_m = h\left(\frac{1}{a}\mathbf{w}_m^\intercal \left(\tilde{\mathbf{x}} - b\mathbf{v}\right) + w_{m0}\right). \tag{5.125}$$

Therefore,

$$\tilde{z}_m = h\left(\tilde{\mathbf{w}}_m^\intercal \mathbf{x} + \tilde{w}_{m0}\right), \tag{5.126}$$

where

$$\begin{aligned} \tilde{\mathbf{w}}_m &= \frac{1}{a}\mathbf{w}_m, \\ \tilde{w}_{m0} &= w_{m0} - \frac{b}{a}\mathbf{w}_m^\intercal \mathbf{v}. \end{aligned} \tag{5.127}$$

### (b)

Let

$$\tilde{y}_n = cy_n + d. \tag{5.128}$$

Then,

$$\frac{\tilde{y}_n - d}{c} = \mathbf{w}_n^\intercal \mathbf{z} + w_{n0}. \tag{5.129}$$

Therefore,

$$\tilde{y}_n = \tilde{\mathbf{w}}_n^\intercal \mathbf{z} + \tilde{w}_{n0}, \tag{5.130}$$

where

$$\begin{aligned} \tilde{\mathbf{w}}_n &= c\mathbf{w}_n, \\ \tilde{w}_{n0} &= cw_{n0} + d. \end{aligned} \tag{5.131}$$

## 5.25

Let $E$ be a quadratic error function such that

$$E(\mathbf{w}) = E_0 + \frac{1}{2}\left(\mathbf{w} - \mathbf{w}^*\right)^\intercal \mathbf{H}^* \left(\mathbf{w} - \mathbf{w}^*\right), \tag{5.132}$$

where $\mathbf{w}^*$ is a stationary point of $E$ and $\mathbf{H}^*$ is a positive definite and constant matrix whose eignevectors and eigenvalues are $\mathbf{u}_1, \cdots, \mathbf{u}_M$ and $\eta_1, \cdots, \eta_M$. Let

$$\begin{aligned} \mathbf{w}^{(\tau)} &= \mathbf{w}^{(\tau-1)} - \rho \nabla E, \\ \mathbf{w}^{(0)} &= \mathbf{0}. \end{aligned} \tag{5.133}$$

### (a)

We have

$$\nabla E = \mathbf{H}^* \left(\mathbf{w} - \mathbf{w}^*\right). \tag{5.134}$$

Then,

$$\mathbf{w}^{(\tau)} = \mathbf{w}^{(\tau-1)} - \rho \mathbf{H} \left(\mathbf{w}^{(\tau-1)} - \mathbf{w}^*\right). \tag{5.135}$$

so that

$$\mathbf{u}_m^\intercal \mathbf{w}^{(\tau)} = \mathbf{u}_m^\intercal \mathbf{w}^{(\tau-1)} - \rho \mathbf{u}_m^\intercal \mathbf{H}^* \left(\mathbf{w}^{(\tau-1)} - \mathbf{w}^*\right). \tag{5.136}$$

Since $\mathbf{H}^*$ is symmetric,

$$\mathbf{u}_m^\intercal \mathbf{H}^* \mathbf{w} = \mathbf{w}^\intercal \mathbf{H}^* \mathbf{u}_m. \tag{5.137}$$

The right hand side can be written as $\eta_m w_m$, where

$$w_m = \mathbf{w}^\intercal \mathbf{u}_m. \tag{5.138}$$

Then,

$$w_m^{(\tau)} = w_m^{(\tau-1)} - \rho \eta_m \left(w_m^{(\tau-1)} - w_m^*\right), \tag{5.139}$$

so that

$$w_m^{(\tau)} - w_m^* = \left(1 - \rho \eta_m\right) \left(w_m^{(\tau-1)} - w_m^*\right). \tag{5.140}$$

Therefore,

$$w_m^{(\tau)} = w_m^* + \left(1 - \rho \eta_m\right)^\tau \left(w_m^{(0)} - w_m^*\right), \tag{5.141}$$

so that

$$w_m^{(\tau)} = \left(1 - \left(1 - \rho \eta_m\right)^\tau\right) w_m^*. \tag{5.142}$$

**(b)**

By (a), if

$$|1 - \rho\eta_m| < 1, \tag{5.143}$$

then

$$\lim_{\tau\to\infty} w_m^{(\tau)} = w_m^*. \tag{5.144}$$

**(c)**

By (a) and the Taylor series

$$(1 - \rho\eta_m)^\tau = 1 - \rho\tau\eta_m + \rho^2\tau^2 O\left(\eta_m^2\right), \tag{5.145}$$

we have

$$w_m^{(\tau)} = \left(\rho\tau\eta_m - \rho^2\tau^2 O\left(\eta_m^2\right)\right) w_m^*, \tag{5.146}$$

so that

$$w_m^{(\tau)} = \rho\tau\eta_m \left(1 - \rho\tau\eta_m O\left(\eta_m\right)\right) w_m^*. \tag{5.147}$$

Therefore,

$$\left|w_m^{(\tau)}\right| \ll |w_m|^*, \quad \text{if} \quad \eta_m \ll (\rho\tau)^{-1}, \tag{5.148}$$

while

$$w_m^{(\tau)} \simeq w_m^*, \quad \text{if} \quad \eta_m \gg (\rho\tau)^{-1}. \tag{5.149}$$

## 5.26 (Incomplete)

Let

$$\tilde{E} = E + \lambda\Omega, \tag{5.150}$$

where

$$\Omega = \frac{1}{2}\sum_{n=1}^{N}\sum_{k=1}^{K}\left(\frac{\partial y_{nk}}{\partial\xi}\right)^2. \tag{5.151}$$

**(a)**

We have

$$\frac{\partial y_{nk}}{\partial\xi} = \mathcal{G} y_{nk}, \tag{5.152}$$

where

$$\mathcal{G} = \sum_{d=1}^{D} \tau_d \frac{\partial}{\partial x_d},$$
$$\tau_d = \frac{\partial x_d}{\partial \xi}. \tag{5.153}$$

Therefore,

$$\Omega = \sum_{n=1}^{N} \Omega_n, \tag{5.154}$$

where

$$\Omega_n = \frac{1}{2} \sum_{k=1}^{K} (\mathcal{G} y_{nk})^2. \tag{5.155}$$

**(b)**

$$\alpha_j = \mathcal{G} z_j = h'(a_j)\beta_j?$$
$$\beta_j = \mathcal{G} a_j = \sum_{d=1}^{D} w_{jd}\alpha_d? \tag{5.156}$$

## 5.27

Let

$$\tilde{E} = \frac{1}{2} \iiint \left(y\left(\mathbf{x} + \boldsymbol{\xi}\right) - t\right)^2 p(t|\mathbf{x})p(\mathbf{x})p(\boldsymbol{\xi}) d\mathbf{x} dt d\boldsymbol{\xi}, \tag{5.157}$$

where

$$\mathrm{E}\,\boldsymbol{\xi} = 0,$$
$$\mathrm{var}\,\boldsymbol{\xi} = \lambda \mathbf{I}. \tag{5.158}$$

**(a)**

We have the Taylor series

$$y(\mathbf{x} + \boldsymbol{\xi}) = y(\mathbf{x}) + \rho(\mathbf{x}, \boldsymbol{\xi}) + O\left(\|\boldsymbol{\xi}\|^3\right), \tag{5.159}$$

where

$$\rho(\mathbf{x}, \boldsymbol{\xi}) = \boldsymbol{\xi}^{\intercal} \nabla y(\mathbf{x}) + \frac{1}{2} \boldsymbol{\xi}^{\intercal} \nabla\nabla y(\mathbf{x}) \boldsymbol{\xi}. \tag{5.160}$$

Then,

$$\left(y\left(\mathbf{x}+\boldsymbol{\xi}\right)-t\right)^2 \simeq \left(y(\mathbf{x})-t\right)^2 + 2\left(y(\mathbf{x})-t\right)\rho(\mathbf{x},\boldsymbol{\xi}) + \rho(\mathbf{x},\boldsymbol{\xi})^2. \tag{5.161}$$

Then,

$$\tilde{E} \simeq E + I_1 + \frac{1}{2} I_2, \tag{5.162}$$

where

$$\begin{aligned}
E &= \frac{1}{2}\iint \left(y(\mathbf{x})-t\right)^2 p(t|\mathbf{x})p(\mathbf{x})d\mathbf{x}dt,\\
I_1 &= \iiint \left(y(\mathbf{x})-t\right)\rho(\mathbf{x},\boldsymbol{\xi})p(t|\mathbf{x})p(\mathbf{x})p(\boldsymbol{\xi})d\mathbf{x}dtd\boldsymbol{\xi},\\
I_2 &= \iint \rho(\mathbf{x},\boldsymbol{\xi})^2 p(\mathbf{x})p(\boldsymbol{\xi})d\mathbf{x}d\boldsymbol{\xi}.
\end{aligned} \tag{5.163}$$

$I_1$ can be written as

$$\begin{aligned}
&\iiint \left(y(\mathbf{x})-t\right)\boldsymbol{\xi}^{\intercal}\nabla y(\mathbf{x})p(t|\mathbf{x})p(\mathbf{x})p(\boldsymbol{\xi})d\mathbf{x}dtd\boldsymbol{\xi}\\
&\quad + \frac{1}{2}\int \left(y(\mathbf{x})-t\right)\boldsymbol{\xi}^{\intercal}\nabla\nabla y(\mathbf{x})\boldsymbol{\xi}p(t|\mathbf{x})p(\mathbf{x})p(\boldsymbol{\xi})d\mathbf{x}dtd\boldsymbol{\xi}\\
&= \left(\int \boldsymbol{\xi}p(\boldsymbol{\xi})d\boldsymbol{\xi}\right)^{\intercal}\iint \left(y(\mathbf{x})-t\right)\nabla y(\mathbf{x})p(t|\mathbf{x})p(\mathbf{x})d\mathbf{x}dt\\
&\quad + \frac{1}{2}\int \boldsymbol{\xi}^{\intercal}\mathbf{M}_1\boldsymbol{\xi}p(\boldsymbol{\xi})d\boldsymbol{\xi},
\end{aligned} \tag{5.164}$$

where

$$\mathbf{M}_1 = \iint \left(y(\mathbf{x})-t\right)\nabla\nabla y(\mathbf{x})p(t|\mathbf{x})p(\mathbf{x})d\mathbf{x}dt. \tag{5.165}$$

Then,

$$I_1 = \frac{\lambda}{2}\operatorname{tr}\mathbf{M}_1. \tag{5.166}$$

$I_2$ can be written as

$$\begin{aligned}
&\iint \left(\boldsymbol{\xi}^{\intercal}\nabla y(\mathbf{x}) + \frac{1}{2}\boldsymbol{\xi}^{\intercal}\nabla\nabla y(\mathbf{x})\boldsymbol{\xi}\right)^2 p(\mathbf{x})p(\boldsymbol{\xi})d\mathbf{x}d\boldsymbol{\xi}\\
&\simeq \int \boldsymbol{\xi}^{\intercal}\mathbf{M}_2\boldsymbol{\xi}p(\boldsymbol{\xi})d\boldsymbol{\xi},
\end{aligned} \tag{5.167}$$

where

$$\mathbf{M}_2 = \int (\nabla y(\mathbf{x})) (\nabla y(\mathbf{x}))^\intercal p(\mathbf{x}) d\mathbf{x}. \tag{5.168}$$

Then,

$$I_2 \simeq \lambda \operatorname{tr} \mathbf{M}_2. \tag{5.169}$$

Therefore,

$$\tilde{E} \simeq E + \frac{\lambda}{2} \operatorname{tr}(\mathbf{M}_1 + \mathbf{M}_2). \tag{5.170}$$

**(b)**

We have

$$\mathbf{M}_1 = \int \left( y(\mathbf{x}) \int p(t|\mathbf{x}) dt - \int t p(t|\mathbf{x}) dt \right) \nabla\nabla y(\mathbf{x}) p(\mathbf{x}) d\mathbf{x}. \tag{5.171}$$

The right hand side can be written as

$$\int (y(\mathbf{x}) - \mathrm{E}(t|\mathbf{x})) \nabla\nabla y(\mathbf{x}) p(\mathbf{x}) d\mathbf{x}. \tag{5.172}$$

Therefore, if

$$y(\mathbf{x}) = \mathrm{E}(t|\mathbf{x}), \tag{5.173}$$

then

$$\tilde{E} \simeq E + \frac{\lambda}{2} \operatorname{tr} \mathbf{M}_2. \tag{5.174}$$

## 5.28 (Incomplete)

## 5.29

Let $\mathbf{w}$ be a variable in $M$ dimensions such that

$$p(w_m) = \sum_{k=1}^{K} \pi_k \mathcal{N}\left(w_m | \mu_k, \sigma_k^2\right). \tag{5.175}$$

Let

$$\tilde{E}(\mathbf{w}) = E(\mathbf{w}) + \lambda \Omega(\mathbf{w}), \tag{5.176}$$

where

$$\Omega(\mathbf{w}) = -\sum_{m=1}^{M} \ln p(w_m). \tag{5.177}$$

We have

$$\frac{\partial \Omega}{\partial w_m} = \frac{\partial \Omega}{\partial p_m} \frac{\partial p_m}{\partial w_m}, \tag{5.178}$$

where

$$p_m = p(w_m). \tag{5.179}$$

We have

$$\begin{aligned} \frac{\partial \Omega}{\partial p_m} &= -\frac{1}{p_m}, \\ \frac{\partial p_m}{\partial w_m} &= -\sum_{k=1}^{K} \frac{w_m - \mu_k}{\sigma_k^2} p_{mk}, \end{aligned} \tag{5.180}$$

where

$$p_{mk} = \pi_k \mathcal{N}\left(w_m | \mu_k, \sigma_k^2\right). \tag{5.181}$$

Therefore,

$$\frac{\partial \tilde{E}}{\partial w_m} = \frac{\partial E}{\partial w_m} + \lambda \sum_{k=1}^{K} \gamma_{mk} \frac{w_m - \mu_k}{\sigma_k^2}, \tag{5.182}$$

where

$$\gamma_{mk} = \frac{p_{mk}}{p_m}. \tag{5.183}$$

## 5.30

Let $\mathbf{w}$ be a variable in $M$ dimensions such that

$$p(w_m) = \sum_{k=1}^{K} \pi_k \mathcal{N}\left(w_m | \mu_k, \sigma_k^2\right). \tag{5.184}$$

Let

$$\tilde{E}(\mathbf{w}) = E(\mathbf{w}) + \lambda \Omega(\mathbf{w}), \tag{5.185}$$

where

$$\Omega(\mathbf{w}) = -\sum_{m=1}^{M} \ln p(w_m). \tag{5.186}$$

We have

$$\frac{\partial \Omega}{\partial \mu_k} = \sum_{m=1}^{M} \frac{\partial \Omega}{\partial p_m} \frac{\partial p_m}{\partial \mu_k}, \tag{5.187}$$

where

$$p_m = p(w_m). \tag{5.188}$$

We have

$$\begin{aligned} \frac{\partial \Omega}{\partial p_m} &= -\frac{1}{p_m}, \\ \frac{\partial p_m}{\partial \mu_k} &= -\frac{\mu_k - w_m}{\sigma_k^2} p_{mk}, \end{aligned} \tag{5.189}$$

where

$$p_{mk} = \pi_k \mathcal{N}\left(w_m | \mu_k, \sigma_k^2\right). \tag{5.190}$$

Therefore,

$$\frac{\partial \tilde{E}}{\partial \mu_k} = \lambda \sum_{m=1}^{M} \gamma_{mk} \frac{\mu_k - w_m}{\sigma_k^2}, \tag{5.191}$$

where

$$\gamma_{mk} = \frac{p_{mk}}{p_m}. \tag{5.192}$$

## 5.31

Let $\mathbf{w}$ be a variable in $M$ dimensions such that

$$p(w_m) = \sum_{k=1}^{K} \pi_k \mathcal{N}\left(w_m | \mu_k, \sigma_k^2\right). \tag{5.193}$$

Let

$$\tilde{E}(\mathbf{w}) = E(\mathbf{w}) + \lambda \Omega(\mathbf{w}), \tag{5.194}$$

where

$$\Omega(\mathbf{w}) = -\sum_{m=1}^{M} \ln p(w_m). \tag{5.195}$$

We have

$$\frac{\partial \Omega}{\partial \sigma_k} = \sum_{m=1}^{M} \frac{\partial \Omega}{\partial p_m} \frac{\partial p_m}{\partial \sigma_k}, \tag{5.196}$$

where

$$p_m = p(w_m). \tag{5.197}$$

We have

$$\begin{aligned}\frac{\partial \Omega}{\partial p_m} &= -\frac{1}{p_m}, \\ \frac{\partial p_m}{\partial \sigma_k} &= \left(-\frac{1}{\sigma_k} + \frac{(w_m - \mu_k)^2}{\sigma_k^3}\right) p_{mk},\end{aligned} \tag{5.198}$$

where

$$p_{mk} = \pi_k \mathcal{N}\left(w_m | \mu_k, \sigma_k^2\right). \tag{5.199}$$

Therefore,

$$\frac{\partial \tilde{E}}{\partial \sigma_k} = \lambda \sum_{m=1}^{M} \gamma_{mk} \left(\frac{1}{\sigma_k} - \frac{(w_m - \mu_k)^2}{\sigma_k^3}\right), \tag{5.200}$$

where

$$\gamma_{mk} = \frac{p_{mk}}{p_m}. \tag{5.201}$$

## 5.32

Let $\mathbf{w}$ be variable in $M$ dimensions such that

$$p(w_m) = \sum_{k=1}^{K} \pi_k \mathcal{N}\left(w_m | \mu_k, \sigma_k^2\right), \tag{5.202}$$

where

$$\pi_k = \frac{\exp(\eta_k)}{\sum_{k'=1}^{K} \exp(\eta_{k'})}. \tag{5.203}$$

Let

$$\tilde{E}(\mathbf{w}) = E(\mathbf{w}) + \lambda \Omega(\mathbf{w}), \tag{5.204}$$

where

$$\Omega(\mathbf{w}) = -\sum_{m=1}^{M} \ln p(w_m). \tag{5.205}$$

### (a)

If $k \neq k'$, then

$$\frac{\partial \pi_k}{\partial \eta_{k'}} = -\pi_k \pi_{k'}. \tag{5.206}$$

We have

$$\frac{\partial \pi_k}{\partial \eta_k} = \pi_k - \pi_k^2. \tag{5.207}$$

Therefore,

$$\frac{\partial \pi_k}{\partial \eta_{k'}} = I_{kk'}\pi_k - \pi_k\pi_{k'}. \tag{5.208}$$

**(b)**

We have

$$\frac{\partial \Omega}{\partial \eta_k} = \sum_{m=1}^{M}\sum_{k'=1}^{K} \frac{\partial \Omega}{\partial p_m}\frac{\partial p_m}{\partial \pi_{k'}}\frac{\partial \pi_{k'}}{\partial \eta_k}, \tag{5.209}$$

where

$$p_m = p(w_m). \tag{5.210}$$

We have

$$\begin{aligned} \frac{\partial \Omega}{\partial p_m} &= -\frac{1}{p_m}, \\ \frac{\partial p_m}{\partial \pi_{k'}} &= \frac{p_{mk'}}{\pi_{k'}}, \end{aligned} \tag{5.211}$$

where

$$p_{mk} = \pi_k \mathcal{N}\left(w_m | \mu_k, \sigma_k^2\right). \tag{5.212}$$

By (a),

$$\frac{\partial \pi_{k'}}{\partial \eta_k} = I_{k'k}\pi_{k'} - \pi_{k'}\pi_{k'}. \tag{5.213}$$

Then,

$$\frac{\partial \Omega}{\partial \eta_k} = -\sum_{m=1}^{M}\sum_{k'=1}^{K} \frac{\gamma_{mk'}}{\pi_{k'}}\left(I_{k'k}\pi_{k'} - \pi_k\pi_{k'}\right), \tag{5.214}$$

where

$$\gamma_{mk} = \frac{p_{mk}}{p_m}. \tag{5.215}$$

The right hand side can be written as

$$-\sum_{m=1}^{M}\sum_{k'=1}^{K} \gamma_{mk'}\left(I_{k'k} - \pi_k\right) = \sum_{m=1}^{M}\left(\pi_k - \gamma_{mk}\right). \tag{5.216}$$

Therefore,

$$\frac{\partial \tilde{E}}{\partial \eta_k} = \lambda \sum_{m=1}^{M}\left(\pi_k - \gamma_{mk}\right). \tag{5.217}$$

## 5.33

Let $(x_1, x_2)$ be the Cartesian coordinates of the end-effector of a two-link robot arm whose joint angles are $\theta_1$ and $\theta_2$ and whose arm lengths are $L_1$ and $L_2$. Let us asuume that the origin of the coordinate system is given by the attachment point of the lower arm. We have

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} L_1 \cos\theta_1 \\ L_1 \sin\theta_1 \end{bmatrix} + \begin{bmatrix} L_2 \cos(\theta_2 - (\pi - \theta_1)) \\ L_2 \sin(\theta_2 - (\pi - \theta_1)) \end{bmatrix}. \tag{5.218}$$

The second term of the right hand side can be written as

$$\begin{bmatrix} L_2 \cos(\theta_1 + \theta_2 - \pi) \\ L_2 \sin(\theta_1 + \theta_2 - \pi) \end{bmatrix} = \begin{bmatrix} -L_2 \cos(\theta_1 + \theta_2) \\ -L_2 \sin(\theta_1 + \theta_2) \end{bmatrix}. \tag{5.219}$$

Therefore,

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} L_1 \cos\theta_1 - L_2 \cos(\theta_1 + \theta_2) \\ L_1 \sin\theta_1 - L_2 \sin(\theta_1 + \theta_2) \end{bmatrix}. \tag{5.220}$$

## 5.34

Let $\mathbf{t}_1, \cdots, \mathbf{t}_N$ be conditionally independent variables such that

$$p(\mathbf{t}_n|\mathbf{w}) = \sum_{k=1}^{K} \pi_k \mathcal{N}_{nk}, \tag{5.221}$$

where

$$\begin{aligned} \pi_k &= \frac{\exp(a_k)}{\sum_{k'=1}^{K} \exp(a_{k'})}, \\ \mathcal{N}_{nk} &= \mathcal{N}\left(\mathbf{t}_n|\boldsymbol{\mu}_k(\mathbf{x}_n, \mathbf{w}), \sigma_k^2(\mathbf{x}_n, \mathbf{w})\mathbf{I}\right). \end{aligned} \tag{5.222}$$

Let

$$E(\mathbf{w}) = \sum_{n=1}^{N} E_n(\mathbf{w}), \tag{5.223}$$

where

$$E_n(\mathbf{w}) = -\ln p(\mathbf{t}_n|\mathbf{w}). \tag{5.224}$$

We have

$$\frac{\partial E_n}{\partial a_k} = \sum_{k'=1}^{K} \frac{\partial E_n}{\partial p_n} \frac{\partial p_n}{\partial \pi_{k'}} \frac{\partial \pi_{k'}}{\partial a_k}, \tag{5.225}$$

where

$$p_n = p(\mathbf{t}_n|\mathbf{w}). \tag{5.226}$$

We have

$$\begin{aligned}\frac{\partial E_n}{\partial p_n} &= -\frac{1}{p_n},\\ \frac{\partial p_n}{\partial \pi_{k'}} &= \frac{p_{nk'}}{\pi_{k'}},\end{aligned} \tag{5.227}$$

where

$$p_{nk} = \pi_k \mathcal{N}_{nk}. \tag{5.228}$$

By 5.32(a),

$$\frac{\partial \pi_{k'}}{\partial a_k} = I_{k'k}\pi_{k'} - \pi_k \pi_{k'}. \tag{5.229}$$

Then,

$$\frac{\partial E_n}{\partial a_k} = -\frac{1}{p_n}\sum_{k'=1}^{K} \frac{p_{nk'}}{\pi_{k'}}(I_{k'k}\pi_{k'} - \pi_k\pi_{k'}). \tag{5.230}$$

The right hand side can be written as

$$-\sum_{k'=1}^{K} \gamma_{nk'}\left(I_{kk'} - \pi_k\right) = \pi_k - \gamma_{nk}, \tag{5.231}$$

where

$$\gamma_{nk} = \frac{p_{nk}}{p_n}, \tag{5.232}$$

Therefore,

$$\frac{\partial E_n}{\partial a_k} = \pi_k - \gamma_{nk}. \tag{5.233}$$

## 5.35

Let $\mathbf{t}_1, \cdots, \mathbf{t}_N$ be conditionally independent variables such that

$$p(\mathbf{t}_n|\mathbf{w}) = \sum_{k=1}^{K} \pi_k \mathcal{N}_{nk}, \tag{5.234}$$

where

$$\mathcal{N}_{nk} = \mathcal{N}\left(\mathbf{t}_n|\boldsymbol{\mu}_k(\mathbf{x}_n, \mathbf{w}), \sigma_k^2(\mathbf{x}_n, \mathbf{w})\mathbf{I}\right). \tag{5.235}$$

Let

$$E(\mathbf{w}) = \sum_{n=1}^{N} E_n(\mathbf{w}), \tag{5.236}$$

where

$$E_n(\mathbf{w}) = -\ln p(\mathbf{t}_n|\mathbf{w}). \tag{5.237}$$

Then,

$$\frac{\partial E_n}{\partial \boldsymbol{\mu}_k} = \frac{\partial E_n}{\partial p_n}\frac{\partial p_n}{\partial \boldsymbol{\mu}_k}, \tag{5.238}$$

where

$$p_n = p(t_n|\mathbf{w}). \tag{5.239}$$

We have

$$\begin{aligned}\frac{\partial E_n}{\partial p_n} &= -\frac{1}{p_n},\\ \frac{\partial p_n}{\partial \boldsymbol{\mu}_k} &= -\sigma_k^{-2}(\boldsymbol{\mu}_k - \mathbf{t}_n)p_{nk},\end{aligned} \tag{5.240}$$

where

$$p_{nk} = \pi_k \mathcal{N}_{nk}. \tag{5.241}$$

Therefore,

$$\frac{\partial E_n}{\partial \boldsymbol{\mu}_k} = \gamma_{nk}\sigma_k^{-2}(\boldsymbol{\mu}_k - \mathbf{t}_n), \tag{5.242}$$

where

$$\gamma_{nk} = \frac{p_{nk}}{p_n}. \tag{5.243}$$

## 5.36

Let $\mathbf{t}_1, \cdots, \mathbf{t}_N$ be conditionally independent variables in $M$ dimensions such that

$$p(\mathbf{t}_n|\mathbf{w}) = \sum_{k=1}^{K} \pi_k \mathcal{N}_{nk}, \tag{5.244}$$

where

$$\begin{aligned}\mathcal{N}_{nk} &= \mathcal{N}\left(\mathbf{t}_n|\boldsymbol{\mu}_k(\mathbf{x}_n, \mathbf{w}), \sigma_k^2(\mathbf{x}_n, \mathbf{w})\mathbf{I}\right),\\ \sigma_k &= \exp\left(a_k^{\sigma}\right).\end{aligned} \tag{5.245}$$

Let

$$E(\mathbf{w}) = \sum_{n=1}^{N} E_n(\mathbf{w}), \tag{5.246}$$

where

$$E_n(\mathbf{w}) = -\ln p(\mathbf{t}_n|\mathbf{w}). \tag{5.247}$$

Then,

$$\frac{\partial E_n}{\partial a_k^\sigma} = \frac{\partial E_n}{\partial p_n}\frac{\partial p_n}{\partial \sigma_k}\frac{\partial \sigma_k}{\partial a_k^\sigma}, \tag{5.248}$$

where

$$p_n = p(\mathbf{t}_n|\mathbf{w}). \tag{5.249}$$

We have

$$\begin{aligned}
\frac{\partial E_n}{\partial p_n} &= -\frac{1}{p_n}, \\
\frac{\partial p_n}{\partial \sigma_k} &= \left(-\frac{M}{\sigma_k} + \frac{\|\mathbf{t}_n - \boldsymbol{\mu}_k\|^2}{\sigma_k^3}\right) p_{nk}, \\
\frac{\partial \sigma_k}{\partial a_k^\sigma} &= \sigma_k,
\end{aligned} \tag{5.250}$$

where

$$p_{nk} = \pi_k \mathcal{N}_{nk}. \tag{5.251}$$

Therefore,

$$\frac{\partial E_n}{\partial a_k^\sigma} = \gamma_{nk}\left(M - \frac{\|\mathbf{t}_n - \boldsymbol{\mu}_k\|^2}{\sigma_k^2}\right), \tag{5.252}$$

where

$$\gamma_{nk} = \frac{p_{nk}}{p_n}. \tag{5.253}$$

## 5.37

Let $\mathbf{t}$ be a variable such that

$$p(\mathbf{t}|\mathbf{x}) = \sum_{k=1}^{K} \pi_k(\mathbf{x}) \mathcal{N}\left(\mathbf{t}|\boldsymbol{\mu}_k(\mathbf{x}), \sigma_k^2(\mathbf{x})\mathbf{I}\right). \tag{5.254}$$

### (a)

We have

$$\mathrm{E}(\mathbf{t}|\mathbf{x}) = \int \mathbf{t} p(\mathbf{t}|\mathbf{x}) d\mathbf{t}. \tag{5.255}$$

The right hand side can be written as

$$\sum_{k=1}^{K} \pi_k(\mathbf{x}) \int \mathbf{t}\mathcal{N}\left(\mathbf{t}|\boldsymbol{\mu}_k(\mathbf{x}), \sigma_k^2(\mathbf{x})\mathbf{I}\right) d\mathbf{t} = \sum_{k=1}^{K} \pi_k(\mathbf{x})\boldsymbol{\mu}_k(\mathbf{x}). \tag{5.256}$$

Therefore,

$$\mathrm{E}(\mathbf{t}|\mathbf{x}) = \sum_{k=1}^{K} \pi_k(\mathbf{x})\boldsymbol{\mu}_k(\mathbf{x}). \tag{5.257}$$

**(b)**

We have

$$\mathrm{var}(\mathbf{t}|\mathbf{x}) = \int \left(\mathbf{t} - \mathrm{E}(\mathbf{t}|\mathbf{x})\right)\left(\mathbf{t} - \mathrm{E}(\mathbf{t}|\mathbf{x})\right)^{\intercal} p(\mathbf{t}|\mathbf{x})d\mathbf{t}. \tag{5.258}$$

The right hand side can be written as

$$\begin{aligned}
&\mathbf{M}_1 - \mathrm{E}(\mathbf{t}|\mathbf{x})\left(\int \mathbf{t}p(\mathbf{t}|\mathbf{x})d\mathbf{t}\right)^{\intercal} - \left(\int \mathbf{t}p(\mathbf{t}|\mathbf{x})d\mathbf{t}\right)\mathrm{E}(\mathbf{t}|\mathbf{x})^{\intercal} \\
&+ \mathrm{E}(\mathbf{t}|\mathbf{x})\,\mathrm{E}(\mathbf{t}|\mathbf{x})^{\intercal} \int p(\mathbf{t}|\mathbf{x})d\mathbf{t} \\
=&\mathbf{M}_1 - \mathrm{E}(\mathbf{t}|\mathbf{x})\,\mathrm{E}(\mathbf{t}|\mathbf{x})^{\intercal},
\end{aligned} \tag{5.259}$$

where

$$\mathbf{M}_1 = \int \mathbf{t}\mathbf{t}^{\intercal} p(\mathbf{t}|\mathbf{x})d\mathbf{t}. \tag{5.260}$$

$\mathbf{M}_1$ can be written as

$$\sum_{k=1}^{K} \pi_k(\mathbf{x}) \int \mathbf{t}\mathbf{t}^{\intercal}\mathcal{N}\left(\mathbf{t}|\boldsymbol{\mu}_k(\mathbf{x}), \sigma_k^2(\mathbf{x})\mathbf{I}\right) d\mathbf{t}. \tag{5.261}$$

The integral can be written as

$$\begin{aligned}
&\int \left(\mathbf{t}' + \boldsymbol{\mu}_k(\mathbf{x})\right)\left(\mathbf{t}' + \boldsymbol{\mu}_k(\mathbf{x})\right)^{\intercal} \mathcal{N}\left(\mathbf{t}|\boldsymbol{\mu}_k(\mathbf{x}), \sigma_k^2(\mathbf{x})\mathbf{I}\right) d\mathbf{t} \\
=&\mathbf{M}_2 + \boldsymbol{\mu}_k(\mathbf{x})\mathbf{v}^{\intercal} + \mathbf{v}\boldsymbol{\mu}_k(\mathbf{x})^{\intercal} + \boldsymbol{\mu}_k(\mathbf{x})\boldsymbol{\mu}_k(\mathbf{x})^{\intercal} I,
\end{aligned} \tag{5.262}$$

where

$$\begin{aligned}
\mathbf{t}' &= \mathbf{t} - \boldsymbol{\mu}_k(\mathbf{x}), \\
\mathbf{M}_2 &= \int \mathbf{t}'\mathbf{t}'^{\mathsf{T}} \mathcal{N}\left(\mathbf{t}|\boldsymbol{\mu}_k(\mathbf{x}), \sigma_k^2(\mathbf{x})\mathbf{I}\right) d\mathbf{t}, \\
\mathbf{v} &= \int \mathbf{t}' \mathcal{N}\left(\mathbf{t}|\boldsymbol{\mu}_k(\mathbf{x}), \sigma_k^2(\mathbf{x})\mathbf{I}\right) d\mathbf{t}, \\
I &= \int \mathcal{N}\left(\mathbf{t}|\boldsymbol{\mu}_k(\mathbf{x}), \sigma_k^2(\mathbf{x})\mathbf{I}\right) d\mathbf{t}.
\end{aligned} \tag{5.263}$$

We have

$$\begin{aligned}
\mathbf{M}_2 &= \int \mathbf{t}'\mathbf{t}'^{\mathsf{T}} \mathcal{N}\left(\mathbf{t}'|\mathbf{0}, \sigma_k^2(\mathbf{x})\mathbf{I}\right) d\mathbf{t}', \\
\mathbf{v} &= \int \mathbf{t}' \mathcal{N}\left(\mathbf{t}'|\mathbf{0}, \sigma_k^2(\mathbf{x})\mathbf{I}\right) d\mathbf{t}'.
\end{aligned} \tag{5.264}$$

Then,

$$\mathbf{M}_1 = \sigma_k^2(\mathbf{x})\mathbf{I} + \boldsymbol{\mu}_k(\mathbf{x})\boldsymbol{\mu}_k(\mathbf{x})^{\mathsf{T}}. \tag{5.265}$$

Therefore, by (a),

$$\begin{aligned}
\mathrm{var}(\mathbf{t}|\mathbf{x}) = &\sum_{k=1}^{K} \pi_k(\mathbf{x}) \left(\sigma_k^2(\mathbf{x})\mathbf{I} + \boldsymbol{\mu}_k(\mathbf{x})\boldsymbol{\mu}_k(\mathbf{x})^{\mathsf{T}}\right) \\
&- \left(\sum_{k=1}^{K} \pi_k(\mathbf{x})\boldsymbol{\mu}_k(\mathbf{x})\right) \left(\sum_{k=1}^{K} \pi_k(\mathbf{x})\boldsymbol{\mu}_k(\mathbf{x})\right)^{\mathsf{T}}.
\end{aligned} \tag{5.266}$$

## 5.38

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned}
p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|y(\mathbf{x}_n, \mathbf{w}), \beta^{-1}\right), \\
p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I}\right).
\end{aligned} \tag{5.267}$$

By marginalisation,

$$p(t_{N+1}|\mathbf{t}) = \int p(t_{N+1}|\mathbf{w})p(\mathbf{w}|\mathbf{t})d\mathbf{w}. \tag{5.268}$$

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{w})p(\mathbf{w}). \tag{5.269}$$

The logarithm of the right hand side except the terms independent of $\mathbf{w}$ can be written as $-E(\mathbf{w})$ where

$$E(\mathbf{w}) = \frac{\beta}{2} \sum_{n=1}^{N} (t_n - y(\mathbf{x}_n, \mathbf{w}))^2 + \frac{\alpha}{2} \|\mathbf{w}\|^2. \tag{5.270}$$

We have the Taylor series

$$\begin{aligned} E(\mathbf{w}) =& E(\mathbf{w}_{\mathrm{MAP}}) + \frac{1}{2}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}})^{\intercal} \mathbf{H}_{\mathrm{MAP}} (\mathbf{w} - \mathbf{w}_{\mathrm{MAP}}) \\ &+ O\left(\|\mathbf{w} - \mathbf{w}_{\mathrm{MAP}}\|^3\right), \end{aligned} \tag{5.271}$$

where $\mathbf{w}_{\mathrm{MAP}}$ is a stationary point of $E$ and

$$\mathbf{H}_{\mathrm{MAP}} = \nabla\nabla E(\mathbf{w}_{\mathrm{MAP}}). \tag{5.272}$$

Then,

$$p(\mathbf{w}|\mathbf{t}) \simeq \mathcal{N}\left(\mathbf{w}|\mathbf{w}_{\mathrm{MAP}}, \mathbf{H}_{\mathrm{MAP}}^{-1}\right). \tag{5.273}$$

We have the Taylor series

$$\begin{aligned} y(\mathbf{x}_{N+1}, \mathbf{w}) =& y_{\mathrm{MAP}\,N+1} + \mathbf{g}_{\mathrm{MAP}\,N+1}{}^{\intercal}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}}) \\ &+ O\left(\|\mathbf{w} - \mathbf{w}_{\mathrm{MAP}}\|^2\right), \end{aligned} \tag{5.274}$$

where

$$\begin{aligned} y_{\mathrm{MAP}\,N+1} &= y(\mathbf{x}_{N+1}, \mathbf{w}_{\mathrm{MAP}}), \\ \mathbf{g}_{\mathrm{MAP}\,N+1} &= \nabla_{\mathbf{w}} y(\mathbf{x}_{N+1}, \mathbf{w})\big|_{\mathbf{w}=\mathbf{w}_{\mathrm{MAP}}}. \end{aligned} \tag{5.275}$$

Then,

$$p(t_{N+1}|\mathbf{w}) \simeq \mathcal{N}\left(t_{N+1}|y_{\mathrm{MAP}\,N+1} + \mathbf{g}_{\mathrm{MAP}\,N+1}{}^{\intercal}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}}), \beta^{-1}\right). \tag{5.276}$$

Then, the logarithm of the integrand except the terms independent of $\mathbf{w}$ can be approximated as

$$\begin{aligned} &-\frac{\beta}{2}\left(t_{N+1} - y_{\mathrm{MAP}\,N+1} - \mathbf{g}_{\mathrm{MAP}\,N+1}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}})\right)^2 \\ &-\frac{1}{2}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}})^{\intercal}\mathbf{H}_{\mathrm{MAP}}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}}) \\ =& -\frac{1}{2}\mathbf{v}^{\intercal}\mathbf{M}\mathbf{v}, \end{aligned} \tag{5.277}$$

where

$$
\begin{aligned}
\mathbf{v} &= \begin{bmatrix} \mathbf{w} - \mathbf{w}_{\text{MAP}} \\ t_{N+1} - y_{\text{MAP}\,N+1}, \end{bmatrix}, \\
\mathbf{M} &= \begin{bmatrix} \mathbf{H}_{\text{MAP}} + \beta \mathbf{g}_{\text{MAP}\,N+1} \mathbf{g}_{\text{MAP}\,N+1}{}^{\intercal} & -\beta \mathbf{g}_{\text{MAP}\,N+1} \\ -\beta \mathbf{g}_{\text{MAP}\,N+1}{}^{\intercal} & \beta \end{bmatrix}.
\end{aligned} \tag{5.278}
$$

By 2.24,

$$
\mathbf{M}^{-1} = \begin{bmatrix} \mathbf{H}_{\text{MAP}}^{-1} & \mathbf{H}_{\text{MAP}}^{-1} \mathbf{g}_{\text{MAP}\,N+1} \\ \mathbf{g}_{\text{MAP}\,N+1}{}^{\intercal} \mathbf{H}_{\text{MAP}}^{-1} & \beta^{-1} + \mathbf{g}_{\text{MAP}\,N+1}{}^{\intercal} \mathbf{H}_{\text{MAP}}^{-1} \mathbf{g}_{\text{MAP}\,N+1} \end{bmatrix}. \tag{5.279}
$$

Therefore,

$$
p(t_{N+1}|\mathbf{t}) \simeq \mathcal{N}\left(t_{N+1} | y_{\text{MAP}\,N+1}, \beta^{-1} + \mathbf{g}_{\text{MAP}\,N+1}{}^{\intercal} \mathbf{H}_{\text{MAP}}^{-1} \mathbf{g}_{\text{MAP}\,N+1}\right). \tag{5.280}
$$

## 5.39

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$
\begin{aligned}
p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n | y(\mathbf{x}_n, \mathbf{w}), \beta^{-1}\right), \\
p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w} | \mathbf{0}, \alpha^{-1}\mathbf{I}\right),
\end{aligned} \tag{5.281}
$$

where $\mathbf{w}$ is in $M$ dimensions. By marginalisation,

$$
p(\mathbf{t}) = \int p(\mathbf{t}|\mathbf{w}) p(\mathbf{w}) d\mathbf{w}. \tag{5.282}
$$

The logarithm of the integrand of the right hand side can be written as

$$
\begin{aligned}
& -\frac{N}{2} \ln\left(2\pi\beta^{-1}\right) - \frac{\beta}{2} \sum_{n=1}^{N} \left(t_n - y(\mathbf{x}_n, \mathbf{w})\right)^2 \\
& -\frac{M}{2} \ln(2\pi) - \frac{1}{2} \ln\left(\det\left(\alpha^{-1}\mathbf{I}\right)\right) - \frac{\alpha}{2} \|\mathbf{w}\|^2 \\
= & -E(\mathbf{w}) - \frac{N+M}{2} \ln(2\pi) + \frac{N}{2} \ln\beta + \frac{M}{2} \ln\alpha,
\end{aligned} \tag{5.283}
$$

where

$$
E(\mathbf{w}) = \frac{\beta}{2} \sum_{n=1}^{N} \left(t_n - y(\mathbf{x}_n, \mathbf{w})\right)^2 + \frac{\alpha}{2} \|\mathbf{w}\|^2. \tag{5.284}
$$

We have the Taylor series

$$
\begin{aligned}
E(\mathbf{w}) =& E(\mathbf{w}_{\mathrm{MAP}}) + \frac{1}{2}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}})^{\intercal}\mathbf{H}_{\mathrm{MAP}}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}}) \\
& + O\left(\|\mathbf{w} - \mathbf{w}_{\mathrm{MAP}}\|^3\right),
\end{aligned}
\tag{5.285}
$$

where $\mathbf{w}_{\mathrm{MAP}}$ is a stationary point of $E$ and

$$
\mathbf{H}_{\mathrm{MAP}} = \nabla\nabla E(\mathbf{w}_{\mathrm{MAP}}). \tag{5.286}
$$

Then, the logarithm of the integrand can be approximated as

$$
\begin{aligned}
& - E(\mathbf{w}_{\mathrm{MAP}}) - \frac{1}{2}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}})^{\intercal}\mathbf{H}_{\mathrm{MAP}}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}}) \\
& - \frac{N+M}{2}\ln(2\pi) + \frac{N}{2}\ln\beta + \frac{M}{2}\ln\alpha.
\end{aligned}
\tag{5.287}
$$

We have

$$
\begin{aligned}
& \int \exp\left(-\frac{1}{2}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}})^{\intercal}\mathbf{H}_{\mathrm{MAP}}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}})\right) d\mathbf{w} \\
=& (2\pi)^{\frac{M}{2}}\left(\det \mathbf{H}_{\mathrm{MAP}}^{-1}\right)^{\frac{1}{2}}.
\end{aligned}
\tag{5.288}
$$

Therefore,

$$
\ln p(\mathbf{t}) \simeq -E(\mathbf{w}_{\mathrm{MAP}}) - \frac{1}{2}\ln|\det \mathbf{H}_{\mathrm{MAP}}| - \frac{N}{2}\ln(2\pi) + \frac{N}{2}\ln\beta + \frac{M}{2}\ln\alpha. \tag{5.289}
$$

## 5.40

Let $\mathbf{t}_1, \cdots, \mathbf{t}_N$ be conditionally independent variables from the standard basis in $K$ dimensions such that

$$
\begin{aligned}
p(\mathbf{t}_n|\mathbf{w}) &= \prod_{k=1}^{K} y_{nk}^{t_{nk}}, \\
p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I}\right),
\end{aligned}
\tag{5.290}
$$

where

$$
\begin{aligned}
y_{nk} &= y_k(\mathbf{a}_n), \\
y_k(\mathbf{a}) &= \frac{\exp(a_k)}{\sum_{k'=1}^{K}\exp(a_{k'})}, \\
\mathbf{a}_n &= \mathbf{a}(\mathbf{x}_n, \mathbf{w}).
\end{aligned}
\tag{5.291}
$$

By marginalisation,

$$p(\mathbf{t}_{N+1}|\mathbf{T}) = \int p(\mathbf{t}_{N+1}|\mathbf{w})p(\mathbf{w}|\mathbf{T})d\mathbf{w}. \tag{5.292}$$

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{T})p(\mathbf{T}) = p(\mathbf{T}|\mathbf{w})p(\mathbf{w}). \tag{5.293}$$

The logarithm of the right hand side except the terms independent of $\mathbf{w}$ can be written as $-E(\mathbf{w})$ where

$$E(\mathbf{w}) = -\sum_{n=1}^{N}\sum_{k=1}^{K} t_{nk} \ln y_{nk} + \frac{\alpha}{2}\|\mathbf{w}\|^2. \tag{5.294}$$

We have the Taylor series

$$\begin{aligned} E(\mathbf{w}) =& E(\mathbf{w}_{\mathrm{MAP}}) + \frac{1}{2}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}})^{\intercal}\mathbf{H}_{\mathrm{MAP}}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}}) \\ &+ O\left(\|\mathbf{w} - \mathbf{w}_{\mathrm{MAP}}\|^3\right), \end{aligned} \tag{5.295}$$

where $\mathbf{w}_{\mathrm{MAP}}$ is a stationary point of $E$ and

$$\mathbf{H}_{\mathrm{MAP}} = \nabla\nabla E(\mathbf{w}_{\mathrm{MAP}}). \tag{5.296}$$

Then,

$$p(\mathbf{w}|\mathbf{T}) \simeq \mathcal{N}\left(\mathbf{w}|\mathbf{w}_{\mathrm{MAP}}, \mathbf{H}_{\mathrm{MAP}}^{-1}\right). \tag{5.297}$$

We have

$$p(\mathbf{t}_{N+1}|\mathbf{w}) = \prod_{k=1}^{K} y_{N+1,k}^{t_{N+1,k}}, \tag{5.298}$$

where

$$y_{N+1,k} = y_k\left(\mathbf{a}(\mathbf{x}_{N+1}, \mathbf{w})\right). \tag{5.299}$$

We have the Taylor series

$$\begin{aligned} \mathbf{a}(\mathbf{x}_{N+1}, \mathbf{w}) =& \mathbf{a}_{\mathrm{MAP}N+1} + \mathbf{J}_{\mathrm{MAP}N+1}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}}) \\ &+ O\left(\|\mathbf{w} - \mathbf{w}_{\mathrm{MAP}}\|^2\right), \end{aligned} \tag{5.300}$$

where

$$\begin{aligned} \mathbf{a}_{\mathrm{MAP}N+1} &= \mathbf{a}(\mathbf{x}_{N+1}, \mathbf{w}_{\mathrm{MAP}}), \\ \mathbf{J}_{\mathrm{MAP}N+1} &= \nabla_{\mathbf{w}}\mathbf{a}(\mathbf{x}_{N+1}, \mathbf{w})\big|_{\mathbf{w}=\mathbf{w}_{\mathrm{MAP}}}. \end{aligned} \tag{5.301}$$

Then,

$$p(\mathbf{t}_{N+1}|\mathbf{w}) \simeq \int \delta\left(\mathbf{a}_{N+1} - \mathbf{c}_{N+1}\right) \left(\prod_{k=1}^{K} y_k(\mathbf{a}_{N+1})^{t_{N+1,k}}\right) d\mathbf{a}_{N+1}, \tag{5.302}$$

where $\delta$ is the Dirac delta function and

$$\mathbf{c}_{N+1} = \mathbf{a}_{\mathrm{MAP}N+1} + \mathbf{J}_{\mathrm{MAP}N+1}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}}). \tag{5.303}$$

Then,

$$\int p(\mathbf{t}_{N+1}|\mathbf{w})p(\mathbf{w}|\mathbf{T})d\mathbf{w} \simeq \int \left(\prod_{k=1}^{K} y_k(\mathbf{a}_{N+1})^{t_{N+1,k}}\right) p(\mathbf{a}_{N+1})d\mathbf{a}_{N+1}, \tag{5.304}$$

where

$$p(\mathbf{a}_{N+1}) = \int \delta\left(\mathbf{a}_{N+1} - \mathbf{c}_{N+1}\right) \mathcal{N}\left(\mathbf{w}|\mathbf{w}_{\mathrm{MAP}}, \mathbf{H}_{\mathrm{MAP}}^{-1}\right) d\mathbf{w}. \tag{5.305}$$

Since $\mathbf{t}_{N+1}$ is a variable from the standard basis in $K$ dimensions, the right hand side of the approximation can be written as

$$\prod_{k=1}^{K} \left(\int y_k(\mathbf{a}_{N+1})p(\mathbf{a}_{N+1})d\mathbf{a}_{N+1}\right)^{t_{N+1,k}}. \tag{5.306}$$

We have

$$\mathrm{E}\,\mathbf{a}_{N+1} = \int \mathbf{a}_{N+1}p(\mathbf{a}_{N+1})d\mathbf{a}_{N+1}. \tag{5.307}$$

The right hand side can be written as

$$\int \mathbf{c}_{N+1}\mathcal{N}\left(\mathbf{w}|\mathbf{w}_{\mathrm{MAP}}, \mathbf{H}_{\mathrm{MAP}}^{-1}\right) d\mathbf{w} = \mathbf{a}_{\mathrm{MAP}N+1}. \tag{5.308}$$

Then,

$$\mathrm{var}\,\mathbf{a}_{N+1} = \int \left(\mathbf{a}_{N+1} - \mathbf{a}_{\mathrm{MAP}N+1}\right)^2 p(\mathbf{a}_{N+1})d\mathbf{a}_{N+1}. \tag{5.309}$$

The right hand side can be written as

$$\begin{aligned}&\int \left(\mathbf{c}_{N+1} - \mathbf{a}_{\mathrm{MAP}N+1}\right)^2 \mathcal{N}\left(\mathbf{w}|\mathbf{w}_{\mathrm{MAP}}, \mathbf{H}_{\mathrm{MAP}}^{-1}\right) d\mathbf{w} \\ =&\mathbf{J}_{\mathrm{MAP}N+1}\mathbf{H}_{\mathrm{MAP}}^{-1}\mathbf{J}_{\mathrm{MAP}N+1}{}^{\intercal}.\end{aligned} \tag{5.310}$$

Then,

$$p(\mathbf{a}_{N+1}) = \mathcal{N}\left(\mathbf{a}_{N+1}|\mathbf{a}_{\mathrm{MAP}N+1}, \mathbf{J}_{\mathrm{MAP}N+1}\mathbf{H}_{\mathrm{MAP}}^{-1}\mathbf{J}_{\mathrm{MAP}N+1}{}^{\intercal}\right). \tag{5.311}$$

Therefore,

$$p(\mathbf{t}_{N+1}|\mathbf{T}) \simeq \prod_{k=1}^{K} \upsilon_{N+1,k}^{t_{N+1,k}}, \tag{5.312}$$

where

$$\upsilon_{N+1,k} = \int y_k(\mathbf{a})\mathcal{N}\left(\mathbf{a}|\mathbf{a}_{\mathrm{MAP}N+1}, \mathbf{J}_{\mathrm{MAP}N+1}\mathbf{H}_{\mathrm{MAP}}^{-1}\mathbf{J}_{\mathrm{MAP}N+1}{}^{\intercal}\right) d\mathbf{a}. \tag{5.313}$$

## 5.41

Let $\mathbf{t}_1, \cdots, \mathbf{t}_N$ be conditionally independent variables from the standard basis in $K$ dimensions such that

$$\begin{aligned} p(\mathbf{t}_n|\mathbf{w}) &= \prod_{k=1}^{K} y_{nk}^{t_{nk}}, \\ p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I}\right), \end{aligned} \tag{5.314}$$

where $\mathbf{w}$ is in $M$ dimensions and

$$y_{nk} = y_k(\mathbf{x}_n, \mathbf{w}). \tag{5.315}$$

By marginalisation,

$$p(\mathbf{T}) = \int p(\mathbf{T}|\mathbf{w})p(\mathbf{w})d\mathbf{w}. \tag{5.316}$$

The logarithm of the integrand of the right hand side can be written as

$$\begin{aligned} &\sum_{n=1}^{N}\sum_{k=1}^{K} t_{nk}\ln y_{nk} - \frac{M}{2}\ln(2\pi) - \frac{1}{2}\ln\left(\det\left(\alpha^{-1}\mathbf{I}\right)\right) - \frac{\alpha}{2}\|\mathbf{w}\|^2 \\ =& -E(\mathbf{w}) - \frac{M}{2}\ln(2\pi) + \frac{M}{2}\ln\alpha, \end{aligned} \tag{5.317}$$

where

$$E(\mathbf{w}) = -\sum_{n=1}^{N}\sum_{k=1}^{K} t_{nk}\ln y_{nk} + \frac{\alpha}{2}\|\mathbf{w}\|^2. \tag{5.318}$$

We have the Taylor series

$$
\begin{aligned}
E(\mathbf{w}) =& E(\mathbf{w}_{\mathrm{MAP}}) + \frac{1}{2}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}})^{\intercal}\mathbf{H}_{\mathrm{MAP}}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}}) \\
& + O\left(\|(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}})\|^{3}\right),
\end{aligned}
\tag{5.319}
$$

where $\mathbf{w}_{\mathrm{MAP}}$ is a stationary point of $E$ and

$$
\mathbf{H}_{\mathrm{MAP}} = \nabla\nabla E(\mathbf{w}_{\mathrm{MAP}}). \tag{5.320}
$$

Then, the logarithm of the integrand can be approximated as

$$
-E(\mathbf{w}_{\mathrm{MAP}}) - \frac{1}{2}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}})^{\intercal}\mathbf{H}_{\mathrm{MAP}}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}}) - \frac{M}{2}\ln(2\pi) + \frac{M}{2}\ln\alpha. \tag{5.321}
$$

We have

$$
\begin{aligned}
& \int \exp\left(-\frac{1}{2}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}})^{\intercal}\mathbf{H}_{\mathrm{MAP}}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}})\right) d\mathbf{w} \\
=& (2\pi)^{\frac{M}{2}} \left(\det \mathbf{H}_{\mathrm{MAP}}^{-1}\right)^{\frac{1}{2}}.
\end{aligned}
\tag{5.322}
$$

Therefore,

$$
\ln p(\mathbf{T}) \simeq -E(\mathbf{w}_{\mathrm{MAP}}) - \frac{1}{2}\ln|\det \mathbf{H}_{\mathrm{MAP}}| + \frac{M}{2}\ln\alpha. \tag{5.323}
$$

# 6 Kernel Methods

## 6.1

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$p(t_n|\mathbf{w}) = \mathcal{N}\left(t_n|\mathbf{w}^\intercal\boldsymbol{\phi}_n, 1\right), \\ p(\mathbf{w}) = \mathcal{N}\left(\mathbf{w}|\mathbf{0}, \lambda^{-1}\mathbf{I}\right). \tag{6.1}$$

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{w})p(\mathbf{w}). \tag{6.2}$$

The logarithm of the right hand side except the terms independent of $\mathbf{w}$ can be written as $-E_{\mathbf{w}}(\mathbf{w})$ where

$$E_{\mathbf{w}}(\mathbf{w}) = \frac{1}{2}\|\mathbf{t} - \boldsymbol{\Phi}\mathbf{w}\|^2 + \frac{\lambda}{2}\|\mathbf{w}\|^2. \tag{6.3}$$

### (a)

Let $\mathbf{w}_{\text{MAP}}$ be a stationary point of $E_{\mathbf{w}}$. Setting the derivative of $E_{\mathbf{w}}$ with respect to $\mathbf{w}$ to zero gives

$$\mathbf{0} = -\boldsymbol{\Phi}^\intercal(\mathbf{t} - \boldsymbol{\Phi}\mathbf{w}) + \lambda\mathbf{w}. \tag{6.4}$$

Therefore,

$$\mathbf{w}_{\text{MAP}} = \boldsymbol{\Phi}^\intercal\mathbf{a}_{\text{MAP}}, \tag{6.5}$$

where

$$\mathbf{a}_{\text{MAP}} = \frac{1}{\lambda}\left(\mathbf{t} - \boldsymbol{\Phi}\mathbf{w}_{\text{MAP}}\right). \tag{6.6}$$

### (b)

Let

$$\mathbf{w} = \boldsymbol{\Phi}^\intercal\mathbf{a}. \tag{6.7}$$

Then,

$$E_{\mathbf{w}}(\mathbf{w}) = E_{\mathbf{a}}(\mathbf{a}), \tag{6.8}$$

where

$$E_{\mathbf{a}}(\mathbf{a}) = \frac{1}{2}\|\mathbf{t} - \boldsymbol{\Phi}\boldsymbol{\Phi}^\intercal\mathbf{a}\|^2 + \frac{\lambda}{2}\mathbf{a}^\intercal\boldsymbol{\Phi}\boldsymbol{\Phi}^\intercal\mathbf{a}. \tag{6.9}$$

The right hand side can be written as

$$\frac{1}{2}\|\mathbf{t}\|^2 - \mathbf{t}^\intercal \mathbf{K}\mathbf{a} + \frac{1}{2}\mathbf{a}^\intercal \mathbf{K}\mathbf{K}\mathbf{a} + \frac{\lambda}{2}\mathbf{a}^\intercal \mathbf{K}\mathbf{a}, \tag{6.10}$$

where

$$\mathbf{K} = \mathbf{\Phi}\mathbf{\Phi}^\intercal. \tag{6.11}$$

Let $\mathbf{a}_{\text{MAP}}$ be a stationary point of $E_{\mathbf{a}}$. Setting the derivative of $E_{\mathbf{a}}$ with respect to $\mathbf{a}$ to zero gives

$$\mathbf{0} = -\mathbf{K}\mathbf{t} + \mathbf{K}\mathbf{K}\mathbf{a} + \lambda\mathbf{K}\mathbf{a}. \tag{6.12}$$

Therefore,

$$\mathbf{a}_{\text{MAP}} = (\mathbf{K} + \lambda\mathbf{I})^{-1}\mathbf{t}. \tag{6.13}$$

**(c)**

By (a) and (b),

$$\frac{1}{\lambda}(\mathbf{t} - \mathbf{\Phi}\mathbf{w}_{\text{MAP}}) = (\mathbf{K} + \lambda\mathbf{I})^{-1}\mathbf{t}, \tag{6.14}$$

so that

$$\mathbf{t} - \mathbf{\Phi}\mathbf{w}_{\text{MAP}} = \mathbf{N}\mathbf{t}, \tag{6.15}$$

where

$$\mathbf{N} = \left(\mathbf{I} + \lambda^{-1}\mathbf{K}\right)^{-1}. \tag{6.16}$$

By 2.26, the right hand side can be written as

$$\mathbf{I} - \mathbf{\Phi}\mathbf{M}\mathbf{\Phi}^\intercal, \tag{6.17}$$

where

$$\mathbf{M} = \left(\lambda\mathbf{I} + \mathbf{\Phi}^\intercal\mathbf{\Phi}\right)^{-1}. \tag{6.18}$$

Then,

$$\mathbf{t} - \mathbf{\Phi}\mathbf{w}_{\text{MAP}} = \left(\mathbf{I} - \mathbf{\Phi}\mathbf{M}\mathbf{\Phi}^\intercal\right)\mathbf{t}, \tag{6.19}$$

so that

$$\mathbf{\Phi}\left(\mathbf{w}_{\text{MAP}} - \mathbf{\Phi}\mathbf{M}\mathbf{\Phi}^\intercal\mathbf{t}\right) = \mathbf{0}. \tag{6.20}$$

Therefore,

$$\mathbf{w}_{\text{MAP}} = \mathbf{M}\mathbf{\Phi}^\intercal\mathbf{t}. \tag{6.21}$$

## 6.2 (Incomplete)

Let $t_1, \cdots, t_N$ be conditionally independent variables from $\{-1, 1\}$ such that

$$p(t_n|\mathbf{w}) = \left(\frac{1+y_n}{2}\right)^{\frac{1+t_n}{2}} \left(\frac{1-y_n}{2}\right)^{\frac{1-t_n}{2}}, \tag{6.22}$$

where

$$y_n = f\left(\mathbf{w}^\intercal \boldsymbol{\phi}_n\right),$$
$$f(a) = \begin{cases} 1, & a \geq 0, \\ -1, & \text{otherwise.} \end{cases} \tag{6.23}$$

Let

$$E(\mathbf{w}) = -\sum_{n \in \mathcal{M}} \mathbf{w}^\intercal \boldsymbol{\phi}_n t_n, \tag{6.24}$$

where $\mathcal{M}$ is the set of all misclassification. We have the Taylor series

$$E(\mathbf{w}^{(\tau)}) = E(\mathbf{w}^{(\tau-1)}) + \mathbf{b}^\intercal \left(\mathbf{w}^{(\tau)} - \mathbf{w}^{(\tau-1)}\right) + O\left(\left\|\mathbf{w}^{(\tau)} - \mathbf{w}^{(\tau-1)}\right\|^2\right), \tag{6.25}$$

where

$$\mathbf{b} = -\sum_{n \in \mathcal{M}} t_n \boldsymbol{\phi}_n. \tag{6.26}$$

## 6.3

We have

$$\|\mathbf{x} - \mathbf{x}_n\|^2 = \|\mathbf{x}\|^2 - 2\mathbf{x}^\intercal \mathbf{x}_n + \|\mathbf{x}_n\|^2. \tag{6.27}$$

The right hand side can be written as

$$k(\mathbf{x}, \mathbf{x}) - 2k(\mathbf{x}, \mathbf{x}_n) + k(\mathbf{x}_n, \mathbf{x}_n), \tag{6.28}$$

where

$$k(\mathbf{x}, \mathbf{x}') = \mathbf{x}^\intercal \mathbf{x}'. \tag{6.29}$$

## 6.4

Let $\mathbf{A}$ be a $2 \times 2$ matrix with positive eigenvalues and with at least one negative element. We have

$$\det(\lambda \mathbf{I} - \mathbf{A}) = (\lambda - A_{11})(\lambda - A_{22}) - A_{12}A_{21}. \tag{6.30}$$

Seeing the left hand side to zero gives

$$0 = \lambda^2 - (A_{11} + A_{22})\lambda + A_{11}A_{22} - A_{12}A_{21}. \tag{6.31}$$

For the equation to have two positive solutions, we have the conditions

$$\begin{aligned} (A_{11} + A_{22})^2 - 4(A_{11}A_{22} - A_{12}A_{21}) &> 0, \\ A_{11} + A_{22} &> 0, \\ A_{11}A_{22} - A_{12}A_{21} &> 0. \end{aligned} \tag{6.32}$$

Therefore, an example of $A$ is given by

$$\begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix}. \tag{6.33}$$

## 6.5

Let

$$k_1(\mathbf{x}, \mathbf{x}') = \boldsymbol{\phi}_1(\mathbf{x})^\intercal \boldsymbol{\phi}_1(\mathbf{x}'). \tag{6.34}$$

### (a)

Let

$$k(\mathbf{x}, \mathbf{x}') = c k_1(\mathbf{x}, \mathbf{x}'), \tag{6.35}$$

where $c$ is a positive constant. The right hand side can be written as

$$c\boldsymbol{\phi}_1(\mathbf{x})^\intercal \boldsymbol{\phi}_1(\mathbf{x}') = \boldsymbol{\phi}(\mathbf{x})^\intercal \boldsymbol{\phi}(\mathbf{x}'), \tag{6.36}$$

where

$$\boldsymbol{\phi} = \sqrt{c}\boldsymbol{\phi}_1. \tag{6.37}$$

Therefore, $k$ is a valid kernel.

### (b)

Let

$$k(\mathbf{x}, \mathbf{x}') = f(\mathbf{x}) k_1(\mathbf{x}, \mathbf{x}') f(\mathbf{x}'). \tag{6.38}$$

The right hand side can be written as

$$f(\mathbf{x})\boldsymbol{\phi}_1(\mathbf{x})^\intercal \boldsymbol{\phi}_1(\mathbf{x}') f(\mathbf{x}') = \boldsymbol{\phi}(\mathbf{x})^\intercal \boldsymbol{\phi}(\mathbf{x}'), \tag{6.39}$$

where

$$\boldsymbol{\phi} = f\boldsymbol{\phi}_1. \tag{6.40}$$

Therefore, $k$ is a valid kernel.

## 6.6

Let

$$k_1(\mathbf{x}, \mathbf{x}') = \boldsymbol{\phi}_1(\mathbf{x})^\intercal \boldsymbol{\phi}_1(\mathbf{x}'), \tag{6.41}$$

where $\boldsymbol{\phi}_1$ is a vector in $M$ dimensions.

### (a)

Let

$$k(\mathbf{x}, \mathbf{x}') = \sum_{j=0}^{J} c_j q_j(\mathbf{x}, \mathbf{x}'), \tag{6.42}$$

where $c_0, \cdots, c_J$ are nonnegative constants and

$$q_j(\mathbf{x}, \mathbf{x}') = \left(k_1(\mathbf{x}, \mathbf{x}')\right)^j. \tag{6.43}$$

We have

$$q_0(\mathbf{x}, \mathbf{x}') = 1. \tag{6.44}$$

Let us asuume that $q_j$ is a valid kernel. Then,

$$q_{j+1}(\mathbf{x}, \mathbf{x}') = q_j(\mathbf{x}, \mathbf{x}') \boldsymbol{\phi}_1(\mathbf{x})^\intercal \boldsymbol{\phi}_1(\mathbf{x}'). \tag{6.45}$$

The right hand side can be written as

$$q_j(\mathbf{x}, \mathbf{x}') \sum_{m=1}^{M} \phi_{1m}(\mathbf{x}) \phi_{1m}(\mathbf{x}') = \sum_{m=1}^{M} \phi_{1m}(\mathbf{x}) q_j(\mathbf{x}, \mathbf{x}') \phi_{1m}(\mathbf{x}'). \tag{6.46}$$

By 6.5(b) and 6.7(a), the right hand side is a valid kernel. Then, $q_{j+1}$ is a valid kernel. Then, the assumption is proved by induction on $j$. Therefore, by 6.5(a) and 6.7(a), $k$ is a valid kernel.

### (b)

Let

$$k(\mathbf{x}, \mathbf{x}') = \exp\left(k_1(\mathbf{x}, \mathbf{x}')\right). \tag{6.47}$$

The right hand side can be written as

$$\sum_{j=0}^{\infty} \frac{1}{j!} \left(k_1(\mathbf{x}, \mathbf{x}')\right)^j. \tag{6.48}$$

Therefore, by (a), $k$ is a valid kernel.

## 6.7

Let

$$\begin{aligned} k_1(\mathbf{x}, \mathbf{x}') &= \boldsymbol{\phi}_1(\mathbf{x})^\intercal \boldsymbol{\phi}_1(\mathbf{x}'), \\ k_2(\mathbf{x}, \mathbf{x}') &= \boldsymbol{\phi}_2(\mathbf{x})^\intercal \boldsymbol{\phi}_2(\mathbf{x}'). \end{aligned} \tag{6.49}$$

### (a)

Let

$$k(\mathbf{x}, \mathbf{x}') = k_1(\mathbf{x}, \mathbf{x}') + k_2(\mathbf{x}, \mathbf{x}'). \tag{6.50}$$

The right hand side can be written as

$$\boldsymbol{\phi}_1(\mathbf{x})^\intercal \boldsymbol{\phi}_1(\mathbf{x}') + \boldsymbol{\phi}_2(\mathbf{x})^\intercal \boldsymbol{\phi}_2(\mathbf{x}') = \boldsymbol{\phi}(\mathbf{x})^\intercal \boldsymbol{\phi}(\mathbf{x}'), \tag{6.51}$$

where

$$\boldsymbol{\phi} = \begin{bmatrix} \boldsymbol{\phi}_1 \\ \boldsymbol{\phi}_2 \end{bmatrix}. \tag{6.52}$$

Therefore, $k$ is a valid kernel.

### (b)

Let

$$k(\mathbf{x}, \mathbf{x}') = k_1(\mathbf{x}, \mathbf{x}') k_2(\mathbf{x}, \mathbf{x}'). \tag{6.53}$$

Then,

$$\ln k(\mathbf{x}, \mathbf{x}') = \ln k_1(\mathbf{x}, \mathbf{x}') + \ln k_2(\mathbf{x}, \mathbf{x}'). \tag{6.54}$$

By (a) and 6.6(b), the right hand side is a valid kernel. Then, $\ln k$ is a valid kernel. Therefore, by 6.6(b), $k$ is a valid kernel.

## 6.8

### (a)

Let

$$k(\mathbf{x}, \mathbf{x}') = k_1\left(\boldsymbol{\phi}(\mathbf{x}), \boldsymbol{\phi}(\mathbf{x}')\right), \tag{6.55}$$

where $k_1$ is a valid kernel. Therefore, $k$ is a valid kernel.

### (b)

Let
$$k(\mathbf{x}, \mathbf{x}') = \mathbf{x}^\intercal \mathbf{A} \mathbf{x}', \tag{6.56}$$
where $\mathbf{A}$ is a symmetric positive semidefinite matrix. Then, there exists $\mathbf{M}$ such that
$$\mathbf{A} = \mathbf{M}^\intercal \mathbf{M}. \tag{6.57}$$
Then,
$$k(\mathbf{x}, \mathbf{x}') = \boldsymbol{\phi}(\mathbf{x})^\intercal \boldsymbol{\phi}(\mathbf{x}'), \tag{6.58}$$
where
$$\boldsymbol{\phi}(\mathbf{x}) = \mathbf{M}\mathbf{x}. \tag{6.59}$$
Therefore, $k$ is a valid kernel.

## 6.9

Let
$$\begin{aligned} k_a(\mathbf{x}_a, \mathbf{x}_a') &= \boldsymbol{\phi}_a(\mathbf{x}_a)^\intercal \boldsymbol{\phi}_a(\mathbf{x}_a'), \\ k_b(\mathbf{x}_b, \mathbf{x}_b') &= \boldsymbol{\phi}_b(\mathbf{x}_b)^\intercal \boldsymbol{\phi}_b(\mathbf{x}_b'). \end{aligned} \tag{6.60}$$

### (a)

Let
$$k(\mathbf{x}, \mathbf{x}') = k_a(\mathbf{x}_a, \mathbf{x}_a') + k_b(\mathbf{x}_b, \mathbf{x}_b'), \tag{6.61}$$
where
$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_a \\ \mathbf{x}_b \end{bmatrix}. \tag{6.62}$$
The right hand side can be written as
$$\boldsymbol{\phi}_a(\mathbf{x}_a)^\intercal \boldsymbol{\phi}_a(\mathbf{x}_a') + \boldsymbol{\phi}_b(\mathbf{x}_b)^\intercal \boldsymbol{\phi}_b(\mathbf{x}_b') = \boldsymbol{\phi}(\mathbf{x})^\intercal \boldsymbol{\phi}(\mathbf{x}'), \tag{6.63}$$
where
$$\boldsymbol{\phi}(\mathbf{x}) = \begin{bmatrix} \boldsymbol{\phi}_a(\mathbf{x}_a) \\ \boldsymbol{\phi}_b(\mathbf{x}_b) \end{bmatrix}. \tag{6.64}$$
Therefore, $k$ is a valid kernel.

**(b)**

Let

$$k(\mathbf{x}, \mathbf{x}') = k_a(\mathbf{x}_a, \mathbf{x}'_a)k_b(\mathbf{x}_b, \mathbf{x}'_b), \tag{6.65}$$

where

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_a \\ \mathbf{x}_b \end{bmatrix}. \tag{6.66}$$

Then,

$$\ln k(\mathbf{x}, \mathbf{x}') = \ln k_a(\mathbf{x}_a, \mathbf{x}'_a) + \ln k_b(\mathbf{x}_b, \mathbf{x}'_b). \tag{6.67}$$

By (a) and 6.6(b), the right hand side is a valid kernel. Then, $\ln k$ is a valid kernel. Therefore, by 6.6(b), $k$ is a valid kernel.

## 6.10 (Incomplete)

Let

$$k(\mathbf{x}, \mathbf{x}') = f(\mathbf{x})f(\mathbf{x}'). \tag{6.68}$$

## 6.11

Let

$$k(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\sigma^2}\right). \tag{6.69}$$

The right hand side can be written as

$$\exp\left(-\frac{\mathbf{x}^\intercal\mathbf{x}}{2\sigma^2}\right)\exp\left(\frac{\mathbf{x}^\intercal\mathbf{x}'}{\sigma^2}\right)\exp\left(-\frac{\mathbf{x}'^\intercal\mathbf{x}'}{2\sigma^2}\right). \tag{6.70}$$

We have

$$\exp\left(\frac{\mathbf{x}^\intercal\mathbf{x}'}{\sigma^2}\right) = \sum_{j=0}^{\infty} q_j(\mathbf{x}, \mathbf{x}'), \tag{6.71}$$

where

$$q_j(\mathbf{x}, \mathbf{x}') = \frac{1}{j!}\left(\frac{\mathbf{x}^\intercal\mathbf{x}'}{\sigma^2}\right)^j. \tag{6.72}$$

By 6.6(a), $q_j$ is a valid kernel. Then, there exists $\boldsymbol{\psi}_j$ such that

$$q_j(\mathbf{x}, \mathbf{x}') = \boldsymbol{\psi}_j(\mathbf{x})^\intercal\boldsymbol{\psi}_j(\mathbf{x}'). \tag{6.73}$$

Then,

$$\exp\left(\frac{\mathbf{x}^\intercal \mathbf{x}'}{\sigma^2}\right) = \sum_{j=0}^{\infty} \boldsymbol{\psi}_j(\mathbf{x})^\intercal \boldsymbol{\psi}_j(\mathbf{x}'). \tag{6.74}$$

The right hand side can be written as $\boldsymbol{\phi}(\mathbf{x})^\intercal \boldsymbol{\phi}(\mathbf{x}')$, where

$$\boldsymbol{\phi} = \begin{bmatrix} \boldsymbol{\psi}_0 \\ \vdots \\ \boldsymbol{\psi}_j \\ \vdots \end{bmatrix}. \tag{6.75}$$

Therefore, $k$ can be written as the inner product of an infinite-dimensional feature vector.

## 6.12

Let $D$ be a set. Let $A_1, \cdots, A_{2^{|D|}}$ be all the subsets of $D$ where $|.|$ is the number of elements in (.). Let

$$k(A_m, A_{m'}) = 2^{|A_m \cap A_{m'}|}. \tag{6.76}$$

Let $\boldsymbol{\phi}$ be a vector such that

$$\phi_m(A) = \begin{cases} 1, & \text{if } A_m \subseteq A, \\ 0, & \text{otherwise.} \end{cases} \tag{6.77}$$

If

$$|A_m \cap A_{m'}| = 0, \tag{6.78}$$

then

$$k(A_m, A_{m'}) = 1. \tag{6.79}$$

The right hand side can be written as

$$\boldsymbol{\phi}(A_m)^\intercal \boldsymbol{\phi}(A_{m'}). \tag{6.80}$$

Let us assume that if

$$|A_m \cap A_{m'}| = n, \tag{6.81}$$

then

$$k(A_m, A_{m'}) = \boldsymbol{\phi}(A_m)^\intercal \boldsymbol{\phi}(A_{m'}). \tag{6.82}$$

Let

$$A_{m''} = A_{m'} \cup \{x\}, \tag{6.83}$$

where $x$ is an element such that

$$x \in A_m - A_{m'}. \tag{6.84}$$

Then,

$$|A_m \cap A_{m''}| = n + 1, \tag{6.85}$$

so that

$$k(A_m, A_{m''}) = 2^{n+1}. \tag{6.86}$$

We have

$$\boldsymbol{\phi}(A_m)^\intercal \boldsymbol{\phi}(A_{m''}) = \boldsymbol{\phi}(A_m)^\intercal \boldsymbol{\phi}(A_{m'}) + \boldsymbol{\phi}(A_m)^\intercal \left(\boldsymbol{\phi}(A_{m''}) - \boldsymbol{\phi}(A_{m'})\right). \tag{6.87}$$

By the assumption, the first term of the right hand side is $2^n$. By the assumption, $\boldsymbol{\phi}(A_m)$ and $\boldsymbol{\phi}(A_{m'})$ have a set of $2^n$ 1s in common. Then, $\boldsymbol{\phi}(A_m)$ and $\boldsymbol{\phi}(A_{m''}) - \boldsymbol{\phi}(A_{m'})$ have another set of $2^n$ 1s in common. Then, the second term of the right hand side is $2^n$. Then,

$$\boldsymbol{\phi}(A_m)^\intercal \boldsymbol{\phi}(A_{m''}) = 2^{n+1}, \tag{6.88}$$

so that

$$k(A_m, A_{m''}) = \boldsymbol{\phi}(A_m)^\intercal \boldsymbol{\phi}(A_{m''}). \tag{6.89}$$

Therefore, the assumption is proved by induction on $n$.

## 6.13

Let

$$k(\mathbf{x}, \mathbf{x}') = \mathbf{g}(\boldsymbol{\theta}, \mathbf{x})^\intercal \mathbf{F}^{-1} \mathbf{g}(\boldsymbol{\theta}, \mathbf{x}'), \tag{6.90}$$

where

$$\begin{aligned} \mathbf{g}(\boldsymbol{\theta}, \mathbf{x}) &= \nabla_{\boldsymbol{\theta}} \ln p(\mathbf{x}|\boldsymbol{\theta}), \\ \mathbf{F} &= \mathrm{E}_{\mathbf{x}}\left(\mathbf{g}(\boldsymbol{\theta}, \mathbf{x}) \mathbf{g}(\boldsymbol{\theta}, \mathbf{x})^\intercal\right). \end{aligned} \tag{6.91}$$

Let $\boldsymbol{\psi}$ be an invertible and differentiable function of $\boldsymbol{\theta}$. Let

$$k_{\boldsymbol{\psi}}(\mathbf{x}, \mathbf{x}') = \mathbf{g}_{\boldsymbol{\psi}}(\boldsymbol{\theta}, \mathbf{x})^\intercal \mathbf{F}_{\boldsymbol{\psi}}^{-1} \mathbf{g}_{\boldsymbol{\psi}}(\boldsymbol{\theta}, \mathbf{x}'), \tag{6.92}$$

where

$$\begin{aligned} \mathbf{g}_{\boldsymbol{\psi}}(\boldsymbol{\theta}, \mathbf{x}) &= \nabla_{\boldsymbol{\psi}} \ln p(\mathbf{x}|\boldsymbol{\theta}), \\ \mathbf{F}_{\boldsymbol{\psi}} &= \mathrm{E}_{\mathbf{x}} \left( \mathbf{g}_{\boldsymbol{\psi}}(\boldsymbol{\theta}, \mathbf{x}) \mathbf{g}_{\boldsymbol{\psi}}(\boldsymbol{\theta}, \mathbf{x})^\intercal \right). \end{aligned} \tag{6.93}$$

We have

$$\mathbf{g}_{\boldsymbol{\psi}}(\boldsymbol{\theta}, \mathbf{x}) = \frac{\partial \boldsymbol{\theta}}{\partial \boldsymbol{\psi}} \nabla_{\boldsymbol{\theta}} \ln p(\mathbf{x}|\boldsymbol{\theta}). \tag{6.94}$$

Then,

$$\mathbf{g}_{\boldsymbol{\psi}}(\boldsymbol{\theta}, \mathbf{x}) = \left( \frac{\partial \boldsymbol{\psi}}{\partial \boldsymbol{\theta}} \right)^{-1} \mathbf{g}(\boldsymbol{\theta}, \mathbf{x}). \tag{6.95}$$

We have

$$\mathbf{F}_{\boldsymbol{\psi}} = \int \mathbf{g}_{\boldsymbol{\psi}}(\boldsymbol{\theta}, \mathbf{x}) \mathbf{g}_{\boldsymbol{\psi}}(\boldsymbol{\theta}, \mathbf{x})^\intercal p(\mathbf{x}) d\mathbf{x}. \tag{6.96}$$

The right hand side can be written as

$$\begin{aligned} &\int \left( \frac{\partial \boldsymbol{\psi}}{\partial \boldsymbol{\theta}} \right)^{-1} \mathbf{g}(\boldsymbol{\theta}, \mathbf{x}) \mathbf{g}(\boldsymbol{\theta}, \mathbf{x})^\intercal \left( \left( \frac{\partial \boldsymbol{\psi}}{\partial \boldsymbol{\theta}} \right)^{-1} \right)^\intercal p(\mathbf{x}) d\mathbf{x} \\ =& \left( \frac{\partial \boldsymbol{\psi}}{\partial \boldsymbol{\theta}} \right)^{-1} \left( \int \mathbf{g}(\boldsymbol{\theta}, \mathbf{x}) \mathbf{g}(\boldsymbol{\theta}, \mathbf{x})^\intercal p(\mathbf{x}) d\mathbf{x} \right) \left( \left( \frac{\partial \boldsymbol{\psi}}{\partial \boldsymbol{\theta}} \right)^{-1} \right)^\intercal. \end{aligned} \tag{6.97}$$

Then,

$$\mathbf{F}_{\boldsymbol{\psi}} = \left( \frac{\partial \boldsymbol{\psi}}{\partial \boldsymbol{\theta}} \right)^{-1} \mathbf{F} \left( \left( \frac{\partial \boldsymbol{\psi}}{\partial \boldsymbol{\theta}} \right)^{-1} \right)^\intercal. \tag{6.98}$$

Then,

$$\begin{aligned} &k_{\boldsymbol{\psi}}(\mathbf{x}, \mathbf{x}') \\ =& \mathbf{g}(\boldsymbol{\theta}, \mathbf{x})^\intercal \left( \left( \frac{\partial \boldsymbol{\psi}}{\partial \boldsymbol{\theta}} \right)^{-1} \right)^\intercal \left( \frac{\partial \boldsymbol{\psi}}{\partial \boldsymbol{\theta}} \right)^\intercal \mathbf{F}^{-1} \left( \frac{\partial \boldsymbol{\psi}}{\partial \boldsymbol{\theta}} \right) \left( \frac{\partial \boldsymbol{\psi}}{\partial \boldsymbol{\theta}} \right)^{-1} \mathbf{g}(\boldsymbol{\theta}, \mathbf{x}'). \end{aligned} \tag{6.99}$$

The right hand side can be written as

$$\mathbf{g}(\boldsymbol{\theta}, \mathbf{x})^\intercal \mathbf{F}^{-1} \mathbf{g}(\boldsymbol{\theta}, \mathbf{x}') = k(\mathbf{x}, \mathbf{x}'). \tag{6.100}$$

Therefore,

$$k_{\boldsymbol{\psi}}(\mathbf{x}, \mathbf{x}') = k(\mathbf{x}, \mathbf{x}'). \tag{6.101}$$

## 6.14

Let $\mathbf{x}$ be a variable such that

$$p(\mathbf{x}|\boldsymbol{\mu}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \mathbf{S}). \tag{6.102}$$

Let

$$k(\mathbf{x}, \mathbf{x}') = \mathbf{g}(\boldsymbol{\mu}, \mathbf{x})^\intercal \mathbf{F}^{-1} \mathbf{g}(\boldsymbol{\mu}, \mathbf{x}'), \tag{6.103}$$

where

$$\begin{aligned} \mathbf{g}(\boldsymbol{\mu}, \mathbf{x}) &= \nabla_{\boldsymbol{\mu}} \ln p(\mathbf{x}|\boldsymbol{\mu}), \\ \mathbf{F} &= \mathrm{E}_{\mathbf{x}} \left( \mathbf{g}(\boldsymbol{\mu}, \mathbf{x}) \mathbf{g}(\boldsymbol{\mu}, \mathbf{x})^\intercal \right). \end{aligned} \tag{6.104}$$

We have

$$\mathbf{g}(\boldsymbol{\mu}, \mathbf{x}) = -\mathbf{S}^{-1}(\mathbf{x} - \boldsymbol{\mu}). \tag{6.105}$$

Then,

$$\mathbf{F} = \int \mathbf{S}^{-1}(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\intercal \mathbf{S}^{-1} \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \mathbf{S}) d\mathbf{x}. \tag{6.106}$$

The right hand side can be written as

$$\mathbf{S}^{-1} \left( \int (\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\intercal \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \mathbf{S}) d\mathbf{x} \right) \mathbf{S}^{-1} = \mathbf{S}^{-1}. \tag{6.107}$$

Then,

$$\mathbf{F} = \mathbf{S}^{-1}, \tag{6.108}$$

so that

$$\mathbf{g}(\boldsymbol{\mu}, \mathbf{x})^\intercal \mathbf{F}^{-1} \mathbf{g}(\boldsymbol{\mu}, \mathbf{x}') = (\mathbf{x} - \boldsymbol{\mu})^\intercal \mathbf{S}^{-1} (\mathbf{x} - \boldsymbol{\mu}). \tag{6.109}$$

Therefore,

$$k(\mathbf{x}, \mathbf{x}') = (\mathbf{x} - \boldsymbol{\mu})^\intercal \mathbf{S}^{-1} (\mathbf{x} - \boldsymbol{\mu}). \tag{6.110}$$

## 6.15

Let $k$ be a positive definite kernel function. Then,

$$\sum_{n=1}^{N} \sum_{n'=1}^{N} c_n c_{n'} k(x_n, x_{n'}) > 0. \tag{6.111}$$

Then,

$$\mathbf{c}^\intercal \mathbf{K} \mathbf{c} \geq 0, \tag{6.112}$$

where

$$\mathbf{c} = \begin{bmatrix} c \\ c' \end{bmatrix}, \mathbf{K} = \begin{bmatrix} k(x,x) & k(x,x') \\ k(x,x') & k(x',x') \end{bmatrix}. \tag{6.113}$$

Then,

$$\det \mathbf{K} \geq 0. \tag{6.114}$$

Therefore,

$$k(x,x')^2 \leq k(x,x)k(x',x'). \tag{6.115}$$

## 6.16

Let

$$J(\mathbf{w}) = f\left(\mathbf{w}^\intercal \boldsymbol{\phi}(\mathbf{x}_1), \cdots, \mathbf{w}^\intercal \boldsymbol{\phi}(\mathbf{x}_N)\right) + g\left(\|\mathbf{w}\|^2\right), \tag{6.116}$$

where $g$ is a monotonically increasing function. Setting the derivative with respect to $\mathbf{w}$ to zero gives

$$\mathbf{0} = \sum_{n=1}^{N} (\nabla_n f) \boldsymbol{\phi}(\mathbf{x}_n) + 2g'\left(\|\mathbf{w}\|^2\right)\mathbf{w}, \tag{6.117}$$

where $\nabla_n$ is the partial derivative with respect to the $n$th argument. Then,

$$\mathbf{w} = -\frac{1}{2g'(\|\mathbf{w}\|^2)} \sum_{n=1}^{N} (\nabla_n f) \boldsymbol{\phi}(\mathbf{x}_n). \tag{6.118}$$

Therefore, $J$ is minimised by $\mathbf{w}$ in the form of a linear combination of $\boldsymbol{\phi}(\mathbf{x}_1), \cdots, \boldsymbol{\phi}(\mathbf{x}_N)$.

## 6.17

Let

$$E = \frac{1}{2} \sum_{n=1}^{N} \int \left(y(\mathbf{x}_n + \boldsymbol{\xi}) - t_n\right)^2 p(\boldsymbol{\xi}) d\boldsymbol{\xi}. \tag{6.119}$$

By the transformation

$$\mathbf{z}_n = \mathbf{x}_n + \boldsymbol{\xi}, \tag{6.120}$$

we have

$$E = \frac{1}{2} \sum_{n=1}^{N} \int \left(y(\mathbf{z}_n) - t_n\right)^2 p(\mathbf{z}_n - \mathbf{x}_n) d\mathbf{z}_n. \tag{6.121}$$

Setting the variation of $E$ with respect to $y$ to zero gives

$$0 = \sum_{n=1}^{N} \int \left(y(\mathbf{z}_n) - t_n\right) p(\mathbf{z}_n - \mathbf{x}_n) \delta y(\mathbf{z}_n) d\mathbf{z}_n. \tag{6.122}$$

Then,

$$\sum_{n=1}^{N} \left(y(\mathbf{z}_n) - t_n\right) p(\mathbf{z}_n - \mathbf{x}_n) = 0, \tag{6.123}$$

so that

$$\sum_{n=1}^{N} y(\mathbf{z}_n) p(\mathbf{z}_n - \mathbf{x}_n) = \sum_{n=1}^{N} t_n p(\mathbf{z}_n - \mathbf{x}_n). \tag{6.124}$$

Therefore,

$$y(\mathbf{x}) = \sum_{n=1}^{N} t_n k(\mathbf{x}, \mathbf{x}_n), \tag{6.125}$$

where

$$k(\mathbf{x}, \mathbf{x}_n) = \frac{p(\mathbf{x} - \mathbf{x}_n)}{\sum_{n'=1}^{N} p(\mathbf{x} - \mathbf{x}_{n'})}. \tag{6.126}$$

## 6.18

Let $x$ be a variable such that

$$p(x, t) = \frac{1}{N} \sum_{n=1}^{N} f(x - x_n, t - t_n), \tag{6.127}$$

where

$$f(x, t) = \mathcal{N}\left(x|0, \sigma^2\right) \mathcal{N}\left(t|0, \sigma^2\right). \tag{6.128}$$

### (a)

By Bayes' theorem,

$$p(t|x) = \frac{p(x, t)}{p(x)}. \tag{6.129}$$

By marginalisation,

$$p(x) = \int p(x, t) dt. \tag{6.130}$$

The right hand side can be written as

$$
\begin{aligned}
&\frac{1}{N}\sum_{n=1}^{N}\mathcal{N}\left(x-x_n|0,\sigma^2\right)\int\mathcal{N}\left(t-t_n|0,\sigma^2\right)dt \\
=&\frac{1}{N}\sum_{n=1}^{N}\mathcal{N}\left(x-x_n|0,\sigma^2\right).
\end{aligned}
\tag{6.131}
$$

Then,

$$
p(t|x)=\frac{\sum_{n=1}^{N}\mathcal{N}\left(x-x_n|0,\sigma^2\right)\mathcal{N}\left(t-t_n|0,\sigma^2\right)}{\sum_{n=1}^{N}\mathcal{N}\left(x-x_n|0,\sigma^2\right)}. \tag{6.132}
$$

Therefore,

$$
p(t|x)=\sum_{n=1}^{N}k(x,x_n)\mathcal{N}\left(t-t_n|0,\sigma^2\right), \tag{6.133}
$$

where

$$
k(x,x_n)=\frac{\mathcal{N}\left(x-x_n|0,\sigma^2\right)}{\sum_{n=1}^{N}\mathcal{N}\left(x-x_n|0,\sigma^2\right)}. \tag{6.134}
$$

**(b)**

We have

$$
\mathrm{E}(t|x)=\int tp(t|x)dt. \tag{6.135}
$$

By (a), the right hand side can be written as

$$
\sum_{n=1}^{N}k(x,x_n)\int t\mathcal{N}\left(t-t_n|0,\sigma^2\right)dt. \tag{6.136}
$$

By the transformation

$$
t'=t-t_n, \tag{6.137}
$$

the integral can be written as

$$
\int(t'+t_n)\mathcal{N}\left(t'|0,\sigma^2\right)dt'=\int t'\mathcal{N}\left(t'|0,\sigma^2\right)dt'+t_n\int\mathcal{N}\left(t'|0,\sigma^2\right)dt'. \tag{6.138}
$$

The right hand side can be written as $t_n$. Therefore,

$$
\mathrm{E}(t|x)=\sum_{n=1}^{N}k(x,x_n)t_n. \tag{6.139}
$$

**(c)**

We have

$$\mathrm{var}(t|x) = \int (t - \mathrm{E}(t|x))^2 \, p(t|x) dt. \tag{6.140}$$

By (a) and (b), the right hand side can be written as

$$\sum_{n=1}^{N} k(x, x_n) \int \left( t - \sum_{n=1}^{N} k(x, x_n) t_n \right)^2 \mathcal{N}\left(t - t_n | 0, \sigma^2\right) dt. \tag{6.141}$$

By the transformation

$$t' = t - t_n, \tag{6.142}$$

the integral can be written as

$$\begin{aligned}
&\int \left( t' + t_n - \sum_{n=1}^{N} k(x, x_n) t_n \right)^2 \mathcal{N}\left(t'|0, \sigma^2\right) dt' \\
&= \int t'^2 \mathcal{N}\left(t'|0, \sigma^2\right) dt' + 2 \left( t_n - \sum_{n=1}^{N} k(x, x_n) t_n \right) \int t' \mathcal{N}\left(t'|0, \sigma^2\right) dt' \\
&\quad + \left( t_n - \sum_{n=1}^{N} k(x, x_n) t_n \right)^2 \int \mathcal{N}\left(t'|0, \sigma^2\right) dt'.
\end{aligned} \tag{6.143}$$

The right hand side can be written as

$$\sigma^2 + \left( t_n - \sum_{n=1}^{N} k(x, x_n) t_n \right)^2. \tag{6.144}$$

Therefore,

$$\mathrm{var}(t|x) = \sum_{n=1}^{N} k(x, x_n) \left( \sigma^2 + \left( t_n - \sum_{n'=1}^{N} k(x, x_{n'}) t_{n'} \right)^2 \right), \tag{6.145}$$

so that

$$\mathrm{var}(t|x) = \sigma^2 + \sum_{n=1}^{N} k(x, x_n) \left( t_n - \sum_{n'=1}^{N} k(x, x_{n'}) t_{n'} \right)^2. \tag{6.146}$$

## 6.19

Let

$$E = \frac{1}{2}\sum_{n=1}^{N}\int \left(y(\mathbf{x}_n - \boldsymbol{\xi}_n) - t_n\right)^2 p(\boldsymbol{\xi}_n) d\boldsymbol{\xi}_n. \tag{6.147}$$

By the transformation

$$\mathbf{z}_n = \mathbf{x}_n - \boldsymbol{\xi}_n, \tag{6.148}$$

we have

$$E = \frac{1}{2}\sum_{n=1}^{N}\int \left(y(\mathbf{z}_n) - t_n\right)^2 p(\mathbf{x}_n - \mathbf{z}_n) d\mathbf{z}_n. \tag{6.149}$$

Setting the variation of $E$ with respect to $y$ to zero gives

$$0 = \sum_{n=1}^{N}\int \left(y(\mathbf{z}_n) - t_n\right) p(\mathbf{x}_n - \mathbf{z}_n)\delta y(\mathbf{z}_n) d\mathbf{z}_n. \tag{6.150}$$

Then,

$$0 = \sum_{n=1}^{N} \left(y(\mathbf{z}_n) - t_n\right) p(\mathbf{x}_n - \mathbf{z}_n), \tag{6.151}$$

so that

$$\sum_{n=1}^{N} y(\mathbf{z}_n) p(\mathbf{x}_n - \mathbf{z}_n) = \sum_{n=1}^{N} t_n p(\mathbf{x}_n - \mathbf{z}_n). \tag{6.152}$$

Therefore,

$$y(\mathbf{x}) = \sum_{n=1}^{N} t_n k(\mathbf{x}, \mathbf{x}_n), \tag{6.153}$$

where

$$k(\mathbf{x}, \mathbf{x}_n) = \frac{p(\mathbf{x}_n - \mathbf{x})}{\sum_{n'=1}^{N} p(\mathbf{x}_{n'} - \mathbf{x})}. \tag{6.154}$$

## 6.20

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(\mathbf{t}|\mathbf{y}) &= \mathcal{N}\left(\mathbf{t}|\mathbf{y}, \beta^{-1}\mathbf{I}\right), \\ p(\mathbf{y}) &= \mathcal{N}(\mathbf{y}|\mathbf{0}, \mathbf{K}), \end{aligned} \tag{6.155}$$

where

$$K_{nn'} = k(\mathbf{x}_n, \mathbf{x}_{n'}). \tag{6.156}$$

By Bayes' theorem,

$$p(t_{N+1}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}'), \tag{6.157}$$

where

$$\mathbf{t}' = \begin{bmatrix} \mathbf{t} \\ t_{N+1} \end{bmatrix}. \tag{6.158}$$

By marginalisation,

$$p(\mathbf{t}) = \int p(\mathbf{t}|\mathbf{y})p(\mathbf{y})d\mathbf{y}. \tag{6.159}$$

The logarighm of the integrand of the right hand side except the terms independent of $\mathbf{y}$ can be written as

$$-\frac{\beta}{2}\|\mathbf{t} - \mathbf{y}\|^2 - \frac{1}{2}\mathbf{y}^\intercal\mathbf{K}^{-1}\mathbf{y} = -\frac{1}{2}\mathbf{v}^\intercal\mathbf{M}\mathbf{v}, \tag{6.160}$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{y} \\ \mathbf{t} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \beta\mathbf{I} + \mathbf{K}^{-1} & -\beta\mathbf{I} \\ -\beta\mathbf{I} & \beta\mathbf{I} \end{bmatrix}. \tag{6.161}$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix} \mathbf{K} & \mathbf{K} \\ \mathbf{K} & \mathbf{C} \end{bmatrix}, \tag{6.162}$$

where

$$\mathbf{C} = \beta^{-1}\mathbf{I} + \mathbf{K}. \tag{6.163}$$

Then,

$$p(\mathbf{t}) = \mathcal{N}\left(\mathbf{t}|\mathbf{0}, \mathbf{C}\right). \tag{6.164}$$

Then,

$$p(\mathbf{t}') = \mathcal{N}\left(\mathbf{t}'|\mathbf{0}, \mathbf{C}'\right), \tag{6.165}$$

where

$$\mathbf{C}' = \begin{bmatrix} \mathbf{C} & \mathbf{k} \\ \mathbf{k}^\intercal & c \end{bmatrix}, \tag{6.166}$$

and

$$\begin{aligned} k_n &= k(\mathbf{x}_n, \mathbf{x}_{N+1}), \\ c &= \beta^{-1} + k(\mathbf{x}_{N+1}, \mathbf{x}_{N+1}). \end{aligned} \tag{6.167}$$

By 2.24,

$$\begin{bmatrix} c & \mathbf{k}^\intercal \\ \mathbf{k} & \mathbf{C} \end{bmatrix}^{-1} = \begin{bmatrix} s^{-1} & -s^{-1}\mathbf{k}^\intercal\mathbf{C}^{-1} \\ -s^{-1}\mathbf{C}^{-1}\mathbf{k} & \mathbf{C}^{-1} + s^{-1}\mathbf{C}^{-1}\mathbf{k}\mathbf{k}^\intercal\mathbf{C}^{-1} \end{bmatrix}, \tag{6.168}$$

where

$$s = c - \mathbf{k}^\intercal\mathbf{C}^{-1}\mathbf{k}. \tag{6.169}$$

Therefore,

$$p(t_{N+1}|\mathbf{t}) = \mathcal{N}\left(t_{N+1}|m, s\right), \tag{6.170}$$

where

$$m = \mathbf{k}^\intercal\mathbf{C}^{-1}\mathbf{t}. \tag{6.171}$$

## 6.21

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal\boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{0}, \mathbf{I}). \end{aligned} \tag{6.172}$$

Then,

$$\begin{aligned} p(\mathbf{t}|\mathbf{y}) &= \mathcal{N}\left(\mathbf{t}|\mathbf{y}, \beta^{-1}\mathbf{I}\right), \\ p(\mathbf{y}) &= \mathcal{N}\left(\mathbf{y}|\mathbf{0}, \mathbf{K}\right), \end{aligned} \tag{6.173}$$

where

$$\mathbf{K} = \boldsymbol{\Phi}\boldsymbol{\Phi}^\intercal. \tag{6.174}$$

By 6.20,

$$p(t_{N+1}|\mathbf{t}) = \mathcal{N}\left(t_{N+1}|m, s\right), \tag{6.175}$$

where

$$\begin{aligned} m &= \mathbf{k}^\intercal\mathbf{C}^{-1}\mathbf{t}, \\ s &= c - \mathbf{k}^\intercal\mathbf{C}^{-1}\mathbf{k}, \end{aligned} \tag{6.176}$$

and

$$\begin{aligned} \mathbf{k} &= \boldsymbol{\Phi}\boldsymbol{\phi}_{N+1}, \\ \mathbf{C} &= \beta^{-1}\mathbf{I} + \mathbf{K}, \\ c &= \beta^{-1} + \boldsymbol{\phi}_{N+1}^\intercal\boldsymbol{\phi}_{N+1}. \end{aligned} \tag{6.177}$$

By 2.26,

$$\mathbf{C}^{-1} = \beta\mathbf{I} - \beta^2\boldsymbol{\Phi}\mathbf{S}\boldsymbol{\Phi}^\intercal, \tag{6.178}$$

where

$$\mathbf{S} = \left(\mathbf{I} + \beta \mathbf{\Phi}^\intercal \mathbf{\Phi}\right)^{-1}. \tag{6.179}$$

Then,

$$\begin{aligned} m &= \boldsymbol{\phi}_{N+1}^\intercal \mathbf{\Phi}^\intercal \left(\beta \mathbf{I} - \beta^2 \mathbf{\Phi} \mathbf{S} \mathbf{\Phi}^\intercal\right) \mathbf{t}, \\ s &= \beta^{-1} + \boldsymbol{\phi}_{N+1}^\intercal \boldsymbol{\phi}_{N+1} - \boldsymbol{\phi}_{N+1}^\intercal \mathbf{\Phi}^\intercal \left(\beta \mathbf{I} - \beta^2 \mathbf{\Phi} \mathbf{S} \mathbf{\Phi}^\intercal\right) \mathbf{\Phi} \boldsymbol{\phi}_{N+1}. \end{aligned} \tag{6.180}$$

We have

$$\beta \mathbf{\Phi}^\intercal \mathbf{\Phi} = \mathbf{S}^{-1} - \mathbf{I}. \tag{6.181}$$

Then,

$$\begin{aligned} m &= \beta \boldsymbol{\phi}_{N+1}^\intercal \left(\mathbf{I} - \left(\mathbf{S}^{-1} - \mathbf{I}\right) \mathbf{S}\right) \mathbf{\Phi}^\intercal \mathbf{t}, \\ s &= \beta^{-1} + \boldsymbol{\phi}_{N+1}^\intercal \left(\mathbf{I} - \left(\mathbf{S}^{-1} - \mathbf{I}\right) + \left(\mathbf{S}^{-1} - \mathbf{I}\right) \mathbf{S} \left(\mathbf{S}^{-1} - \mathbf{I}\right)\right) \boldsymbol{\phi}_{N+1}. \end{aligned} \tag{6.182}$$

Therefore,

$$\begin{aligned} m &= \beta \boldsymbol{\phi}_{N+1}^\intercal \mathbf{S} \mathbf{\Phi}^\intercal \mathbf{t}, \\ s &= \beta^{-1} + \boldsymbol{\phi}_{N+1}^\intercal \mathbf{S} \boldsymbol{\phi}_{N+1}. \end{aligned} \tag{6.183}$$

## 6.22

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(\mathbf{t}|\mathbf{y}) &= \mathcal{N}\left(\mathbf{t}|\mathbf{y}, \beta^{-1}\mathbf{I}\right), \\ p(\mathbf{y}) &= \mathcal{N}\left(\mathbf{y}|\mathbf{0}, \mathbf{K}\right), \end{aligned} \tag{6.184}$$

where

$$K_{nn'} = k(\mathbf{x}_n, \mathbf{x}_{n'}). \tag{6.185}$$

### (a)

By Bayes' theorem,

$$p(t_{N+1}, \cdots, t_{N+L}|\mathbf{t}) p(\mathbf{t}) = p(\mathbf{t}'), \tag{6.186}$$

where

$$\mathbf{t}' = \begin{bmatrix} \mathbf{t} \\ t_{N+1} \\ \vdots \\ t_{N+L} \end{bmatrix}. \tag{6.187}$$

By 6.20,

$$p(\mathbf{t}) = \mathcal{N}\left(\mathbf{t}|\mathbf{0}, \mathbf{C}\right), \tag{6.188}$$

where

$$\mathbf{C} = \beta^{-1}\mathbf{I} + \mathbf{K}. \tag{6.189}$$

Let $\mathbf{K}'$ and $\mathbf{\Gamma}$ be matrices where

$$\begin{aligned} K'_{nl} &= k(\mathbf{x}_n, \mathbf{x}_{N+l}), \\ \Gamma_{ll'} &= \beta^{-1}I_{ll'} + k(\mathbf{x}_{N+l}, \mathbf{x}_{N+l'}). \end{aligned} \tag{6.190}$$

Then,

$$p(\mathbf{t}') = \mathcal{N}\left(\mathbf{t}'|\mathbf{0}, \mathbf{C}'\right), \tag{6.191}$$

where

$$\mathbf{C}' = \begin{bmatrix} \mathbf{C} & \mathbf{K}' \\ \mathbf{K}'^{\intercal} & \mathbf{\Gamma} \end{bmatrix}. \tag{6.192}$$

By 2.24,

$$\begin{bmatrix} \mathbf{\Gamma} & \mathbf{K}'^{\intercal} \\ \mathbf{K}' & \mathbf{C} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{S}^{-1} & -\mathbf{S}^{-1}\mathbf{K}'^{\intercal}\mathbf{C}^{-1} \\ -\mathbf{C}^{-1}\mathbf{K}'\mathbf{S}^{-1} & \mathbf{C}^{-1} + \mathbf{C}^{-1}\mathbf{K}'\mathbf{S}^{-1}\mathbf{K}'^{\intercal}\mathbf{C}^{-1} \end{bmatrix}, \tag{6.193}$$

where

$$\mathbf{S} = \mathbf{\Gamma} - \mathbf{K}'^{\intercal}\mathbf{C}^{-1}\mathbf{K}'. \tag{6.194}$$

Therefore,

$$p(t_{N+1}, \cdots, t_{N+L}|\mathbf{t}) = \mathcal{N}\left(t_{N+1}, \cdots, t_{N+L}|\mathbf{m}, \mathbf{S}\right), \tag{6.195}$$

where

$$\mathbf{m} = \mathbf{K}'^{\intercal}\mathbf{C}^{-1}\mathbf{t}. \tag{6.196}$$

**(b)**

By (a),

$$p(t_{N+1}|\mathbf{t}) = \mathcal{N}\left(t_{N+1}|m, s\right), \tag{6.197}$$

where

$$\begin{aligned} m &= \mathbf{k}^{\intercal}\mathbf{C}^{-1}\mathbf{t}, \\ s &= c - \mathbf{k}^{\intercal}\mathbf{C}^{-1}\mathbf{k}, \end{aligned} \tag{6.198}$$

and

$$\begin{aligned} k_n &= k(\mathbf{x}_n, \mathbf{x}_{N+1}), \\ c &= \beta^{-1} + k(\mathbf{x}_{N+1}, \mathbf{x}_{N+1}). \end{aligned} \tag{6.199}$$

## 6.23

Let $\mathbf{t}_1, \cdots, \mathbf{t}_N$ be conditionally independent variables such that

$$\begin{aligned} p(\mathbf{t}_n|\mathbf{y}_n) &= \mathcal{N}\left(\mathbf{t}_n|\mathbf{y}_n, \beta^{-1}\mathbf{I}\right), \\ p(\mathbf{Y}) &= \mathcal{N}\left(\mathbf{Y}|\mathbf{O}, \mathbf{K}\right), \end{aligned} \tag{6.200}$$

where

$$K_{nn'} = k(\mathbf{x}_n, \mathbf{x}_{n'}). \tag{6.201}$$

By Bayes' theorem,

$$p(\mathbf{t}_{N+1}|\mathbf{T})p(\mathbf{T}) = p(\mathbf{T}'), \tag{6.202}$$

where

$$\mathbf{T}' = \begin{bmatrix} \mathbf{T} \\ \mathbf{t}_{N+1}^{\intercal} \end{bmatrix}. \tag{6.203}$$

By marginalisation,

$$p(\mathbf{T}) = \int p(\mathbf{T}|\mathbf{Y})p(\mathbf{Y})d\mathbf{Y}. \tag{6.204}$$

The logarithm of the integrannd of the right hand side except the terms independent of $\mathbf{T}$ and $\mathbf{Y}$ can be written as

$$-\frac{\beta}{2}\sum_{n=1}^{N} \|\mathbf{t}_n - \mathbf{y}_n\|^2 - \frac{1}{2}\operatorname{tr}\left(\mathbf{Y}^{\intercal}\mathbf{K}^{-1}\mathbf{Y}\right) = -\frac{1}{2}\operatorname{tr}\mathbf{M}, \tag{6.205}$$

where

$$\mathbf{M} = \begin{bmatrix} \mathbf{Y} \\ \mathbf{T} \end{bmatrix}^{\intercal} \begin{bmatrix} \beta\mathbf{I} + \mathbf{K}^{-1} & -\beta\mathbf{I} \\ -\beta\mathbf{I} & \beta\mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{Y} \\ \mathbf{T} \end{bmatrix}. \tag{6.206}$$

By 2.24,

$$\begin{bmatrix} \beta\mathbf{I} + \mathbf{K}^{-1} & -\beta\mathbf{I} \\ -\beta\mathbf{I} & \beta\mathbf{I} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{K} & \mathbf{K} \\ \mathbf{K} & \mathbf{C} \end{bmatrix}, \tag{6.207}$$

where

$$\mathbf{C} = \beta^{-1}\mathbf{I} + \mathbf{K}. \tag{6.208}$$

Then,

$$p(\mathbf{T}) = \mathcal{N}\left(\mathbf{T}|\mathbf{O}, \mathbf{C}\right). \tag{6.209}$$

Then,

$$p(\mathbf{T}') = \mathcal{N}\left(\mathbf{T}'|\mathbf{O}, \mathbf{C}'\right), \tag{6.210}$$

where

$$\mathbf{C}' = \begin{bmatrix} \mathbf{C} & \mathbf{k} \\ \mathbf{k}^\intercal & c \end{bmatrix}, \tag{6.211}$$

and

$$\begin{aligned} k_n &= k(\mathbf{x}_n, \mathbf{x}_{N+1}), \\ c &= \beta^{-1} + k(\mathbf{x}_{N+1}, \mathbf{x}_{N+1}). \end{aligned} \tag{6.212}$$

By 2.24,

$$\begin{bmatrix} c & \mathbf{k}^\intercal \\ \mathbf{k} & \mathbf{C} \end{bmatrix}^{-1} = \begin{bmatrix} s^{-1} & -s^{-1}\mathbf{k}^\intercal\mathbf{C}^{-1} \\ -s^{-1}\mathbf{C}^{-1}\mathbf{k} & \mathbf{C}^{-1} + s^{-1}\mathbf{C}^{-1}\mathbf{k}\mathbf{k}^\intercal\mathbf{C}^{-1} \end{bmatrix}, \tag{6.213}$$

where

$$s = c - \mathbf{k}^\intercal\mathbf{C}^{-1}\mathbf{k}. \tag{6.214}$$

Therefore,

$$p(\mathbf{t}_{N+1}|\mathbf{T}) = \mathcal{N}\left(\mathbf{t}_{N+1}|\mathbf{m}, s\mathbf{I}\right), \tag{6.215}$$

where

$$\mathbf{m} = \mathbf{T}^\intercal\mathbf{C}^{-1}\mathbf{k}. \tag{6.216}$$

## 6.24

### (a)

Let $\mathbf{M}$ be a $D \times D$ diagonal matrix such that $0 < M_{dd} < 1$. For any vector $\mathbf{v}$ in $D$ dimensions,

$$\mathbf{v}^\intercal\mathbf{M}\mathbf{v} = \sum_{d=1}^{D}\sum_{d'=1}^{D} v_d M_{dd'} v_{d'}. \tag{6.217}$$

The right hand side can be written as

$$\sum_{d=1}^{D} M_{dd} v_d^2 > 0. \tag{6.218}$$

Therefore, $\mathbf{M}$ is positive definite.

### (b)

Let $\mathbf{M}_1$ and $\mathbf{M}_2$ be $D \times D$ positive definite matrices. Then, for any vector $\mathbf{v}$ in $D$ dimensions,

$$\begin{aligned}\mathbf{v}^\intercal \mathbf{M}_1 \mathbf{v} &> 0, \\ \mathbf{v}^\intercal \mathbf{M}_2 \mathbf{v} &> 0.\end{aligned} \tag{6.219}$$

Then,

$$\mathbf{v}^\intercal (\mathbf{M}_1 + \mathbf{M}_2) \mathbf{v} > 0. \tag{6.220}$$

Therefore, $\mathbf{M}_1 + \mathbf{M}_2$ is positive definite.

## 6.25

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned}p(t_n|a_n) &= \sigma(a_n)^{t_n} \left(1 - \sigma(a_n)\right)^{1-t_n}, \\ p(\mathbf{a}) &= \mathcal{N}\left(\mathbf{a}|\mathbf{0}, \mathbf{C}\right),\end{aligned} \tag{6.221}$$

where

$$\begin{aligned}\sigma(a) &= \frac{1}{1 + \exp(-a)}, \\ C_{nn'} &= k(\mathbf{x}_n, \mathbf{x}_{n'}) + \nu I_{nn'}.\end{aligned} \tag{6.222}$$

### (a)

By Bayes' theorem,

$$p(\mathbf{a}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{a})p(\mathbf{a}). \tag{6.223}$$

The logarithm of the right hand side except the terms independent of $\mathbf{t}$ and $\mathbf{a}$ can be written as $-E(\mathbf{a})$ where

$$E(\mathbf{a}) = -\sum_{n=1}^{N} \left(t_n \ln \sigma(a_n) + (1 - t_n) \ln\left(1 - \sigma(a_n)\right)\right) + \frac{1}{2}\mathbf{a}^\intercal \mathbf{C}^{-1}\mathbf{a}. \tag{6.224}$$

We have the Taylor series

$$\begin{aligned}E(\mathbf{a}) =& E(\mathbf{a}_{\mathrm{MAP}}) + \frac{1}{2}(\mathbf{a} - \mathbf{a}_{\mathrm{MAP}})^\intercal \mathbf{H}_{\mathrm{MAP}}(\mathbf{a} - \mathbf{a}_{\mathrm{MAP}}) \\ &+ O\left(\|\mathbf{a} - \mathbf{a}_{\mathrm{MAP}}\|^3\right),\end{aligned} \tag{6.225}$$

where $\mathbf{a}_{\mathrm{MAP}}$ is a stationary point of $E$ and

$$\mathbf{H}_{\mathrm{MAP}} = \nabla\nabla E(\mathbf{a}_{\mathrm{MAP}}). \tag{6.226}$$

Therefore,

$$p(\mathbf{a}|\mathbf{t}) \simeq \mathcal{N}\left(\mathbf{a}|\mathbf{a}_{\mathrm{MAP}}, \mathbf{H}_{\mathrm{MAP}}^{-1}\right). \tag{6.227}$$

**(b)**

We have the Taylor series

$$\nabla E(\mathbf{a}_{\mathrm{MAP}}) = \nabla E(\mathbf{a}) + \nabla\nabla E(\mathbf{a})(\mathbf{a}_{\mathrm{MAP}} - \mathbf{a}) + O\left(\|\mathbf{a}_{\mathrm{MAP}} - \mathbf{a}\|^2\right). \tag{6.228}$$

We have

$$(\nabla E(\mathbf{a}))_n = -\left(t_n\left(1 - \sigma(a_n)\right) - (1 - t_n)\sigma(a_n)\right) + \left(\mathbf{C}^{-1}\mathbf{a}\right)_n, \tag{6.229}$$

so that

$$\nabla E(\mathbf{a}) = \mathbf{C}^{-1}\mathbf{a} + \boldsymbol{\sigma} - \mathbf{t}, \tag{6.230}$$

where

$$\sigma_n = \sigma(a_n). \tag{6.231}$$

Then,

$$\nabla\nabla E(\mathbf{a}) = \mathbf{C}^{-1} + \mathbf{W}, \tag{6.232}$$

where

$$W_{nn'} = \sigma(a_n)\left(1 - \sigma(a_n)\right) I_{nn'}. \tag{6.233}$$

Then,

$$\mathbf{0} \simeq \mathbf{C}^{-1}\mathbf{a} + \boldsymbol{\sigma} - \mathbf{t} + \left(\mathbf{C}^{-1} + \mathbf{W}\right)(\mathbf{a}_{\mathrm{MAP}} - \mathbf{a}). \tag{6.234}$$

Then,

$$\left(\mathbf{C}^{-1} + \mathbf{W}\right)\mathbf{a}_{\mathrm{MAP}} \simeq \mathbf{t} - \boldsymbol{\sigma} + \mathbf{W}\mathbf{a}. \tag{6.235}$$

Therefore,

$$\mathbf{a}_{\mathrm{MAP}} \simeq \left(\mathbf{C}^{-1} + \mathbf{W}\right)^{-1}\left(\mathbf{t} - \boldsymbol{\sigma} + \mathbf{W}\mathbf{a}\right). \tag{6.236}$$

**(c)**

By (b),

$$\mathbf{a}_{\mathrm{MAP}} = \left(\mathbf{C}^{-1} + \mathbf{W}_{\mathrm{MAP}}\right)^{-1} \left(\mathbf{t} - \boldsymbol{\sigma}_{\mathrm{MAP}} + \mathbf{W}_{\mathrm{MAP}}\mathbf{a}_{\mathrm{MAP}}\right), \tag{6.237}$$

where

$$\begin{aligned} W_{\mathrm{MAP}nn'} &= \sigma(a_{\mathrm{MAP}n}) \left(1 - \sigma(a_{\mathrm{MAP}n})\right) I_{nn'}, \\ \sigma_{\mathrm{MAP}n} &= \sigma(a_{\mathrm{MAP}n}). \end{aligned} \tag{6.238}$$

Then,

$$\left(\mathbf{C}^{-1} + \mathbf{W}_{\mathrm{MAP}}\right) \mathbf{a}_{\mathrm{MAP}} = \mathbf{t} - \boldsymbol{\sigma}_{\mathrm{MAP}} + \mathbf{W}_{\mathrm{MAP}}\mathbf{a}_{\mathrm{MAP}}. \tag{6.239}$$

Therefore,

$$\mathbf{a}_{\mathrm{MAP}} = \mathbf{C}(\mathbf{t} - \boldsymbol{\sigma}_{\mathrm{MAP}}). \tag{6.240}$$

## 6.26

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|a_n) &= \sigma(a_n)^{t_n} \left(1 - \sigma(a_n)\right)^{1-t_n}, \\ p(\mathbf{a}) &= \mathcal{N}\left(\mathbf{a}|\mathbf{0}, \mathbf{C}\right), \end{aligned} \tag{6.241}$$

where

$$\begin{aligned} \sigma(a) &= \frac{1}{1 + \exp(-a)}, \\ C_{nn'} &= k(\mathbf{x}_n, \mathbf{x}_{n'}) + \nu I_{nn'}. \end{aligned} \tag{6.242}$$

By marginalisation,

$$p(a_{N+1}|\mathbf{t}) = \int p(a_{N+1}|\mathbf{a})p(\mathbf{a}|\mathbf{t})d\mathbf{a}. \tag{6.243}$$

We have

$$p(\mathbf{a}') = \mathcal{N}\left(\mathbf{a}'|\mathbf{0}, \mathbf{C}'\right), \tag{6.244}$$

where

$$\mathbf{a}' = \begin{bmatrix} \mathbf{a} \\ a_{N+1} \end{bmatrix}, \mathbf{C}' = \begin{bmatrix} \mathbf{C} & \mathbf{k} \\ \mathbf{k}^{\intercal} & c \end{bmatrix}, \tag{6.245}$$

and

$$\begin{aligned} k_n &= k(\mathbf{x}_n, \mathbf{x}_{N+1}), \\ c &= k(\mathbf{x}_{N+1}, \mathbf{x}_{N+1}) + \nu. \end{aligned} \tag{6.246}$$

By 2.24,

$$\begin{bmatrix} c & \mathbf{k}^\intercal \\ \mathbf{k} & \mathbf{C} \end{bmatrix}^{-1} = \begin{bmatrix} s^{-1} & -s^{-1}\mathbf{k}^\intercal \mathbf{C}^{-1} \\ -s^{-1}\mathbf{C}^{-1}\mathbf{k} & \mathbf{C}^{-1} + s^{-1}\mathbf{C}^{-1}\mathbf{k}\mathbf{k}^\intercal \mathbf{C}^{-1} \end{bmatrix}, \tag{6.247}$$

where

$$s = c - \mathbf{k}^\intercal \mathbf{C}^{-1}\mathbf{k}. \tag{6.248}$$

Then,

$$p(a_{N+1}|\mathbf{a}) = \mathcal{N}\left(a_{N+1}|m, s\right), \tag{6.249}$$

where

$$m = \mathbf{k}^\intercal \mathbf{C}^{-1}\mathbf{a}. \tag{6.250}$$

By 6.25(a),

$$p(\mathbf{a}|\mathbf{t}) \simeq \mathcal{N}\left(\mathbf{a}|\mathbf{a}_{\mathrm{MAP}}, \mathbf{H}_{\mathrm{MAP}}^{-1}\right), \tag{6.251}$$

where $\mathbf{a}_{\mathrm{MAP}}$ is a stationary point of $E$ and

$$\mathbf{H}_{\mathrm{MAP}} = \nabla\nabla E(\mathbf{a}_{\mathrm{MAP}}), \tag{6.252}$$

where

$$E(\mathbf{a}) = -\sum_{n=1}^{N}\left(t_n \ln \sigma(a_n) + (1-t_n)\ln\left(1-\sigma(a_n)\right)\right) + \frac{1}{2}\mathbf{a}^\intercal \mathbf{C}^{-1}\mathbf{a}. \tag{6.253}$$

Then, the logarithm of the integrand except the terms independent of $\mathbf{t}$ and $\mathbf{a}$ can be approximated as

$$\begin{aligned} &-\frac{1}{2s}\left(a_{N+1} - \mathbf{k}^\intercal \mathbf{C}^{-1}\mathbf{a}\right)^2 - \frac{1}{2}\left(\mathbf{a} - \mathbf{a}_{\mathrm{MAP}}\right)^\intercal \mathbf{H}_{\mathrm{MAP}}\left(\mathbf{a} - \mathbf{a}_{\mathrm{MAP}}\right) \\ =& -\frac{1}{2}\mathbf{a}'^\intercal \mathbf{M}\mathbf{a}' + \mathbf{a}'^\intercal \mathbf{v} - \frac{1}{2}\mathbf{a}_{\mathrm{MAP}}^\intercal \mathbf{H}_{\mathrm{MAP}}\mathbf{a}_{\mathrm{MAP}}. \end{aligned} \tag{6.254}$$

where

$$\begin{aligned} \mathbf{M} &= \begin{bmatrix} \mathbf{H}_{\mathrm{MAP}} + s^{-1}\mathbf{C}^{-1}\mathbf{k}\mathbf{k}^\intercal \mathbf{C}^{-1} & -s^{-1}\mathbf{C}^{-1}\mathbf{k} \\ -s^{-1}\mathbf{k}^\intercal \mathbf{C}^{-1} & s^{-1} \end{bmatrix}, \\ \mathbf{v} &= \begin{bmatrix} \mathbf{H}_{\mathrm{MAP}}\mathbf{a}_{\mathrm{MAP}} \\ 0 \end{bmatrix}. \end{aligned} \tag{6.255}$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix} \mathbf{H}_{\mathrm{MAP}}^{-1} & \mathbf{H}_{\mathrm{MAP}}^{-1}\mathbf{C}^{-1}\mathbf{k} \\ \mathbf{k}^\intercal \mathbf{C}^{-1}\mathbf{H}_{\mathrm{MAP}}^{-1} & \sigma^2 \end{bmatrix}, \tag{6.256}$$

so that

$$\mathbf{M}^{-1}\mathbf{v} = \begin{bmatrix} \mathbf{a}_{\text{MAP}} \\ \mu \end{bmatrix}, \tag{6.257}$$

where

$$\begin{aligned} \mu &= \mathbf{k}^\intercal \mathbf{C}^{-1}\mathbf{a}_{\text{MAP}}, \\ \sigma^2 &= s + \mathbf{k}^\intercal \mathbf{C}^{-1}\mathbf{H}_{\text{MAP}}^{-1}\mathbf{C}^{-1}\mathbf{k}. \end{aligned} \tag{6.258}$$

Then,

$$p(a_{N+1}|\mathbf{t}) \simeq \mathcal{N}\left(a_{N+1}|\mu, \sigma^2\right). \tag{6.259}$$

By 6.25(c),

$$\mathbf{a}_{\text{MAP}} = \mathbf{C}(\mathbf{t} - \boldsymbol{\sigma}_{\text{MAP}}), \tag{6.260}$$

where

$$\sigma_{\text{MAP}n} = \sigma(a_{\text{MAP}n}). \tag{6.261}$$

Therefore,

$$\begin{aligned} \mu &= \mathbf{k}^\intercal (\mathbf{t} - \boldsymbol{\sigma}_{\text{MAP}}), \\ \sigma^2 &= c - \mathbf{k}^\intercal \mathbf{C}^{-1}\left(\mathbf{C} - \mathbf{H}_{\text{MAP}}^{-1}\right)\mathbf{C}^{-1}\mathbf{k}. \end{aligned} \tag{6.262}$$

## 6.27

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|a_n) &= \sigma(a_n)^{t_n}\left(1 - \sigma(a_n)\right)^{1-t_n}, \\ p(\mathbf{a}|\boldsymbol{\theta}) &= \mathcal{N}\left(\mathbf{a}|\mathbf{0}, \mathbf{C}\right), \end{aligned} \tag{6.263}$$

where

$$\sigma(a) = \frac{1}{1 + \exp(-a)} \tag{6.264}$$

and $\mathbf{C}$ is dependent on $\boldsymbol{\theta}$ in $M$ dimensions.

### (a)

By marginalisation,

$$p(\mathbf{t}|\boldsymbol{\theta}) = \int p(\mathbf{t}|\mathbf{a})p(\mathbf{a}|\boldsymbol{\theta})d\mathbf{a}. \tag{6.265}$$

The logarithm of the integrand of the right hand side except the terms independent of $\mathbf{a}$ can be written as $-E(\mathbf{a})$ where

$$E(\mathbf{a}) = -\sum_{n=1}^{N}\left(t_n \ln\sigma(a_n) - (1 - t_n)\ln\left(1 - \sigma(a_n)\right)\right) + \frac{1}{2}\mathbf{a}^\intercal \mathbf{C}^{-1}\mathbf{a}. \tag{6.266}$$

We have the Taylor series

$$\begin{aligned} E(\mathbf{a}) =& E(\mathbf{a}_{\mathrm{MAP}}) + \frac{1}{2} (\mathbf{a} - \mathbf{a}_{\mathrm{MAP}})^{\intercal} \mathbf{H}_{\mathrm{MAP}} (\mathbf{a} - \mathbf{a}_{\mathrm{MAP}}) \\ & + O\left(\|\mathbf{a} - \mathbf{a}_{\mathrm{MAP}}\|^3\right), \end{aligned} \tag{6.267}$$

where $\mathbf{a}_{\mathrm{MAP}}$ is a statrionary point of $E$ and

$$\mathbf{H}_{\mathrm{MAP}} = \nabla\nabla E(\mathbf{a}_{\mathrm{MAP}}). \tag{6.268}$$

Then,

$$p(\mathbf{t}|\boldsymbol{\theta}) \simeq \exp\left(-E(\mathbf{a}_{\mathrm{MAP}})\right) I, \tag{6.269}$$

where

$$I = \int \exp\left(-\frac{1}{2} (\mathbf{a} - \mathbf{a}_{\mathrm{MAP}})^{\intercal} \mathbf{H}_{\mathrm{MAP}} (\mathbf{a} - \mathbf{a}_{\mathrm{MAP}})\right) d\mathbf{a}. \tag{6.270}$$

We have

$$I = (2\pi)^{\frac{N}{2}} (\det \mathbf{H}_{\mathrm{MAP}}^{-1})^{\frac{1}{2}}. \tag{6.271}$$

Therefore,

$$\ln p(\mathbf{t}|\boldsymbol{\theta}) \simeq -E(\mathbf{a}_{\mathrm{MAP}}) + \frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln(\det \mathbf{H}_{\mathrm{MAP}}). \tag{6.272}$$

**(b)**

By (a),

$$(\nabla E(\mathbf{a}))_n = -\left(t_n (1 - \sigma(a_n)) + (1 - t_n)\sigma(a_n)\right) + \left(\mathbf{C}^{-1}\mathbf{a}\right)_n. \tag{6.273}$$

Then,

$$\nabla E(\mathbf{a}) = \boldsymbol{\sigma} - \mathbf{t} + \mathbf{C}^{-1}\mathbf{a}, \tag{6.274}$$

where

$$\sigma_n = \sigma(a_n). \tag{6.275}$$

Then,

$$\nabla\nabla E(\mathbf{a}) = \mathbf{W} + \mathbf{C}^{-1}, \tag{6.276}$$

where

$$W_{nn'} = \sigma(a_n) (1 - \sigma(a_n)) I_{nn'}. \tag{6.277}$$

Therefore,

$$\mathbf{H}_{\mathrm{MAP}} = \mathbf{W}_{\mathrm{MAP}} + \mathbf{C}^{-1}, \tag{6.278}$$

where

$$W_{\mathrm{MAP}nn'} = \sigma(a_{\mathrm{MAP}n}) (1 - \sigma(a_{\mathrm{MAP}n})) I_{nn'}. \tag{6.279}$$

**(c)**

By (a),

$$\frac{\partial \ln p(\mathbf{t}|\boldsymbol{\theta})}{\partial \theta_m} \simeq -\left(\frac{\partial E(\mathbf{a}_{\mathrm{MAP}})}{\partial \theta_m} + \frac{1}{2}\frac{\partial \ln(\det \mathbf{H}_{\mathrm{MAP}})}{\partial \theta_m}\right). \tag{6.280}$$

We have

$$\frac{\partial E(\mathbf{a}_{\mathrm{MAP}})}{\partial \theta_m} = \operatorname{tr}\left(\frac{\partial E(\mathbf{a}_{\mathrm{MAP}})}{\partial \mathbf{C}}\frac{\partial \mathbf{C}}{\partial \theta_m}\right). \tag{6.281}$$

We have

$$\frac{\partial E(\mathbf{a}_{\mathrm{MAP}})}{\partial \mathbf{C}} = \nabla E(\mathbf{a}_{\mathrm{MAP}}) \cdot \frac{\partial \mathbf{a}_{\mathrm{MAP}}}{\partial \mathbf{C}} - \frac{1}{2}\mathbf{a}_{\mathrm{MAP}}\left(\mathbf{C}^{-1}\right)^2 \mathbf{a}_{\mathrm{MAP}}^{\intercal}, \tag{6.282}$$

where $(\cdot)$ denotes contraction. We have

$$\nabla E(\mathbf{a}_{\mathrm{MAP}}) = \mathbf{0}. \tag{6.283}$$

Then,

$$\frac{\partial E(\mathbf{a}_{\mathrm{MAP}})}{\partial \theta_m} = -\frac{1}{2}\operatorname{tr}\left(\mathbf{a}_{\mathrm{MAP}}\mathbf{C}^{-1}\frac{\partial \mathbf{C}}{\partial \theta_m}\mathbf{C}^{-1}\mathbf{a}_{\mathrm{MAP}}^{\intercal}\right). \tag{6.284}$$

We have

$$\frac{\partial \ln(\det \mathbf{H}_{\mathrm{MAP}})}{\partial \theta_m} = \operatorname{tr}\left(\frac{\partial \ln(\det \mathbf{H}_{\mathrm{MAP}})}{\partial \mathbf{H}_{\mathrm{MAP}}}\frac{\partial \mathbf{H}_{\mathrm{MAP}}}{\partial \theta_m}\right). \tag{6.285}$$

By 3.21 and (b),

$$\frac{\partial \ln(\det \mathbf{H}_{\mathrm{MAP}})}{\partial \mathbf{H}_{\mathrm{MAP}}}\frac{\partial \mathbf{H}_{\mathrm{MAP}}}{\partial \theta_m} = \mathbf{H}_{\mathrm{MAP}}^{-1}\left(\frac{\partial \mathbf{W}_{\mathrm{MAP}}}{\partial \theta_m} + \frac{\partial \mathbf{C}^{-1}}{\partial \theta_m}\right). \tag{6.286}$$

We have

$$\frac{\partial \mathbf{W}_{\mathrm{MAP}}}{\partial \theta_m} = \mathbf{T}_{\mathrm{MAP}} \cdot \frac{\partial \mathbf{a}_{\mathrm{MAP}}}{\partial \theta_m}, \tag{6.287}$$

where

$$T_{\mathrm{MAP}nn'n''} = \sigma(a_{\mathrm{MAP}n})\left(1 - \sigma(a_{\mathrm{MAP}n})\right)\left(1 - 2\sigma(a_{\mathrm{MAP}n})\right) I_{nn'} I_{n'n''}. \tag{6.288}$$

Then,

$$\frac{\partial \ln(\det \mathbf{H}_{\mathrm{MAP}})}{\partial \theta_m} = \operatorname{tr}\left(\mathbf{H}_{\mathrm{MAP}}^{-1}\left(\mathbf{T}_{\mathrm{MAP}} \cdot \frac{\partial \mathbf{a}_{\mathrm{MAP}}}{\partial \theta_m} + \frac{\partial \mathbf{C}^{-1}}{\partial \theta_m}\right)\right). \tag{6.289}$$

Therefore,

$$\frac{\partial \ln p(\mathbf{t}|\boldsymbol{\theta})}{\partial \theta_m} \simeq \frac{1}{2} \operatorname{tr} \mathbf{M}, \tag{6.290}$$

where

$$\mathbf{M} = \mathbf{a}_{\mathrm{MAP}} \mathbf{C}^{-1} \frac{\partial \mathbf{C}}{\partial \theta_m} \mathbf{C}^{-1} \mathbf{a}_{\mathrm{MAP}}^{\intercal} - \mathbf{H}_{\mathrm{MAP}}^{-1} \left( \mathbf{T}_{\mathrm{MAP}} \cdot \frac{\partial \mathbf{a}_{\mathrm{MAP}}}{\partial \theta_m} + \frac{\partial \mathbf{C}^{-1}}{\partial \theta_m} \right). \tag{6.291}$$

**(d)**

By 6.25(c),

$$\mathbf{a}_{\mathrm{MAP}} = \mathbf{C}(\mathbf{t} - \boldsymbol{\sigma}_{\mathrm{MAP}}). \tag{6.292}$$

Then,

$$\frac{\partial \mathbf{a}_{\mathrm{MAP}}}{\partial \theta_m} = \frac{\partial \mathbf{C}}{\partial \theta_m}(\mathbf{t} - \boldsymbol{\sigma}_{\mathrm{MAP}}) - \mathbf{C} \frac{\partial \boldsymbol{\sigma}_{\mathrm{MAP}}}{\partial \theta_m}. \tag{6.293}$$

We have

$$\frac{\partial \boldsymbol{\sigma}_{\mathrm{MAP}}}{\partial \theta_m} = \mathbf{W}_{\mathrm{MAP}} \frac{\partial \mathbf{a}_{\mathrm{MAP}}}{\partial \theta_m}. \tag{6.294}$$

Therefore,

$$\frac{\partial \mathbf{a}_{\mathrm{MAP}}}{\partial \theta_m} = (\mathbf{I} + \mathbf{C}\mathbf{W}_{\mathrm{MAP}})^{-1} \frac{\partial \mathbf{C}}{\partial \theta_m}(\mathbf{t} - \boldsymbol{\sigma}_{\mathrm{MAP}}). \tag{6.295}$$

# 7 Sparse Kernel Machines

## 7.1 (Incomplete)

## 7.2

Let $t_1, \cdots, t_N$ be variables such that

$$t_n \left(\mathbf{w}^\intercal \boldsymbol{\phi}_n + b\right) \geq \gamma, \tag{7.1}$$

for some positive $\gamma$. In order to minimise $\|\mathbf{w}\|$, let

$$L = \frac{1}{2}\|\mathbf{w}\|^2 - \sum_{n=1}^{N} a_n \left(t_n \left(\mathbf{w}^\intercal \boldsymbol{\phi}_n + b\right) - \gamma\right), \tag{7.2}$$

where $a_n > 0$. Setting the derivatives of $L$ with respect to $\mathbf{w}$ and $b$ to zero gives

$$\begin{aligned} \mathbf{0} &= \mathbf{w} - \sum_{n=1}^{N} a_n t_n \boldsymbol{\phi}_n, \\ 0 &= -\sum_{n=1}^{N} a_n t_n. \end{aligned} \tag{7.3}$$

Therefore,

$$\begin{aligned} \mathbf{w}^* &= \sum_{n=1}^{N} a_n t_n \boldsymbol{\phi}_n, \\ 0 &= \sum_{n=1}^{N} a_n t_n, \end{aligned} \tag{7.4}$$

where $\mathbf{w}^*$ is an optimal solution to $\mathbf{w}$, which is independent of $\gamma$.

## 7.3 (Incomplete)

## 7.4

Let $t_1, \cdots, t_N$ be conditionally independent variables from $\{-1, 1\}$ such that

$$p(t_n|\mathbf{w}) = \sigma(y_n)^{\frac{1+t_n}{2}} \left(1 - \sigma(y_n)\right)^{\frac{1-t_n}{2}}, \tag{7.5}$$

where

$$\begin{aligned} y_n &= \mathbf{w}^\intercal \boldsymbol{\phi}_n + b, \\ \sigma(a) &= \frac{1}{1+\exp(-a)}. \end{aligned} \tag{7.6}$$

**(a)**

Let

$$\rho = \max_{\mathbf{w},b}\left(\min_n d_n\right), \tag{7.7}$$

where

$$d_n = \|\boldsymbol{\phi}_n - \boldsymbol{\phi}_{n\perp}\|, \tag{7.8}$$

and $\boldsymbol{\phi}_{n\perp}$ is a vector such that

$$\begin{aligned} \mathbf{w}^\intercal \boldsymbol{\phi}_{n\perp} + b &= 0, \\ \boldsymbol{\phi}_n - \boldsymbol{\phi}_{n\perp} &= c\mathbf{w}, \end{aligned} \tag{7.9}$$

for some $c$. Then,

$$y_n = c\|\mathbf{w}\|^2, \tag{7.10}$$

so that

$$d_n = \frac{|y_n|}{\|\mathbf{w}\|}. \tag{7.11}$$

Focusing on the correctly classified points gives

$$t_n y_n = |y_n|. \tag{7.12}$$

Then,

$$d_n = \frac{t_n y_n}{\|\mathbf{w}\|}, \tag{7.13}$$

so that

$$\rho = \max_{\mathbf{w},b}\left(\|\mathbf{w}\|^{-1}\min_n t_n y_n\right). \tag{7.14}$$

Rescaling $\mathbf{w}$ and $b$ gives

$$\rho = \max \|\mathbf{w}\|^{-1}, \tag{7.15}$$

under the constraint

$$\min_n t_n y_n = 1. \tag{7.16}$$

Therefore,

$$\rho = \|\mathbf{w}^*\|^{-1}, \tag{7.17}$$

where $\mathbf{w}^*$ is an optimal solution to $\mathbf{w}$.

**(b)**

Let

$$L(\mathbf{w}, b, \mathbf{a}) = \frac{1}{2}\|\mathbf{w}\|^2 - \sum_{n=1}^{N} a_n \left(t_n y_n - 1\right), \tag{7.18}$$

where $a_n > 0$. Setting the derivatives of $L$ with respect to $\mathbf{w}$ and $b$ to zero gives

$$\begin{aligned} \mathbf{0} &= \mathbf{w} - \sum_{n=1}^{N} a_n t_n \boldsymbol{\phi}_n, \\ 0 &= -\sum_{n=1}^{N} a_n t_n. \end{aligned} \tag{7.19}$$

We have

$$L(\mathbf{w}^*, b^*, \mathbf{a}) = \frac{1}{2}\|\mathbf{w}^*\|^2 - \sum_{n=1}^{N} a_n \left(t_n \left(\mathbf{w}^{*\intercal}\boldsymbol{\phi}_n + b^*\right) - 1\right), \tag{7.20}$$

where $b^*$ is an optimal solutions to $b$. The right hand side can be written as

$$\begin{aligned} &\frac{1}{2}\|\mathbf{w}^*\|^2 - \mathbf{w}^{*\intercal}\sum_{n=1}^{N} a_n t_n \boldsymbol{\phi}_n - b^* \sum_{n=1}^{N} a_n t_n + \sum_{n=1}^{N} a_n \\ =&\frac{1}{2}\left\|\sum_{n=1}^{N} a_n t_n \boldsymbol{\phi}_n\right\|^2 - \left(\sum_{n=1}^{N} a_n t_n \boldsymbol{\phi}_n\right)^\intercal \sum_{n=1}^{N} a_n t_n \boldsymbol{\phi}_n + \sum_{n=1}^{N} a_n. \end{aligned} \tag{7.21}$$

The right hand side can be written as

$$\tilde{L}(\mathbf{a}) = \sum_{n=1}^{N} a_n - \frac{1}{2}\left\|\sum_{n=1}^{N} a_n t_n \boldsymbol{\phi}_n\right\|^2. \tag{7.22}$$

Setting the derivative of $\tilde{L}$ with respect to $a_n$ to zero gives

$$0 = 1 - t_n \boldsymbol{\phi}_n^\intercal \sum_{n'=1}^{N} a_{n'} t_{n'} \boldsymbol{\phi}_{n'}. \tag{7.23}$$

Then,

$$\sum_{n=1}^{N} a_n^* = \left\|\sum_{n=1}^{N} a_n^* t_n \boldsymbol{\phi}_n\right\|^2, \tag{7.24}$$

where $\mathbf{a}^*$ is an optimal solution to $\mathbf{a}$. By (a), the right hand side can be written as

$$\|\mathbf{w}^*\|^2 = \rho^{-2}. \tag{7.25}$$

Therefore,

$$\rho = \left(\sum_{n=1}^{N} a_n^*\right)^{-\frac{1}{2}}. \tag{7.26}$$

**(c)**

By (b),

$$\tilde{L}(\mathbf{a}^*) = \sum_{n=1}^{N} a_n^* - \frac{1}{2}\left\|\sum_{n=1}^{N} a_n^* t_n \boldsymbol{\phi}_n\right\|^2. \tag{7.27}$$

By (b), the right hand side can be written as

$$\frac{1}{2}\sum_{n=1}^{N} a_n^* = \frac{1}{2}\rho^{-2}. \tag{7.28}$$

Therefore,

$$\rho = \left(2\tilde{L}(\mathbf{a}^*)\right)^{-\frac{1}{2}}. \tag{7.29}$$

## 7.5

Refer to 7.4.

## 7.6

Let $t_1, \cdots, t_N$ be conditionally independent variables from $\{-1, 1\}$ such that

$$p(t_n|\mathbf{w}) = \sigma(y_n)^{\frac{1+t_n}{2}} \left(1 - \sigma(y_n)\right)^{\frac{1-t_n}{2}}, \tag{7.30}$$

where

$$\begin{aligned} \sigma(a) &= \frac{1}{1 + \exp(-a)}, \\ y_n &= \mathbf{w}^\intercal \boldsymbol{\phi}_n + b. \end{aligned} \tag{7.31}$$

We have

$$1 - \sigma(a) = \sigma(-a). \tag{7.32}$$

Then,

$$p(t_n|\mathbf{w}) = \sigma(y_n t_n). \tag{7.33}$$

Then,

$$p(\mathbf{t}|\mathbf{w}) = \prod_{n=1}^{N} \sigma(y_n t_n), \tag{7.34}$$

so that

$$-\ln p(\mathbf{t}|\mathbf{w}) = -\sum_{n=1}^{N} \ln \sigma(y_n t_n). \tag{7.35}$$

Therefore,

$$-\ln p(\mathbf{t}|\mathbf{y}) = \sum_{n=1}^{N} \ln\left(1 + \exp(-y_n t_n)\right). \tag{7.36}$$

## 7.7

Let $t_1, \cdots, t_N$ be variables. Let

$$\begin{aligned} \xi_n &= \begin{cases} 0, & t_n - y_n < \epsilon, \\ t_n - y_n - \epsilon, & \text{otherwise}, \end{cases} \\ \hat{\xi}_n &= \begin{cases} 0, & y_n - t_n < \epsilon, \\ y_n - t_n - \epsilon, & \text{otherwise}, \end{cases} \end{aligned} \tag{7.37}$$

where

$$\begin{aligned} y_n &= \mathbf{w}^\intercal \boldsymbol{\phi}_n + b, \\ \boldsymbol{\phi}_n &= \boldsymbol{\phi}(\mathbf{x}_n), \end{aligned} \tag{7.38}$$

and $\epsilon > 0$. Let

$$E = \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{n=1}^{N} E_\epsilon(y_n - t_n), \tag{7.39}$$

where

$$E_\epsilon(a) = \begin{cases} 0, & |a| < \epsilon, \\ |a| - \epsilon, & \text{otherwise}. \end{cases} \tag{7.40}$$

**(a)**

We have

$$E = \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{n=1}^{N}\left(\xi_n + \hat{\xi}_n\right). \tag{7.41}$$

In order to minimise $E$, let

$$\begin{aligned} L(\mathbf{w}, b, \boldsymbol{\xi}, \hat{\boldsymbol{\xi}}, \boldsymbol{\mu}, \hat{\boldsymbol{\mu}}, \mathbf{a}, \hat{\mathbf{a}}) =& E - \sum_{n=1}^{N}\mu_n\xi_n - \sum_{n=1}^{N}\hat{\mu}_n\hat{\xi}_n \\ & - \sum_{n=1}^{N} a_n(\xi_n - t_n + y_n + \epsilon) \\ & - \sum_{n=1}^{N} \hat{a}_n(\hat{\xi}_n - y_n + t_n + \epsilon), \end{aligned} \tag{7.42}$$

where $\mu_n \geq 0$, $\hat{\mu}_n \geq 0$, $a_n \geq 0$ and $\hat{a}_n \geq 0$, . Setting the derivatives of $L$ with respect to $\mathbf{w}$, $b$, $\xi_n$ and $\hat{\xi}_n$ to zero gives

$$\begin{aligned} \mathbf{0} &= \mathbf{w} - \sum_{n=1}^{N} a_n\boldsymbol{\phi}_n + \sum_{n=1}^{N}\hat{a}_n\boldsymbol{\phi}_n, \\ 0 &= -\sum_{n=1}^{N} a_n + \sum_{n=1}^{N}\hat{a}_n, \\ 0 &= C - \mu_n - a_n, \\ 0 &= C - \hat{\mu}_n - \hat{a}_n. \end{aligned} \tag{7.43}$$

We have

$$\begin{aligned} L(\mathbf{w}^*, b^*, \boldsymbol{\xi}^*, \hat{\boldsymbol{\xi}}^*, \boldsymbol{\mu}, \hat{\boldsymbol{\mu}}, \mathbf{a}, \hat{\mathbf{a}}) =& \frac{1}{2}\|\mathbf{w}^*\|^2 + C\sum_{n=1}^{N}\left(\xi_n^* + \hat{\xi}_n^*\right) \\ & - \sum_{n=1}^{N}\mu_n\xi_n^* - \sum_{n=1}^{N}\hat{\mu}_n\hat{\xi}_n^* \\ & - \sum_{n=1}^{N} a_n\left(\xi_n^* - t_n + \mathbf{w}^{*\mathsf{T}}\boldsymbol{\phi}_n + b^* + \epsilon\right) \\ & - \sum_{n=1}^{N}\hat{a}_n\left(\hat{\xi}_n^* - \mathbf{w}^{*\mathsf{T}}\boldsymbol{\phi}_n - b^* + t_n + \epsilon\right), \end{aligned} \tag{7.44}$$

where $\mathbf{w}^*$, $b^*$, $\boldsymbol{\xi}^*$ and $\hat{\boldsymbol{\xi}}^*$ are optimal solutions to $\mathbf{w}$, $b$, $\boldsymbol{\xi}$ and $\hat{\boldsymbol{\xi}}$. The right hand side can be written as

$$\begin{aligned}
&\frac{1}{2}\|\mathbf{w}^*\|^2 - \sum_{n=1}^{N} a_n \left(-t_n + \mathbf{w}^{*\intercal}\boldsymbol{\phi}_n + b^* + \epsilon\right) \\
&- \sum_{n=1}^{N} \hat{a}_n \left(-\mathbf{w}^{*\intercal}\boldsymbol{\phi}_n - b^* + t_n + \epsilon\right) \\
=&\frac{1}{2}\left\|\sum_{n=1}^{N} \left(a_n - \hat{a}_n\right)\boldsymbol{\phi}_n\right\|^2 + \sum_{n=1}^{N} \left(a_n - \hat{a}_n\right) t_n \\
&- \left(\sum_{n=1}^{N} \left(a_n - \hat{a}_n\right)\boldsymbol{\phi}_n\right)^\intercal \sum_{n=1}^{N} \left(a_n - \hat{a}_n\right)\boldsymbol{\phi}_n - \epsilon \sum_{n=1}^{N} \left(a_n + \hat{a}_n\right).
\end{aligned} \tag{7.45}$$

The right hand side can be written as

$$\begin{aligned}
\tilde{L}(\mathbf{a}, \hat{\mathbf{a}}) =& -\frac{1}{2}\sum_{n=1}^{N}\sum_{n'=1}^{N} (a_n - \hat{a}_n)(a_{n'} - \hat{a}_{n'}) k(\mathbf{x}_n, \mathbf{x}_{n'}) \\
&+ \sum_{n=1}^{N} \left(a_n - \hat{a}_n\right) t_n - \epsilon \sum_{n=1}^{N} \left(a_n + \hat{a}_n\right),
\end{aligned} \tag{7.46}$$

where

$$k(\mathbf{x}_n, \mathbf{x}_{n'}) = \boldsymbol{\phi}_n^\intercal \boldsymbol{\phi}_{n'}. \tag{7.47}$$

Therefore, the minimisation of $E$ is equivalent to the maximisation of $\tilde{L}$.

### (b)

By (a), the corresponding KKT conditions are given by

$$\begin{aligned}
(C - a_n)\xi_n &= 0, \\
(C - \hat{a}_n)\hat{\xi}_n &= 0, \\
a_n(\xi_n - t_n + y_n + \epsilon) &= 0, \\
\hat{a}_n(\hat{\xi}_n - y_n + t_n + \epsilon) &= 0.
\end{aligned} \tag{7.48}$$

Therefore,

$$C = \begin{cases} a_n, & \xi_n > 0, \\ \hat{a}_n, & \hat{\xi}_n < 0. \end{cases} \tag{7.49}$$

## 7.8

Refer to 7.7.

## 7.9

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal \boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{0}, \mathbf{A}^{-1}\right). \end{aligned} \tag{7.50}$$

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{w})p(\mathbf{w}). \tag{7.51}$$

The logarithm of the right hand side except the terms independent of $\mathbf{t}$ and $\mathbf{w}$ can be written as

$$-\frac{\beta}{2}\|\mathbf{t} - \boldsymbol{\Phi}\mathbf{w}\|^2 - \frac{1}{2}\mathbf{w}^\intercal \mathbf{A}\mathbf{w} = -\frac{1}{2}\mathbf{v}^\intercal \mathbf{M}\mathbf{v}, \tag{7.52}$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ \mathbf{t} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \beta\boldsymbol{\Phi}^\intercal\boldsymbol{\Phi} + \mathbf{A} & -\beta\boldsymbol{\Phi}^\intercal \\ -\beta\boldsymbol{\Phi} & \beta\mathbf{I} \end{bmatrix}. \tag{7.53}$$

Therefore,

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}\left(\mathbf{w}|\mathbf{m}, \mathbf{S}\right), \tag{7.54}$$

where

$$\begin{aligned} \mathbf{m} &= \beta\mathbf{S}\boldsymbol{\Phi}^\intercal\mathbf{t}, \\ \mathbf{S} &= \left(\beta\boldsymbol{\Phi}^\intercal\boldsymbol{\Phi} + \mathbf{A}\right)^{-1}. \end{aligned} \tag{7.55}$$

## 7.10

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal \boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{0}, \mathbf{A}^{-1}\right). \end{aligned} \tag{7.56}$$

By marginalisation,

$$p(\mathbf{t}) = \int p(\mathbf{t}|\mathbf{w})p(\mathbf{w})d\mathbf{w}. \tag{7.57}$$

The logarithm of the integrand of the right hand side except the terms independent of $\mathbf{t}$ and $\mathbf{w}$ can be written as

$$-\frac{\beta}{2}\|\mathbf{t}-\mathbf{\Phi w}\|^2-\frac{1}{2}\mathbf{w}^\intercal\mathbf{A}\mathbf{w}=-\frac{1}{2}\mathbf{v}^\intercal\mathbf{M}\mathbf{v}, \tag{7.58}$$

where

$$\mathbf{v}=\begin{bmatrix}\mathbf{w}\\ \mathbf{t}\end{bmatrix}, \mathbf{M}=\begin{bmatrix}\mathbf{S}^{-1} & -\beta\mathbf{\Phi}^\intercal\\ -\beta\mathbf{\Phi} & \beta\mathbf{I}\end{bmatrix}, \tag{7.59}$$

and

$$\mathbf{S}=\left(\beta\mathbf{\Phi}^\intercal\mathbf{\Phi}+\mathbf{A}\right)^{-1}. \tag{7.60}$$

The right hand side can be written as

$$-\frac{1}{2}\mathbf{u}^\intercal\mathbf{S}^{-1}\mathbf{u}-\frac{1}{2}\mathbf{t}^\intercal\mathbf{N}\mathbf{t}, \tag{7.61}$$

where

$$\begin{aligned}\mathbf{u}&=\mathbf{w}-\beta\mathbf{S}\mathbf{\Phi}^\intercal\mathbf{t},\\ \mathbf{N}&=\beta\mathbf{I}-\beta^2\mathbf{\Phi}\mathbf{S}\mathbf{\Phi}^\intercal.\end{aligned} \tag{7.62}$$

By 2.26,

$$\mathbf{N}^{-1}=\beta^{-1}\mathbf{I}+\mathbf{\Phi}\mathbf{A}^{-1}\mathbf{\Phi}^\intercal. \tag{7.63}$$

Therefore,

$$p(\mathbf{t})=\mathcal{N}\left(\mathbf{t}|\mathbf{0},\beta^{-1}\mathbf{I}+\mathbf{\Phi}\mathbf{A}^{-1}\mathbf{\Phi}^\intercal\right). \tag{7.64}$$

## 7.11

Let $t_1,\cdots,t_N$ be conditionally independent variables such that

$$\begin{aligned}p(t_n|\mathbf{w})&=\mathcal{N}\left(t_n|\mathbf{w}^\intercal\boldsymbol{\phi}_n,\beta^{-1}\right),\\ p(\mathbf{w})&=\mathcal{N}\left(\mathbf{w}|\mathbf{0},\mathbf{A}^{-1}\right).\end{aligned} \tag{7.65}$$

By marginalisation,

$$p(\mathbf{t})=\int p(\mathbf{t}|\mathbf{w})p(\mathbf{w})d\mathbf{w}. \tag{7.66}$$

The logarithm of the integrand of the right hand side except the terms independent of $\mathbf{t}$ and $\mathbf{w}$ can be written as

$$-\frac{\beta}{2}\|\mathbf{t}-\mathbf{\Phi w}\|^2-\frac{1}{2}\mathbf{w}^\intercal\mathbf{A}\mathbf{w}=-\frac{1}{2}\mathbf{v}^\intercal\mathbf{M}\mathbf{v}, \tag{7.67}$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ \mathbf{t} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \beta\boldsymbol{\Phi}^\intercal\boldsymbol{\Phi} + \mathbf{A} & -\beta\boldsymbol{\Phi}^\intercal \\ -\beta\boldsymbol{\Phi} & \beta\mathbf{I} \end{bmatrix}. \tag{7.68}$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix} \mathbf{A}^{-1} & \mathbf{A}^{-1}\boldsymbol{\Phi}^\intercal \\ \boldsymbol{\Phi}\mathbf{A}^{-1} & \beta^{-1}\mathbf{I} + \boldsymbol{\Phi}\mathbf{A}^{-1}\boldsymbol{\Phi}^\intercal \end{bmatrix}. \tag{7.69}$$

Therefore,

$$p(\mathbf{t}) = \mathcal{N}\left(\mathbf{t}|\mathbf{0}, \beta^{-1}\mathbf{I} + \boldsymbol{\Phi}\mathbf{A}^{-1}\boldsymbol{\Phi}^\intercal\right). \tag{7.70}$$

## 7.12

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal\boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{0}, \mathbf{A}^{-1}\right), \end{aligned} \tag{7.71}$$

where

$$\mathbf{A} = \mathrm{diag}(\alpha_m). \tag{7.72}$$

By marginalisation,

$$p(\mathbf{t}) = \int p(\mathbf{t}|\mathbf{w})p(\mathbf{w})d\mathbf{w}. \tag{7.73}$$

The logarithm of the integrand of the right hand side can be written as

$$\begin{aligned} &-\frac{N}{2}\ln(2\pi) - \frac{1}{2}\ln\left(\det\left(\beta^{-1}\mathbf{I}\right)\right)\frac{\beta}{2}\|\mathbf{t} - \boldsymbol{\Phi}\mathbf{w}\|^2 \\ &-\frac{M}{2}\ln(2\pi) - \frac{1}{2}\ln\left(\det\mathbf{A}^{-1}\right) - \frac{1}{2}\mathbf{w}^\intercal\mathbf{A}\mathbf{w} \\ =&-\frac{N+M}{2}\ln(2\pi) + \frac{N}{2}\ln\beta + \frac{1}{2}\ln(\det\mathbf{A}) - E(\mathbf{w}), \end{aligned} \tag{7.74}$$

where

$$E(\mathbf{w}) = \frac{\beta}{2}\|\mathbf{t} - \boldsymbol{\Phi}\mathbf{w}\|^2 + \frac{1}{2}\mathbf{w}^\intercal\mathbf{A}\mathbf{w}. \tag{7.75}$$

Then,

$$p(\mathbf{t}) = (2\pi)^{-\frac{N+M}{2}}\beta^{\frac{N}{2}}(\det\mathbf{A})^{\frac{1}{2}}\int\exp(-E(\mathbf{w}))\,d\mathbf{w}. \tag{7.76}$$

We have

$$\begin{aligned} \nabla E(\mathbf{w}) &= \mathbf{S}^{-1}(\mathbf{w} - \mathbf{m}), \\ \nabla\nabla E(\mathbf{w}) &= \mathbf{S}^{-1}, \end{aligned} \tag{7.77}$$

where

$$
\begin{aligned}
\mathbf{m} &= \beta \mathbf{S} \boldsymbol{\Phi}^{\intercal} \mathbf{t}, \\
\mathbf{S} &= \left(\beta \boldsymbol{\Phi}^{\intercal} \boldsymbol{\Phi} + \mathbf{A}\right)^{-1}.
\end{aligned} \tag{7.78}
$$

Then,

$$
E(\mathbf{w}) = E(\mathbf{m}) + \frac{1}{2}(\mathbf{w} - \mathbf{m})^{\intercal} \mathbf{S}^{-1} (\mathbf{w} - \mathbf{m}), \tag{7.79}
$$

so that

$$
\int \exp\left(-E(\mathbf{w})\right) d\mathbf{w} = (2\pi)^{\frac{M}{2}} \left(\det \mathbf{S}\right)^{\frac{1}{2}} \exp\left(-E(\mathbf{m})\right). \tag{7.80}
$$

Then,

$$
\ln p(\mathbf{t}) = -\frac{N}{2} \ln(2\pi) + \frac{N}{2} \ln \beta + \frac{1}{2} \ln(\det \mathbf{A}) + \frac{1}{2} \ln(\det \mathbf{S}) - E(\mathbf{m}). \tag{7.81}
$$

By 3.21, setting the derivatives of $\ln p(\mathbf{t})$ with respect to $\alpha_m$ and $\beta$ to zero gives

$$
\begin{aligned}
0 &= \frac{1}{2\alpha_m} + \frac{1}{2} \operatorname{tr}\left(\mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \alpha_m}\right) - \frac{\partial E(\mathbf{m})}{\partial \alpha}, \\
0 &= \frac{N}{2\beta} + \frac{1}{2} \operatorname{tr}\left(\mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \beta}\right) - \frac{\partial E(\mathbf{m})}{\partial \beta}.
\end{aligned} \tag{7.82}
$$

We have

$$
\begin{aligned}
\mathbf{O} &= \frac{\partial \mathbf{S}^{-1} \mathbf{S}}{\partial \alpha_m}, \\
\mathbf{O} &= \frac{\partial \mathbf{S}^{-1} \mathbf{S}}{\partial \beta}.
\end{aligned} \tag{7.83}
$$

The right hand sides can be written as

$$
\begin{aligned}
\mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \alpha_m} + \frac{\partial \mathbf{S}^{-1}}{\partial \alpha_m} \mathbf{S} &= \mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \alpha_m} + \mathbf{e}_m \mathbf{e}_m^{\intercal} \mathbf{S}, \\
\mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \beta} + \frac{\partial \mathbf{S}^{-1}}{\partial \beta} \mathbf{S} &= \mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \beta} + \boldsymbol{\Phi}^{\intercal} \boldsymbol{\Phi} \mathbf{S},
\end{aligned} \tag{7.84}
$$

where $\mathbf{e}_m$ is the $m$-th vector of the standard basis in $M$ dimensions. Then,

$$
\begin{aligned}
\mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \alpha_m} &= -\mathbf{e}_m \mathbf{e}_m^{\intercal} \mathbf{S}, \\
\mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \beta} &= -\boldsymbol{\Phi}^{\intercal} \boldsymbol{\Phi} \mathbf{S}.
\end{aligned} \tag{7.85}
$$

We have

$$
\begin{aligned}
\frac{\partial E(\mathbf{m})}{\partial \alpha} &= \nabla E(\mathbf{m})\frac{\partial \mathbf{m}}{\partial \alpha} + \frac{m_m^2}{2}, \\
\frac{\partial E(\mathbf{m})}{\partial \beta} &= \nabla E(\mathbf{m})\frac{\partial \mathbf{m}}{\partial \beta} + \frac{\|\mathbf{t}-\mathbf{\Phi}\mathbf{m}\|^2}{2}.
\end{aligned} \tag{7.86}
$$

Then,

$$
\begin{aligned}
0 &= \frac{1}{2\alpha_m} - \frac{1-\gamma_m}{2\alpha_m} - \frac{m_m^2}{2}, \\
0 &= \frac{N}{2\beta} - \frac{\sum_{m=1}^{M}\gamma_m}{2\beta} - \frac{\|\mathbf{t}-\mathbf{\Phi}\mathbf{m}\|^2}{2},
\end{aligned} \tag{7.87}
$$

where

$$
\gamma_m = 1 - S_{mm}\alpha_m. \tag{7.88}
$$

Therefore, the maximum likelihood solutions to $\alpha_m$ and $\beta$ satisfy

$$
\begin{aligned}
(\alpha_m)_{\mathrm{ML}} &= \frac{\gamma_m}{m_m^2}, \\
\beta_{\mathrm{ML}} &= \frac{N - \sum_{m=1}^{M}\gamma_m}{\|\mathbf{t}-\mathbf{\Phi}\mathbf{m}\|^2}.
\end{aligned} \tag{7.89}
$$

## 7.13

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$
\begin{aligned}
p(t_n|\mathbf{w},\beta) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal\boldsymbol{\phi}_n, \beta^{-1}\right), \\
p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{0}, \mathbf{A}^{-1}\right), \\
p(\beta) &= \mathrm{Gam}(\beta|a_0, b_0),
\end{aligned} \tag{7.90}
$$

where

$$
\mathbf{A} = \mathrm{diag}(\alpha_m). \tag{7.91}
$$

By Bayes' theorem,

$$
p(\mathbf{t},\beta) = p(\mathbf{t}|\beta)p(\beta). \tag{7.92}
$$

By marginalisation,

$$
p(\mathbf{t}|\beta) = \int p(\mathbf{t}|\mathbf{w},\beta)p(\mathbf{w})d\mathbf{w}. \tag{7.93}
$$

The logarithm of the integrand of the right hand side can be written as

$$\begin{aligned}
&-\frac{N}{2}\ln(2\pi)-\frac{N}{2}\ln\beta^{-1}-\frac{\beta}{2}\|\mathbf{t}-\mathbf{\Phi}\mathbf{w}\|^2\\
&-\frac{M}{2}\ln(2\pi)-\frac{1}{2}\ln\left(\det\mathbf{A}^{-1}\right)-\frac{1}{2}\mathbf{w}^\intercal\mathbf{A}\mathbf{w}\\
=&-\frac{N+M}{2}\ln(2\pi)+\frac{N}{2}\ln\beta+\frac{1}{2}\ln(\det\mathbf{A})-E(\mathbf{w}),
\end{aligned} \tag{7.94}$$

where

$$E(\mathbf{w})=\frac{\beta}{2}\|\mathbf{t}-\mathbf{\Phi}\mathbf{w}\|^2+\frac{1}{2}\mathbf{w}^\intercal\mathbf{A}\mathbf{w}. \tag{7.95}$$

We have

$$\begin{aligned}
\nabla E(\mathbf{w})&=\mathbf{S}^{-1}(\mathbf{w}-\mathbf{m}),\\
\nabla\nabla E(\mathbf{w})&=\mathbf{S}^{-1},
\end{aligned} \tag{7.96}$$

where

$$\begin{aligned}
\mathbf{S}&=\left(\beta\mathbf{\Phi}^\intercal\mathbf{\Phi}+\mathbf{A}\right)^{-1},\\
\mathbf{m}&=\beta\mathbf{S}\mathbf{\Phi}^\intercal\mathbf{t}.
\end{aligned} \tag{7.97}$$

Then,

$$E(\mathbf{w})=E(\boldsymbol{\mu})+\frac{1}{2}(\mathbf{w}-\mathbf{m})^\intercal\mathbf{S}^{-1}(\mathbf{w}-\mathbf{m}). \tag{7.98}$$

Then,

$$\int\exp\left(-E(\mathbf{w})\right)d\mathbf{w}=(2\pi)^{\frac{M}{2}}(\det\mathbf{S})^{\frac{1}{2}}\exp\left(-E(\mathbf{m})\right). \tag{7.99}$$

Then,

$$\begin{aligned}
\ln p(\mathbf{t}|\beta)=&-\frac{N}{2}\ln(2\pi)+\frac{N}{2}\ln\beta+\frac{1}{2}\ln(\det\mathbf{A})+\frac{1}{2}\ln(\det\mathbf{S})\\
&-E(\mathbf{m}),
\end{aligned} \tag{7.100}$$

so that

$$\begin{aligned}
\ln p(\mathbf{t},\beta)=&-\frac{N}{2}\ln(2\pi)+\frac{N}{2}\ln\beta+\frac{1}{2}\ln(\det\mathbf{A})+\frac{1}{2}\ln(\det\mathbf{S})\\
&-E(\mathbf{m})+(a_0-1)\ln\beta-b_0\beta,
\end{aligned} \tag{7.101}$$

By 3.21, setting the derivatives of $\ln p(\mathbf{t},\beta)$ with respect to $\alpha_m$ and $\beta$ to zero gives

$$\begin{aligned}
0&=\frac{1}{2\alpha_m}+\frac{1}{2}\operatorname{tr}\left(\mathbf{\Sigma}^{-1}\frac{\partial\mathbf{S}}{\partial\alpha_m}\right)-\frac{\partial E(\mathbf{m})}{\partial\alpha},\\
0&=\frac{N}{2\beta}+\frac{1}{2}\operatorname{tr}\left(\mathbf{S}^{-1}\frac{\partial\mathbf{S}}{\partial\beta}\right)-\frac{\partial E(\mathbf{m})}{\partial\beta}+\frac{a_0-1}{\beta}-b_0.
\end{aligned} \tag{7.102}$$

We have

$$\begin{aligned}\mathbf{O} &= \frac{\partial \mathbf{S}^{-1}\mathbf{S}}{\partial \alpha_m},\\ \mathbf{O} &= \frac{\partial \mathbf{S}^{-1}\mathbf{S}}{\partial \beta}.\end{aligned} \tag{7.103}$$

The right hand side can be written as

$$\begin{aligned}\mathbf{S}^{-1}\frac{\partial \mathbf{S}}{\partial \alpha_m} + \frac{\partial \mathbf{S}^{-1}}{\partial \alpha_m}\mathbf{S} &= \mathbf{S}^{-1}\frac{\partial \mathbf{S}}{\partial \alpha_m} + \mathbf{e}_m\mathbf{e}_m^\intercal\mathbf{S},\\ \mathbf{S}^{-1}\frac{\partial \mathbf{S}}{\partial \beta} + \frac{\partial \mathbf{S}^{-1}}{\partial \beta}\mathbf{S} &= \mathbf{S}^{-1}\frac{\partial \mathbf{S}}{\partial \beta} + \mathbf{\Phi}^\intercal\mathbf{\Phi}\mathbf{S},\end{aligned} \tag{7.104}$$

where $\mathbf{e}_m$ is the $m$-th vector of the standard basis in $M$ dimensions. Then,

$$\begin{aligned}\mathbf{S}^{-1}\frac{\partial \mathbf{S}}{\partial \alpha_m} &= -\mathbf{e}_m\mathbf{e}_m^\intercal\mathbf{S},\\ \mathbf{S}^{-1}\frac{\partial \mathbf{S}}{\partial \beta} &= -\mathbf{\Phi}^\intercal\mathbf{\Phi}\mathbf{S}.\end{aligned} \tag{7.105}$$

We have

$$\begin{aligned}\frac{\partial E(\mathbf{m})}{\partial \alpha} &= \nabla E(\mathbf{m})\frac{\partial \mathbf{m}}{\partial \alpha} + \frac{m_m^2}{2},\\ \frac{\partial E(\mathbf{m})}{\partial \beta} &= \nabla E(\mathbf{m})\frac{\partial \mathbf{m}}{\partial \alpha} + \frac{\|\mathbf{t}-\mathbf{\Phi}\mathbf{m}\|^2}{2}.\end{aligned} \tag{7.106}$$

Then,

$$\begin{aligned}0 &= \frac{1}{2\alpha_m} - \frac{1-\gamma_m}{2\alpha_m} - \frac{m_m^2}{2},\\ 0 &= \frac{N}{2\beta} - \frac{\sum_{m=1}^M \gamma_m}{2\beta} - \frac{\|\mathbf{t}-\mathbf{\Phi}\mathbf{m}\|^2}{2} + \frac{a_0-1}{\beta} - b_0.\end{aligned} \tag{7.107}$$

where

$$\gamma_m = 1 - S_{mm}\alpha_m. \tag{7.108}$$

Therefore, the maximum likelihood solutions to $\alpha_m$ and $\beta$ satisfy

$$\begin{aligned}(\alpha_m)_{\mathrm{ML}} &= \frac{\gamma_m}{m_m^2},\\ \beta_{\mathrm{ML}} &= \frac{N - \sum_{m=1}^M \gamma_m + 2a_0 - 2}{\|\mathbf{t}-\mathbf{\Phi}\mathbf{m}\|^2 + 2b_0}.\end{aligned} \tag{7.109}$$

## 7.14

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal \boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{0}, \mathbf{A}^{-1}\right). \end{aligned} \tag{7.110}$$

By marginalisation,

$$p(t_{N+1}|\mathbf{t}) = \int p(t_{N+1}|\mathbf{w})p(\mathbf{w}|\mathbf{t})d\mathbf{w}. \tag{7.111}$$

By 7.9,

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}\left(\mathbf{w}|\mathbf{m}, \mathbf{S}\right), \tag{7.112}$$

where

$$\begin{aligned} \mathbf{m} &= \beta \mathbf{S} \boldsymbol{\Phi}^\intercal \mathbf{t}, \\ \mathbf{S} &= \left(\beta \boldsymbol{\Phi}^\intercal \boldsymbol{\Phi} + \mathbf{A}\right)^{-1}. \end{aligned} \tag{7.113}$$

Then, the logarithm of the integrand except the terms independent of $\mathbf{w}$ can be written as

$$\begin{aligned} &-\frac{\beta}{2}\left(t_{N+1} - \mathbf{w}^\intercal \boldsymbol{\phi}_{N+1}\right)^2 - \frac{1}{2}(\mathbf{w}-\mathbf{m})^\intercal \mathbf{S}^{-1}(\mathbf{w}-\mathbf{m}) \\ =&-\frac{1}{2}\mathbf{v}^\intercal \mathbf{M}\mathbf{v} + \mathbf{v}^\intercal \mathbf{u} - \frac{1}{2}\mathbf{m}^\intercal \mathbf{S}^{-1}\mathbf{m}, \end{aligned} \tag{7.114}$$

where

$$\begin{aligned} \mathbf{v} &= \begin{bmatrix} \mathbf{w} \\ t_{N+1} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \mathbf{S}^{-1} + \beta \boldsymbol{\phi}_{N+1}\boldsymbol{\phi}_{N+1}^\intercal & -\beta \boldsymbol{\phi}_{N+1} \\ -\beta \boldsymbol{\phi}_{N+1}^\intercal & \beta \end{bmatrix}, \\ \mathbf{u} &= \begin{bmatrix} \mathbf{S}^{-1}\mathbf{m} \\ 0 \end{bmatrix}. \end{aligned} \tag{7.115}$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix} \mathbf{S} & \mathbf{S}\boldsymbol{\phi}_{N+1} \\ \boldsymbol{\phi}_{N+1}^\intercal \mathbf{S} & \beta^{-1} + \boldsymbol{\phi}_{N+1}^\intercal \mathbf{S} \boldsymbol{\phi}_{N+1} \end{bmatrix}, \tag{7.116}$$

so that

$$\mathbf{M}^{-1}\mathbf{u} = \begin{bmatrix} \mathbf{m} \\ \mathbf{m}^\intercal \boldsymbol{\phi}_{N+1} \end{bmatrix}. \tag{7.117}$$

Therefore,

$$p(t_{N+1}|\mathbf{t}) = \mathcal{N}\left(t_{N+1}|\mathbf{m}^\intercal \boldsymbol{\phi}_{N+1}, \beta^{-1} + \boldsymbol{\phi}_{N+1}^\intercal \mathbf{S} \boldsymbol{\phi}_{N+1}\right). \tag{7.118}$$

## 7.15

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal\boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{0}, \mathbf{A}^{-1}\right), \end{aligned} \tag{7.119}$$

where

$$\mathbf{A} = \operatorname{diag}(\alpha_m). \tag{7.120}$$

By 7.11,

$$p(\mathbf{t}) = \mathcal{N}\left(\mathbf{t}|\mathbf{0}, \mathbf{C}\right), \tag{7.121}$$

where

$$\mathbf{C} = \beta^{-1}\mathbf{I} + \mathbf{\Phi}\mathbf{A}^{-1}\mathbf{\Phi}^\intercal. \tag{7.122}$$

Then,

$$\ln p(\mathbf{t}) = -\frac{N}{2}\ln(2\pi) - \frac{1}{2}\ln(\det\mathbf{C}) - \frac{1}{2}\mathbf{t}^\intercal\mathbf{C}^{-1}\mathbf{t}. \tag{7.123}$$

We have

$$\mathbf{C} = \mathbf{C}_{-m} + \alpha_m^{-1}\boldsymbol{\varphi}_m\boldsymbol{\varphi}_m^\intercal, \tag{7.124}$$

where $\mathbf{C}_{-m}$ is independent of $\alpha_m$ and $\boldsymbol{\varphi}_m$ is the $m^{\text{th}}$ column of $\mathbf{\Phi}$. By 2.26,

$$\mathbf{C}^{-1} = \mathbf{C}_{-m}^{-1} - \left(\alpha_m + s_m\right)^{-1}\mathbf{C}_{-m}^{-1}\boldsymbol{\varphi}_m\boldsymbol{\varphi}_m^\intercal\mathbf{C}_{-m}^{-1}, \tag{7.125}$$

where

$$s_m = \boldsymbol{\varphi}_m^\intercal\mathbf{C}_{-m}^{-1}\boldsymbol{\varphi}_m. \tag{7.126}$$

Then, $\ln p(\mathbf{t})$ except the terms independent of $\alpha_m$ can be written as

$$\lambda(\alpha_m) = \frac{1}{2}\left(\ln\alpha_m - \ln\left(\alpha_m + s_m\right) + \left(\alpha_m + s_m\right)^{-1}q_m^2\right), \tag{7.127}$$

where

$$q_m = \boldsymbol{\varphi}_m^\intercal\mathbf{C}_{-m}^{-1}\mathbf{t}. \tag{7.128}$$

## 7.16

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal\boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{0}, \mathbf{A}^{-1}\right), \end{aligned} \tag{7.129}$$

where

$$\mathbf{A} = \mathrm{diag}(\alpha_m). \tag{7.130}$$

By 7.11,

$$p(\mathbf{t}) = \mathcal{N}\left(\mathbf{t}|\mathbf{0}, \mathbf{C}\right), \tag{7.131}$$

where

$$\mathbf{C} = \beta^{-1}\mathbf{I} + \mathbf{\Phi}\mathbf{A}^{-1}\mathbf{\Phi}^\intercal. \tag{7.132}$$

By 7.15, $\ln p(\mathbf{t})$ except the terms independent of $\alpha_m$ can be written as

$$\lambda(\alpha_m) = \frac{1}{2}\left(\ln \alpha_m - \ln\left(\alpha_m + s_m\right) + \left(\alpha_m + s_m\right)^{-1} q_m^2\right), \tag{7.133}$$

where

$$\begin{aligned} s_m &= \boldsymbol{\varphi}_m^\intercal \mathbf{C}_{-m}^{-1} \boldsymbol{\varphi}_m, \\ q_m &= \boldsymbol{\varphi}_m^\intercal \mathbf{C}_{-m}^{-1} \mathbf{t}, \\ \mathbf{C}_{-m} &= \mathbf{C} - \alpha_m^{-1} \boldsymbol{\varphi}_m \boldsymbol{\varphi}_m^\intercal, \end{aligned} \tag{7.134}$$

and $\boldsymbol{\varphi}_m$ is the $m^{\text{th}}$ column of $\mathbf{\Phi}$. Setting the derivative of $\lambda(\alpha_m)$ to zero gives

$$0 = \frac{1}{2}\left(\alpha_m^{-1} - (\alpha_m + s_m)^{-1} - (\alpha_m + s_m)^{-2} q_m^2\right). \tag{7.135}$$

Then,

$$0 = (\alpha_m + s_m)^2 - (\alpha_m + s_m)\alpha_m - \alpha_m q_m^2. \tag{7.136}$$

Therefore, the maximum likelihood solution to $\alpha_m$ is given by

$$(\alpha_m)_{\mathrm{ML}} = \frac{s_m^2}{q_m^2 - s_m}. \tag{7.137}$$

## 7.17

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal \boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{0}, \mathbf{A}^{-1}\right), \end{aligned} \tag{7.138}$$

where

$$\mathbf{A} = \mathrm{diag}(\alpha_m). \tag{7.139}$$

**(a)**

By 7.11,

$$p(\mathbf{t}) = \mathcal{N}\left(\mathbf{t}|\mathbf{0}, \mathbf{C}\right), \tag{7.140}$$

where

$$\mathbf{C} = \beta^{-1}\mathbf{I} + \mathbf{\Phi}\mathbf{A}^{-1}\mathbf{\Phi}^{\intercal}. \tag{7.141}$$

By 7.15, $\ln p(\mathbf{t})$ except the terms independent of $\alpha_m$ can be written as

$$\lambda(\alpha_m) = \frac{1}{2}\left(\ln \alpha_m - \ln\left(\alpha_m + s_m\right) + \left(\alpha_m + s_m\right)^{-1} q_m^2\right), \tag{7.142}$$

where

$$\begin{aligned} s_m &= \boldsymbol{\varphi}_m^{\intercal}\mathbf{C}_{-m}^{-1}\boldsymbol{\varphi}_m, \\ q_m &= \boldsymbol{\varphi}_m^{\intercal}\mathbf{C}_{-m}^{-1}\mathbf{t}, \\ \mathbf{C}_{-m} &= \mathbf{C} - \alpha_m^{-1}\boldsymbol{\varphi}_m\boldsymbol{\varphi}_m^{\intercal}, \end{aligned} \tag{7.143}$$

and $\boldsymbol{\varphi}_m$ is the $m^{\text{th}}$ column of $\mathbf{\Phi}$. By 2.26,

$$\mathbf{C}_{-m}^{-1} = \mathbf{C}^{-1} + \left(\alpha_m - S_m\right)^{-1}\mathbf{C}^{-1}\boldsymbol{\varphi}_m\boldsymbol{\varphi}_m^{\intercal}\mathbf{C}^{-1}, \tag{7.144}$$

where

$$S_m = \boldsymbol{\varphi}_m^{\intercal}\mathbf{C}^{-1}\boldsymbol{\varphi}_m. \tag{7.145}$$

Then,

$$\begin{aligned} s_m &= S_m + \left(\alpha_m - S_m\right)^{-1} S_m^2, \\ q_m &= Q_m + \left(\alpha_m - S_m\right)^{-1} S_m Q_m, \end{aligned} \tag{7.146}$$

where

$$Q_m = \boldsymbol{\varphi}_m^{\intercal}\mathbf{C}^{-1}\mathbf{t}. \tag{7.147}$$

Therefore,

$$\begin{aligned} s_m &= (\alpha_m - S_m)^{-1}\alpha_m S_m, \\ q_m &= (\alpha_m - S_m)^{-1}\alpha_m Q_m. \end{aligned} \tag{7.148}$$

**(b)**

By 2.26,

$$\mathbf{C}^{-1} = \beta\mathbf{I} - \beta^2\mathbf{\Phi}\mathbf{\Sigma}\mathbf{\Phi}^{\intercal}, \tag{7.149}$$

where

$$\mathbf{\Sigma} = \left(\mathbf{A} + \beta\mathbf{\Phi}^{\intercal}\mathbf{\Phi}\right)^{-1}. \tag{7.150}$$

Therefore,

$$\begin{aligned} S_m &= \beta\|\boldsymbol{\varphi}_m\|^2 - \beta^2\boldsymbol{\varphi}_m^{\intercal}\mathbf{\Phi}\mathbf{\Sigma}\mathbf{\Phi}^{\intercal}\boldsymbol{\varphi}_m, \\ Q_m &= \beta\boldsymbol{\varphi}_m^{\intercal}\mathbf{t} - \beta^2\boldsymbol{\varphi}_m^{\intercal}\mathbf{\Phi}\mathbf{\Sigma}\mathbf{\Phi}^{\intercal}\mathbf{t}. \end{aligned} \tag{7.151}$$

## 7.18

Let $t_1, \cdots, t_N$ be conditionally independent binary variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= y_n^{t_n}(1-y_n)^{1-t_n}, \\ p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{0}, \mathbf{A}^{-1}\right), \end{aligned} \tag{7.152}$$

where

$$\begin{aligned} y_n &= \sigma\left(\mathbf{w}^\intercal \boldsymbol{\phi}_n\right), \\ \sigma(a) &= \frac{1}{1+\exp(-a)}, \\ \mathbf{A} &= \operatorname{diag}(\alpha_m). \end{aligned} \tag{7.153}$$

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{w})p(\mathbf{w}). \tag{7.154}$$

The logarithm of the right hand side except the terms independent of $\mathbf{w}$ and $\mathbf{t}$ can be written as

$$\sum_{n=1}^{N}\left(t_n \ln y_n + (1-t_n)\ln(1-y_n)\right) - \frac{1}{2}\mathbf{w}^\intercal \mathbf{A}\mathbf{w}. \tag{7.155}$$

Then,

$$\begin{aligned} \nabla \ln p(\mathbf{w}|\mathbf{t}) &= \sum_{n=1}^{N}\left(t_n - y_n\right)\boldsymbol{\phi}_n - \mathbf{A}\mathbf{w}, \\ \nabla\nabla p(\mathbf{w}|\mathbf{t}) &= -\sum_{n=1}^{N} y_n(1-y_n)\boldsymbol{\phi}_n\boldsymbol{\phi}_n^\intercal - \mathbf{A}. \end{aligned} \tag{7.156}$$

Therefore,

$$\begin{aligned} \nabla \ln p(\mathbf{w}|\mathbf{t}) &= \boldsymbol{\Phi}(\mathbf{t}-\mathbf{y}) - \mathbf{A}\mathbf{w}, \\ \nabla\nabla \ln p(\mathbf{w}|\mathbf{t}) &= -\boldsymbol{\Phi}^\intercal \mathbf{B}\boldsymbol{\Phi} - \mathbf{A}, \end{aligned} \tag{7.157}$$

where

$$\mathbf{B} = \operatorname{diag}\left(y_n(1-y_n)\right). \tag{7.158}$$

## 7.19

Let $t_1, \cdots, t_N$ be conditionally independent binary variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= y_n^{t_n}(1-y_n)^{1-t_n}, \\ p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{0}, \mathbf{A}^{-1}\right), \end{aligned} \tag{7.159}$$

where $\mathbf{w}$ is in $M$ dimensions and

$$
\begin{aligned}
y_n &= \sigma\left(\mathbf{w}^\intercal \boldsymbol{\phi}_n\right), \\
\sigma(a) &= \frac{1}{1+\exp(-a)}, \\
\mathbf{A} &= \operatorname{diag}(\alpha_m).
\end{aligned}
\tag{7.160}
$$

By marginalisation,

$$
p(\mathbf{t}) = \int p(\mathbf{t}|\mathbf{w})p(\mathbf{w})d\mathbf{w}. \tag{7.161}
$$

The logarithm of the integrand of the right hand side except the terms independent of $\mathbf{w}$ and $\mathbf{t}$ can be written as

$$
E(\mathbf{w}) = \sum_{n=1}^{N} \left(t_n \ln y_n + (1-t_n)\ln(1-y_n)\right) - \frac{1}{2}\mathbf{w}^\intercal \mathbf{A}\mathbf{w}. \tag{7.162}
$$

We have

$$
\begin{aligned}
\nabla E(\mathbf{w}) &= \boldsymbol{\Phi}(\mathbf{t}-\mathbf{y}) - \mathbf{A}\mathbf{w}, \\
\nabla\nabla E(\mathbf{w}) &= -\boldsymbol{\Phi}^\intercal \mathbf{B}\boldsymbol{\Phi} - \mathbf{A},
\end{aligned}
\tag{7.163}
$$

where

$$
\mathbf{B} = \operatorname{diag}\left(y_n(1-y_n)\right). \tag{7.164}
$$

Then, we have the Taylor series

$$
\begin{aligned}
E(\mathbf{w}) =& (\mathbf{w}_{\mathrm{MAP}}) - \frac{1}{2}\left(\mathbf{w}-\mathbf{w}_{\mathrm{MAP}}\right)^\intercal \boldsymbol{\Sigma}^{-1}\left(\mathbf{w}-\mathbf{w}_{\mathrm{MAP}}\right) \\
&+ O\left(\|\mathbf{w}-\mathbf{w}_{\mathrm{MAP}}\|^3\right),
\end{aligned}
\tag{7.165}
$$

where

$$
\begin{aligned}
\mathbf{w}_{\mathrm{MAP}} &= \mathbf{A}^{-1}\boldsymbol{\Phi}(\mathbf{t}-\mathbf{y}), \\
\boldsymbol{\Sigma} &= \left(\mathbf{A} + \boldsymbol{\Phi}^\intercal \mathbf{B}\boldsymbol{\Phi}\right)^{-1}.
\end{aligned}
\tag{7.166}
$$

We have

$$
\int \exp\left(-\frac{1}{2}\left(\mathbf{w}-\mathbf{w}_{\mathrm{MAP}}\right)\boldsymbol{\Sigma}^{-1}\left(\mathbf{w}-\mathbf{w}_{\mathrm{MAP}}\right)\right) d\mathbf{w} = (2\pi)^{\frac{M}{2}}(\det\boldsymbol{\Sigma})^{\frac{1}{2}}. \tag{7.167}
$$

Then,

$$
p(\mathbf{t}) \simeq (2\pi)^{\frac{M}{2}}(\det\boldsymbol{\Sigma})^{\frac{1}{2}} p(\mathbf{t}|\mathbf{w}_{\mathrm{MAP}})p(\mathbf{w}_{\mathrm{MAP}}). \tag{7.168}
$$

Then, $\ln p(\mathbf{t})$ can be approximated as

$$l(\mathbf{t}) = \frac{1}{2}\ln(\det \boldsymbol{\Sigma}) - \frac{1}{2}\ln\left(\det \mathbf{A}^{-1}\right) + E(\mathbf{w}_{\mathrm{MAP}}). \tag{7.169}$$

By 3.21, setting the derivative of $l(\mathbf{t})$ with respect to $\alpha_m$ to zero gives

$$0 = \frac{1}{2}\operatorname{tr}\left(\boldsymbol{\Sigma}^{-1}\frac{\partial \boldsymbol{\Sigma}}{\partial \alpha_m}\right) + \frac{1}{2\alpha_m} - \frac{1}{2}(w_m)^2_{\mathrm{MAP}}. \tag{7.170}$$

We have

$$\mathbf{O} = \frac{\partial \boldsymbol{\Sigma}^{-1}\boldsymbol{\Sigma}}{\partial \alpha_m}. \tag{7.171}$$

The right hand side can be written as

$$\frac{\partial \boldsymbol{\Sigma}^{-1}}{\partial \alpha_m}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}^{-1}\frac{\partial \boldsymbol{\Sigma}}{\partial \alpha_m} = \mathbf{v}_m\mathbf{v}_m^{\intercal}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}^{-1}\frac{\partial \boldsymbol{\Sigma}}{\partial \alpha_m}, \tag{7.172}$$

where $\mathbf{v}_m$ is the $m$-th vector of the standard basis in $M$ dimensions. Then,

$$\boldsymbol{\Sigma}^{-1}\frac{\partial \boldsymbol{\Sigma}}{\partial \alpha_m} = -\mathbf{v}_m\mathbf{v}_m^{\intercal}\boldsymbol{\Sigma}. \tag{7.173}$$

Then,

$$0 = -\frac{1}{2}\Sigma_{mm} + \frac{1}{2\alpha_m} - \frac{1}{2}(w_m)^2_{\mathrm{MAP}}. \tag{7.174}$$

Therefore, the maximum likelihood solution to $\alpha_m$ satisfies

$$(\alpha_m)_{\mathrm{ML}} \simeq \left(\Sigma_{mm} + (w_m)^2_{\mathrm{MAP}}\right)^{-1}. \tag{7.175}$$

# 8 Graphical Models

## 8.1

Let $\mathbf{x}$ be a graph of $K$ nodes such that

$$p(\mathbf{x}) = \prod_{k=1}^{K} p(x_k|\mathrm{pa}_k), \tag{8.1}$$

where $\mathrm{pa}_k$ is the set of parents of $x_k$. For $K = 1$, we have

$$\int p(x_1)dx_1 = 1. \tag{8.2}$$

Let us assume that

$$\int p(\mathbf{x})d\mathbf{x} = 1. \tag{8.3}$$

Let $\mathbf{x}'$ be a graph of $K + 1$ nodes such that

$$p(\mathbf{x}') = \prod_{k=1}^{K+1} p(x_k|\mathrm{pa}_k). \tag{8.4}$$

Then,

$$\int p(\mathbf{x}')d\mathbf{x}' = \int p(\mathbf{x}) \left( \int p(x_{K+1}|\mathrm{pa}_{K+1})dx_{K+1} \right) d\mathbf{x}. \tag{8.5}$$

We have

$$\int p(x_{K+1}|\mathrm{pa}_{K+1})dx_{K+1} = 1. \tag{8.6}$$

Then,

$$\int p(\mathbf{x}')d\mathbf{x}' = \int p(\mathbf{x})d\mathbf{x} = 1. \tag{8.7}$$

Therefore, the asuumption is proved by induction on $K$.

# 9 Mixture Models and EM

# 10 Approximate Inference

# 11 Sampling Methods

# 12 Continuous Latent Variables

# 13 Sequential Data

# 14 Combining Models

# 15 Appendix

## 15.1

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$p(t_n|\mathbf{w}) = \mathcal{N}\left(t_n|\mathbf{w}^\intercal \boldsymbol{\phi}_n, \beta^{-1}\right),$$
$$p(\mathbf{w}) = \mathcal{N}\left(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0\right).$$

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{w})p(\mathbf{w}).$$

The logarithm of the right hand side except the terms independent of $\mathbf{t}$ and $\mathbf{w}$ can be written as

$$\begin{aligned} & -\frac{\beta}{2}\|\mathbf{t} - \boldsymbol{\Phi}\mathbf{w}\|^2 - \frac{1}{2}(\mathbf{w} - \mathbf{m}_0)^\intercal \mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{m}_0) \\ = & -\frac{1}{2}\mathbf{v}^\intercal \mathbf{M}\mathbf{v} + \mathbf{v}^\intercal \mathbf{u} - \frac{1}{2}\mathbf{m}_0^\intercal \mathbf{S}_0^{-1}\mathbf{m}_0, \end{aligned}$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ \mathbf{t} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \mathbf{S}_0^{-1} + \beta\boldsymbol{\Phi}^\intercal\boldsymbol{\Phi} & -\beta\boldsymbol{\Phi}^\intercal \\ -\beta\boldsymbol{\Phi} & \beta\mathbf{I} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \mathbf{S}_0^{-1}\mathbf{m}_0 \\ \mathbf{0} \end{bmatrix}.$$

Therefore,

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}\left(\mathbf{w}|\mathbf{m}, \mathbf{S}\right),$$

where

$$\mathbf{m} = \mathbf{S}\left(\mathbf{S}_0^{-1}\mathbf{m}_0 + \beta\boldsymbol{\Phi}^\intercal\mathbf{t}\right),$$
$$\mathbf{S} = \left(\mathbf{S}_0^{-1} + \beta\boldsymbol{\Phi}^\intercal\boldsymbol{\Phi}\right)^{-1}.$$

## 15.2

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal \boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0\right). \end{aligned}$$

By Bayes' theorem,

$$p(\mathbf{t}) = \int p(\mathbf{t}|\mathbf{w})p(\mathbf{w})d\mathbf{w}.$$

The logarithm of the integrand of the right hand side except the terms independent of $\mathbf{t}$ and $\mathbf{w}$ can be written as

$$\begin{aligned} &-\frac{\beta}{2}\|\mathbf{t}-\boldsymbol{\Phi}\mathbf{w}\|^2 - \frac{1}{2}(\mathbf{w}-\mathbf{m}_0)^\intercal \mathbf{S}_0^{-1}(\mathbf{w}-\mathbf{m}_0) \\ =&-\frac{1}{2}\mathbf{v}^\intercal \mathbf{M}\mathbf{v} + \mathbf{v}^\intercal \mathbf{u} - \frac{1}{2}\mathbf{m}_0^\intercal \mathbf{S}_0^{-1}\mathbf{m}_0, \end{aligned}$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ \mathbf{t} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \mathbf{S}_0^{-1} + \beta\boldsymbol{\Phi}^\intercal\boldsymbol{\Phi} & -\beta\boldsymbol{\Phi}^\intercal \\ -\beta\boldsymbol{\Phi} & \beta\mathbf{I} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \mathbf{S}_0^{-1}\mathbf{m}_0 \\ \mathbf{0} \end{bmatrix}.$$

We have

$$\mathbf{M}^{-1} = \begin{bmatrix} \mathbf{S}_0 & \mathbf{S}_0\boldsymbol{\Phi}^\intercal \\ \boldsymbol{\Phi}\mathbf{S}_0 & \beta^{-1}\mathbf{I} + \boldsymbol{\Phi}\mathbf{S}_0\boldsymbol{\Phi}^\intercal \end{bmatrix},$$

so that

$$\mathbf{M}^{-1}\mathbf{v} = \begin{bmatrix} \mathbf{m}_0 \\ \boldsymbol{\Phi}\mathbf{m}_0 \end{bmatrix}.$$

Therefore,

$$p(\mathbf{t}) = \mathcal{N}\left(\mathbf{t}|\boldsymbol{\Phi}\mathbf{m}_0, \beta^{-1}\mathbf{I} + \boldsymbol{\Phi}\mathbf{S}_0\boldsymbol{\Phi}^\intercal\right).$$

## 15.3

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal\boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0\right). \end{aligned}$$

By marginalisation,

$$p(t_{N+1}|\mathbf{t}) = \int p(t_{N+1}|\mathbf{w})p(\mathbf{w}|\mathbf{t})d\mathbf{w}.$$

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{w})p(\mathbf{w}).$$

The logarithm of the right hand side except the terms independent of $\mathbf{t}$ and $\mathbf{w}$ can be written as

$$\begin{aligned} & -\frac{\beta}{2}\|\mathbf{t}-\boldsymbol{\Phi}\mathbf{w}\|^2 - \frac{1}{2}(\mathbf{w}-\mathbf{m}_0)^\intercal\mathbf{S}_0^{-1}(\mathbf{w}-\mathbf{m}_0) \\ = & -\frac{1}{2}\mathbf{v}^\intercal\mathbf{M}\mathbf{v} + \mathbf{v}^\intercal\mathbf{u} - \frac{1}{2}\mathbf{m}_0^\intercal\mathbf{S}_0^{-1}\mathbf{m}_0, \end{aligned}$$

where

$$\mathbf{v} = \begin{bmatrix}\mathbf{w}\\ \mathbf{t}\end{bmatrix}, \mathbf{M} = \begin{bmatrix}\mathbf{S}_0^{-1}+\beta\boldsymbol{\Phi}^\intercal\boldsymbol{\Phi} & -\beta\boldsymbol{\Phi}^\intercal \\ -\beta\boldsymbol{\Phi} & \beta\mathbf{I}\end{bmatrix}, \mathbf{u} = \begin{bmatrix}\mathbf{S}_0^{-1}\mathbf{m}_0 \\ \mathbf{0}\end{bmatrix}.$$

Then,

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}\left(\mathbf{w}|\mathbf{m}, \mathbf{S}\right),$$

where

$$\begin{aligned} \mathbf{m} &= \mathbf{S}\left(\mathbf{S}_0^{-1}\mathbf{m}_0 + \beta\boldsymbol{\Phi}^\intercal\mathbf{t}\right), \\ \mathbf{S} &= \left(\mathbf{S}_0^{-1} + \beta\boldsymbol{\Phi}^\intercal\boldsymbol{\Phi}\right)^{-1}. \end{aligned}$$

Then, the logarithm of the integrand of the right hand side except the terms independent of $t_{N+1}$ and $\mathbf{w}$ can be written as

$$\begin{aligned} & -\frac{\beta}{2}\left(t_{N+1}-\mathbf{w}^\intercal\boldsymbol{\phi}_{N+1}\right)^2 - \frac{1}{2}(\mathbf{w}-\mathbf{m})^\intercal\mathbf{S}^{-1}(\mathbf{w}-\mathbf{m}) \\ = & -\frac{1}{2}\mathbf{v}'^\intercal\mathbf{M}'\mathbf{v}' + \mathbf{v}'^\intercal\mathbf{u}' - \frac{1}{2}\mathbf{m}^\intercal\mathbf{S}^{-1}\mathbf{m}, \end{aligned}$$

where

$$\mathbf{v}' = \begin{bmatrix} \mathbf{w} \\ t_{N+1} \end{bmatrix}, \mathbf{M}' = \begin{bmatrix} \mathbf{S}^{-1} + \beta\boldsymbol{\phi}_{N+1}\boldsymbol{\phi}_{N+1}^\intercal & -\beta\boldsymbol{\phi}_{N+1} \\ -\beta\boldsymbol{\phi}_{N+1}^\intercal & \beta \end{bmatrix}, \mathbf{u}' = \begin{bmatrix} \mathbf{S}^{-1}\mathbf{m} \\ 0 \end{bmatrix}.$$

We have

$$\mathbf{M}'^{-1} = \begin{bmatrix} \mathbf{S} & \mathbf{S}\boldsymbol{\phi}_{N+1} \\ \boldsymbol{\phi}_{N+1}^\intercal\mathbf{S} & \beta^{-1} + \boldsymbol{\phi}_{N+1}^\intercal\mathbf{S}\boldsymbol{\phi}_{N+1} \end{bmatrix},$$

so that

$$\mathbf{M}'^{-1}\mathbf{u}' = \begin{bmatrix} \mathbf{m} \\ \mathbf{m}^\intercal\boldsymbol{\phi}_{N+1} \end{bmatrix}.$$

Therefore,

$$p(t_{N+1}|\mathbf{t}) = \mathcal{N}\left(t_{N+1}|\mathbf{m}^\intercal\boldsymbol{\phi}_{N+1}, \beta^{-1} + \boldsymbol{\phi}_{N+1}^\intercal\mathbf{S}\boldsymbol{\phi}_{N+1}\right).$$

## 15.4

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w},\beta) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal\boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}|\beta) &= \mathcal{N}\left(\mathbf{w}|\mathbf{m}_0, \beta^{-1}\mathbf{S}_0\right), \\ p(\beta) &= \mathrm{Gam}(\beta|a_0, b_0), \end{aligned}$$

where $\mathbf{w}$ is a vector in $M$ dimensions. By marginalisation,

$$\begin{aligned} p(\mathbf{t}) &= \int_0^\infty p(\mathbf{t}|\beta)p(\beta)d\beta, \\ p(\mathbf{t}|\beta) &= \int p(\mathbf{t}|\mathbf{w},\beta)p(\mathbf{w}|\beta)d\mathbf{w}. \end{aligned}$$

The logaritm of $p(\mathbf{t}|\mathbf{w},\beta)p(\mathbf{w}|\beta)$ except the terms independent of $\mathbf{w}$ can be written as

$$-\frac{\beta}{2}\|\mathbf{t}-\boldsymbol{\Phi}\mathbf{w}\|^2 - \frac{\beta}{2}(\mathbf{w}-\mathbf{m}_0)^\intercal\mathbf{S}_0^{-1}(\mathbf{w}-\mathbf{m}_0) = -\frac{1}{2}\mathbf{v}^\intercal\mathbf{M}\mathbf{v} + \mathbf{v}^\intercal\mathbf{u},$$

where

$$\mathbf{v} = \begin{bmatrix}\mathbf{w}\\ \mathbf{t}\end{bmatrix}, \mathbf{M} = \begin{bmatrix}\beta\left(\mathbf{S}_0^{-1}+\boldsymbol{\Phi}^\intercal\boldsymbol{\Phi}\right) & -\beta\boldsymbol{\Phi}^\intercal \\ -\beta\boldsymbol{\Phi} & \beta\mathbf{I}\end{bmatrix}, \mathbf{u} = \begin{bmatrix}\mathbf{S}_0^{-1}\mathbf{m}_0 \\ \mathbf{0}\end{bmatrix}.$$

We have

$$\mathbf{M}^{-1} = \begin{bmatrix}\beta^{-1}\mathbf{S}_0 & \beta^{-1}\mathbf{S}_0\boldsymbol{\Phi}^\intercal \\ \beta^{-1}\boldsymbol{\Phi}\mathbf{S}_0 & \beta^{-1}\left(\mathbf{I}+\boldsymbol{\Phi}\mathbf{S}_0\boldsymbol{\Phi}^\intercal\right)\end{bmatrix},$$

so that

$$\mathbf{M}^{-1}\mathbf{v} = \begin{bmatrix}\mathbf{m}_0 \\ \boldsymbol{\Phi}\mathbf{m}_0\end{bmatrix}.$$

Then,

$$p(\mathbf{t}|\beta) = \mathcal{N}\left(\mathbf{t}|\boldsymbol{\Phi}\mathbf{m}_0, \beta^{-1}\left(\mathbf{I}+\boldsymbol{\Phi}\mathbf{S}_0\boldsymbol{\Phi}^\intercal\right)\right).$$

Then, the logarithm of $p(\mathbf{t}|\beta)p(\beta)$ except the terms independent of $\beta$ can be written as

$$\begin{aligned} &-\frac{N}{2}\ln\beta^{-1} - \frac{\beta}{2}(\mathbf{t}-\boldsymbol{\Phi}\mathbf{m}_0)^\intercal\left(\mathbf{I}+\boldsymbol{\Phi}\mathbf{S}_0\boldsymbol{\Phi}^\intercal\right)^{-1}(\mathbf{t}-\boldsymbol{\Phi}\mathbf{m}_0) \\ &+(a_0-1)\ln\beta - b_0\beta \\ =&(a-1)\ln\beta - b\beta, \end{aligned}$$

where

$$
\begin{aligned}
a &= a_0 + \frac{N}{2}, \\
b &= b_0 + \frac{1}{2}(\mathbf{t} - \mathbf{\Phi}\mathbf{m}_0)^\intercal \left(\mathbf{I} + \mathbf{\Phi}\mathbf{S}_0\mathbf{\Phi}^\intercal\right)^{-1} (\mathbf{t} - \mathbf{\Phi}\mathbf{m}_0).
\end{aligned}
$$

We have

$$
\int_0^\infty \beta^{a-1} \exp(-b\beta) d\beta = b^{-a}\Gamma(a).
$$

We have

$$
b^{-a} = \left(1 + (\mathbf{t} - \boldsymbol{\mu})^\intercal (\nu\mathbf{\Sigma})^{-1} (\mathbf{t} - \boldsymbol{\mu})\right)^{-\frac{\nu+N}{2}} b_0^{-a},
$$

where

$$
\begin{aligned}
\boldsymbol{\mu} &= \mathbf{\Phi}\mathbf{m}_0, \\
\mathbf{\Sigma} &= \frac{b_0}{a_0} \left(\mathbf{I} + \mathbf{\Phi}\mathbf{S}_0\mathbf{\Phi}^\intercal\right), \\
\nu &= 2a_0.
\end{aligned}
$$

Therefore,

$$
p(\mathbf{t}) = \mathrm{St}\left(\mathbf{t} | \boldsymbol{\mu}, \mathbf{\Sigma}, \nu\right).
$$

## 15.5

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}, \beta) &= \mathcal{N}\left(t_n|\mathbf{w}^\intercal \boldsymbol{\phi}_n, \beta^{-1}\right), \\ p(\mathbf{w}|\beta) &= \mathcal{N}\left(\mathbf{w}|\mathbf{m}_0, \beta^{-1}\mathbf{S}_0\right), \\ p(\beta) &= \mathrm{Gam}(\beta|a_0, b_0), \end{aligned}$$

where $\mathbf{w}$ is a vector in $M$ dimensions.

### (a)

By marginalisation,

$$p(t_{N+1}|\beta, \mathbf{t}) = \int p(t_{N+1}|\mathbf{w}, \beta) p(\mathbf{w}|\beta, \mathbf{t}) d\mathbf{w}.$$

By Bayes' theorem,

$$p(\mathbf{w}|\beta, \mathbf{t}) p(\mathbf{t}|\beta) = p(\mathbf{t}|\mathbf{w}, \beta) p(\mathbf{w}|\beta).$$

The logarithm of the right hand side except the terms independent of $\mathbf{w}$ can be written as

$$\begin{aligned} & -\frac{\beta}{2}\|\mathbf{t} - \boldsymbol{\Phi}\mathbf{w}\|^2 - \frac{\beta}{2}(\mathbf{w} - \mathbf{m}_0)^\intercal \mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{m}_0) \\ = & -\frac{1}{2}\mathbf{v}^\intercal \mathbf{M}\mathbf{v} + \mathbf{v}^\intercal \mathbf{u} - \frac{\beta}{2}\mathbf{m}_0^\intercal \mathbf{S}_0^{-1}\mathbf{m}_0, \end{aligned}$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ \mathbf{t} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \beta\left(\mathbf{S}_0^{-1} + \boldsymbol{\Phi}^\intercal \boldsymbol{\Phi}\right) & -\beta\boldsymbol{\Phi}^\intercal \\ -\beta\boldsymbol{\Phi} & \beta\mathbf{I} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \beta\mathbf{S}_0^{-1}\mathbf{m}_0 \\ \mathbf{0} \end{bmatrix}.$$

Then,

$$p(\mathbf{w}|\beta, \mathbf{t}) = \mathcal{N}\left(\mathbf{w}|\mathbf{m}, \beta^{-1}\mathbf{S}\right),$$

where

$$\begin{aligned} \mathbf{m} &= \mathbf{S}\left(\mathbf{S}_0^{-1}\mathbf{m}_0 + \boldsymbol{\Phi}^\intercal \mathbf{t}\right), \\ \mathbf{S} &= \left(\mathbf{S}_0^{-1} + \boldsymbol{\Phi}^\intercal \boldsymbol{\Phi}\right)^{-1}. \end{aligned}$$

Then, the logarithm of $p(t_{N+1}|\mathbf{w},\beta)p(\mathbf{w}|\beta,\mathbf{t})$ except the terms independent of $\mathbf{w}$ can be written as

$$
\begin{aligned}
&-\frac{\beta}{2}\left(t_{N+1}-\mathbf{w}^\intercal\boldsymbol{\phi}_{N+1}\right)^2-\frac{\beta}{2}(\mathbf{w}-\mathbf{m})^\intercal\mathbf{S}^{-1}(\mathbf{w}-\mathbf{m})\\
=&-\frac{1}{2}\mathbf{v}'^\intercal\mathbf{M}'\mathbf{v}'+\mathbf{v}'^\intercal\mathbf{u}'-\frac{\beta}{2}\mathbf{m}^\intercal\mathbf{S}^{-1}\mathbf{m},
\end{aligned}
$$

where

$$
\mathbf{v}'=\begin{bmatrix}\mathbf{w}\\ t_{N+1}\end{bmatrix},\mathbf{M}'=\begin{bmatrix}\beta\left(\mathbf{S}^{-1}+\boldsymbol{\phi}_{N+1}\boldsymbol{\phi}_{N+1}^\intercal\right) & -\beta\boldsymbol{\phi}_{N+1}\\ -\beta\boldsymbol{\phi}_{N+1}^\intercal & \beta\end{bmatrix},\mathbf{u}'=\begin{bmatrix}\beta\mathbf{S}^{-1}\mathbf{m}\\ 0\end{bmatrix}.
$$

We have

$$
\mathbf{M}'^{-1}=\begin{bmatrix}\beta^{-1}\mathbf{S} & \beta^{-1}\mathbf{S}\boldsymbol{\phi}_{N+1}\\ \beta^{-1}\boldsymbol{\phi}_{N+1}^\intercal\mathbf{S} & \beta^{-1}\left(1+\boldsymbol{\phi}_{N+1}^\intercal\mathbf{S}\boldsymbol{\phi}_{N+1}\right)\end{bmatrix},
$$

so that

$$
\mathbf{M}'^{-1}\mathbf{u}'=\begin{bmatrix}\mathbf{m}\\ \mathbf{m}^\intercal\boldsymbol{\phi}_{N+1}\end{bmatrix}.
$$

Therefore,

$$
p(t_{N+1}|\beta,\mathbf{t})=\mathcal{N}\left(t_{N+1}|\mathbf{m}^\intercal\boldsymbol{\phi}_{N+1},\beta^{-1}\left(1+\boldsymbol{\phi}_{N+1}^\intercal\mathbf{S}\boldsymbol{\phi}_{N+1}\right)\right).
$$

**(b)**

By Bayes' theorem,

$$
p(\beta|\mathbf{t})p(\mathbf{t})=p(\mathbf{t}|\beta)p(\beta).
$$

We have

$$
\mathbf{M}^{-1}=\begin{bmatrix}\beta^{-1}\mathbf{S}_0 & \beta^{-1}\mathbf{S}_0\boldsymbol{\Phi}^\intercal\\ \beta^{-1}\boldsymbol{\Phi}\mathbf{S}_0 & \beta^{-1}\left(\mathbf{I}+\boldsymbol{\Phi}\mathbf{S}_0\boldsymbol{\Phi}^\intercal\right)\end{bmatrix},
$$

so that

$$
\mathbf{M}^{-1}\mathbf{u}=\begin{bmatrix}\mathbf{m}_0\\ \boldsymbol{\Phi}\mathbf{m}_0\end{bmatrix}.
$$

Then,

$$
p(\mathbf{t}|\beta)=\mathcal{N}\left(\mathbf{t}|\boldsymbol{\Phi}\mathbf{m}_0,\beta^{-1}\left(\mathbf{I}+\boldsymbol{\Phi}\mathbf{S}_0\boldsymbol{\Phi}^\intercal\right)\right).
$$

The logarithm of the right hand side except the terms independent of $\beta$ can be written as

$$
-\frac{N}{2}\ln\beta^{-1}-\frac{\beta}{2}(\mathbf{t}-\boldsymbol{\Phi}\mathbf{m}_0)^\intercal\left(\mathbf{I}+\boldsymbol{\Phi}\mathbf{S}_0\boldsymbol{\Phi}^\intercal\right)^{-1}(\mathbf{t}-\boldsymbol{\Phi}\mathbf{m}_0).
$$

Then the logarithm of $p(\mathbf{t}|\beta)p(\beta)$ except the terms independent of $\beta$ can be written as

$$(a-1)\ln\beta - b\beta,$$

where

$$\begin{aligned} a &= a_0 + \frac{N}{2}, \\ b &= b_0 + \frac{1}{2}(\mathbf{t}-\mathbf{\Phi}\mathbf{m}_0)^\intercal \left(\mathbf{I}+\mathbf{\Phi}\mathbf{S}_0\mathbf{\Phi}^\intercal\right)^{-1}(\mathbf{t}-\mathbf{\Phi}\mathbf{m}_0). \end{aligned}$$

Therefore,

$$p(\beta|\mathbf{t}) = \mathrm{Gam}(\beta|a,b).$$

### (c)

By marginalisation,

$$p(t_{N+1}|\mathbf{t}) = \int_0^\infty p(t_{N+1}|\beta,\mathbf{t})p(\beta|\mathbf{t})d\beta,$$

By (a) and (b), the logarithm of $p(t_{N+1}|\beta,\mathbf{t})p(\beta|\mathbf{t})$ except the terms independent of $\beta$ can be written as

$$\begin{aligned} & -\frac{1}{2}\ln\beta^{-1} - \frac{(t_{N+1}-\mathbf{m}^\intercal\boldsymbol{\phi}_{N+1})^2}{2(1+\boldsymbol{\phi}_{N+1}^\intercal\mathbf{S}\boldsymbol{\phi}_{N+1})}\beta + (a-1)\ln\beta - b\beta \\ =&(a'-1)\ln\beta - b'\ln\beta, \end{aligned}$$

where

$$\begin{aligned} a' &= a + \frac{1}{2}, \\ b' &= b + \frac{(t_{N+1}-\mathbf{m}^\intercal\boldsymbol{\phi}_{N+1})^2}{2(1+\boldsymbol{\phi}_{N+1}^\intercal\mathbf{S}\boldsymbol{\phi}_{N+1})}. \end{aligned}$$

We have

$$\int_0^\infty \beta^{a'-1}\exp(-b'\beta)d\beta = \Gamma(a')b'^{-a'}.$$

We have

$$b'^{-a'} = \left(1+\frac{(t_{N+1}-\mu)^2}{\nu\sigma^2}\right)^{-\frac{\nu+1}{2}} b^{-a'},$$

where

$$\begin{aligned}\mu &= \mathbf{m}^\intercal \boldsymbol{\phi}_{N+1}, \\ \sigma^2 &= \frac{b}{a}(1 + \boldsymbol{\phi}_{N+1}^\intercal \mathbf{S} \boldsymbol{\phi}_{N+1}), \\ \nu &= 2a.\end{aligned}$$

Therefore,

$$p(t_{N+1}|\mathbf{t}) = \mathrm{St}(t_{N+1}|\mu, \sigma^2, \nu).$$

## 15.6

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$
\begin{aligned}
p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|y(\mathbf{x}_n,\mathbf{w}),\beta^{-1}\right), \\
p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{m}_0,\mathbf{S}_0\right).
\end{aligned}
$$

By Bayes' theorem,

$$
p(\mathbf{w}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{w})p(\mathbf{w}).
$$

The logarithm of the right hand side except the terms independent of $\mathbf{t}$ and $\mathbf{w}$ can be written as $-E(\mathbf{w})$ where

$$
E(\mathbf{w}) = \frac{\beta}{2}\sum_{n=1}^{N}\left(t_n - y(\mathbf{x}_n,\mathbf{w})\right)^2 + \frac{1}{2}(\mathbf{w}-\mathbf{m}_0)^\intercal\mathbf{S}_0^{-1}(\mathbf{w}-\mathbf{m}_0).
$$

We have the Taylor series

$$
E(\mathbf{w}) = E(\mathbf{w}_{\text{MAP}}) + \frac{1}{2}(\mathbf{w}-\mathbf{w}_{\text{MAP}})^\intercal\mathbf{H}_{\text{MAP}}(\mathbf{w}-\mathbf{w}_{\text{MAP}}) + O\left(\|\mathbf{w}-\mathbf{w}_{\text{MAP}}\|^3\right),
$$

where $\mathbf{w}_{\text{MAP}}$ is a stationary point of $E$ and

$$
\mathbf{H}_{\text{MAP}} = \nabla\nabla E(\mathbf{w}_{\text{MAP}}).
$$

Therefore,

$$
p(\mathbf{w}|\mathbf{t}) \simeq \mathcal{N}\left(\mathbf{w}|\mathbf{w}_{\text{MAP}},\mathbf{H}_{\text{MAP}}^{-1}\right).
$$

## 15.7

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|y(\mathbf{x}_n, \mathbf{w}), \beta^{-1}\right), \\ p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0\right). \end{aligned}$$

By marginalisation,

$$p(\mathbf{t}) = \int p(\mathbf{t}|\mathbf{w})p(\mathbf{w})d\mathbf{w}.$$

The logarithm of the right hand side except the terms independent of $\mathbf{t}$ and $\mathbf{w}$ can be written as $-E(\mathbf{w})$ where

$$E(\mathbf{w}) = \frac{\beta}{2}\sum_{n=1}^{N}(t_n - y(\mathbf{x}_n, \mathbf{w}))^2 + \frac{1}{2}(\mathbf{w} - \mathbf{m}_0)^\intercal \mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{m}_0).$$

We have the Taylor series

$$y(\mathbf{x}_n, \mathbf{w}) = y_{0n} + \mathbf{g}_{0n}{}^\intercal(\mathbf{w} - \mathbf{m}_0) + O\left(\|\mathbf{w} - \mathbf{m}_0\|^2\right),$$

where

$$\begin{aligned} y_{0n} &= y(\mathbf{x}_n, \mathbf{m}_0). \\ \mathbf{g}_{0n} &= \nabla_{\mathbf{w}} y(\mathbf{x}_n, \mathbf{w})\big|_{\mathbf{w}=\mathbf{m}_0}. \end{aligned}$$

Then, $E(\mathbf{w})$ can be approximated as

$$\begin{aligned} &\frac{\beta}{2}\sum_{n=1}^{N}(t_n - y_{0n} - \mathbf{g}_{0n}{}^\intercal(\mathbf{w} - \mathbf{m}_0))^2 + \frac{1}{2}(\mathbf{w} - \mathbf{m}_0)^\intercal \mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{m}_0) \\ =&\frac{1}{2}\mathbf{v}^\intercal \mathbf{M}\mathbf{v} - \mathbf{v}^\intercal \mathbf{u}, \end{aligned}$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} - \mathbf{m}_0 \\ \mathbf{t} - \mathbf{y}_0 \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \mathbf{S}_0^{-1} + \beta \mathbf{G}_0{}^\intercal \mathbf{G}_0 & -\beta \mathbf{G}_0{}^\intercal \\ -\beta \mathbf{G}_0 & \beta \mathbf{I} \end{bmatrix}.$$

We have

$$\mathbf{M}^{-1} = \begin{bmatrix} \mathbf{S}_0 & \mathbf{S}_0 \mathbf{G}_0{}^\intercal \\ \mathbf{G}_0 \mathbf{S}_0 & \beta^{-1}\mathbf{I} + \mathbf{G}_0 \mathbf{S}_0 \mathbf{G}_0{}^\intercal \end{bmatrix}.$$

Therefore,

$$p(\mathbf{t}) \simeq \mathcal{N}\left(\mathbf{t}|\mathbf{y}_0, \beta^{-1}\mathbf{I} + \mathbf{G}_0 \mathbf{S}_0 \mathbf{G}_0{}^\intercal\right).$$

## 15.8

Let $t_1, \cdots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}\left(t_n|y(\mathbf{x}_n, \mathbf{w}), \beta^{-1}\right), \\ p(\mathbf{w}) &= \mathcal{N}\left(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0\right). \end{aligned}$$

By marginalisation,

$$p(t_{N+1}|\mathbf{t}) = \int p(t_{N+1}|\mathbf{w})p(\mathbf{w}|\mathbf{t})d\mathbf{w}.$$

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{w})p(\mathbf{w}).$$

The logarithm of the right hand side except the terms independent of $\mathbf{t}$ and $\mathbf{w}$ can be written as $-E(\mathbf{w})$ where

$$E(\mathbf{w}) = \frac{\beta}{2}\sum_{n=1}^{N}\left(t_n - y(\mathbf{x}_n, \mathbf{w})\right)^2 + \frac{1}{2}(\mathbf{w} - \mathbf{m}_0)^\intercal \mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{m}_0).$$

We have the Taylor series

$$E(\mathbf{w}) = E(\mathbf{w}_{\text{MAP}}) + \frac{1}{2}(\mathbf{w} - \mathbf{w}_{\text{MAP}})^\intercal \mathbf{H}_{\text{MAP}}(\mathbf{w} - \mathbf{w}_{\text{MAP}}) + O\left(\|\mathbf{w} - \mathbf{w}_{\text{MAP}}\|^3\right),$$

where $\mathbf{w}_{\text{MAP}}$ is a stationary point of $E$ and

$$\mathbf{H}_{\text{MAP}} = \nabla\nabla E(\mathbf{w}_{\text{MAP}}).$$

Then,

$$p(\mathbf{w}|\mathbf{t}) \simeq \mathcal{N}\left(\mathbf{w}|\mathbf{w}_{\text{MAP}}, \mathbf{H}_{\text{MAP}}^{-1}\right).$$

We have the Taylor series

$$\begin{aligned} y(\mathbf{x}_{N+1}, \mathbf{w}) =& y_{\text{MAP}\,N+1} + \mathbf{g}_{\text{MAP}\,N+1}{}^\intercal(\mathbf{w} - \mathbf{w}_{\text{MAP}}) \\ &+ O\left(\|\mathbf{w} - \mathbf{w}_{\text{MAP}}\|^2\right), \end{aligned}$$

where

$$\begin{aligned} y_{\text{MAP}\,N+1} &= y(\mathbf{x}_{N+1}, \mathbf{w}_{\text{MAP}}), \\ \mathbf{g}_{\text{MAP}\,N+1} &= \nabla_{\mathbf{w}} y(\mathbf{x}_{N+1}, \mathbf{w})\big|_{\mathbf{w}=\mathbf{w}_{\text{MAP}}}. \end{aligned}$$

Then, the logarithm of the integrand except the terms independent of $\mathbf{w}$ can be approximated as

$$
\begin{aligned}
&-\frac{\beta}{2}\left(t_{N+1} - y_{\mathrm{MAP}\,N+1} - \mathbf{g}_{\mathrm{MAP}\,N+1}{}^{\mathsf{T}}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}})\right)^2 \\
&-\frac{1}{2}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}})^{\mathsf{T}}\mathbf{H}_{\mathrm{MAP}}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}}) \\
=&-\frac{1}{2}\mathbf{v}^{\mathsf{T}}\mathbf{M}\mathbf{v},
\end{aligned}
$$

where

$$
\begin{aligned}
\mathbf{v} &= \begin{bmatrix} \mathbf{w} - \mathbf{w}_{\mathrm{MAP}} \\ t_{N+1} - y_{\mathrm{MAP}\,N+1} \end{bmatrix}, \\
\mathbf{M} &= \begin{bmatrix} \mathbf{H}_{\mathrm{MAP}} + \beta\mathbf{g}_{\mathrm{MAP}\,N+1}\mathbf{g}_{\mathrm{MAP}\,N+1}{}^{\mathsf{T}} & -\beta\mathbf{g}_{\mathrm{MAP}\,N+1} \\ -\beta\mathbf{g}_{\mathrm{MAP}\,N+1}{}^{\mathsf{T}} & \beta \end{bmatrix}.
\end{aligned}
$$

We have

$$
\mathbf{M}^{-1} = \begin{bmatrix} \mathbf{H}_{\mathrm{MAP}}^{-1} & \mathbf{H}_{\mathrm{MAP}}^{-1}\mathbf{g}_{\mathrm{MAP}\,N+1} \\ \mathbf{g}_{\mathrm{MAP}\,N+1}{}^{\mathsf{T}}\mathbf{H}_{\mathrm{MAP}}^{-1} & \beta^{-1} + \mathbf{g}_{\mathrm{MAP}\,N+1}{}^{\mathsf{T}}\mathbf{H}_{\mathrm{MAP}}^{-1}\mathbf{g}_{\mathrm{MAP}\,N+1} \end{bmatrix}.
$$

Therefore,

$$
p(t_{N+1}|\mathbf{t}) \simeq \mathcal{N}\left(t_{N+1}|y_{\mathrm{MAP}\,N+1}, \beta^{-1} + \mathbf{g}_{\mathrm{MAP}\,N+1}{}^{\mathsf{T}}\mathbf{H}_{\mathrm{MAP}}^{-1}\mathbf{g}_{\mathrm{MAP}\,N+1}\right).
$$

## 15.9

Let $\mathbf{t}_1, \cdots, \mathbf{t}_N$ be conditionally independent variables such that

$$p(\mathbf{t}_n|\mathbf{W}) = \prod_{k=1}^{K} y_{nk}^{t_{nk}},$$
$$p(\mathbf{w}_k) = \mathcal{N}\left(\mathbf{w}_k|\mathbf{0}, \alpha_k^{-1}\mathbf{I}\right).$$

where

$$y_{nk} = \frac{\exp(a_{nk})}{\sum_{k'=1}^{K}\exp(a_{nk'})},$$
$$a_{nk} = \mathbf{w}_k^{\intercal}\boldsymbol{\phi}_n.$$

By Bayes' theorem,

$$p(\mathbf{W}|\mathbf{T})p(\mathbf{T}) = p(\mathbf{T}|\mathbf{W})p(\mathbf{W}).$$

The logarithm of the right hand side except the terms independent of $\mathbf{W}$ can be written as

$$-\sum_{k=1}^{K} E_k(\mathbf{w}_k),$$

where

$$E_k(\mathbf{w}_k) = -\sum_{n=1}^{N} t_{nk}\ln y_{nk} + \frac{\alpha_k}{2}\|\mathbf{w}_k\|^2.$$

We have the Taylor series

$$E_k(\mathbf{w}_k) = E_k(\mathbf{w}_{k\mathrm{MAP}}) + \frac{1}{2}(\mathbf{w}_k - \mathbf{w}_{k\mathrm{MAP}})^{\intercal}\mathbf{H}_k(\mathbf{w}_k - \mathbf{w}_{k\mathrm{MAP}}) + O\left(\|\mathbf{w}_k - \mathbf{w}_{k\mathrm{MAP}}\|^3\right),$$

where $\mathbf{w}_{k\mathrm{MAP}}$ is a stationary point of $E_k$ and

$$\mathbf{H}_k = \nabla\nabla E_k(\mathbf{w}_{k\mathrm{MAP}}).$$

Then,

$$p(\mathbf{w}_k|\mathbf{T}) \simeq \mathcal{N}\left(\mathbf{w}_k|\mathbf{w}_{k\mathrm{MAP}}, \mathbf{H}_k^{-1}\right).$$

## 15.10

Let $\mathbf{t}_1, \cdots, \mathbf{t}_N$ be conditionally independent variables such that

$$p(\mathbf{t}_n|\mathbf{W}) = \prod_{k=1}^{K} y_{nk}^{t_{nk}},$$
$$p(\mathbf{w}_k) = \mathcal{N}\left(\mathbf{w}_k|\mathbf{0}, \alpha_k^{-1}\mathbf{I}\right).$$

where

$$y_{nk} = y_k(\mathbf{a}_n),$$
$$y_k(\mathbf{a}) = \frac{\exp(a_k)}{\sum_{k'=1}^{K}\exp(a_{k'})},$$
$$a_{nk} = \mathbf{w}_k^\intercal \boldsymbol{\phi}_n.$$

By Bayes' theorem,

$$p(\mathbf{T}) = \int p(\mathbf{T}|\mathbf{W})p(\mathbf{W})d\mathbf{W}.$$

The logarithm of the integrand of the right hand side except the terms independent of $\mathbf{W}$ can be written as $-E(\mathbf{W})$ where

$$E(\mathbf{W}) = \sum_{k=1}^{K} E_k(\mathbf{w}_k),$$
$$E_k(\mathbf{w}_k) = -\sum_{n=1}^{N} t_{nk}\ln y_{nk} + \frac{\alpha_k}{2}\|\mathbf{w}_k\|^2.$$

Then, $E(\mathbf{W})$ can be approximated as $\frac{1}{2}\operatorname{tr}\mathbf{M}$ where

$$\mathbf{M} = \begin{bmatrix}\mathbf{W}\\ \mathbf{T}\end{bmatrix}^\intercal \begin{bmatrix} & \end{bmatrix} \begin{bmatrix}\mathbf{W}\\ \mathbf{T}\end{bmatrix}.$$